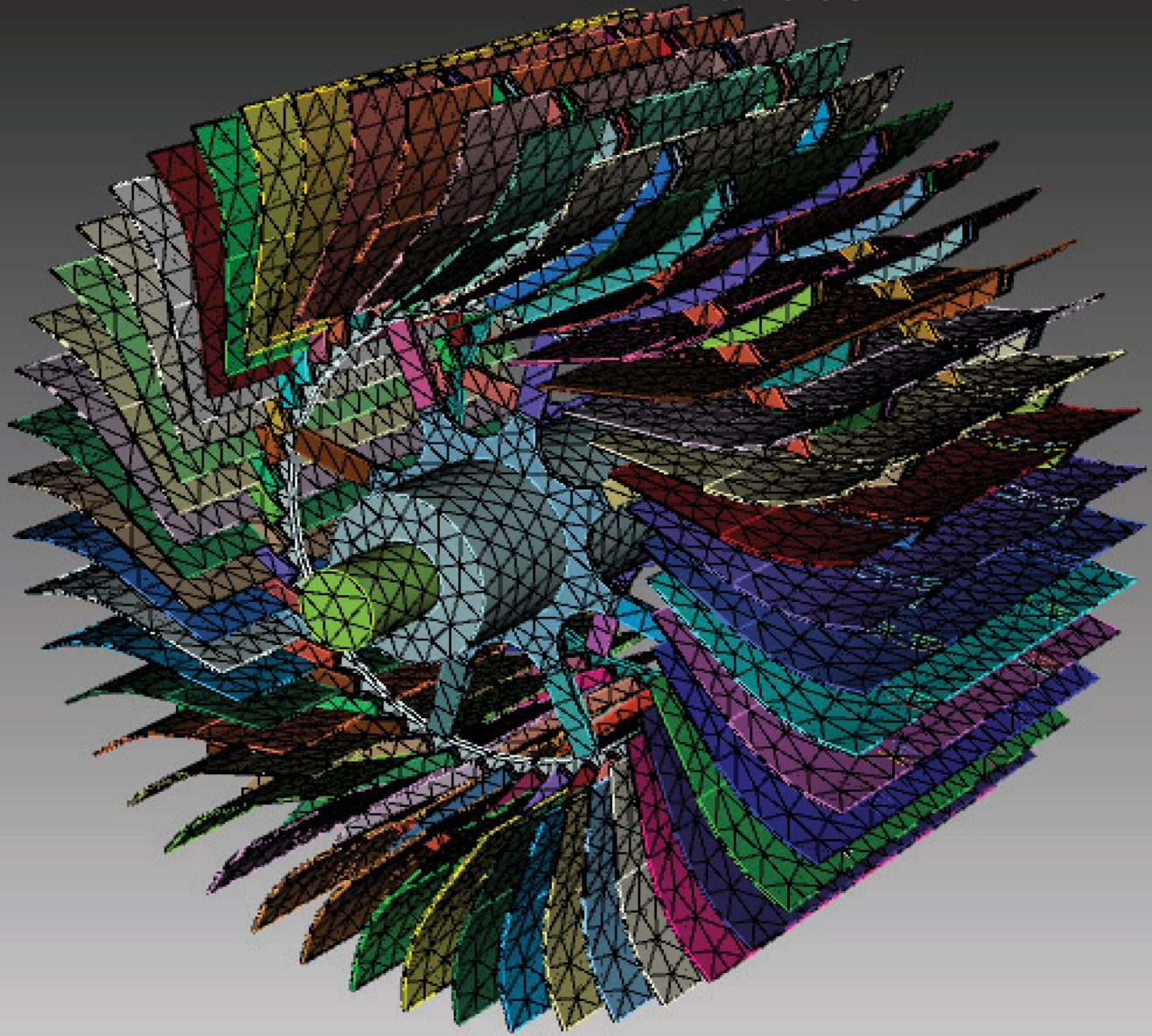


Numerical Methods

2nd Edition



3G E-LEARNING

NUMERICAL METHODS

2nd Edition



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NUMERICAL METHODS

2nd Edition



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PREFACE

In numerical analysis, a numerical method is a mathematical tool designed to solve numerical problems. Numerical methods provide approximations to the problems in question. No matter how accurate they are they do not, in most cases, provide the exact answer. In some instances working out the exact answer by a different approach may not be possible or may be too time consuming and it is in these cases where numerical methods are most often used. The implementation of a numerical method with an appropriate convergence check in a programming language is called a numerical algorithm.

The Second Edition of *Numerical Methods* is designed to explore the coverage of algorithms, polynomial approximation, numerical solution of differential equations, exponential approximation, testing of hypothesis, and other topics.

ORGANIZATION OF THE BOOK

This new edition provides an accessible and self-contained introduction, an overview for students who need to know the methods without having to perform the analysis.

As laid out in the Table of Contents, Second Edition of *Numerical Methods* presents the coverage in eight chapters.

Chapter 1 presents an illustration of iterative methods for simple roots, multiple roots, nonlinear equations, and polynomial equations.

Chapter 2 focuses on polynomial interpolation, a method of estimating values between known data points.

Chapter 3 covers the numerical differentiation and differentiation methods.

Chapter 4 starts with first-order ordinary differential equations and then higher order linear differential equations with constant coefficients are elaborated. Cauchy's and Legendre's linear equations are also described. Finally, simultaneous first order linear equations with constant coefficients are described.

Chapter 5 delivers the numerical solution of ordinary differential equations with Euler's method, Taylor's method, Runge-kutta methods, and predictor-corrector methods.

Chapter 6 explains the numerical solution of partial differential equations.

Chapter 7 covers the approximate analytical methods, such as least squares method, perturbation method, and WKB method

Chapter 8 presents the coverage of testing of hypothesis, a statistical test, used to determine whether there is enough evidence in a sample of data to infer that a certain condition is true for the entire population.

A PEDAGOGICAL FRAMEWORK FOR STUDYING AND LEARNING

The book contains numerous features to increase your learning experience:

Learning Objectives - These Highlight the learning you will gain the diligent study of the material in each chapter.

Key-terms – Important terms and concepts are highlighted with boldfaces and italic type in the text.

Examples: The book cabinets' examples to illustrate specific ideas in each chapter.

Case Study - This reveals what students need to create and provide an opportunity for the development of key skills such as communication, group working and problem solving.

End of Chapter Summaries - A brief version of main ideas, theories, and strategies are provided at the end of each chapter.

Review Questions - These questions test students understanding of the chapter material and gain deeper insight.

Knowledge Check - This is given to the students for progress check at the end of each chapter.

Role Model: A biography of someone who has/had acquired remarkable success in their respective field as Role Models are important because they give us the ability to imagine our future selves.

Chapter

1

TRANSCENDENTAL AND POLYNOMIAL EQUATIONS

LEARNING OBJECTIVES

After studying this chapter, you will be able to:

1. Define the Iterative Methods for Simple Roots
2. Explain the Iterative Methods for Multiple Roots
3. Understand the Iterative Methods for a System of Nonlinear Equations
4. Discuss about Complex Roots
5. Drive the Iterative Methods for Polynomial Equations

INTRODUCTION

We consider the methods for determining the roots of the equation

$$f(x) = 0 \quad (1)$$

which may be given explicitly as a polynomial of degree n in x or $f(x)$ may be defined implicitly as a transcendental function. A transcendental equation (1) may have no root, a finite or an infinite number of real and / or complex roots while a polynomial equation (1) has exactly n (real and / or complex) roots. If the function $f(x)$ changes sign in any one of the intervals $[x^* - \varepsilon, x^*]$, $[x^*, x^* + \varepsilon]$, then x^* defines an approximation to the root of $f(x)$ with accuracy ε . This is known as intermediate value theorem. Hence, if the interval $[a, b]$ containing x^* and ξ where ξ is the exact root of (1), is sufficiently small, then

$$|x^* - \xi| \leq b - a$$

can be used as a measure of the error.

There are two types of methods that can be used to find the roots of the above equation (1).

- Direct methods: These methods give the exact value of the roots (in the absence of round off errors) in a finite number of steps. These methods determine all the roots at the same time.
- Iterative methods: These methods are based on the idea of successive

approximations. Starting with one or more initial approximations to the root, we obtain a sequence of iterates $\{x_k\}$ which in the limit converges to the root. These methods determine one or two roots at a time.

Definition 1 A sequence of iterates $\{x_k\}$ is said to converge to the root ξ if

$$\lim_{k \rightarrow \infty} |x_k - \xi| = 0.$$

If $x_k, x_{k-1}, \dots, x_{k-m+1}$ are m approximates to a root, then we write an iteration method in the form

$$x_{k+1} = \phi(x_k, x_{k-1}, \dots, x_{k-m+1}) \quad (2)$$

where we have written the equation (1) in the equivalent form

$$x = \phi(x).$$

The function ϕ is called the iteration function. For $m = 1$, we get the one-point iteration method

$$x_{k+1} = \phi(x_k), \quad k = 0, 1, \dots \quad (3)$$

If $\phi(x)$ is continuous in the interval $[a, b]$ that contains the root and $|\phi'(x)| \leq c < 1$ in this interval, then for any choice of $x_0 \in [a, b]$, the sequence of iterates $\{x_k\}$ obtained from (3) converges to the root of $x = \phi(x)$ or $f(x) = 0$.



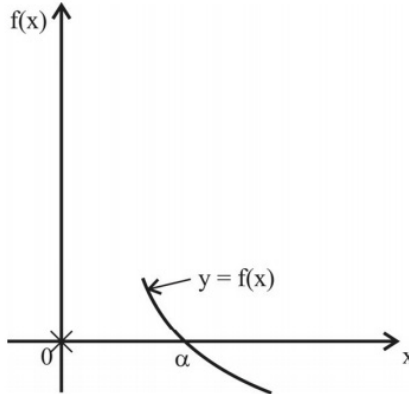


Figure 1.1: Geometrical Interpretation of a root of $f(x) = 0$.

Thus, for any iterative method of the form (2) or (3), we need the iteration function $\phi(x)$ and one or more initial approximations to the root. In practical applications, it is not always possible to find ξ exactly. We therefore attempt to obtain an approximate root x_{k+1} such that

$$|f(x_{k+1})| < \varepsilon \quad (4)$$

and / or

$$|x_{k+1} - x_k| < \varepsilon \quad (5)$$

where x_k and x_{k+1} are two consecutive iterates and ε is the prescribed error tolerance.

Definition 2 An iterative method is said to be of order p or has the rate of convergence p , if p is the largest positive real number for which

$$|\varepsilon_{k+1}| \leq c |\varepsilon_k|^p \quad (6)$$

where $\varepsilon_k = x_k - \xi$ is the error in the k th iterate.

The constant c is called the asymptotic error constant. It depends on various order derivatives of $f(x)$ evaluated at ξ and is independent of k . The relation

$$\varepsilon_{k+1} = c\varepsilon_k^p + O(\varepsilon_k^{p+1})$$

is called the error equation.

By substituting $x_i = \xi + \varepsilon_i$ for all i in any iteration method and simplifying we obtain the error equation for that method. The value of p thus obtained is called the order of this method.

1.1 ITERATIVE METHODS FOR SIMPLE ROOTS

➔ A root ξ is called a simple root of $f(x) = 0$, if $f(\xi) = 0$ and $f'(\xi) \neq 0$. Then, we can also write $f(x) = (x - \xi)g(x)$, where $g(x)$ is bounded and $g(\xi) \neq 0$.

LEARNING OBJECTIVE

1. Define the Iterative Methods for Simple Roots

1.1.1 Bisection Method

If the function $f(x)$ satisfies $f(a_0)f(b_0) < 0$, then the equation $f(x) = 0$ has at least one real root or an odd number of real roots in the interval (a_0, b_0) . If $m_1 = 1/2(a_0 + b_0)$ is the **midpoint** of this interval, then the root will lie either in the interval (a_0, m_1) or in the interval (m_1, b_0) provided that $f(m_1) \neq 0$. If $f(m_1) = 0$, then m_1 is the required root.

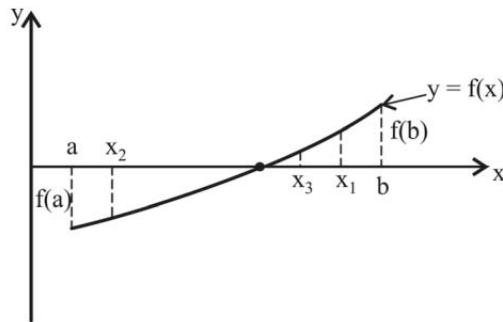


Figure 1.2: Bisection Method.

Repeating this procedure a number of times, we obtain the bisection method.

$$m_{k+1} = a_k + \frac{1}{2}(b_k - a_k), \quad k = 0, 1, \dots \quad (7)$$

Where

$$(a_{k+1}, b_{k+1}) = \begin{cases} (a_k, m_{k+1}), & \text{if } f(a_k)f(m_{k+1}) < 0, \\ (m_{k+1}, b_k), & \text{if } f(m_{k+1})f(b_k) < 0. \end{cases}$$

We take the midpoint of the last interval as an approximation to the root. This method always converges, if $f(x)$ is continuous in

KEY TERMS

Midpoint is the middle point of a line segment. It is equidistant from both endpoints, and it is the centroid both of the segment and of the endpoints.

the interval $[a, b]$ which contains the root. If an error tolerance ε is prescribed, then the approximate number of the iterations required may be determined from the relation

$$n \geq [\log(b_0 - a_0) - \log \varepsilon] / \log 2.$$

1.1.2 Secant Method

In this method, we approximate the graph of the function $y = f(x)$ in the neighborhood of the root by a straight line (secant) passing through the points (x_{k-1}, f_{k-1}) and (x_k, f_k) , where $f_k = f(x_k)$ and take the point of intersection of this line with the x -axis as the next iterate. We thus obtain

$$x_{k+1} = x_k - \frac{x_k - x_{k-1}}{f_k - f_{k-1}} f_k, \quad k = 1, 2, \dots$$

Or

$$x_{k+1} = \frac{x_{k-1} f_k - x_k f_{k-1}}{f_k - f_{k-1}}, \quad k = 1, 2, \dots \quad (8)$$

where x_{k-1} and x_k are two consecutive iterates. In this method, we need two initial approximations x_0 and x_1 . This method is also called the chord method. The order of the method (8) is obtained as

$$p = \frac{1}{2} (1 + \sqrt{5}) \approx 1.62.$$

If the approximations are chosen such that $f(x_{k-1}) f(x_k) < 0$ for each k , then the method is known as Regula-Falsi method and has linear (first order) rate of convergence. Both these methods require one function evaluation per iteration.

1.1.3 Newton-Raphson method

In this method, we approximate the graph of the function $y = f(x)$ in the neighborhood of the root by the tangent to the curve at the point (x_k, f_k) and take its point of intersection with the x -axis as the next iterate. We have the Newton-



REMEMBER

A polynomial of N th order should have N roots which could be real, or, complex conjugate pair if the coefficients of the polynomial are all real.

KEY TERMS

Function is a relation between a set of inputs and a set of permissible outputs with the property that each input is related to exactly one output.

Raphson method as

$$x_{k+1} = x_k - \frac{f_k}{f'_k}, \quad k = 0, 1, \dots \quad (9)$$

and its order is $p = 2$. This method requires one **function** evaluation and one first derivative evaluation per iteration.

In problems in cylindrical coordinates where $x = -1$ would correspond to the axis. On the other hand, for the solution of boundary value problems, the property of the set of collocation points held by the Gauss-Lobatto points to contain the boundaries is indispensable.

1.1.4 Chebyshev Method

In numerical linear algebra, the Chebyshev iteration is an iterative method for determining the solutions of a system of linear equations. The method is named after Russian mathematician Pafnuty Chebyshev. Chebyshev iteration avoids the computation of inner products as is necessary for the other nonstationary methods. For some distributed-memory architectures these inner products are a bottleneck with respect to efficiency. The price one pays for avoiding inner products is that the method requires enough knowledge about spectrum of the coefficient matrix A , that is an upper estimate for the upper eigenvalue and lower estimate for the lower eigenvalue. There are modifications of the method for non-symmetric matrices A .



REMEMBER

If the eigenvalues had been repeated, it would imply that the length of the two axes of the ellipse are the same.

Writing $f(x) = f(x_k + x - x_k)$ and approximating $f(x)$ by a second degree Taylor series expansion about the point x_k , we obtain the method

$$x_{k+1} = x_k - \frac{f_k}{f'_k} - \frac{1}{2}(x_{k+1} - x_k)^2 \frac{f''_k}{f'_k}$$

Replacing $x_{k+1} - x_k$ on the right hand side by $(-f_k/f'_k)$, we get the Chebyshev method



$$x_{k+1} = x_k - \frac{f_k}{f'_k} - \frac{1}{2} \left(\frac{f_k}{f'_k} \right)^2 \frac{f''_k}{f'_k}, \quad k = 0, 1, \dots \quad (10)$$

1.1.5 Multipoint Iteration Methods

It is possible to modify the Chebyshev method and obtain third order iterative methods which do not require the evaluation of the second order derivative. We give below two multipoint iteration methods.

$$x_{k+1}^* = x_k - \frac{1}{2} \frac{f_k}{f'_k}$$

$$x_{k+1} = x_k - \frac{f_k}{f'(x_{k+1}^*)}$$

$$\text{order } p = 3. \quad (11)$$

This method requires one function and two first derivative evaluations per iteration.

$$x_{k+1}^* = x_k - \frac{f_k}{f'_k}$$

$$x_{k+1} = x_{k+1}^* - \frac{f(x_{k+1}^*)}{f'_k}$$

$$\text{order } p = 3. \quad (12)$$

This method requires two functions and one first derivative evaluation per iteration.

1.1.6 Müller Method

Muller's method is a root-finding algorithm, a **numerical method** for solving equations of the form $f(x) = 0$. It was first presented by David E. Muller in 1956. Muller's method is based on the secant method, which constructs at every iteration a line through two points on the graph of f . Instead, Muller's method uses three points, constructs the parabola through these three

KEY TERMS

Numerical method is a mathematical tool designed to solve numerical problems.

points, and takes the intersection of the x -axis with the parabola to be the next approximation.

This method is a **generalization** of the secant method. In this method, we approximate the graph of the function $y = f(x)$ in the neighborhood of the root by a second degree curve and take one of its points of intersection with the x axis as the next approximation.

KEY TERMS

Generalization is the formulation of general concepts from specific instances by abstracting common properties.

We have the method as

$$x_{k+1} = x_k + (x_k - x_{k-1}) \lambda_{k+1}, \quad k = 2, 3, \dots \quad (13)$$

Where

$$\begin{aligned} h_k &= x_k - x_{k-1}, \quad h_{k-1} = x_{k-1} - x_{k-2}, \\ \lambda_k &= h_k / h_{k-1}, \quad \delta_k = 1 + \lambda_k, \\ g_k &= \lambda_k^2 f(x_{k-2}) - \delta_k^2 f(x_{k-1}) + (\lambda_k + \delta_k) f(x_k), \\ c_k &= \lambda_k (\lambda_k f(x_{k-2}) - \delta_k f(x_{k-1}) + f(x_k)), \\ \lambda_{k+1} &= - \frac{2 \delta_k f(x_k)}{g_k \pm \sqrt{g_k^2 - 4 \delta_k c_k f(x_k)}}. \end{aligned}$$

The sign in the denominator is chosen so that λ_{k+1} has the smallest absolute value, i.e., the sign of the square root in the denominator is that of g_k .

Alternative

We have the method as

$$x_{k+1} = x_k - \frac{2a_2}{a_1 \pm \sqrt{a_1^2 - 4a_0a_2}}, \quad k = 2, 3, \dots \quad (14)$$

Where

$$\begin{aligned} a_2 &= f_k, \quad h_1 = x_k - x_{k-2}, \quad h_2 = x_k - x_{k-1}, \quad h_3 = x_{k-1} - x_{k-2}, \\ a_1 &= \frac{1}{D} [h_1^2 (f_k - f_{k-1}) - h_2^2 (f_k - f_{k-2})], \\ a_0 &= \frac{1}{D} [h_1 (f_k - f_{k-1}) - h_2 (f_k - f_{k-2})], \\ D &= h_1 h_2 h_3. \end{aligned}$$



The sign in the denominator is chosen so that λ_{k+1} has the smallest absolute value, i.e., the sign of the square root in the denominator is that of a_1 . This method requires three initial approximations to the root and one function evaluation per iteration. The order of the method is $p = 1.84$.

1.1.7 Derivative Free Methods

In many practical applications, only the data regarding the function $f(x)$ is available. In these cases, methods which do not require the evaluation of the derivatives can be applied. We give below two such methods.

$$x_{k+1} = x_k - \frac{f_k}{g_k}, \quad k = 0, 1, \dots$$

$$g_k = \frac{f(x_k + f_k) - f_k}{f_k},$$

$$\text{order } p = 2. \quad (15)$$

This method requires two function evaluations per iteration.

$$x_{k+1} = x_k - w_1(x_k) - w_2(x_k), \quad k = 0, 1, \dots$$

$$w_1(x_k) = \frac{f_k}{g_k}$$

$$w_2(x_k) = \frac{f(x_k - w_1(x_k))}{g_k}$$

$$g_k = \frac{f(x_k + \beta f_k) - f_k}{\beta f_k} \quad (16)$$

where $\beta \neq 0$ is arbitrary and order $p = 3$.

This method requires three function evaluations per iteration.

1.1.8 Aitken Δ^2 -process

If x_{k+1} and x_{k+2} are two approximations obtained from a general linear iteration method

$$x_{k+1} = \phi(x_k), \quad k = 0, 1, \dots$$

then, the error in two successive approximations is given by

$$\epsilon_{k+1} = a_1 \epsilon_k$$

$$\epsilon_{k+2} = a_1 \epsilon_{k+1}, \quad a_1 = \phi'(\xi).$$

Eliminating a_1 from the above equations, we get

$$\varepsilon_{k+1}^2 = \varepsilon_k \varepsilon_{k+2}$$

Using $\varepsilon_k = \xi - x_k$, we obtain

$$\begin{aligned}\xi \approx x_k^* &= x_k - \frac{(x_{k+1} - x_k)^2}{x_{k+2} - 2x_{k+1} + x_k} \\ &= x_k - \frac{(\Delta x_k)^2}{\Delta^2 x_k}\end{aligned}\quad (17)$$

which has second order convergence.

1.1.9 A Sixth Order Method

A one-parameter family of sixth order methods for finding simple zeros of $f(x)$, which require three evaluations of $f(x)$ and one evaluation of the derivative $f'(x)$ are given by

$$\begin{aligned}w_n &= x_n - \frac{f(x_n)}{f'(x_n)} \\ z_n &= w_n - \frac{f(w_n)}{f'(x_n)} \left[\frac{f(x_n) + A f(w_n)}{f(x_n) + (A - 2) f(w_n)} \right] \\ x_{n+1} &= z_n - \frac{f(z_n)}{f'(x_n)} \left[\frac{f(x_n) - f(w_n) + D f(z_n)}{f(x_n) - 3 f(w_n) + D f(z_n)} \right] \quad n = 0, 1, \dots\end{aligned}$$

with error term

$$\varepsilon_{n+1} = \frac{1}{144} [2F_3^2 F_2 - 3(2A + 1)F_2^3 F_3] \varepsilon_n^6 + \dots$$

where $F^{(i)} = f^{(i)}(\xi) / f'(\xi)$.

The order of the methods does not depend on D and the error term is simplified when $A = -1/2$. The simplified formula for $D = 0$ and $A = -1/2$ is

$$\begin{aligned}
 w_n &= x_n - \frac{f(x_n)}{f'(x_n)} \\
 z_n &= w_n - \frac{f(w_n)}{f'(x_n)} \left[\frac{2f(x_n) - f(w_n)}{2f(x_n) - 5f(w_n)} \right] \\
 x_{n+1} &= z_n - \frac{f(z_n)}{f'(x_n)} \left[\frac{f(x_n) - f(w_n)}{f(x_n) - 3f(w_n)} \right], n = 0, 1, \dots
 \end{aligned}$$

1.2 ITERATIVE METHODS FOR MULTIPLE ROOTS

 2. If the root ξ of (1) is a repeated root, then we may write (1) as

$$f(x) = (x - \xi)^m g(x) = 0$$

where $g(x)$ is bounded and $g(\xi) \neq 0$. The root ξ is called a multiple root of multiplicity m . We obtain from the above equation

$$f(\xi) = f'(\xi) = \dots = f^{(m-1)}(\xi) = 0, f^{(m)}(\xi) \neq 0.$$

The methods listed in above section do not retain their order while determining a multiple root and the order is reduced at least by one. If the **multiplicity** m of the root is known in advance, then some of these methods can be modified so that they have the same rate of convergence as that for determining simple roots. We list some of the modified methods.

Newton-Raphson method

$$x_{k+1} = x_k - m \frac{f_k}{f'_k}, \quad k = 0, 1, \dots$$

order $p = 2$.

(18)

LEARNING OBJECTIVE

 2. Describe the basic home electrical system

KEY TERMS

Multiplicity of a member of a multiset is the number of times it appears in the multiset.

Chebyshev method

$$x_{k+1} = x_k - \frac{m(3-m)}{2} \frac{f_k}{f_k'} - \frac{m^2}{2} \left[\frac{f_k}{f_k'} \right]^2 \frac{f_k''}{f_k'}, \quad k = 0, 1, \dots$$

$$\text{order } p = 3. \quad (19)$$

Alternatively, we apply the methods given in above section to the equation

$$G(x) = 0 \quad (20)$$

Where

$$G(x) = \frac{f(x)}{f'(x)}$$

has a simple root ξ regardless of the multiplicity of the root of $f(x) = 0$. Thus, the Newton Raphson method (9), when applied to (20) becomes

$$x_{k+1} = x_k - \frac{G(x_k)}{G'(x_k)}$$

Or

$$x_{k+1} = x_k - \frac{f(x_k) f'(x_k)}{[f'(x_k)]^2 - f(x_k) f''(x_k)}, \quad k = 0, 1, 2, \dots \quad (21)$$

The secant method for (20) can be written as

$$x_{k+1} = \frac{x_{k-1} f_k f_{k-1}' - x_k f_{k-1} f_k'}{f_k f_{k-1}' - f_{k-1} f_k'}$$

Derivative free method

$$x_{k+1} = x_k - W_1(x_k) - W_2(x_k) \quad (22)$$

$$\begin{aligned}
 W_1(x_k) &= \frac{F(x_k)}{g(x_k)} \\
 W_2(x_k) &= \frac{F(x_k - W_1(x_k))}{g(x_k)} \\
 g(x_k) &= \frac{F(x_k + \beta F(x_k)) - F(x_k)}{\beta F(x_k)}
 \end{aligned}$$

Where

$$F(x) = -\frac{f^2(x)}{f(x - f(x)) - f(x)}$$

and $\beta \neq 0$ is an arbitrary constant. The method requires six function evaluations per iteration and has order 3.

1.3 ITERATIVE METHODS FOR A SYSTEM OF NONLINEAR EQUATIONS



Let the given system of equations be

$$\begin{aligned}
 f_1(x_1, x_2, \dots, x_n) &= 0, \\
 f_2(x_1, x_2, \dots, x_n) &= 0, \\
 &\dots\dots\dots \\
 f_n(x_1, x_2, \dots, x_n) &= 0.
 \end{aligned} \tag{23}$$

Starting with the initial approximations $\mathbf{x}^{(0)} = (x_1^{(0)}, x_2^{(0)}, \dots, x_n^{(0)})$, we obtain the sequence of iterates, using the Newton-Raphson method as

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \mathbf{J}^{-1} \mathbf{f}^{(k)}, \quad k = 0, 1, \dots \tag{24}$$

Where

$$\begin{aligned}
 \mathbf{x}^{(k)} &= (x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)})^T \\
 \mathbf{f}^{(k)} &= (f_1^{(k)}, f_2^{(k)}, \dots, f_n^{(k)})^T \\
 f_i^{(k)} &= f_i(x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)})
 \end{aligned}$$

LEARNING OBJECTIVE

3. Understand the Iterative Methods for a System of Nonlinear Equations

**REMEMBER**

The coefficients of an n th-order $p(x)$ should be ordered in descending powers of x into a row matrix of order $n + 1$.

and J is the Jacobian matrix of the functions $f_1, f_2, \dots, f_n$ evaluated at $(x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)})$. The method has second order rate of convergence.

Alternatively, we may write (24) in the form

$$\mathbf{J}(\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}) = -\mathbf{f}^{(k)}$$

and may solve it as a linear system of equations. Very often, for systems which arise while solving ordinary and partial differential equations, J is of some special form like a tridiagonal, five diagonal or a banded matrix.

1.4 COMPLEX ROOTS



The Fundamental Theorem of Algebra states that every polynomial of degree one or greater has at least one root in the complex number system (keep in mind that a complex number can be real if the imaginary part of the complex root is zero). A further theorem, in some cases referred to as the Linear Factorization Theorem, states that a polynomial of degree n has exactly n linear factors, and each can be written in the form $(x - c)$, where c is a root. These n complex roots (possibly including some real roots) are counted with multiplicity. This much was explained in Roots of Higher Degree Polynomials.

In Roots of Higher Degree Polynomials, we discussed how a polynomial can be resolved into linear factors irreducible over the reals. The factors that are first-degree polynomials are real roots of the original polynomial. The factors that are second-degree polynomials cannot be reduced using real numbers. With complex numbers, however, we can solve those quadratic equations which are irreducible over the reals, and we can then find each of the n roots of a polynomial of degree n .

We write the given equation

$$f(z) = 0, \quad z = x + iy$$

in the form $u(x, y) + iv(x, y) = 0$,

where $u(x, y)$ and $v(x, y)$ are the real and imaginary parts of $f(z)$ respectively. The problem of finding a complex root

LEARNING OBJECTIVE

4. Discuss about Complex Roots

of $f(z) = 0$ is equivalent to finding a solution (x, y) of the system of two equations

$$\begin{aligned} u(x, y) &= 0, \\ v(x, y) &= 0. \end{aligned}$$

Starting with $(x^{(0)}, y^{(0)})$, we obtain a sequence of iterates $\{x^{(k)}, y^{(k)}\}$ using the Newton Raphson method as

$$\begin{pmatrix} x^{(k+1)} \\ y^{(k+1)} \end{pmatrix} = \begin{pmatrix} x^{(k)} \\ y^{(k)} \end{pmatrix} - \mathbf{J}^{-1} \begin{pmatrix} u(x^{(k)}, y^{(k)}) \\ v(x^{(k)}, y^{(k)}) \end{pmatrix}, \quad k = 0, 1, \dots \quad (25)$$

Where

$$\mathbf{J} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}_{(x^{(k)}, y^{(k)})}$$

is the Jacobian matrix of $u(x, y)$ and $v(x, y)$ evaluated at $(x^{(k)}, y^{(k)})$.

Alternatively, we can apply directly the Newton-Raphson method (9) to solve $f(z) = 0$ in the form

$$z_{k+1} = z_k - \frac{f(z_k)}{f'(z_k)}, \quad k = 0, 1, \dots, \quad (26)$$

and use complex arithmetic. The initial approximation z_0 must also be complex. The secant method can also be applied using complex arithmetic.

After one root z_1 is obtained, Newton's method should be applied on the deflated polynomial

$$f^*(z) = \frac{f(z)}{z - z_1}.$$

This procedure can be repeated after finding every root. If k roots are already obtained, then the new iteration can be applied on the function

$$f^*(z) = \frac{f(z)}{(z - z_1)(z - z_2) \dots (z - z_k)}.$$

The new iteration is

$$z_{k+1} = z_k - \frac{f^*(z_k)}{f^{*'}(z_k)}.$$

The computation of $f^*(z_k) / f^{*'}(z_k)$ can be easily performed as follows

$$\begin{aligned} \frac{f^{*'}}{f^*} &= \frac{d}{dz} (\log f^*) = \frac{d}{dz} [\log f(z) - \log (z - z_1)] \\ &= \frac{f'}{f} - \frac{1}{z - z_1}. \end{aligned}$$



REMEMBER

Thus, breakers should not be used in place of switches unless they are specially listed for the purpose.

Hence, computations are carried out with

$$\frac{f^{*'}(z_k)}{f^*(z_k)} = \frac{f'(z_k)}{f(z_k)} - \frac{1}{z_k - z_1}$$

Further, the following precautions may also be taken:

- Any zero obtained by using the deflated polynomial should be refined by applying Newton's method to the original polynomial with this zero as the starting approximation.
- The zeros should be computed in the increasing order of magnitude.



We will find the roots of the polynomial $x^5 - x^4 + x^3 - x^2 - 12x + 12$.

Complex Roots

$$x^5 - x^4 + x^3 - x^2 - 12x + 12 = 0$$

$$(x - 1)(x^4 + x^2 - 12) = 0$$

$$(x - 1)(x^2 - 3)(x^2 + 4) = 0$$

$$(x - 1)(x + \sqrt{3})(x - \sqrt{3})(x^2 + 4) = 0$$

$$(x - 1)(x + \sqrt{3})(x - \sqrt{3})(x + 2i)(x - 2i) = 0$$

The fifth-degree polynomial does indeed have five roots; three real, and two complex.



1.5 ITERATIVE METHODS FOR POLYNOMIAL EQUATIONS



The methods discussed in the above sections can be directly applied to obtain the roots of a polynomial of degree n

$$P_n(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_{n-1} x + a_n = 0 \quad (27)$$

where $a_0, a_1, \dots, a_n$ are real numbers. Most often, we are interested to determine all the roots (real or complex, simple or multiple) of the polynomial and we need to know.

- the exact number of real and complex roots along with their multiplicities.
- the interval in which each real roots lies.

We can obtain this information using Sturm sequences.

Let $f(x)$ be the given polynomial of degree n and let $f_1(x)$ denote its first order derivative. Denote by $f_2(x)$ the remainder of $f(x)$ divided by $f_1(x)$ taken with reverse sign and by $f_3(x)$ the remainder of $f_1(x)$ divided by $f_2(x)$ with the reverse sign and so on until a constant remainder is obtained. The sequence of the functions $f(x), f_1(x), f_2(x), \dots, f_n(x)$ is called the Sturm sequence. The number of real roots of the equation $f(x) = 0$ in (a, b) equals the difference between the number of sign changes in the Sturm sequence at $x = a$ and $x = b$ provided $f(a) \neq 0$ and $f(b) \neq 0$.

We note that if any function in the Sturm sequence becomes 0 for some value of x , we give to it the sign of the immediate preceding term. If $f(x) = 0$ has a multiple root, we obtain the Sturm sequence $f(x), f_1(x), \dots, f_r(x)$ where $f_{r-1}(x)$ is exactly divisible by $f_r(x)$. In this case, $f_r(x)$ will not be a constant. Since $f_r(x)$ gives the greatest common divisor of $f(x)$ and $f'(x)$, the multiplicity of the root of $f(x) = 0$ is one more than that of the root of $f_r(x) = 0$. We obtain a new Sturm sequence by dividing all the functions $f(x), f_1(x), \dots, f_r(x)$ by $f_r(x)$. Using this sequence, we determine the number of real roots of the equation $f(x) = 0$ in the same way, without taking their multiplicity into account. While obtaining the Sturm sequence, any positive constant common factor in any Sturm function $f_i(x)$ can be neglected. Since a polynomial of degree n has

LEARNING OBJECTIVE



5. Drive the Iterative Methods for Polynomial Equations

exactly n roots, the number of complex roots equals $(n - \text{number of real roots})$, where a real root of multiplicity m is counted m times.

If $x = \xi$ is a real root of $P_n(x) = 0$ then $x - \xi$ must divide $P_n(x)$ exactly. Also, if $x = \alpha + i\beta$ is a complex root of $P_n(x) = 0$, then its complex conjugate $\alpha - i\beta$ is also a root. Hence

$$\begin{aligned} \{x - (\alpha + i\beta)\} \{x - (\alpha - i\beta)\} &= (x - \alpha)^2 + \beta^2 \\ &= x^2 - 2\alpha x + \alpha^2 + \beta^2 \\ &= x^2 + px + q \end{aligned}$$

for some real p and q must divide $P_n(x)$ exactly.

The quadratic factor $x^2 + px + q = 0$ may have a pair of real roots or a pair of complex roots. Hence, the iterative methods for finding the real and complex roots of $P_n(x) = 0$ are based on the philosophy of extracting linear and quadratic factors of $P_n(x)$. We assume that the polynomial $P_n(x)$ is complete, that is, it has $(n + 1)$ terms. If some term is not present, we introduce it at the proper place with zero coefficient.

1.5.1 Birge-Vieta Method

In this method, we seek to determine a real number p such that $x - p$ is a factor of $P_n(x)$. Starting with p_0 , we obtain a sequence of iterates $\{p_k\}$ from

$$p_{k+1} = p_k - \frac{P_n(p_k)}{P_n'(p_k)}, \quad k = 0, 1, \dots \quad (28)$$

Or

$$p_{k+1} = p_k - \frac{b_n}{c_{n-1}}, \quad k = 0, 1, \dots \quad (29)$$

which is same as the Newton-Raphson method.

The values of b_n and c_{n-1} are obtained from the recurrence relations

$$\begin{aligned} b_i &= a_i + p_k b_{i-1}, \quad i = 0, 1, \dots, n \\ c_i &= b_i + p_k c_{i-1}, \quad i = 0, 1, \dots, n-1 \end{aligned}$$

with

$$c_0 = b_0 = a_0, \quad b_{-1} = 0 = c_{-1}.$$

We can also obtain b_i 's and c_i 's by using synthetic division method as given below:

p_k	a_0	a_1	a_2	$\dots$	a_{n-1}	a_n
		$p_k b_0$	$p_k b_1$	$\dots$	$p_k b_{n-2}$	$p_k b_{n-1}$
	b_0	b_1	b_2	$\dots$	b_{n-1}	b_n
		$p_k c_0$	$p_k c_1$	$\dots$	$p_k c_{n-2}$	
	c_0	c_1	c_2	$\dots$	c_{n-1}	

where $b_0 = a_0$ and $c_0 = b_0 = a_0$.

We have

$$\lim_{k \rightarrow \infty} b_n = 0 \quad \text{and} \quad \lim_{k \rightarrow \infty} p_k = p.$$

The order of this method is 2.

When p has been determined to the desired accuracy, we extract the next linear factor from the deflated polynomial

$$Q_{n-1}(x) = \frac{P_n(x)}{x - p} = b_0 x^{n-1} + b_1 x^{n-2} + \dots + b_{n-1}$$

which can also be obtained from the first part of the synthetic division.

Synthetic division procedure for obtaining b_n is same as Horner's method for evaluating the polynomial $P_n(p_k)$, which is the most efficient way of evaluating a polynomial. We can extract a multiple root of multiplicity m , using the Newton-Raphson method

$$p_{k+1} = p_k - m \frac{b_n}{c_{n-1}}, \quad k = 0, 1, 2, \dots$$

In this case, care should be taken while finding the deflated polynomial. For example, if $m = 2$, then as $k \rightarrow \infty$, $f(x) \approx b_n \rightarrow 0$ and $f'(x) \approx c_{n-1} \rightarrow 0$. Hence, the deflated polynomial is given by

$$c_0 x^{n-2} + c_1 x^{n-3} + \dots + c_{n-2} = 0.$$

1.5.2 Bairstow Method

In numerical analysis, Bairstow's method is an efficient algorithm for finding the roots of a real polynomial of arbitrary degree. A well-known and widely-used process for determining the roots of a given polynomial with real coefficients. The method determines a second-degree divisor of the given polynomial iteratively, and hence by using the formula for the roots of second-degree polynomials one can calculate an approximation of two roots of the given polynomial. An advantage of the method is that it uses real arithmetic only.

This method is used to find two real numbers p and q such that $x^2 + px + q$ is a factor of $P_n(x)$. Starting with p_0, q_0 , we obtain a sequence of iterates $\{(p_k, q_k)\}$ from

$$\begin{aligned} p_{k+1} &= p_k + \Delta p_k, \\ q_{k+1} &= q_k + \Delta q_k, \quad k = 0, 1, \dots \end{aligned} \quad (30)$$

Where

$$\begin{aligned} \Delta p_k &= -\frac{b_n c_{n-3} - b_{n-1} c_{n-2}}{c_{n-2}^2 - c_{n-3} (c_{n-1} - b_{n-1})} \\ \Delta q_k &= -\frac{b_{n-1} (c_{n-1} - b_{n-1}) - b_n c_{n-2}}{c_{n-2}^2 - c_{n-3} (c_{n-1} - b_{n-1})} \end{aligned}$$

The values of b_i 's and c_i 's are obtained from the recurrence relations

$$\begin{aligned} b_i &= a_i - p_k b_{i-1} - q_k b_{i-2}, \quad i = 1, 2, \dots, n, \\ c_i &= b_i - p_k c_{i-1} - q_k c_{i-2}, \quad i = 1, 2, \dots, n-1, \end{aligned}$$

With

$$c_0 = b_0 = a_0, \quad c_{-1} = b_{-1} = 0.$$

We can also obtain the values of b_i 's and c_i 's using the synthetic division method as given below:

$-p_k$	a_0	a_1	a_2	$\dots$	a_{n-1}	a_n
		$-p_k b_0$	$-p_k b_1$	$\dots$	$-p_k b_{n-2}$	$-p_k b_{n-1}$
$-q_k$			$-q_k b_0$	$\dots$	$-q_k b_{n-3}$	$-q_k b_{n-2}$
	b_0	b_1	b_2	$\dots$	b_{n-1}	b_n
		$-p_k c_0$	$-p_k c_1$	$\dots$	$-p_k c_{n-2}$	
			$-q_k c_0$	$\dots$	$-q_k c_{n-3}$	
	c_0	c_1	c_2	$\dots$	c_{n-1}	

where $b_0 = a_0$ and $c_0 = b_0 = a_0$.

We have

$$\lim_{k \rightarrow \infty} b_n = 0, \quad \lim_{k \rightarrow \infty} b_{n-1} = 0,$$

$$\lim_{k \rightarrow \infty} p_k = p, \quad \lim_{k \rightarrow \infty} q_k = q.$$

The order of this method is 2.

When p and q have been obtained to the desired accuracy we obtain the next quadratic factor from the deflated polynomial

$$Q_{n-2}(x) = b_0 x^{n-2} + b_1 x^{n-3} + \dots + b_{n-3} x + b_{n-2}$$

which can be obtained from the first part of the above synthetic division method.

1.5.3 Laguerre Method

In numerical analysis, Laguerre's method is a root-finding algorithm tailored to polynomials. In other words, Laguerre's method can be used to numerically solve the equation $p(x) = 0$ for a given polynomial $p(x)$. One of the most useful properties of this method is that it is, from extensive empirical study, very close to being a "sure-fire" method, meaning that it is almost guaranteed to always converge to *some* root of the polynomial, no matter what initial guess is chosen. This method is named in honor of Edmond Laguerre, a French mathematician.

Define

$$A = -P'_n(x_k) / P_n(x_k),$$

$$B = A^2 - P''_n(x_k) / P_n(x_k).$$

Then, the method is given by

$$x_{k+1} = x_k + \frac{n}{A \pm \sqrt{(n-1)(nB - A^2)}} \quad (31)$$

The values $P_n(x_k)$, $P'_n(x_k)$ and $P''_n(x_k)$ can be obtained using the synthetic division method. The sign in the denominator on the right hand side of (31) is taken as the sign of A to make the denominator largest in magnitude. The order of the method is 2.

1.5.4 Graeffe's Root Squaring Method

In mathematics, Graeffe's method or Dandelin–Lobachesky–Graeffe method is an algorithm for finding all of the roots of a polynomial. It was developed independently by Germinal Pierre Dandelin in 1826 and Lobachevsky in 1834. In 1837 Karl Heinrich Gräffe also discovered the principal idea of the method. The method separates the roots of a polynomial by squaring them repeatedly. This squaring of the roots is done implicitly, that is, only working on the coefficients of the polynomial. Finally, Viète's formulas are used in order to approximate the roots.

Graeffe's method is one of the root finding method of a polynomial with real coefficients. This method gives all the roots approximated in each iteration also this is one of the direct root finding method. Because this method does not require any initial guesses for roots. It was invented independently by Graeffe Dandelin and Lobachevsky. Which was the most popular method for finding roots of polynomials in the 19th and 20th centuries. This is a direct method and is used to find all the roots of a polynomial with real coefficients. The roots may be real and distinct, real and equal or complex. We separate the roots of the equation (27) by forming another equation, whose roots are very high powers of the roots of (27) with the help of root squaring process.

Let $\xi_1, \xi_2, \dots, \xi_n$ be the roots of (27). Separating the even and odd powers of x in (27) and squaring we get

$$(a_0 x^n + a_2 x^{n-2} + \dots)^2 = (a_1 x^{n-1} + a_3 x^{n-3} + \dots)^2.$$

Simplifying, we obtain

$$a_0^2 x^{2n} - (a_1^2 - 2a_0 a_2) x^{2n-2} + \dots + (-1)^n a_n^2 = 0.$$

Substituting $z = -x^2$, we get

$$b_0 z^n + b_1 z^{n-1} + \dots + b_{n-1} z + b_n = 0 \quad (32)$$

which has roots $-\xi_1^2, -\xi_2^2, \dots, -\xi_n^2$. The coefficients b_k 's are obtained from:

a_0	a_1	a_2	a_3	...	a_n
a_0^2	a_1^2	a_2^2	a_3^2	...	a_n^2
	$-2a_0a_2$	$-2a_1a_3$	$-2a_2a_4$	...	
		$+2a_0a_4$	$+2a_1a_5$	...	
			$\vdots$		
b_0	b_1	b_2	b_3	...	b_n

The $(k+1)$ th column in the above table is obtained as explained below:

The terms in each column alternate in sign starting with a positive sign. The first term is square of the $(k+1)$ th coefficient a_k . The second term is twice the product of the nearest neighboring pair a_{k-1} and a_{k+1} . The next term is twice the product of the next neighboring pair a_{k-2} and a_{k+2} . This procedure is continued until there are no available coefficients to form the cross products.

After repeating this procedure m times we obtain the equation

$$B_0 x^n + B_1 x^{n-1} + \dots + B_{n-1} x + B_n = 0 \quad (33)$$

whose roots are $R_1, R_2, \dots, R_n$, where

$$R_i = -\xi_i^{2m}, \quad i = 1, 2, \dots, n.$$

If we assume

$$|\xi_1| > |\xi_2| > \dots > |\xi_n|,$$

Then

$$|R_1| \gg |R_2| \gg \dots \gg |R_n|.$$

We obtain from (33)

$$|R_i| \approx \frac{|B_i|}{|B_{i-1}|} = |\xi_i|^{2m}$$

Or

$$\log (|\xi_i|) = 2^{-m} [\log |B_i| - \log |B_{i-1}|].$$

This determines the magnitude of the roots and substitution in the original equation (27) will give the sign of the roots.

We stop the squaring process when another squaring process produces new coefficients that are almost the squares of the corresponding coefficients B_k 's, i.e., when the cross product terms become negligible in comparison to square terms.

After few squaring, if the magnitude of the coefficient B_k is half the square of the magnitude of the corresponding coefficient in the above equation, then it indicates that ξ_k is a double root. We can find this double root by using the following procedure. We have

$$R_k \simeq -\frac{B_k}{B_{k-1}} \quad \text{and} \quad R_{k+1} \simeq -\frac{B_{k+1}}{B_k}$$

$$R_k R_{k+1} \simeq R_k^2 \simeq \left| \frac{B_{k+1}}{B_{k-1}} \right|$$

Or

$$|R_k|^2 = |\xi_k|^{2(2^m)} = \left| \frac{B_{k+1}}{B_{k-1}} \right|.$$

This gives the magnitude of the double root. Substituting in the given equation, we can find its sign. This double root can also be found directly since R_k and R_{k+1} converge to the same root after sufficient squaring. Usually, this convergence to the double root is slow. By making use of the above observation, we can save a number of squaring.

If ξ_k and ξ_{k+1} form a complex pair, then this would cause the coefficients of x^{n-k} in the successive squarings to fluctuate both in magnitude and sign. If $\xi_k, \xi_{k+1} = \beta_k \exp (\pm i\phi_k)$ is the complex pair, then the coefficients would fluctuate in magnitude and sign by an amount $2\beta_k^m \cos (m\phi_k)$. A complex pair can be spotted by such an oscillation. For m sufficiently large, $2\beta_k$ can be determined from the relation

$$\beta_k^{2(2^m)} \simeq \left| \frac{B_{k+1}}{B_{k-1}} \right|$$

and ϕ is suitably determined from the relation

$$2\beta_k^m \cos (m\phi_k) \simeq \frac{B_{k+1}}{B_{k-1}}.$$

If the equation has only one complex pair, then we can first determine all the real roots. The complex pair can be written as $\xi_k, \xi_{k+1} = p \pm iq$. The sum of the roots then gives

$$\xi_1 + \xi_2 + \dots + \xi_{k-1} + 2p + \xi_{k+2} + \dots + \xi_n = -a_1.$$

This determines p . We also have $|\beta_k|^2 = p^2 + q^2$. Since $|\beta_k|$ is already determined, this equation gives q .

Attributes of n^{th} order polynomial

- There will be n roots. Sometimes all the roots may be real, all the roots may be complex and sometimes roots may be combination of real and complex values.
 - All the roots are real - ($x^2 - 5x + 6 = 0$)



- All the roots are complex - $(5x^2 + x + 3 = 0)$
- Combination of real and complex roots - $(5x^3 - 4x^2 - x - 3 = 0)$
- There must be at least one real root, if n is odd. Because complex roots are occur in pairs.
- Discartes' rule of sign will be true for any n th order polynomial.

Discartes' rule of sign

If we get n th order polynomial as $f(x) = 0$, Discartes' rule of sign says that maximum number of positive roots of the polynomial $f(x) = 0$, is equal to the number of sign changes of the polynomial $f(x)$. Also maximum number of negative roots of the polynomial $f(x)$, is equal to the number of sign changes of the polynomial $f(-x)$.

EX : let $f(x) = x^2 - 5x + 6 = 0$. Then $f(-x) = x^2 + 5x + 6 = 0$

Number of sign changes of $f(x) = 0$ is 2. Because sign change (+ to -) and (- to +). So, $f(x)$ has maximum 2 positive roots. Number of sign changes of $f(-x) = 0$ is 0. Because sign does not changed. So, $f(x) = 0$ does not have any negative roots.

Let's get n th order polynomial as $f(x) = 0$. Then get $\{f(\sqrt{x}) * f(-\sqrt{x})\} = g(x) = 0$. Then Graeffe's method says that square root of the division of successive co-efficient of polynomial $g(x)$ becomes the first iteration roots of the polynomial $f(x)$. Also second iteration roots can get by multiplying $\{g(\sqrt{x}) * g(-\sqrt{x})\} = h(x) = 0$ and getting quadratic root of the polynomial $g(x)$. Likewise we can reach exact solutions for the polynomial $f(x)$.

EX : If $f(x) = x^2 - 5x + 6 = 0$ then $f(\sqrt{x}) = x - 5\sqrt{x} + 6 = 0$ and $f(-\sqrt{x}) = x + 5\sqrt{x} + 6 = 0$.

$$g(x) = x^2 - 13x + 36 = 0.$$

First iteration roots are,

$$R1 = (36/13)^{1/2} = 1.664100 \quad \text{and} \quad R2 = (13/1)^{1/2} = 3.605551$$

$$h(x) = g(\sqrt{x}) * g(-\sqrt{x}) = x^2 - 97x + 1296 = 0.$$

Second iteration roots are,

$$R1 = (1296/97)^{1/4} = 1.911869 \quad R2 = (97/1)^{1/4} = 3.138288$$

Likewise for the third iteration, $R1 = (c3/c2)^{1/8}$ and $R1 = (c2/c1)^{1/8}$. We can get any number of iterations and when iteration increases roots converge in to the exact roots.



ROLE MODEL

PAFNUTY CHEBYSHEV: RUSSIAN MATHEMATICIAN

Pafnuty Chebyshev, in full Pafnuty Lvovich Chebyshev (born May 4 [May 16, New Style], 1821, Okatovo, Russia—died November 26 [December 8], 1894, St. Petersburg), founder of the St. Petersburg mathematical school (sometimes called the Chebyshev school), who is remembered primarily for his work on the theory of prime numbers and on the approximation of functions.

Chebyshev became assistant professor of mathematics at the University of St. Petersburg (now St. Petersburg State University) in 1847. In 1860 he became a correspondent and in 1874 a foreign associate of the Institut de France. He developed a basic inequality of probability theory called Chebyshev's inequality, a generalized form of the Bienaymé-Chebyshev inequality, and used the latter inequality to give a very simple and precise demonstration of the generalized law of large numbers—i.e., the average value for a large sample of identically distributed random variables converges to the average for individual variables. (*See* probability theory: The law of large numbers.)

Chebyshev proved Joseph Bertrand's conjecture that for any $n > 3$ there must exist a prime between n and $2n$. He also contributed to the proof of the prime number theorem, a formula for determining the number of primes below a given number. He studied theoretical mechanics and devoted much attention to the problem of obtaining rectilinear motion from rotary motion by mechanical linkage. The Chebyshev parallel motion is a three-bar linkage that gives a very close approximation to exact rectilinear motion. His mathematical writings covered a wide range of subjects, including the theory of probabilities, quadratic forms, orthogonal functions, the theory of integrals, gearings, the construction of geographic maps, and formulas for the computation of volumes. His important work on the approximation of functions by means of Chebyshev polynomials advanced applied mathematics. His *Teoria sravneny* (1849; "Theory of Congruences") made him widely known in the mathematical world and was used as a textbook in Russian universities for many years.



CASE STUDY

ADVANCED STUDY CASES FOR NUMERICAL ANALYSIS

A course on Numerical Methods typically covers introductory topics in numerical analysis for students of engineering, science, mathematics, and computer science who have completed elementary calculus, linear algebra and matrix theory. The course is usually limited to exploring basic algorithms for solving traditional simple problems in sciences and engineering. As a result, the students have become inadequately prepared to construct and explore more sophisticated algorithms of modern technological challenges. To reduce this gap, the author has offered a series of study cases, which provide concrete examples of the ways numerical methods lead to solutions of some scientific problems. The similar approach was promoted in.

Various problems of applied mathematics, thermophysics, and aerodynamics (e.g., stability of mechanical systems and flow boundary layers, fuel combustion, and heat protection of space probes) come to solving differential equations with small coefficients at the highest derivatives. This phenomenon leads to the formation of regions with small linear dimensions where gradients of functions are large. The numerical analysis of such problems by traditional box-schemes is restricted by non-uniform convergence or even divergence of numerical solutions. In this study case, the numerical solutions of the model singular ordinary differential equation have been evaluated for the linear boundary value problem. The developed numerical method was used for the analysis of gas flow parameters in boundary and viscous shock layers under the conditions of blowing on the body surface and non-equilibrium chemical reactions.

From a mathematical point of view, the increase of the flow rate of blowing gas or chemical-reaction rates is equivalent to the existence of a small coefficient at the highest derivative in the boundary-layer equations. A sub layer (of uncertain location) with large gradients of functions is created. The gas flow in the boundary layer was studied using a two-point exponential box-scheme and an effective regularization algorithm. The uniform second-order convergence was obtained for functions and derivatives in the full range of small parameters such as blowing factors and inverse chemical-reaction rates. The approach could be applied to boundary layers with gas injection and combustion.

Many technological problems (e.g., calculations of transport coefficients and chemical-reaction rates) require estimations of multi-fold integrals under the conditions when classical Newton-Cotes formulas for numerical integration could not be applied. In the second case, students studied the rotational relaxation of a gas of homonuclear diatomic molecules. The realistic models of the intermolecular interaction have been used for calculating the redistribution of the rotational and translational energies upon

collision. To estimate the rotational relaxation time, the sixfold integrals were calculated at 200 points over the range of temperatures from 200 K to 10,000 K, using the Monte Carlo simulation technique, with 4000 tests at each point. The data for intermediate points were determined by means of the interpolation technique using cubic splines of defect 1 with smoothing. The accuracy of the calculations was estimated as 1.5%. The experimental data for molecular nitrogen correlates well with the numerical results.

The purpose of the third study case was to analyze heat transfer processes at the catalytic materials of the shock-tube end after shock wave refraction in terms of the model of the nonsteady-state nonequilibrium thermal boundary layer. The major challenge was to develop the algorithm that could correctly and uniformly calculate flow parameters under all possible states (chemically “frozen”, nonequilibrium, and equilibrium ones) right behind the moving reflected shock wave. The types of governing differential equations are significantly different in these three cases, and the algorithm should automatically “adapt” to these changes across the flow.

The system of algebraic-differential equations was solved at each point of the flowfield behind the incident and reflected shock waves. A modified Newton’s method with optimal choice of iteration step was used for numerical solution of the equations. In the case of the initial pressure $p_1 = 100$ Pa, the flow behind the reflected wave is close to a state of local thermodynamic equilibrium, while in the other case $p_1 = 1$ Pa, the flow is significantly in nonequilibrium. The data were used as external boundary conditions for the flow parameters in the thermal viscous layer under the considered conditions.

The special topics covered mathematical foundations of evolution of dynamic systems that could be described in terms of attractors. The study cases of numerical modeling of the attractors of chaotic dynamical systems (turbulence, weather forecast, and economic system development) also became a part of the course. They were introduced through classical examples of bifurcations of systems modeling equilibrium in chemical reactions (Rössler attractor) and atmospheric dynamics (Lorenz oscillator). After brief in-class discussions of the study cases, each student continued working on a selected case as a project, created computer codes, run them at various parameters, compared the results with experimental data, and presented the findings to the classmates. In the course evaluations, students stated that they became deeply engaged in the course activities because they were examining the challenging problems.



SUMMARY

1. If the function $f(x)$ satisfies $f(a_0) f(b_0) < 0$, then the equation $f(x) = 0$ has at least one real root or an odd number of real roots in the interval (a_0, b_0) .
2. In numerical linear algebra, the Chebyshev iteration is an iterative method for determining the solutions of a system of linear equations.
3. Chebyshev iteration avoids the computation of inner products as is necessary for the other nonstationary methods.
4. Muller's method is based on the secant method, which constructs at every iteration a line through two points on the graph of f .
5. The Fundamental Theorem of Algebra states that every polynomial of degree one or greater has at least one root in the complex number system (keep in mind that a complex number can be real if the imaginary part of the complex root is zero).
6. The quadratic factor $x^2 + px + q = 0$ may have a pair of real roots or a pair of complex roots. Hence, the iterative methods for finding the real and complex roots of $P_n(x) = 0$ are based on the philosophy of extracting linear and quadratic factors of $P_n(x)$.
7. In numerical analysis, Bairstow's method is an efficient algorithm for finding the roots of a real polynomial of arbitrary degree. A well-known and widely-used process for determining the roots of a given polynomial with real coefficients.

KNOWLEDGE CHECK

1. An equation of form $ax^2 + bx + c = 0$, $a \neq 0$ is called the
 - a. Quadratic equation
 - b. cubic equation
 - c. linear equation
 - d. polynomial of degree n
2. If $a < 0$, then function $f(x) = ax^2 + bx + c$ has
 - a. maximum value
 - b. minimum value

- c. constant value
 - d. positive value
3. If roots of $x^2 - 5x + a = 0$ are equal, then $a =$
- a. $25/5$
 - b. $\pm 25/4$
 - c. $25/4$
 - d. None of Above
4. An example of an algebraic equation is
- a. $\tan x = e^x$
 - b. $x = \log x$
 - c. $x^3 - 5x + 3 = 0$
 - d. None
5. An example of a transcendental equation is
- a. $x^3 - 2x - 10 = 0$
 - b. $x^3 e^x = 5$
 - c. $x^2 + 11x - 1 = 0$
 - d. None
6. In finding a real root of the equation $x^3 - x - 10 = 0$ by bisection, if the root lies between $x_0 = 2$ and $x_1 = 3$, then, $x_2 =$
- a. 2.5
 - b. 2.75
 - c. 2.60
 - d. None
7. Which of the following methods has linear rate of convergence
- a. Regular false
 - b. Bisection
 - c. Newton-Raphson
 - d. None

Check Your Result

1. (a) 2. (a) 3. (c) 4. (c) 5. (b) 6. (a)
7. (a)

REVIEW QUESTIONS

- Find the interval in which the smallest positive root of the following equations lies:

- $\tan x + \tanh x = 0$

- $x^3 - x - 4 = 0$.

Determine the roots correct to two decimal places using the bisection method.

- Find the iterative methods based on the Newton-Raphson method for finding $\sqrt{N}$, $1/N$, $N^{1/3}$, where N is a positive real number. Apply the methods to $N = 18$ to obtain the results correct to two decimal places.

- Perform two iterations using the sixth order method, to find a root of the following equations:

- $x^4 - x - 10 = 0$, $x_0 = 2.0$;

- $x - e^{-x} = 0$, $x_0 = 1.0$

- The equation $x^2 + ax + b = 0$ has two real roots α and β . Show that the iteration method

- $x_{k+1} = -(ax_k + b)/x_k$ is convergent near $x = \alpha$ if $|\alpha| > |\beta|$.

- $x_{k+1} = -b/(x_k + a)$ is convergent near $x = \alpha$ if $|\alpha| < |\beta|$.

- $x_{k+1} = -(x_k^2 + b)/a$ is convergent near $x = \alpha$ if $2|\alpha| < |\alpha + \beta|$.

- Obtain the number of real roots between 0 and 3 of the equation

$$P(x) = x^4 - 4x^3 + 3x^2 + 4x - 4 = 0$$

using Sturm's sequence.

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Chapter

2

POLYNOMIAL INTERPOLATION

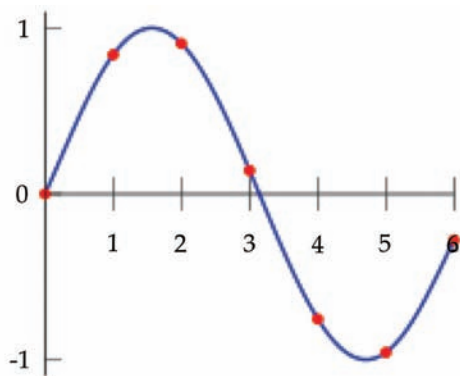
LEARNING OBJECTIVES

After studying this chapter, you will be able to:

1. Explain Taylor polynomial with remainder
2. Define the interpolation and the Newton form
3. Discuss about divided difference
4. Describe the computing with the Newton form
5. Understand the interpolation error

INTRODUCTION

Polynomial interpolation is a method of estimating values between known data points. When graphical data contains a gap, but data is available on either side of the gap or at a few specific points within the gap, an estimate of values within the gap can be made by interpolation.



A fundamental mathematical technique is to approximate something complicated by something simple, or at least less complicated, in the hope that the simple can capture some of the essential information in the complicated. This is the core idea of approximation with Taylor polynomials, a tool that has been central to mathematics since the calculus was first discovered.

The wide-spread use of computers has made the idea of approximation even more important. Computers are basically good at

doing very simple operations many times over. Effective use of computers therefore means that a problem must be broken up into (possibly very many) simple sub-problems. The result may provide only an approximation to the original problem, but this does not matter as long as the approximation is sufficiently good.

The idea of approximation is often useful when it comes to studying functions. Most mathematical functions only exist in quite abstract mathematical terms and cannot be expressed as combinations of the elementary functions we know from school. In spite of this, virtually all functions of practical interest can be approximated arbitrarily well by simple functions like polynomials, trigonometric or exponential functions. Polynomials in particular are very appealing for use on a computer since the value of a polynomial at a point can be computed by utilizing simple operations like addition and multiplication that computers can perform extremely quickly.

A classical example is Taylor polynomials which is a central tool in calculus. A Taylor polynomial is a simple approximation to a function that is based on information about the function at a single point only. In practice, the degree of a Taylor polynomial is often low, perhaps only degree one (linear), but by increasing the degree the approximation can in many cases become arbitrarily good over large intervals.



2.1 THE TAYLOR POLYNOMIAL WITH REMAINDER

1 A discussion of Taylor polynomials involves two parts: The Taylor polynomial itself, and the error, the remainder, committed in approximating a function by a **polynomial**. Let us consider each of these in turn.

LEARNING OBJECTIVE

1. Explain Taylor polynomial with remainder

2.1.1 The Taylor Polynomial

Taylor polynomials are discussed extensively, so the description here is brief. The essential feature of a Taylor polynomial is that it approximates a given function well at a single point.

Definition (Taylor polynomial). Suppose that the first n derivatives of the function f exist at $x = a$. The Taylor polynomial of f of degree n at a is written $T_n(f; a)$ (sometimes shortened to $T_n(x)$) and satisfies the conditions

$$T_n(f; a)^{(i)}(a) = f^{(i)}(a), \quad \text{for } i = 0, 1, \dots, n. \quad (1)$$

The conditions (1) mean that $T_n(f; a)$ and f have the same value and first n derivatives at a . This makes it quite easy to derive an explicit formula for the Taylor polynomial.

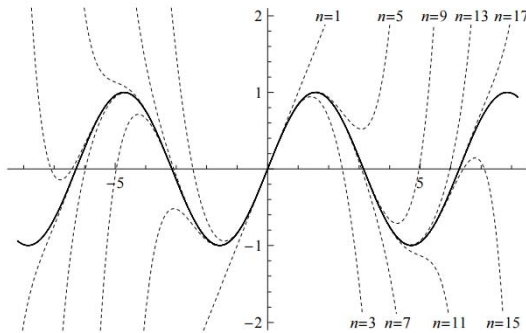


Figure 2.1: The Taylor polynomials of $\sin x$ (around $a = 0$) for degrees 1 to 17.

Theorem. The Taylor polynomial of f of degree n at a is unique and can be written as

KEY TERMS

Polynomial is a mathematical expression consisting of a sum of terms, each term including a variable or variables raised to a power and multiplied by a coefficient.

$$T_n(f; a)(x) = f(a) + (x - a)f'(a) + \frac{(x - a)^2}{2}f''(a) + \cdots + \frac{(x - a)^n}{n!}f^{(n)}(a). \quad (2)$$

Figure 2.1 shows the Taylor polynomials of $\sin x$, generated about $a = 0$, for degrees up to 17. Note that the even degree terms for these Taylor polynomials are 0, so there are only 9 such Taylor polynomials. We observe that as the degree increases, the approximation improves on an ever larger interval.

Observation. In the derivation of the Taylor polynomial, the manipulations simplify if polynomials of degree n are written as

$$p_n(x) = c_0 + c_1(x - a) + c_2(x - a)^2 + \cdots + c_n(x - a)^n.$$

This is an important observation: It is wise to adapt the form of the polynomial to the problem that is to be solved.

The elementary exponential and trigonometric functions have very simple and important Taylor polynomials.

Example (The Taylor polynomial of e^x). The function $f(x) = e^x$ has the nice property that $f^{(n)}(x) = e^x$ for all integers $n \geq 0$. The Taylor polynomial about $a = 0$ is therefore very simple since $f^{(n)}(0) = 1$ for all n . The general term in the Taylor polynomial then becomes

$$\frac{(x - a)^k f^{(k)}(a)}{k!} = \frac{x^k}{k!}.$$

This means that the Taylor polynomial of degree n about $a = 0$ for $f(x) = e^x$ is given by

$$T_n(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}.$$

For the exponential function the Taylor polynomials will be very good approximations for large values of n . More specifically, it can be shown that for any value of x , the difference between $T_n(x)$ and e^x can be made as small as we wish if we just let n be big enough. This is often expressed by writing

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots.$$



REMEMBER

The polynomial's graph can be thought of as "filling in the curve" to account for data between the known points. This methodology, known as polynomial interpolation, often (but not always) provides more accurate results than linear interpolation.



It turns out that the Taylor polynomials of the trigonometric functions $\sin x$ and $\cos x$ converge in a similar way.

2.1.2 The Remainder

The Taylor polynomial $T_n(f)$ is an approximation to f , and in many situations it will be important to control the error in the **approximation**. The error can be expressed in a number of ways, and the following two are the most common.

Theorem. Suppose that f is a function whose derivatives up to order $n + 1$ exist and are continuous. Then the remainder in the Taylor expansion $R_n(f; a)(x) = f(x) - T_n(f; a)(x)$ is given by

$$R_n(f; a)(x) = \frac{1}{n!} \int_a^x f^{(n+1)}(t)(x-t)^n dt. \quad (3)$$

The remainder may also be written as

$$R_n(f; a)(x) = \frac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(\xi), \quad (4)$$

where ξ is a number in the interval (a, x) (the interval (x, a) if $x < a$).

We are going to make use of Taylor polynomials with remainder in future chapters to analyse the error in a number of numerical methods. Here we just consider one example of how we can use the remainder to control how well a function is approximated by a polynomial.

Example. We want to determine a polynomial approximation of the function $\sin x$ on the interval $[-1, 1]$ with error smaller than 10^{-5} . We want to use Taylor polynomials about the point $a = 0$; the question is how high the degree needs to be in order to get the error to be small.

If we look at the error term (4), there is one factor that looks rather difficult to control, namely $f^{(n+1)}(\xi)$: Since we do not know the degree, we do not really know what this derivative is, and on top of this we do not know at which point it should be evaluated either. The solution is not so difficult if we realize that we do not need to control the error exactly, it is sufficient to make sure that the error is smaller than 10^{-5} .

KEY TERMS

Approximation is anything that is similar but not exactly equal to something else.

We want to find the smallest n such that

$$\left| \frac{x^{n+1}}{(n+1)!} f^{(n+1)}(\xi) \right| \leq 10^{-5}, \quad (5)$$

where the function $f(x) = \sin x$ and ξ is a number in the interval $(0, x)$. Here we demand that the absolute value of the error should be smaller than 10^{-5} . This is important since otherwise we could make the error small by making it negative, with large absolute value. The main ingredient in achieving what we want is the observation that since $f(x) = \sin x$, any derivative of f is either $\cos x$ or $\sin x$ (possibly with a minus sign which disappears when we take absolute values). But then we certainly know that

$$\left| f^{(n+1)}(\xi) \right| \leq 1. \quad (6)$$

This may seem like a rather crude estimate, which may be the case, but it was certainly very easy to derive; to estimate the correct value of ξ would be much more difficult. If we insert the estimate (6) on the left in (5), we can also change our required inequality,

$$\left| \frac{x^{n+1}}{(n+1)!} f^{(n+1)}(\xi) \right| \leq \frac{|x|^{n+1}}{(n+1)!} \leq 10^{-5}.$$

If we manage to find an n such that this last inequality is satisfied, then (5) will also be satisfied. Since $x \in [-1, 1]$ we know that $|x| \leq 1$ so this last inequality will be satisfied if

$$\frac{1}{(n+1)!} \leq 10^{-5}. \quad (7)$$

The left-hand side of this inequality decreases with increasing n , so we can just determine n by computing $1/(n+1)!$ for the first few values of n , and use the first value of n for which the inequality holds. If we do this, we find that $1/8! \approx 2.5 \times 10^{-5}$ and $1/9! \approx 2.8 \times 10^{-6}$. This means that the smallest value of n for which (7) will be satisfied is $n = 8$. The Taylor polynomial we are looking for is therefore

$$p_8(x) = T_8(\sin; 0)(x) = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040},$$

since the term of degree 8 is zero.

If we check the approximation at $x = 1$, we find $p_8(1) \approx 0.8414682$. Comparing with the exact value $\sin 1 \approx 0.8414710$, we find that the error is roughly 2.73×10^{-6} , which is



close to the upper bound $1/9!$ which we computed above.

Figure 2.1 shows the Taylor polynomials of $\sin x$ about $a = 0$ of degree up to 17. In particular we see that for degree 7, the approximation is indistinguishable from the original in the plot, at least up to $x = 2$.

The error formula (4) will be most useful for us, and for easy reference we record the complete Taylor expansion in a corollary

Corollary. Any function f whose first $n + 1$ derivatives are continuous at $x = a$ can be expanded in a Taylor polynomial of degree n at $x = a$ with a corresponding error term,

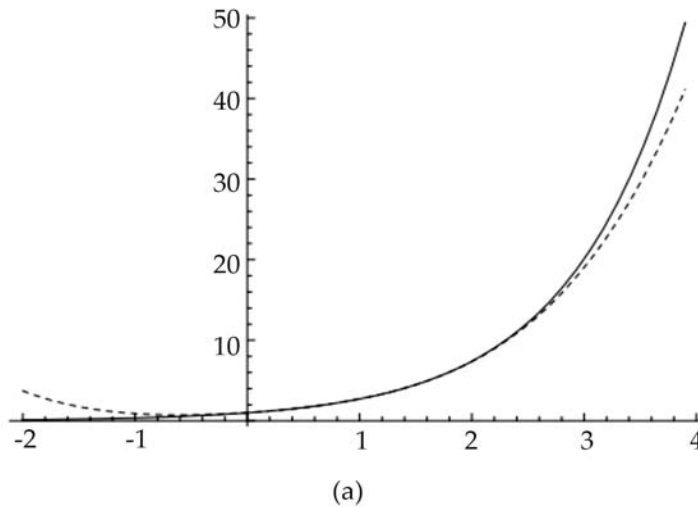
$$f(x) = f(a) + (x-a)f'(a) + \cdots + \frac{(x-a)^n}{n!} f^{(n)}(a) + \frac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(\xi_x), \quad (8)$$

where ξ_x is a number in the interval (a, x) (the interval (x, a) if $x < a$) that depends on x . This is called a Taylor expansion of f .

The remainder term in (8) lets us control the error in the Taylor approximation. It turns out that the error behaves quite differently for different functions.

Example (Taylor polynomials for $f(x) = \sin x$). If we go back to figure 2.1, it seems like the Taylor polynomials approximate $\sin x$ well on larger intervals as we increase the degree. Let us see if this observation can be derived from the error term

$$e(x) = \frac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(\xi). \quad (9)$$



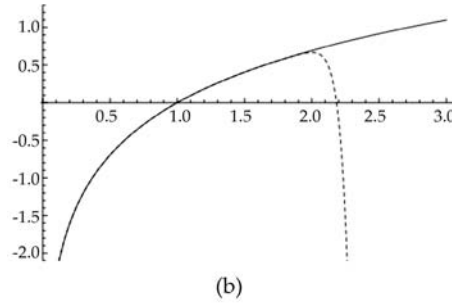


Figure 2.2: In (a) the Taylor polynomial of degree 4 about the point $a = 1$ for the function $f(x) = e^x$ is shown. Figure (b) shows the Taylor polynomial of degree 20 for the function $f(x) = \ln x$, also about the point $a = 1$.

When $f(x) = \sin x$ we know that $|f^{(n+1)}(\xi)| \leq 1$, so the error is bounded by

$$|e(x)| \leq \frac{|x|^{n+1}}{(n+1)!}$$

where we have also inserted $a = 0$ which was used in figure 2.1. Suppose we want the error to be small on the interval $[-b, b]$. Then $|x| \leq b$, so on this interval the error is bounded by

$$|e(x)| \leq \frac{b^{n+1}}{(n+1)!}.$$

The question is what happens to the expression on the right when n becomes large; does it tend to 0 or does it not? It is not difficult to show that regardless of what the value of b is, the factorial $(n+1)!$ will tend to infinity more quickly, so

$$\lim_{n \rightarrow \infty} \frac{b^{n+1}}{(n+1)!} = 0.$$

In other words, if we just choose the degree n to be high enough, we can get the Taylor polynomial to be an arbitrarily good approximation to $\sin x$ on an interval $[-b, b]$, regardless of what the value of b is.

Example (Taylor polynomials for $f(x) = \ln x$). The plot in figure 2.2 shows the logarithm function $f(x) = \ln x$ and its Taylor polynomial of degree 20, expanded at $a = 1$. The Taylor polynomial seems to be very close to $\ln x$ as long as x is a bit smaller

than 2, but for $x > 2$ it seems to diverge quickly. The error term involves the derivative $f^{(n+1)}(\xi)$ of $f(x) = \ln x$, so we need a formula for this. Since $f(x) = \ln x$, we have

$$f'(x) = \frac{1}{x} = x^{-1}, \quad f''(x) = -x^{-2}, \quad f'''(x) = 2x^{-3}$$

and from this we find that the general formula is

$$f^{(k)}(x) = (-1)^{k+1} (k-1)! x^{-k}, \quad k \geq 1. \quad (10)$$

Since $a = 1$, this means that the general term in the Taylor polynomial is

$$\frac{(x-1)^k}{k!} f^{(k)}(1) = (-1)^{k+1} \frac{(x-1)^k}{k}.$$

The Taylor expansion (8) therefore becomes

$$\ln x = \sum_{k=1}^n (-1)^{k+1} \frac{(x-1)^k}{k} + \frac{(x-1)^{n+1}}{n+1} \xi^{-n-1},$$

where ξ is some number in the interval $(1, x)$ (in $(x, 1)$ if $0 < x < 1$). The problematic area seems to be to the right of $x = 1$, so let us assume that $x > 1$. In this case $\xi > 1$, so therefore $\xi^{-n-1} < 1$. The error is then bounded by

$$\left| \frac{(x-1)^{n+1}}{n+1} \xi^{-n-1} \right| \leq \frac{(x-1)^{n+1}}{n+1}.$$

When $x-1 < 1$, i.e., when $x < 2$, we know that $(x-1)^{n+1}$ will tend to zero when n tends to infinity, and the denominator $n+1$ will just contribute to this happening even more quickly

For $x > 2$, one can try and analyse the error term, and if one uses the integral form of the remainder (3) it is in fact possible to find an exact formula for the error. However, it is much simpler to consider the Taylor polynomial directly,

$$p_n(x) = T(\ln; 1)(x) = \sum_{k=1}^n (-1)^{k+1} \frac{(x-1)^k}{k}.$$

Note that for $x > 2$, the absolute value of the terms in the sum will become arbitrarily

large since

$$\lim_{k \rightarrow \infty} \frac{c^k}{k} = \infty$$

when $c > 1$. This means that the sum will jump around more and more, so there is no way it can converge for $x > 2$, and it is this effect we see in figure 2.2 (b).

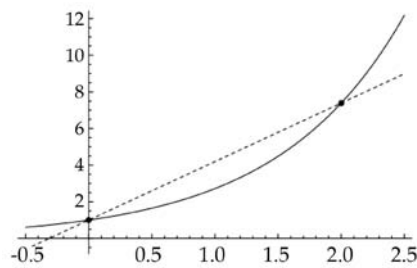
2.2 INTERPOLATION AND THE NEWTON FORM

LEARNING OBJECTIVE

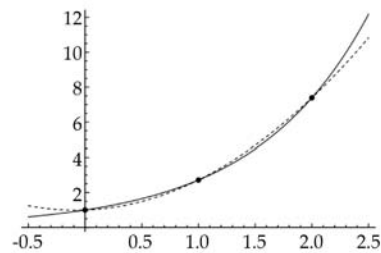
2. Define the interpolation and the Newton form



A Taylor polynomial based at a point $x = a$ usually provides a very good approximation near a , but as we move away from this point, the error will increase. If we want a good approximation to a function f across a whole interval, it seems natural that we ought to utilize information about f from different parts of the interval. Polynomial interpolation lets us do just that.



(a)



(b)

Figure 2.3: Interpolation of e^x at two points with a secant (a), and at three points with a parabola (b).

2.2.1 The Interpolation Problem

Just like Taylor approximation is a generalization of the tangent, interpolation is a generalization of the secant, see figure 2.3.

The idea behind polynomial interpolation is simple: We approximate a function f by a polynomial p by forcing p to have the same function values as f at a number of points. A general parabola has three free coefficients, and we should therefore expect to be able to force a parabola through three arbitrary points. More generally, suppose we have $n+1$ distinct numbers $\{x_i\}_{i=0}^n$ scattered throughout an interval $[a,b]$ where f is defined. Since a polynomial of degree n has $n+1$ free coefficients it is natural to try and find a polynomial of degree n with the same values as f at the numbers $\{x_i\}_{i=0}^n$.

Problem (Polynomial interpolation). Let f be a given function defined on an interval $[a,b]$, and let $\{x_i\}_{i=0}^n$ be $n+1$ distinct numbers in $[a,b]$. The polynomial interpolation problem is to find a polynomial $p_n = P(f; x_0, \dots, x_n)$ of degree n that matches f at each x_i ,

$$p_n(x_i) = f(x_i), \quad \text{for } i = 0, 1, \dots, n. \quad (11)$$

The numbers $\{x_i\}_{i=0}^n$ are called interpolation points, the conditions (11) are called the interpolation conditions, and the polynomial $p_n = P(f; x_0, \dots, x_n)$ is called a polynomial interpolant.

The notation $P(f; x_0, \dots, x_n)$ for a polynomial interpolant is similar to the notation $T_n(f; a)$ for the Taylor polynomial. However, it is a bit cumbersome, so we will often just use p_n when no confusion is possible.

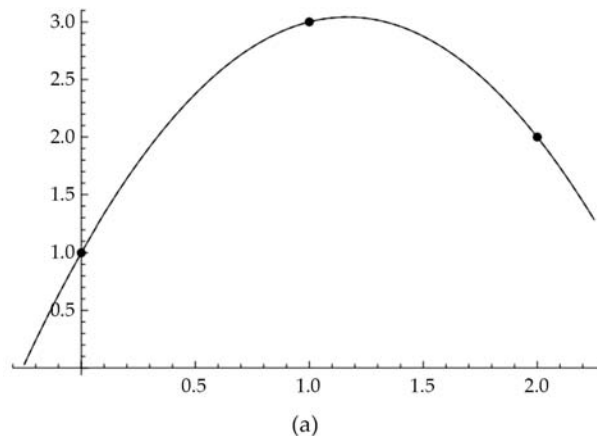


Figure 2.4: Three interpolation points and the corresponding quadratic interpolating polynomial.

In many situations the function f may not be known, just its function values at the points $\{x_i\}_{i=0}^n$, as in the following example.

Example. Suppose we want to find a polynomial that passes through the three points $(0, 1)$, $(1, 3)$, and $(2, 2)$. In other words, we want to find a polynomial p such that.

$$p(0) = 1, \quad p(1) = 3, \quad p(2) = 2. \quad (12)$$

Since there are three points it is natural to try and accomplish this with a quadratic polynomial, i.e., we assume that $p(x) = c_0 + c_1x + c_2x^2$. If we insert this in the conditions (12) we obtain the three equations

$$\begin{aligned} 1 &= p(0) = c_0, \\ 3 &= p(1) = c_0 + c_1 + c_2, \\ 2 &= p(2) = c_0 + 2c_1 + 4c_2. \end{aligned}$$

We solve these and find $c_0 = 1$, $c_1 = 7/2$, and $c_2 = -3/2$, so p is given by

$$p(x) = 1 + \frac{7}{2}x - \frac{3}{2}x^2.$$

A plot of this polynomial and the interpolation points is shown in figure 2.4.

There are at least four questions raised by above problem: Is there a polynomial of degree n that satisfies the interpolation conditions? How many such polynomials are there? How can we find one such polynomial? What is a convenient way to write an interpolating polynomial?

2.2.2 The Newton form of the Interpolating Polynomial

We start by considering the last of the four questions above. We have already seen that by writing polynomials in a particular form, the computations of the Taylor polynomial simplified. This is also the case for interpolating polynomials.

Definition (Newton form). Let $\{x_i\}_{i=0}^n$ be $n+1$ distinct real numbers. The Newton form of a polynomial of degree n is an expression in the form

$$\begin{aligned} p_n(x) &= c_0 + c_1(x - x_0) + c_2(x - x_0)(x - x_1) + \cdots \\ &\quad + c_n(x - x_0)(x - x_1) \cdots (x - x_{n-1}). \end{aligned} \quad (13)$$



The advantage of the Newton form will become evident when we consider some examples.

Example (Newton form for $n = 0$). Suppose we have only one interpolation point x_0 . Then the Newton form is just $p_0(x) = c_0$. To interpolate f at x_0 we have to choose $c_0 = f(x_0)$,

$$p_0(x) = f(x_0).$$

Example (Newton form for $n = 1$). With two points x_0 and x_1 the Newton form is $p_1(x) = c_0 + c_1(x - x_0)$. Interpolation at x_0 means that $f(x_0) = p_1(x_0) = c_0$, while interpolation at x_1 yields

$$f(x_1) = p_1(x_1) = f(x_0) + c_1(x_1 - x_0).$$

Together this means that

$$c_0 = f(x_0), \quad c_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}. \quad (14)$$

We note that c_0 remains the same as in the case $n = 0$.

Example (Newton form for $n = 2$). We add another point and consider interpolation with a quadratic polynomial

$$p_2(x) = c_0 + c_1(x - x_0) + c_2(x - x_0)(x - x_1).$$

at the three points x_0, x_1, x_2 . **Interpolation** at x_0 and x_1 gives the equations

$$\begin{aligned} f(x_0) &= p_2(x_0) = c_0, \\ f(x_1) &= p_2(x_1) = c_0 + c_1(x_1 - x_0), \end{aligned}$$

which we note are the same equations as we solved in the case $n = 1$. From the third condition

$$f(x_2) = p_2(x_2) = c_0 + c_1(x_2 - x_0) + c_2(x_2 - x_0)(x_2 - x_1),$$

we obtain

$$c_2 = \frac{f(x_2) - f(x_0) - \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x_2 - x_0)}{(x_2 - x_0)(x_2 - x_1)}.$$

KEY TERMS

Interpolation is a method of constructing new data points within the range of a discrete set of known data points.

Playing around a bit with this expression one finds that it can also be written as

$$c_2 = \frac{\frac{f(x_2)-f(x_1)}{x_2-x_1} - \frac{f(x_1)-f(x_0)}{x_1-x_0}}{x_2-x_0}. \quad (15)$$

It is easy to see that what happened in the quadratic case happens in the general case: The equation that results from the interpolation condition at x_k involves only the points $(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_k, f(x_k))$. This becomes clear if we write down all the equations,

$$\begin{aligned} f(x_0) &= c_0, \\ f(x_1) &= c_0 + c_1(x_1 - x_0), \\ f(x_2) &= c_0 + c_1(x_2 - x_0) + c_2(x_2 - x_0)(x_2 - x_1), \\ &\vdots \\ f(x_k) &= c_0 + c_1(x_k - x_0) + c_2(x_k - x_0)(x_k - x_1) + \dots \\ &\quad + c_{k-1}(x_k - x_0) \dots (x_k - x_{k-2}) + c_k(x_k - x_0) \dots (x_k - x_{k-1}). \end{aligned} \quad (16)$$

This is an example of a triangular system where each new equation introduces one new variable and one new point. This means that each coefficient c_k only depends on the data $(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_k, f(x_k))$, so the following theorem is immediate.

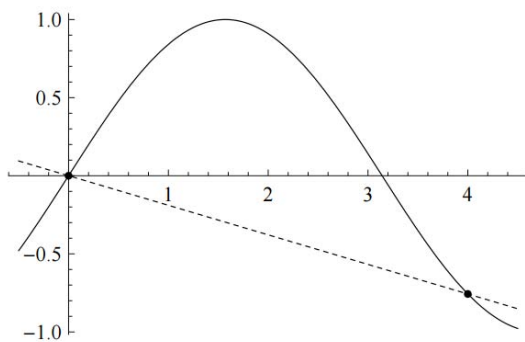
Theorem. Let f be a given function and $x_0, \dots, x_n$ given and distinct interpolation points. There is a unique polynomial of degree n which interpolates f at these points. If the interpolating polynomial is expressed in Newton form,

$$p_n(x) = c_0 + c_1(x - x_0) + \dots + c_n(x - x_0)(x - x_1) \dots (x - x_{n-1}), \quad (17)$$

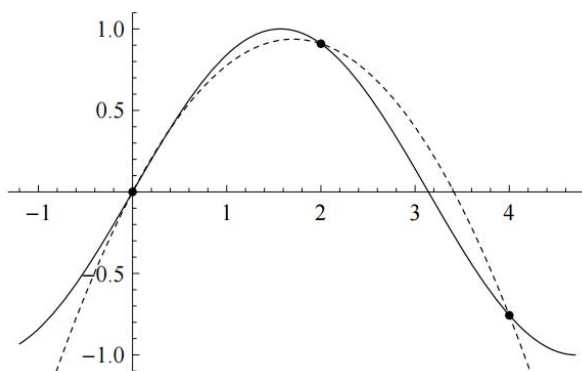
then c_k depends only on $(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_k, f(x_k))$ which is indicated by the notation

$$c_k = f[x_0, \dots, x_k] \quad (18)$$

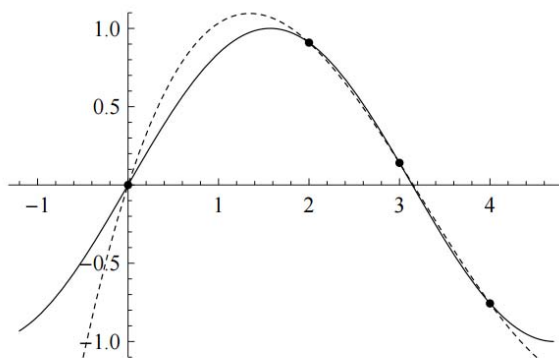




(a)



(b)



(c)

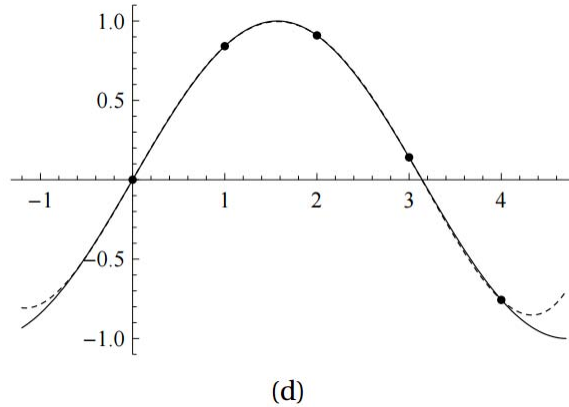


Figure 2.5: Interpolation of $\sin x$ with a line (a), a parabola (b), a cubic (c), and a quartic polynomial (d).

for $k = 0, 1, \dots, n$. The interpolating polynomials p_n and p_{n-1} are related by

$$p_n(x) = p_{n-1}(x) + f[x_0, \dots, x_n](x - x_0) \cdots (x - x_{n-1}).$$

Some examples of interpolation are shown in figure 2.5. Note how the quality of the approximation improves with increasing degree.

Proof. Most of this theorem is a direct consequence of writing the interpolating polynomial in Newton form, which becomes

$$p_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + \cdots + f[x_0, \dots, x_n](x - x_0) \cdots (x - x_{n-1}) \quad (19)$$

when we write the coefficients as in (18). The coefficients can be computed, one by one, from the equations (16), starting with c_0 . The uniqueness follows since there is no choice in solving the equations (16); there is one and only one solution.

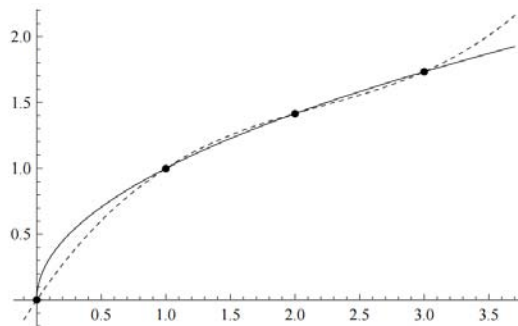


Figure 2.6: The function $f(x) = \sqrt{x}$ (solid) and its cubic interpolant at the four points 0, 1, 2, and 3 (dashed).

Theorem answers the questions raised above: above problem has a solution and it is unique. The theorem itself does not tell us directly how to find the solution, but in the text preceding the theorem we showed how it could be constructed. One concrete example will illustrate the procedure.

Example. Suppose we have the four points $x_i = i$, for $i = 0, \dots, 3$, and we want to interpolate the function $\sqrt{x}$ at these points. In this case the Newton form becomes

$$p_3(x) = c_0 + c_1x + c_2x(x-1) + c_3x(x-1)(x-2).$$

The interpolation conditions become

$$\begin{aligned} 0 &= c_0, \\ 1 &= c_0 + c_1, \\ \sqrt{2} &= c_0 + 2c_1 + 2c_2, \\ \sqrt{3} &= c_0 + 3c_1 + 6c_2 + 6c_3. \end{aligned}$$

Not surprisingly, the equations are triangular and we find

$$c_0 = 0, \quad c_1 = 1, \quad c_2 = -(1 - \sqrt{2}/2), \quad c_3 = (3 + \sqrt{3} - 3\sqrt{2})/6$$

Figure 2.6 shows a plot of this interpolant.

We emphasize that the Newton form is just one way to write the interpolating polynomial — there are many alternatives.

Merits and Demerits of Lagrange's Formula

The formula is simple and easy to remember.

- There is no need to construct the divided difference table and we can directly interpolate the unknown value with the help of given observations.
- The calculations in the formula are more complicated than in the divided difference formula.
- The application of the formula is not speedy
- There is always a chance of committing some error due to a number of (+)ve and (-)ve sign in the denominator and numerator of each term.
- The calculations provide no check whether the functional values used are taken correctly or not, whereas the differences used in a difference formula provide a check on the functional values.



This time we are looking for the Lagrange form of the interpolation polynomial, $Q_2(x)$, that interpolates three points: $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_2, f(x_2))$. Unfortunately, the Lagrange form of the interpolation polynomial does not let us use the interpolation polynomial through the first two points, $Q_1(x)$, as a building block for $Q_2(x)$. This means that we have to compute all the polynomials $l_j^m(x)$ from scratch. We start with

$$\begin{aligned} l_0^2(x) &= \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}, \\ l_1^2(x) &= \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}, \\ l_2^2(x) &= \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}. \end{aligned}$$



REMEMBER

Interpolation nodes should be calculated anew for each new function $f(x)$, but the algorithm is hard to be implemented numerically.

The interpolation polynomial is therefore given by

$$\begin{aligned} Q_2(x) &= f(x_0)l_0^2(x) + f(x_1)l_1^2(x) + f(x_2)l_2^2(x) \\ &= f(x_0)\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} + f(x_1)\frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} + f(x_2)\frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}. \end{aligned}$$

It is easy to verify that indeed $Q_2(x_j) = f(x_j)$ for $j = 0, 1, 2$, as desired.

2.3 DIVIDED DIFFERENCES



The coefficients $c_k = f[x_0, \dots, x_k]$ have certain properties that are useful both for computation and understanding.

When doing interpolation at the points $x_0, \dots, x_k$ we can consider two smaller problems, namely interpolation at the points $x_0, \dots, x_{k-1}$ as well as interpolation at the points $x_1, \dots, x_k$.

Suppose that the polynomial q_0 interpolates f at the points $x_0, \dots, x_{k-1}$ and that q_1 interpolates f at the points $x_1, \dots, x_k$, and consider the polynomial defined by the formula

$$p(x) = \frac{x_k - x}{x_k - x_0} q_0(x) + \frac{x - x_0}{x_k - x_0} q_1(x). \quad (20)$$

Our claim is that $p(x)$ interpolates f at the points $x_0, \dots, x_k$, which means that $p = p_k$ since a polynomial interpolant of degree k which interpolates $k+1$ points is unique.

We first check that p interpolates f at an interior point x_i with

LEARNING OBJECTIVE



3. Discuss about divided difference



$0 < i < k$. In this case $q_0(x_i) = q_1(x_i) = f(x_i)$ so

$$p(x_i) = \frac{x_k - x}{x_k - x_0} f(x_i) + \frac{x - x_0}{x_k - x_0} f(x_i) = f(x_i).$$

At $x = x_0$ we have

$$p(x_0) = \frac{x_k - x_0}{x_k - x_0} q_0(x_0) + \frac{x_0 - x_0}{x_k - x_0} q_1(x_0) = q_0(x_0) = f(x_0),$$

as required, and in a similar way we also find that $p(x_k) = f(x_k)$.

Lemma. Let $P(f; x_0, \dots, x_k)$ denote the polynomial of degree $k - 1$ that interpolates the function f at the points $x_0, \dots, x_k$. Then

$$P(f; x_0, \dots, x_k)(x) = \frac{x_k - x}{x_k - x_0} P(f; x_0, \dots, x_{k-1})(x) + \frac{x - x_0}{x_k - x_0} P(f; x_1, \dots, x_k)(x).$$

From this lemma we can deduce a useful formula for the coefficients of the interpolating polynomial. We first recall that the term of highest degree in a polynomial is referred to as the leading term, and the coefficient of the leading term is referred to as the leading coefficient. From equation (17) we see that the leading coefficient of the interpolating polynomial p_k is $f[x_0, \dots, x_k]$. This observation combined with above Lemma leads to a useful formula for $f[x_0, \dots, x_k]$.

Theorem. Let $ck = f[x_0, \dots, x_k]$ denote the leading coefficient of the interpolating polynomial $P(f; x_0, \dots, x_k)$. This is called a k th order divided difference of f and satisfies the relations $f[x_0] = f(x_0)$, and

$$f[x_0, \dots, x_k] = \frac{f[x_1, \dots, x_k] - f[x_0, \dots, x_{k-1}]}{x_k - x_0} \quad (21)$$

for $k > 0$.

Proof. The relation (21) follows from the relation in above lemma if we consider the leading coefficients on both sides. On the left the leading coefficient is $f[x_0, \dots, x_k]$. The right-hand side has the form

$$\begin{aligned} & \frac{x_k - x}{x_k - x_0} (f[x_0, \dots, x_{k-1}] x^{k-1} + \text{lower degree terms}) + \\ & \frac{x - x_0}{x_k - x_0} (f[x_1, \dots, x_k] x^{k-1} + \text{lower degree terms}) \\ & = \frac{f[x_1, \dots, x_k] - f[x_0, \dots, x_{k-1}]}{x_k - x_0} x^k + \text{lower degree terms}, \end{aligned}$$

and from this (21) follows.

The significance of above theorem is that it provides a simple formula for computing the coefficients of the interpolating polynomial in Newton form. The relation (21) also explains the name 'divided difference', and it should not come as a surprise that $f[x_0, \dots, x_k]$ is related to the k th derivative of f .

It is helpful to organize the computations of divided differences in a table,

$$\begin{array}{ccccccc}
 x_0 & f[x_0] & & & & & \\
 x_1 & f[x_1] & f[x_0, x_1] & & & & \\
 x_2 & f[x_2] & f[x_1, x_2] & f[x_0, x_1, x_2] & & & \\
 x_3 & f[x_3] & f[x_2, x_3] & f[x_1, x_2, x_3] & f[x_0, x_1, x_2, x_3] & & \\
 \vdots & & & & & &
 \end{array} \quad (22)$$

Here an entry in the table (except for the first two columns) is computed by subtracting the entry to the left and above from the entry to the left, and dividing by the last minus the first x_i involved. Then all the coefficients of the Newton form can be read off from the diagonal. An example will illustrate how this is used.

Example. Suppose we have the data

x	0	1	2	3
$f(x)$	0	1	1	2

We want to compute the divided differences using (21) and organize the computations as in (22),

x	$f(x)$			
0	0			
1	1	1		
2	1	0	-1/2	
3	2	1	1/2	1/3

This means that the interpolating polynomial is

$$\begin{aligned}
 p_3(x) &= 0 + 1(x-0) - \frac{1}{2}(x-0)(x-1) + \frac{1}{3}(x-0)(x-1)(x-2) \\
 &= x - \frac{1}{2}x(x-1) + \frac{1}{3}x(x-1)(x-2).
 \end{aligned}$$

A plot of this polynomial with the interpolation points is shown in figure 2.7.

There is one more important property of divided differences that we need to discuss. If we look back on equation (14), we see that

$$c_1 = f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}.$$



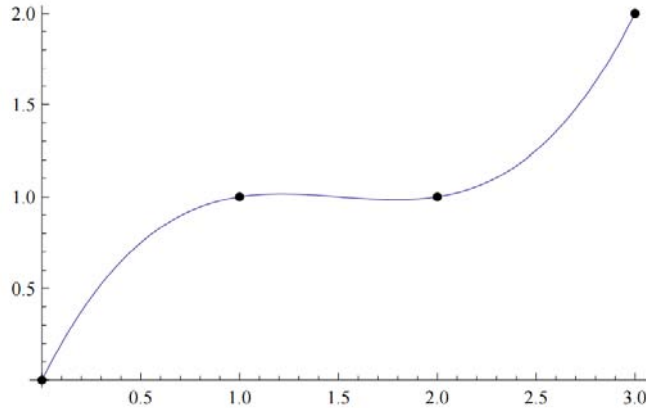


Figure 2.7: The data points and the interpolant in above example.

From the mean value theorem for derivatives we can conclude from this that $f[x_0, x_1] = f'(\xi)$ for some number ξ in the interval (x_0, x_1) , provided f' is continuous in this interval. The relation (21) shows that higher order divided differences are built from lower order ones in a similar way, so it should come as no surprise that divided differences can be related to derivatives in general.

Theorem. Let f be a function whose first k derivatives are continuous in the smallest interval $[a, b]$ that contains all the numbers $x_0, \dots, x_k$. Then

$$f[x_0, \dots, x_k] = \frac{f^{(k)}(\xi)}{k!} \quad (23)$$

where ξ is some number in the interval (a, b) .

We skip the proof of this theorem, but return to the Newton form of the interpolating polynomial,

$$p_n = f[x_0] + f[x_0, x_1](x - x_0) + \dots + f[x_0, \dots, x_n](x - x_0) \cdots (x - x_{n-1}).$$

Above theorem shows that divided differences can be associated with derivatives, so this formula is very similar to the formula for a Taylor polynomial. In fact, if we let all the interpolation points x_i approach a common number z , it is quite easy to show that the interpolating polynomial p_n approaches the Taylor polynomial

$$T_n(f; z)(x) = f(z) + f'(z)(x - z) + \dots + f^{(n)}(z) \frac{(x - z)^n}{n!}. \quad (24)$$

2.3.1 Properties of Divided Differences

1. The divided differences are symmetrical in their arguments, i.e., independent of the order of arguments.

$$[x_0, x_1] = \frac{y_1}{x_1 - x_0} + \frac{y_0}{x_0 - x_1} = [x_1, x_0]$$

Also,

$$\begin{aligned} [x_0, x_1, x_2] &= \frac{y_0}{(x_0 - x_1)(x_0 - x_2)} + \frac{y_1}{(x_1 - x_0)(x_1 - x_2)} + \frac{y_2}{(x_2 - x_0)(x_2 - x_1)} \\ &= [x_2, x_0, x_1] \text{ or } [x_1, x_2, x_0] \end{aligned}$$

2. The n th divided differences of a polynomial of n th degree are constant.

Let the arguments be equally spaced so that

$$x_1 - x_0 = x_2 - x_1 = \dots = x_n - x_{n-1} = h$$

then,

$$\begin{aligned} [x_0, x_1] &= \frac{y_1 - y_0}{x_1 - x_0} = \frac{\Delta y_0}{h} \\ [x_0, x_1, x_2] &= \frac{[x_1, x_2] - [x_0, x_1]}{(x_2 - x_0)} \\ &= \frac{1}{2h} \left(\frac{\Delta y_1}{h} - \frac{\Delta y_0}{h} \right) = \frac{1}{2!} \cdot \frac{1}{h^2} (\Delta^2 y_0) \end{aligned}$$

In general,

$$[x_0, x_1, x_2, \dots, x_n] = \frac{1}{n!} \cdot \frac{1}{h^n} \Delta^n y_0$$

If tabulated function is a n th degree polynomial.

$$\therefore \Delta^n y_0 = \text{constant}$$

$\therefore$ n th divided differences will also be constant.

2.4 COMPUTING WITH THE NEWTON FORM

4 Our use of polynomial interpolation will primarily be as a tool for developing numerical methods for differentiation, integration, and solution of differential equations. For such purposes the interpolating polynomial is just a step on the way to a final computational formula, and is not computed explicitly. There are situations though where one may need to determine the interpolating polynomial explicitly, and then the Newton form is usually preferred.

LEARNING OBJECTIVE

4. Describe the computing with the Newton form

To use the Newton form in practice, we need two algorithms: One for determining the divided differences involved in the formula (19), and one for computing the value $p_n(x)$ of the interpolating polynomial for a given number x . We consider each of these in turn.

The Newton form of the polynomial that interpolates a given function f at the $n+1$ points $x_0, \dots, x_n$ is given by

$$p_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + \dots + f[x_0, \dots, x_n](x - x_0) \cdots (x - x_{n-1}),$$

and to represent this polynomial we need to compute the divided differences $f[x_0], f[x_0, x_1], \dots, f[x_0, \dots, x_n]$. The obvious way to do this is as indicated in the table (22) which we repeat here for convenience,

x_0	$f[x_0]$			
x_1	$f[x_1]$	$f[x_0, x_1]$		
x_2	$f[x_2]$	$f[x_1, x_2]$	$f[x_0, x_1, x_2]$	
x_3	$f[x_3]$	$f[x_2, x_3]$	$f[x_1, x_2, x_3]$	$f[x_0, x_1, x_2, x_3]$
$\vdots$				

We start with the interpolation points $x_0, \dots, x_n$ and the function f , and then compute the values in the table, column by column. Let us denote the entries in the table by $(d_{i,k})_{i=0,k=0}^n$, where i runs down the columns and k indicates the column number, starting with 0 for the column with the function values. The first column of function values is special and must be computed separately. Otherwise we note that a value $d_{i,k}$ in column k is given by the two neighboring values in column $k-1$,

$$d_{i,k} = \frac{d_{i,k-1} - d_{i-1,k-1}}{x_i - x_{i-k}}, \quad (25)$$

for $i \geq k$, while the other entries in column k are not defined. We start by computing the first column of function values. Then we can use the formula (25) to compute the next column. It would be natural to start by computing the diagonal entry and then proceed down the column. However, it is better to start with the last entry in the column, and proceed up, to the diagonal entry. The reason is that once an entry $d_{i,k}$ has been computed, the entry $d_{i,k-1}$ immediately to the left is not needed any more. Therefore, there is no need to use a two dimensional array; we can just start with the one dimensional array of function values and let every new column overwrite the one to the left. Since no column contains values above the diagonal, we end up with the correct divided differences in the one dimensional array at the end.

Algorithm (Computing divided differences). Let f be a given function, and $x_0, \dots, x_n$ given interpolation points for some nonnegative integer n . After the code

```

for  $i = 0, 1, \dots, n$ 
     $f_i = f(x_i)$ ;
for  $k = 1, 2, \dots, n$ 
    for  $i = n, n-1, \dots, k$ 
         $f_i = (f_i - f_{i-1}) / (x_i - x_{i-k})$ ;

```

has been performed, the array f contains the divided differences needed for the Newton form of the interpolating polynomial, so

$$p_n = f_0 + f_1(x - x_0) + f_2(x - x_0)(x - x_1) + \dots + f_n(x - x_0) \cdots (x - x_{n-1}). \quad (26)$$

Note that this algorithm has two nested for-loops, so the number of subtractions is

$$\sum_{k=1}^n \sum_{i=k}^n 2 = \sum_{k=1}^n 2(n - k + 1) = 2 \sum_{k=1}^n k = n(n + 1) = n^2 + n$$

which follows from the formula for the sum on the first n integers. We note that this grows with the square of the degree n , which is a consequence of the double for-loop. This is much faster than linear growth, which is what we would have if there was only the outer for-loop. However, for this problem the quadratic growth is not usually a problem since the degree tends to be low — rarely more than 10. If one has more points than this the general advice is to use some other approximation method which avoids high degree polynomials since these are also likely to lead to considerable rounding-errors.

The second algorithm that is needed is evaluation of the interpolation polynomial (26). Let us consider a specific example,



$$p_3(x) = f_0 + f_1(x - x_0) + f_2(x - x_0)(x - x_1) + f_3(x - x_0)(x - x_1)(x - x_2). \quad (27)$$

Given a number x , there is an elegant algorithm for computing the value of the polynomial which is based on rewriting (27) slightly as

$$p_3(x) = f_0 + (x - x_0) \left(f_1 + (x - x_1) (f_2 + (x - x_2) f_3) \right). \quad (28)$$

To compute $p_3(x)$ we start from the inner-most parenthesis and then repeatedly multiply and add,

$$s_3 = f_3,$$

$$s_2 = (x - x_2) s_3 + f_2,$$

$$s_1 = (x - x_1) s_2 + f_1,$$

$$s_0 = (x - x_0) s_1 + f_0.$$

After this we see that $s_0 = p_3(x)$. This can easily be generalized to a more formal algorithm. Note that there is no need to keep the different s_i -values; we can just use one variable s and accumulate the calculations in this.

Algorithm (Horner's rule). Let $x_0, \dots, x_n$ be given numbers,

and let $(f_k)_{k=0}^n$ be the coefficients of the polynomial

$$p_n(x) = f_0 + f_1(x - x_0) + \dots + f_n(x - x_0) \cdots (x - x_{n-1}).$$

(29)

After the code

```
s = fn;
for k = n - 1, n - 2, ... 0
    s = (x - xk) * s + fk;
```

the variable s will contain the value of $p_n(x)$.



REMEMBER

Formally, if $r(x)$ is any non-zero polynomial, it must be writable as $r(x) = A(x - x_0)(x - x_1) \cdots (x - x_n)$, for some constant A .

2.5 INTERPOLATION ERROR

 The interpolating polynomial p_n is an approximation to f , but unless f itself is a polynomial of degree n , there will be a nonzero error $e(x) = f(x) - p_n(x)$. At times it is useful to have an explicit expression for the error.

LEARNING OBJECTIVE

 5. Understand the interpolation error

Theorem. Suppose f is interpolated by a polynomial of degree n at $n + 1$ distinct points $x_0, \dots, x_n$. Let $[a, b]$ be the smallest interval that contains all the interpolation points as well as the number x , and suppose that the function f has continuous derivatives up to order $n + 1$ in $[a, b]$. Then the error $e(x) = f(x) - p_n(x)$ is given by

$$e(x) = f[x_0, \dots, x_n, x](x - x_0) \cdots (x - x_n) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!} (x - x_0) \cdots (x - x_n), \quad (30)$$

where ξ_x is a number in the interval (a, b) that depends on x .

Proof. The second equality in (30) follows from (23), so our job is to prove the first equality. For this we add the (arbitrary) number x as an interpolation point and consider interpolation with a polynomial of degree $n + 1$ at the points $x_0, \dots, x_n, x$. We use t as the free variable to avoid confusion with x . Then we know that

$$p_{n+1}(t) = p_n(t) + f[x_0, \dots, x_n, x](t - x_0) \cdots (t - x_n).$$

Since p_{n+1} interpolates f at $t = x$ we have $p_{n+1}(x) = f(x)$ so

$$f(x) = p_n(x) + f[x_0, \dots, x_n, x](x - x_0) \cdots (x - x_n)$$

which proves the first relation in (30).

The error term in (30) is very similar to the error term in the Taylor expansion (8). A natural question to ask is therefore: Which approximation method will give the smallest error, Taylor expansion or interpolation? Since the only essential difference between the two error terms is the factor $(x - a)^{n+1}$ in the Taylor case and the factor $(x - x_0) \cdots (x - x_n)$ in the interpolation case, a reasonable way to compare the methods is to compare the two polynomials $(x - a)^{n+1}$ and $(x - x_0) \cdots (x - x_n)$.

In reality, we do not just have two approximation methods but infinitely many, since there are infinitely many ways to choose the interpolation points. In figure 2.8 we compare the two most obvious choices in the case $n = 3$ for the interval $[0, 3]$: Taylor expansion about the midpoint $a = 3/2$ and interpolation at the integers 0, 1, 2, and 3. In the Taylor case, the polynomial $(x - 3/2)^4$ is nonnegative and small in the interval $[1,$



2], but outside this interval it grows quickly and soon becomes larger than the polynomial $x(x-1)(x-2)(x-3)$ corresponding to interpolation at the integers. We have also included a plot of a third polynomial which corresponds to the best possible interpolation points in the sense that the maximum value of this polynomial is as small as possible in the interval $[0, 3]$, given that its leading **coefficient** should be 1.

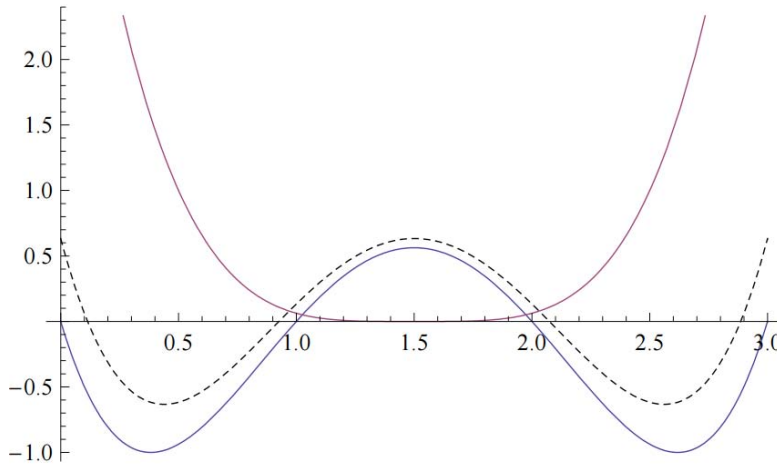


Figure 2.8: The solid, nonnegative graph is the polynomial factor $(x - 3/2)^4$ in the error term for Taylor expansion of degree 3 about the $a = 3/2$, while the other solid graph is the polynomial part $x(x-1)(x-2)(x-3)$ of the error term for interpolation at 0, 1, 2, 3. The dashed graph is the smallest possible polynomial part of an error terms for interpolation at 4 points in $[0, 3]$.

If used sensibly, polynomial interpolation will usually provide a good approximation to the underlying data. As the distance between the data points decreases, either by increasing the number of points or by moving the points closer together, the approximation can be expected to become better. However, we saw that there are functions for which Taylor approximation does not work well, and the same may happen with interpolation. As for Taylor approximation, the problem arises when the derivatives of the function to be approximated become large. A famous example is the so-called Runge function $1/(1+x^2)$ on the interval $[-5, 5]$. Figure 2.9 shows the interpolants for degree 10 and degree 20. In

KEY TERMS

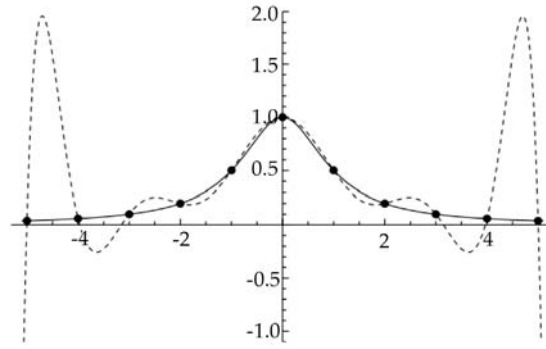
Coefficient is a multiplicative factor in some term of a polynomial, a series or any expression; it is usually a number, but may be any expression.



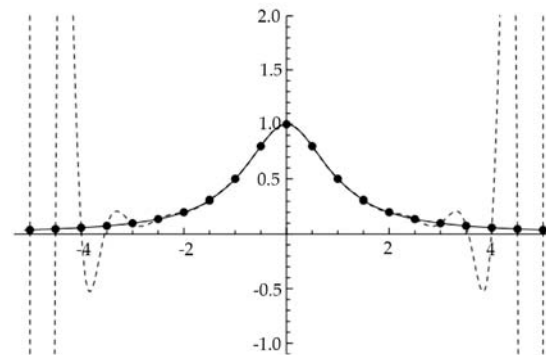
REMEMBER

Runge's phenomenon shows that for high values of n , the interpolation polynomial may oscillate wildly between the data points. This problem is commonly resolved by the use of spline interpolation.

the middle of the interval, the error becomes smaller when the degree is increased, but towards the ends of the interval the error becomes larger when the degree increases.



(a)



(b)

Figure 2.9: Interpolation of the function $f(x) = 1/(1+x^2)$ on the interval $[-5, 5]$ with polynomials of degree 10 in (a), and degree 20 in (b). The points are uniformly distributed in the interval in each case.





ROLE MODEL

JAMES GREGORY: SCOTTISH MATHEMATICIAN AND ASTRONOMER

James Gregory is the son of Gregory and Janet pair John Anderson was born in a small town, Drumoak, about 15 miles from Aberdeen, Scotland. John Gregory was an abbot at Drumoak because of her educational background in the field of theology and graduated from St. Joseph's University. Andrews. Her sister, Alexander Anderson was editing work, and once a student Viete Viete while studying in Paris. Gregory is the youngest child who has two older brothers, Alexander and David. Age differences with Gregory David was ten years old.

Early recognition of mathematical Gregory is the mother who taught him geometry. But when she was 13 years old, his father died and the task of educating left to David Gregory. David gave the book Elements of Euclid's work to learn her sister, who quickly mastered. Before entering the university had studied at Marischal College in Aberdeen. Later, through connections David, Gregory can get acquainted with John Collins (1625-1683), pusatakawan Royal Society. Collins was also a mathematician with pleasure as [Marin] Mersenne in France, namely the central correspondence other mathematicians.

Making a telescope

Although Gregory weak health conditions, but this does not prevent him to study fields other than mathematics. Studying Optiks and build telescopes are areas of concern. With the encouragement of an older brother, David, Gregory wrote a book Optima Promota postulates that contains 5, 37 definitions and 59 theorem (systematics similar to the book Elements of Euclid) of the theory of reflection and the reflection of light.

Light theory presented in the book is used as a basis for making a telescope that has the effect of reflection. By using a mirror konkave (convex / concave) parabolic capable of making light converged at one focus of the mirror konkave elipsodial. Refelsi light coming from the surface will be converged at the second focus located behind the mirror.

There is a hole in the primary mirror is made so that light can pass through and the dinar which may be captured by the lens of the eye. Gregorian telescope's tube to be shorter than the total width of the focal points of the second mirror. The idea to use mirrors and lenses to telescopes is new, and it turned out this way more effective

than using mirrors or lenses separately. Ways of making it a model telescope can not be done.

In 1663, Gregory went to London. Met with Collins and a true friend. Through Optima Promota Collins book can be published and create a telescope design in the book. Collins suggested that Gregory met optician named Reive, but again failed. Gregorian telescope is, ultimately, can be made by Hooke (read: Newton and Halley) ten years later. While in London, Gregory met with the President of the Royal Society, Robert Moray, who then tried to bring Gregory to Huygens in Paris, because it has the same interests (read: Huygens).

Working in Italy

As late as 1663, Gregory went to Italy and became acquainted with the successors of Torricelli, especially Stefano Angeli. Angeli's works include infinitesimal method with an emphasis on the quadratic equation applicable to the form of a spiral, parabola and hyperbola and tried to find breadth apparently appealed to Gregory and immediately learn the Angeli in the period 1664 - 1668 at the University of Padua. Keep learning apparently captivated Gregory then study two variations of the beginning of calculus, methods of tangents (differentiation) and squaring (integration).

In Padua, Gregory stayed home philosophy professor, Caddenhead from Scotland. Their cooperation to fruition, namely *Vera circuli circuli et hyperbolae quadratura* (1667) and *geometriae pars universalis* (1668) before returning to the UK.

Through the two posts above, Gregory gave an important basis for the geometry infinitesimal that would become very important. More than a decade later, when the analysis was progressing very quickly, before it was completed by the discovery of the next mathematician, Newton and Leibniz (including Huygens, Barrow). The work was also berusaha prove that π and e are transcendental numbers, but the reasons cited Gregory is still wrong, but the major breakthrough idea: convergent, determining function, aljabarik functions, transcendental functions, and others.

Two works of this magnitude, the contents of which are the findings, published simultaneously in France, Italy, the Netherlands and the UK. His first book Cartesian remodel that still distinguish between the curves "geometrical: and" mechanical ". Gregory would rather divide mathematics into the theorem 'general' and unproved 'specials', not divided into functions of algebraic and transcendental.

Conflict with the Huygens

Gregory returned to London in mid-1668, right into St. Andrew, and send a copy of the book *Vera calculi circuli et hyperbolae quadratura* he sent a letter to ask for feedback from Huygens. Instead of responding, Huygens published a review of the book was published in July 1668. Mentioned that he had developed some findings and stated



that he was the person who first gave evidence against the results. Knowing Huygens comments discrediting himself makes Gregory mentioned that the ideas were stolen by the Huygens without notice.

Relations between the two mathematicians lead to conflict, although at that time the conflict between common mathematicians. It is just like the conflict between Newton and Leibniz that occurred during this period as well. This conflict can never be reconciled because each stick with their respective establishments. Gregory prove himself and verification conducted by Huygens is also valid. In the 1930's, Turnbull conducted a study of the papers in the library of St. Gregory's work. Andrew. Finally, the findings stated that Gregory is genuine and can be said to be brilliant for the period.

Gregory Series

Towards the end of 1668, Gregory completed what is known as the series sin, cos, and tg. Gregory gave the formula that:

$$\int \sec x \, dx = \ln (\sec x + \tan x)$$

This formula is used to solve the classical problem in manufacturing tables for a cruise. Issuing Exercitationes Geometricae as counter-attack (counter-attack) on Huygens. Although the methods used are not disclosed, the booklet includes many series, the logarithmic integral function and other ideas that are related. One series is called by the name of Gregory in a series like this.

$$\int \frac{dx}{1+x^2} = \arctg x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

Gregory found separately with Newton, binomial theorem for fractional rank, the result was known earlier by Newton (but not yet published), and through various processes of differentiation, which was previously known as the Taylor series. Developed so well known Maclaurin series for sec x and tg x and arctg arc sec x and x., Before being formulated by Cavalieri into the above formula.

Do some research

At St. Andrew, Gregory taught only two times in one week and the time used for tasks metematikal and astronomical. Recent research has always presented to Collins by letter. Correspondence for 6 years, until his death, between Gregory and Collins is still stored in university libraries St. Andrew to find out how the leading mathematicians of the time of discovery.

Send the book by Collins [Isaac] Barrow to Gregory developed the idea of the book before it is sent to Collins. Year 1671, Gregory discovered Taylor theorem (never published by Taylor until the year 1715), in which the theorem is contained in the letter

to Collins. Nevertheless Collins a letter stating that Newton also found similar results, Gregory decided that Newton published his findings before he himself published. Apparently conflicts with the Huygens still made an impression on his heart and did not want it to happen between him and Newton as well.

In the library of St. atad space. Andrew is not blocked by any object, Gregory put the telescope. Pendulum clocks hang next to a window for observation. Pendulum clock purchased in London in 1673, but the idea had been patented by Huygens in 1656, before Huygens wrote pendulum theory.

Moving to Edinburgh

In 1674, Gregory collaborate with colleagues from Paris to simultaneously observe the lunar eclipse. Only a year earlier, Gregory allowed by the university to purchase equipment for the observatory, and brought Gregory to raise funds to build an observatory. Returned to Aberdeen and knocked on the door of the church to help build the observatory. First obtaining the advice of Flamsteed, the astronomer royal, before start buying equipment observation.

Gregory left St. Andrew to Edinburgh in 1674. Aasan move back because jealousy is triggered by other mathematicians. In Edinbergh, Gregory became the first to occupy the chair of mathematics department there. This is not a long-held position, because it is almost a year later Gregory died.

At the time of this death, Gregory was still actively doing research in the field of astronomy and mathematics. In mathematics, Gregory was peeling a cubic equation to solve the problem within five (quintic) in algebra and find interesting things on Diophantine problems. Died suddenly. Night while observing satellites of Jupiter with his students by using telescopes, suddenly had a stroke and went blind. A few days later Gregory died young, aged 36 years.

Contribution

Pursue complementary fields, mathematics and astronomy, without losing focus. Coming to his light and initiated, although crude, the basics of what was later known as calculus. Starting as an attempt to calculate the area of irregular areas such as parabolic, hyperbolic, but it is not alleged to be the forerunner of calculus.



SUMMARY

1. Polynomials in particular are very appealing for use on a computer since the value of a polynomial at a point can be computed by utilizing simple operations like addition and multiplication that computers can perform extremely quickly.
2. A Taylor polynomial is a simple approximation to a function that is based on information about the function at a single point only.
3. The Taylor polynomial itself, and the error, the remainder, committed in approximating a function by a polynomial.
4. The coefficients $c_k = f[x_0, \dots, x_k]$ have certain properties that are useful both for computation and understanding.
5. Our use of polynomial interpolation will primarily be as a tool for developing numerical methods for differentiation, integration, and solution of differential equations.
6. The interpolating polynomial p_n is an approximation to f , but unless f itself is a polynomial of degree n , there will be a nonzero error $e(x) = f(x) - p_n(x)$. At times it is useful to have an explicit expression for the error.

KNOWLEDGE CHECK

1. If a polynomial of degree n has more than n zeros, then the polynomial is
 - a. oscillatory
 - b. zero everywhere
 - c. quadratic
 - d. not defined
2. Image linear interpolation is given by formula
 - a. $v(x,y) = ax+by+cxy+d$.
 - b. $v(x,y) = ax+by+cxy$.
 - c. $v(x,y) = ax+by+d$.
 - d. $v(x,y) = by+cxy+d$.
3. The following x - y data is given

x	15	18	22
y	24	37	25

The Newton's divided difference second order polynomial for the above data is given by

$$f_2(x) = b_0 + b_1(x-15) + b_2(x-15)(x-22)$$

The value of b_1 is

- a. -1.048
- b. 0.1433
- c. 4.333
- d. 24.00

4. The following data of the velocity of a body is given as a function of time.

Time (s)	0	15	18	22	24
Velocity (m/s)	22	24	37	25	123

If you were going to use quadratic interpolation to find the value of the velocity at seconds, the three data points of time you would choose for interpolation are

- a. 0, 15, 18
- b. 15, 18, 22
- c. 0, 15, 22
- d. 0, 18, 24

5. Polynomial interpolation is a method of estimating values between known data points.

- a. True
- b. False

6. The polynomial that passes through the following x - y data

x	18	22	24
y	?	25	123

is given by

$$8.125x^2 - 324.75x + 3237, 18 \leq x \leq 24.$$

The corresponding polynomial using Newton's divided difference polynomial is given by

$$f_2(x) = b_0 + b_1(x-18) + b_2(x-18)(x-22)$$

The value of b_2 is



- a. 0.2500
 - b. 8.125
 - c. 24.00
 - d. not obtainable with the information given
7. **Velocity vs. time data for a body is approximated by a second order Newton's divided difference polynomial as**

$$v(t) = b_0 + 39.622(t - 20) + 0.5540(t - 20)(t - 15), 10 \leq t \leq 20$$
- The acceleration in m/s^2 at is
- a. 0.5540 m/s^2
 - b. 39.622 m/s^2
 - c. 36.852 m/s^2
 - d. not obtainable with the given information
8. **The Taylor polynomial $T_n(f)$ is an approximation to f , and in many situations it will be important to control the error in the approximation.**
- a. True
 - b. False

Check Your Result

- | | | | | | |
|--------|--------|--------|--------|--------|--------|
| 1. (a) | 2. (c) | 3. (a) | 4. (a) | 5. (a) | 6. (b) |
| 7. (c) | 8. (a) | | | | |

REVIEW QUESTIONS

1. Find the second order Taylor approximation of the following functions at the given point a .
 - a) $f(x) = x^3, a = 1$
 - b) $f(x) = 12x^2 + 3x + 1, a = 0$
 - c) $f(x) = 2^x, a = 0$
2. Determine a polynomial approximation of the function $\sin x$ on the interval $[-1, 1]$ with error smaller than 10^{-5} .
3. In many contexts, the approximation $\sin x \approx x$ is often used. Explain why this approximation is reasonable.

- a) Estimate the error in the approximation for x in the interval $[0, 0.1]$
4. Define the Newton form of the interpolating polynomial.
5. Construct a divided difference table for the following:

$x:$	1	2	4	7	12
$f(x):$	22	30	82	106	216.

6. a) The data

x	0	1	2	3
$f(x)$	0	1	4	9

is sampled from the function $f(x) = x^2$. Determine the third order divided difference $f[0, 1, 2, 3]$.

- b) Explain why we always have $f[x_0, \dots, x_n] = 0$ if f is a polynomial of degree at most $n - 1$.

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Chapter

3

NUMERICAL INTEGRATION AND DIFFERENTIATION

LEARNING OBJECTIVES

After studying this chapter, you will be able to:

1. Define the numerical differentiation
2. Explain the numerical integration

INTRODUCTION

Consider a function of a single variable $y = f(x)$. If $f(x)$ is defined as an expression, its derivative or integral may often be determined using the techniques of calculus.

However, when $f(x)$ is a complicated function or when it is given in a tabular form, numerical methods are used.

The numerical methods for approximating the derivative(s) $f^{(r)}(x)$, $r \geq 1$ of a given function $f(x)$ and for the evaluation of the integral $\int_a^b f(x) dx$ where a , b may be finite or infinite.

The accuracy attainable by these methods would depend on the given function and the order of the polynomial used. If the polynomial fitted is exact then the error would be, theoretically, zero. In practice, however, rounding errors will introduce errors in the calculated values. The error introduced in obtaining derivatives is, in general, much worse than that introduced in determining integrals. It may be observed that any errors in approximating a function are amplified while taking the derivative whereas they are smoothed out in integration.

3.1 NUMERICAL DIFFERENTIATION

LEARNING OBJECTIVE

1. Define the numerical differentiation

KEY TERMS

Interpolation is a method of constructing new data points within the range of a discrete set of known data points.

 In the case of numerical data, the functional form of $f(x)$ is not known in general. First we have to find an appropriate form of $f(x)$ and then obtain its derivatives. So “Numerical Differentiation” is concerned with the method of finding the successive derivatives of a function at a given argument, using the given table of entries corresponding to a set of arguments, equally or unequally spaced. Using the theory of **interpolation**, a suitable interpolating polynomial can be chosen to represent the function to a good degree of approximation in the given interval of the argument.

The criterion is the same as in the case of interpolation problems. In the case of equidistant values of x , if the derivative is to be found at a point near the beginning or the end of the given set of values, Newton’s forward or backward difference formula should be used accordingly. Also if the derivative is to be found at a point near the middle of the given set of values, then any one of the central difference formulae should be used. However, if the values of the function are not known at equidistant values of x , Newton’s divided difference or Lagrange’s formula should be used.



3.1.1 Formulae for Derivatives

(1) Newton's forward difference interpolation formula is

$$y = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \dots \quad (1)$$

where

$$u = \frac{x-a}{h} \quad (2)$$

Differentiating equation (1) with respect to u, we get

$$\frac{dy}{du} = \Delta y_0 + \frac{2u-1}{2} \Delta^2 y_0 + \frac{3u^2-6u+2}{6} \Delta^3 y_0 + \dots \quad (3)$$

Differentiating equation (2) with respect to x, we get

$$\frac{du}{dx} = \frac{1}{h} \quad (4)$$

We know that

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{1}{h} \left[\Delta y_0 + \left(\frac{2u-1}{2} \right) \Delta^2 y_0 + \left(\frac{3u^2-6u+2}{6} \right) \Delta^3 y_0 + \dots \right] \quad (5)$$

Expression (5) provides the value of $\frac{dy}{dx}$ at any x which is not tabulated. Formula (5) becomes simple for tabulated values of x, in particular when $x = a$ and $u = 0$. Putting $u = 0$ in (5), we get

$$\left(\frac{dy}{dx} \right)_{x=a} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \frac{1}{5} \Delta^5 y_0 - \dots \right] \quad (6)$$

Differentiating equation (5) with respect to x, we get

$$\begin{aligned}
\frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{du} \left(\frac{dy}{dx} \right) \frac{du}{dx} \\
&= \frac{1}{h} \left[\Delta^2 y_0 + (u-1) \Delta^3 y_0 + \left(\frac{6u^2 - 18u + 11}{12} \right) \Delta^4 y_0 + \dots \right] \frac{1}{h} \\
&= \frac{1}{h^2} \left[\Delta^2 y_0 + (u-1) \Delta^3 y_0 + \left(\frac{6u^2 - 18u + 11}{12} \right) \Delta^4 y_0 + \dots \right]
\end{aligned} \tag{7}$$

Putting $u = 0$ in (7), we get

$$\left(\frac{d^2 y}{dx^2} \right)_{x=a} = \frac{1}{h^2} \left(\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 + \dots \right) \tag{8}$$

Similarly, we get

$$\left(\frac{d^3 y}{dx^3} \right)_{x=a} = \frac{1}{h^3} \left(\Delta^3 y_0 - \frac{3}{2} \Delta^4 y_0 + \dots \right) \tag{9}$$

and so on.

Formulae for computing higher derivatives may be obtained by successive differentiation.

(2) Newton's backward difference interpolation formula is

$$y = y_n + u \nabla y_n + \frac{u(u+1)}{2!} \nabla^2 y_n + \frac{u(u+1)(u+2)}{3!} \nabla^3 y_n + \dots \tag{10}$$

where

$$u = \frac{x - x_n}{h} \tag{11}$$

Differentiating (10) with respect to, u , we get

$$\frac{dy}{du} = \nabla y_n + \left(\frac{2u+1}{2} \right) \nabla^2 y_n + \left(\frac{3u^2 + 6u + 2}{6} \right) \nabla^3 y_n + \dots \tag{12}$$



Differentiating (11) with respect to x , we get

$$\frac{du}{dx} = \frac{1}{h} \quad (13)$$

Now,

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= \frac{1}{h} \left[\nabla y_n + \left(\frac{2u+1}{2} \right) \nabla^2 y_n + \left(\frac{3u^2+6u+2}{6} \right) \nabla^3 y_n + \dots \right] \end{aligned} \quad (14)$$

Expression (14) provides us the value of $\frac{dy}{dx}$ at any x which is not tabulated.

At $x = x_n$, we have $u = 0$

$\therefore$ Putting $u = 0$ in (14), we get

$$\left(\frac{dy}{dx} \right)_{x=x_n} = \frac{1}{h} \left(\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n + \dots \right) \quad (15)$$

Differentiating (14) with respect to x , we get

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{du} \left(\frac{dy}{dx} \right) \frac{du}{dx} \\ &= \frac{1}{h^2} \left[\nabla^2 y_n + (u+1) \nabla^3 y_n + \left(\frac{6u^2+18u+11}{12} \right) \nabla^4 y_n + \dots \right] \end{aligned} \quad (16)$$

Putting $u = 0$ in (16), we get

$$\left(\frac{d^2 y}{dx^2} \right)_{x=x_n} = \frac{1}{h^2} \left(\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \dots \right) \quad (17)$$

Similarly, we get

$$\left(\frac{d^3 y}{dx^3} \right)_{x=x_n} = \frac{1}{h^3} \left(\nabla^3 y_n + \frac{3}{2} \nabla^4 y_n + \dots \right) \quad (18)$$

and so on.

Formulae for computing higher derivatives may be obtained by successive differentiation.

(3) Stirling's central difference interpolation formula is

$$y = y_0 + \frac{u}{1!} \left(\frac{\Delta y_0 + \Delta y_{-1}}{2} \right) + \frac{u^2}{2!} \Delta^2 y_{-1} + \frac{u(u^2 - 1^2)}{3!} \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) \\ + \frac{u^2(u^2 - 1^2)}{4!} \Delta^4 y_{-2} + \frac{u(u^2 - 1^2)(u^2 - 2^2)}{5!} \left(\frac{\Delta^5 y_{-2} + \Delta^5 y_{-3}}{2} \right) + \dots \quad (19)$$

where

$$u = \frac{x - a}{h} \quad (20)$$

Differentiating equation (19) with respect to u , we get

$$\frac{dy}{du} = \frac{\Delta y_0 + \Delta y_{-1}}{2} + u \Delta^2 y_{-1} + \left(\frac{3u^2 - 1}{6} \right) \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) \\ + \left(\frac{4u^3 - 2u}{4!} \right) \Delta^4 y_{-2} + \left(\frac{5u^4 - 15u^2 + 4}{5!} \right) \left(\frac{\Delta^5 y_{-2} + \Delta^5 y_{-3}}{2} \right) + \dots \quad (21)$$

Differentiating (20) with respect to x , we get

$$\frac{du}{dx} = \frac{1}{h} \quad (22)$$

Now,

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \\ = \frac{1}{h} \left[\frac{\Delta y_0 + \Delta y_{-1}}{2} + u \Delta^2 y_{-1} + \left(\frac{3u^2 - 1}{6} \right) \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) \right. \\ \left. + \left(\frac{4u^3 - 2u}{4!} \right) \Delta^4 y_{-2} + \left(\frac{5u^4 - 15u^2 + 4}{5!} \right) \left(\frac{\Delta^5 y_{-2} + \Delta^5 y_{-3}}{2} \right) + \dots \right] \quad (23)$$

Expression (23) provides the value of $\frac{dy}{dx}$ at any x which is not tabulated.
Given $x = a$, we have $u = 0$

$\therefore$ Given $u = 0$ in (23), we get

$$\left(\frac{dy}{dx}\right)_{x=a} = \frac{1}{h} \left[\left(\frac{\Delta y_0 + \Delta y_{-1}}{2} \right) - \frac{1}{6} \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) + \frac{1}{30} \left(\frac{\Delta^5 y_{-2} + \Delta^5 y_{-3}}{2} \right) - \dots \right] \quad (24)$$

Differentiating (23) with respect to x , we get

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{du} \left(\frac{dy}{dx} \right) \frac{du}{dx} \\ &= \frac{1}{h^2} \left[\Delta^2 y_{-1} + u \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) + \left(\frac{6u^2 - 1}{12} \right) \Delta^4 y_{-2} \right. \\ &\quad \left. + \left(\frac{2u^3 - 3u}{12} \right) \left(\frac{\Delta^5 y_{-2} + \Delta^5 y_{-3}}{2} \right) + \dots \right] \end{aligned} \quad (25)$$

Given $u = 0$ in (25), we get

$$\left(\frac{d^2 y}{dx^2}\right)_{x=a} = \frac{1}{h^2} \left(\Delta^2 y_{-1} - \frac{1}{12} \Delta^4 y_{-2} + \frac{1}{90} \Delta^6 y_{-3} - \dots \right) \quad (26)$$

and so on.

Formulae for computing higher derivatives may be obtained by successive differentiation.

(4) Bessel's central difference interpolation formula is

$$\begin{aligned} y &= \left(\frac{y_0 + y_1}{2} \right) + \left(u - \frac{1}{2} \right) \Delta y_0 + \frac{u(u-1)}{2!} \left(\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} \right) \\ &\quad + \frac{u(u-1) \left(u - \frac{1}{2} \right)}{3!} \Delta^3 y_{-1} + \frac{(u+1) u(u-1)(u-2)}{4!} \left(\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} \right) \\ &\quad + \frac{(u+1) u(u-1)(u-2) \left(u - \frac{1}{2} \right)}{5!} \Delta^5 y_{-2} \\ &\quad + \frac{(u+2)(u+1) u(u-1)(u-2)(u-3)}{6!} \left(\frac{\Delta^6 y_{-3} + \Delta^6 y_{-2}}{2} \right) + \dots \end{aligned} \quad (27)$$

where

$$u = \frac{x - a}{h} \quad (28)$$

Differentiating equation (27) with respect to u , we get

$$\begin{aligned} \frac{dy}{du} = & \Delta y_0 + \left(\frac{2u-1}{2!} \right) \left(\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} \right) + \left(\frac{3u^2 - 3u + \frac{1}{2}}{3!} \right) \Delta^3 y_{-1} \\ & + \left(\frac{4u^3 - 6u^2 - 2u + 2}{4!} \right) \left(\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} \right) + \left(\frac{5u^4 - 10u^3 + 5u - 1}{5!} \right) \Delta^5 y_{-2} \\ & + \left(\frac{6u^5 - 15u^4 - 20u^3 + 45u^2 + 8u - 12}{6!} \right) \left(\frac{\Delta^6 y_{-3} + \Delta^6 y_{-2}}{2} \right) + \dots \end{aligned}$$

Differentiating (28) with respect to x , we get

$$\frac{du}{dx} = \frac{1}{h}$$

Now,

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= \frac{1}{h} \left[\Delta y_0 + \left(\frac{2u-1}{2!} \right) \left(\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} \right) + \left(\frac{3u^2 - 3u + \frac{1}{2}}{3!} \right) \Delta^3 y_{-1} \right. \\ &+ \left(\frac{4u^3 - 6u^2 - 2u + 2}{4!} \right) \left(\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} \right) + \left(\frac{5u^4 - 10u^3 + 5u - 1}{5!} \right) \Delta^5 y_{-2} \\ &\left. + \left(\frac{6u^5 - 15u^4 - 20u^3 + 45u^2 + 8u - 12}{6!} \right) \left(\frac{\Delta^6 y_{-3} + \Delta^6 y_{-2}}{2} \right) + \dots \right] \quad (29) \end{aligned}$$

Expression (30) provides us the value of $\frac{dy}{dx}$ at any x which is not tabulated.

Given $x = a$, we have $u = 0$

$\therefore$ Given $u = 0$ in (30), we get

$$\begin{aligned} \left(\frac{dy}{dx}\right)_{x=a} &= \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \left(\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} \right) + \frac{1}{12} \Delta^3 y_{-1} + \frac{1}{12} \left(\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} \right) \right. \\ &\quad \left. - \frac{1}{120} \Delta^5 y_{-2} - \frac{1}{60} \left(\frac{\Delta^6 y_{-3} + \Delta^6 y_{-2}}{2} \right) + \dots \right] \end{aligned} \quad (30)$$

Differentiating (30) with respect to x, we get

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{du} \left(\frac{dy}{dx} \right) \frac{du}{dx} \\ &= \frac{1}{h^2} \left[\left(\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} \right) + \left(\frac{2u-1}{2} \right) \Delta^3 y_{-1} + \left(\frac{6u^2-6u-1}{12} \right) \left(\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} \right) \right. \\ &\quad \left. + \left(\frac{4u^3-6u^2+1}{24} \right) \Delta^5 y_{-2} \right. \\ &\quad \left. + \left(\frac{15u^4-30u^3-30u^2+45u+4}{360} \right) \left(\frac{\Delta^6 y_{-3} + \Delta^6 y_{-2}}{2} \right) + \dots \right] \end{aligned} \quad (32)$$

Given $u = 0$ in (32), we get

$$\begin{aligned} \left(\frac{d^2 y}{dx^2}\right)_{x=a} &= \frac{1}{h^2} \left[\left(\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} \right) - \frac{1}{2} \Delta^3 y_{-1} - \frac{1}{12} \left(\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} \right) \right. \\ &\quad \left. + \frac{1}{24} \Delta^5 y_{-2} + \frac{1}{90} \left(\frac{\Delta^6 y_{-3} + \Delta^6 y_{-2}}{2} \right) + \dots \right] \end{aligned} \quad (33)$$

and so on

(5) For unequally spaced values of the argument

(i) Newton's divided difference formula is

$$\begin{aligned} f(x) &= f(x_0) + (x-x_0) \Delta f(x_0) + (x-x_0)(x-x_1) \Delta^2 f(x_0) + (x-x_0)(x-x_1) \\ &\quad (x-x_2) \Delta^3 f(x_0) + (x-x_0)(x-x_1)(x-x_2)(x-x_3) \Delta^4 f(x_0) + \dots \end{aligned} \quad (34)$$

$f'(x)$ is given by

$$\begin{aligned} f'(x) &= \Delta f(x_0) + \{2x - (x_0 + x_1)\} \Delta^2 f(x_0) + \{3x^2 - 2x(x_0 + x_1 + x_2) \\ &\quad + (x_0 x_1 + x_1 x_2 + x_2 x_0)\} \Delta^3 f(x_0) + \dots \end{aligned} \quad (35)$$

(ii) Lagrange's interpolation formula is

$$f(x) = \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} f(x_0) + \frac{(x-x_0)(x-x_2)\dots(x-x_n)}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)} f(x_1) + \dots \quad (36)$$

$f'(x)$ can be obtained by differentiating $f(x)$ in equation (36).

3.1.2 Maxima and Minima of a Tabulated Function

Since maxima and minima of $y = f(x)$ can be found by equating $\frac{dy}{dx}$ to zero and solving the equation for the argument x , the same method can be used to determine maxima and minima of tabulated function by differentiating the interpolating polynomial.

For example, if Newton's forward difference formula is used, we have

$$y = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \dots \quad (37)$$

Differentiating (37) with respect to u , we get

$$\frac{dy}{du} = \Delta y_0 + \frac{2u-1}{2!} \Delta^2 y_0 + \frac{3u^2-6u+2}{3!} \Delta^3 y_0 + \dots$$

For maxima or minima,

$$\frac{dy}{du} = 0$$

$$\Rightarrow \Delta y_0 + \frac{2u-1}{2!} \Delta^2 y_0 + \frac{3u^2-6u+2}{3!} \Delta^3 y_0 + \dots = 0 \quad (38)$$

If we terminate L.H.S. series after third differences for convenience, equation (38) being a quadratic in u gives two values of u . Corresponding to these values, $x = a + uh$ will give the corresponding x at which function may be maximum or minimum.

For maximum, $\frac{d^2y}{du^2} = (-)\text{ve}$

For minimum, $\frac{d^2y}{du^2} = (+)\text{ve}$.

Example

Find $\frac{dy}{dx}$ at $x = 0.1$ from the following table:

$x:$	0.1	0.2	0.3	0.4
$y:$	0.9975	0.9900	0.9776	0.9604.



Sol. Take $h = 0.1$. The difference table is:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
0.1	0.9975			
0.2	0.9900	- 0.0075		
0.3	0.9776	- 0.0124	- 0.0049	
0.4	0.9604	- 0.0172	- 0.0048	0.0001

Here

$$h = 0.1 \quad \text{and} \quad y_0 = 0.9975$$

$$\begin{aligned}
 \left[\frac{dy}{dx} \right]_{x=0.1} &= \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 \right] \\
 &= \frac{1}{0.1} \left[-0.0075 - \frac{1}{2}(-0.0049) + \frac{1}{3}(0.0001) \right] \\
 &= -0.050167.
 \end{aligned}$$

3.2 NUMERICAL INTEGRATION

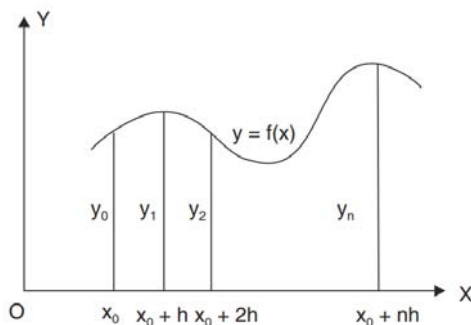
2. Given a set of tabulated values of the integrand $f(x)$, determining the value of $\int_{x_0}^{x_n} f(x) dx$ is called **numerical integration**. The given interval of integration is subdivided into a large number of subintervals of equal width h and the function tabulated at the points of subdivision is replaced by any one of the interpolating polynomials like Newton-Gregory's, Stirling's, Bessel's over each of the subintervals and the integral is evaluated. There are several formulae for numerical integration which we shall derive in the sequel.

LEARNING OBJECTIVE

2. Explain the numerical integration

KEY TERMS

Numerical integration is the approximate computation of an integral using numerical techniques.



3.2.1 Newton-Cote's Quadrature Formula

Let $I = \int_a^b y \, dx$, where y takes the values $y_0, y_1, y_2, \dots, y_n$ for $x = x_0, x_1, x_2, \dots, x_n$.

Let the interval of integration (a, b) be divided into n equal sub-intervals, each of width $h = \frac{b-a}{n}$ so that

$$x_0 = a, x_1 = x_0 + h, x_2 = x_0 + 2h, \dots, x_n = x_0 + nh = b.$$

$$\therefore I = \int_{x_0}^{x_0 + nh} f(x) \, dx$$

Since any x is given by $x = x_0 + rh$ and $dx = h \, dr$

$$\begin{aligned} \therefore I &= h \int_0^n f(x_0 + rh) \, dr \\ &= h \int_0^n \left[y_0 + r \Delta y_0 + \frac{r(r-1)}{2!} \Delta^2 y_0 + \frac{r(r-1)(r-2)}{3!} \Delta^3 y_0 + \dots \right] dr \\ &\quad \text{[by Newton's forward interpolation formula]} \\ &= h \left[r y_0 + \frac{r^2}{2} \Delta y_0 + \frac{1}{2} \left(\frac{r^3}{3} - \frac{r^2}{2} \right) \Delta^2 y_0 \right. \\ &\quad \left. + \frac{1}{6} \left(\frac{r^4}{4} - r^3 + r^2 \right) \Delta^3 y_0 + \dots \right]_0^n \end{aligned}$$

$$= nh \left[y_0 + \frac{n}{2} \Delta y_0 + \frac{n(2n-3)}{12} \Delta^2 y_0 + \frac{n(n-2)^2}{24} \Delta^3 y_0 + \dots \right] \quad (39)$$

This is a general quadrature formula and is known as Newton-Cote's quadrature formula. A number of important deductions viz. Trapezoidal rule, Simpson's one-third and three-eighth rules, **Weddle's rule** can be immediately deduced by putting $n = 1, 2, 3$, and 6 , respectively, in formula (39).

3.2.2 Trapezoidal Rule ($n = 1$)

Putting $n = 1$ in formula (39) and taking the curve through (x_0, y_0) and (x_1, y_1) as a polynomial of degree one so that differences of an order higher than one vanish, we get

$$\int_{x_0}^{x_0+h} f(x) dx = h \left(y_0 + \frac{1}{2} \Delta y_0 \right) = \frac{h}{2} [2y_0 + (y_1 - y_0)] = \frac{h}{2} (y_0 + y_1)$$

Similarly, for the next sub-interval $(x_0 + h, x_0 + 2h)$, we get

$$\int_{x_0+h}^{x_0+2h} f(x) dx = \frac{h}{2} (y_1 + y_2), \dots, \int_{x_0+(n-1)h}^{x_0+nh} f(x) dx = \frac{h}{2} (y_{n-1} + y_n)$$

Adding the above integrals, we get

$$\int_{x_0}^{x_0+nh} f(x) dx = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$$

which is known as **Trapezoidal rule**. By increasing the number of subintervals, thereby making h very small, we can improve the accuracy of the value of the given integral.

3.2.3 Simpson's One-Third Rule ($N = 2$)

Putting $n = 2$ in formula (39) and taking the curve through (x_0, y_0) , (x_1, y_1) and (x_2, y_2) as a polynomial of degree two so that differences of order higher than two vanish, we get

KEY TERMS

Weddle's Rule is a method of integration, the Newton-Cotes formula with $N=6$.

KEY TERMS

The trapezoidal rule is a method for approximating a definite integral by evaluating the integrand at two points.

$$\begin{aligned}
 \int_{x_0}^{x_0+2h} f(x) dx &= 2h \left[y_0 + \Delta y_0 + \frac{1}{6} \Delta^2 y_0 \right] \\
 &= \frac{2h}{6} [6y_0 + 6(y_1 - y_0) + (y_2 - 2y_1 + y_0)] \\
 &= \frac{h}{3} (y_0 + 4y_1 + y_2)
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 \int_{x_0+2h}^{x_0+4h} f(x) dx &= \frac{h}{3} (y_2 + 4y_3 + y_4), \dots, \\
 \int_{x_0+(n-2)h}^{x_0+nh} f(x) dx &= \frac{h}{3} (y_{n-2} + 4y_{n-1} + y_n)
 \end{aligned}$$

Adding the above integrals, we get

$$\begin{aligned}
 \int_{x_0}^{x_0+nh} f(x) dx &= \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \dots + y_{n-1}) \\
 &\quad + 2(y_2 + y_4 + \dots + y_{n-2})]
 \end{aligned}$$



REMEMBER

The given interval of integration must be divided into an even number of sub-intervals, since we find the area over two sub-intervals at a time.

which is known as Simpson's one-third rule.

While using this formula.

3.2.4 Simpson's Three-Eighth Rule (n = 3)

Putting $n = 3$ in formula (39) and taking the curve through (x_0, y_0) , (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) as a polynomial of degree three so that differences of order higher than three vanish, we get

$$\begin{aligned}
 \int_{x_0}^{x_0+3h} f(x) dx &= 3h \left(y_0 + \frac{3}{2} \Delta y_0 + \frac{3}{4} \Delta^2 y_0 + \frac{1}{8} \Delta^3 y_0 \right) \\
 &= \frac{3h}{8} [8y_0 + 12(y_1 - y_0) + 6(y_2 - 2y_1 + y_0) + (y_3 - 3y_2 + 3y_1 - y_0)] \\
 &= \frac{3h}{8} [y_0 + 3y_1 + 3y_2 + y_3]
 \end{aligned}$$

Similarly,

$$\int_{x_0+3h}^{x_0+6h} f(x) dx = \frac{3h}{8} [y_3 + 3y_4 + 3y_5 + y_6], \dots$$

$$\int_{x_0+(n-3)h}^{x_0+6h} f(x) dx = \frac{3h}{8} [y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n]$$

Adding the above integrals, we get

$$\begin{aligned} \int_{x_0}^{x_0+nh} f(x) dx &= \frac{3h}{8} [(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 \\ &\quad + \dots + y_{n-2} + y_{n-1}) + 2(y_3 + y_6 + \dots + y_{n-3})] \end{aligned}$$

which is known as Simpson's three-eighth rule.

While using this formula, the given interval of integration must be divided into sub-intervals whose number n is a multiple of 3.

3.2.5 Boole's Rule

Putting $n = 4$ in formula (39) and neglecting all differences of order higher than four, we get

$$\begin{aligned} \int_{x_0}^{x_0+4h} f(x) dx &= h \int_0^4 \left[y_0 + r\Delta y_0 + \frac{r(r-1)}{2!} \Delta^2 y_0 + \frac{r(r-1)(r-2)}{3!} \Delta^3 y_0 \right. \\ &\quad \left. + \frac{r(r-1)(r-2)(r-3)}{4!} \Delta^4 y_0 \right] dr \\ &\quad | \text{ By Newton's forward interpolation formula} \\ &= 4h \left[y_0 + \frac{n}{2} \Delta y_0 + \frac{n(2n-3)}{12} \Delta^2 y_0 + \frac{n(n-2)^2}{24} \Delta^3 y_0 \right. \\ &\quad \left. + \left(\frac{n^4}{5} - \frac{3n^3}{2} + \frac{11n^2}{3} - 3n \right) \frac{\Delta^4 y_0}{4!} \right]_0^4 \\ &= 4h \left[y_0 + 2\Delta y_0 + \frac{5}{3} \Delta^2 y_0 + \frac{3}{2} \Delta^3 y_0 + \frac{7}{90} \Delta^4 y_0 \right] \\ &= \frac{2h}{45} (7y_0 + 32y_1 + 12y_2 + 32y_3 + 7y_4) \end{aligned}$$

Similarly, $\int_{x_0+4h}^{x_0+8h} f(x) dx = \frac{2h}{45} (7y_4 + 32y_5 + 12y_6 + 32y_7 + 7y_8)$ and so on.

Adding all these integrals from x_0 to $x_0 + nh$, where n is a multiple of 4, we get

$$\int_{x_0}^{x_0+nh} f(x) dx = \frac{2h}{45} [7y_0 + 32y_1 + 12y_2 + 32y_3 + 14y_4 + 32y_5 + 12y_6 + 32y_7 + 14y_8 + \dots]$$



REMEMBER

While applying Boole's rule, the number of sub-intervals should be taken as a multiple of 4.

This is known as Boole's rule.

3.2.6 Weddle's Rule (n = 6)

Putting $n = 6$ in formula (39) and neglecting all differences of order higher than six, we get

$$\begin{aligned} \int_{x_0}^{x_0+6h} f(x) dx &= h \int_0^6 \left[y_0 + r\Delta y_0 + \frac{r(r-1)}{2!} \Delta^2 y_0 + \frac{r(r-1)(r-2)}{3!} \Delta^3 y_0 \right. \\ &\quad + \frac{r(r-1)(r-2)(r-3)}{4!} \Delta^4 y_0 + \frac{r(r-1)(r-2)(r-3)(r-4)}{5!} \Delta^5 y_0 \\ &\quad \left. + \frac{r(r-1)(r-2)(r-3)(r-4)(r-5)}{6!} \Delta^6 y_0 \right] dr \\ &= h \left[ry_0 + \frac{r^2}{2} \Delta y_0 + \frac{1}{2} \left(\frac{r^3}{3} - \frac{r^2}{2} \right) \Delta^2 y_0 + \frac{1}{6} \left(\frac{r^4}{4} - r^3 + r^2 \right) \Delta^3 y_0 \right. \\ &\quad + \frac{1}{24} \left(\frac{r^5}{5} - \frac{3r^4}{2} + \frac{11r^3}{3} - 3r^2 \right) \Delta^4 y_0 \\ &\quad \left. + \frac{1}{120} \left(\frac{r^6}{6} - 2r^5 + \frac{35r^4}{4} - \frac{50r^3}{3} + 12r^2 \right) \Delta^5 y_0 \right] \end{aligned}$$



$$\begin{aligned}
& + \frac{1}{720} \left(\frac{r^7}{7} - \frac{5r^6}{2} + 17r^5 - \frac{225r^4}{4} + \frac{274r^3}{3} - 60r^2 \right) \Delta^6 y_0 \Bigg]_0^6 \\
& = 6h \left[y_0 + 3\Delta y_0 + \frac{9}{2} \Delta^2 y_0 + 4\Delta^3 y_0 + \frac{41}{20} \Delta^4 y_0 \right. \\
& \quad \left. + \frac{11}{20} \Delta^5 y_0 + \frac{41}{840} \Delta^6 y_0 \right] \\
& = \frac{6h}{20} \left[20y_0 + 60\Delta y_0 + 90\Delta^2 y_0 + 80\Delta^3 y_0 + 41\Delta^4 y_0 \right. \\
& \quad \left. + 11\Delta^5 y_0 + \frac{41}{42} \Delta^6 y_0 \right] \\
& = \frac{3h}{10} [20y_0 + 60(y_1 - y_0) + 90(y_2 - 2y_1 + y_0) \\
& \quad + 80(y_3 - 3y_2 + 3y_1 - y_0) \\
& \quad + 41(y_4 - 4y_3 + 6y_2 - 4y_1 + y_0) \\
& \quad + 11(y_5 - 5y_4 + 10y_3 - 10y_2 + 5y_1 - y_0) \\
& \quad + (y_6 - 6y_5 + 15y_4 - 20y_3 \\
& \quad + 15y_2 - 6y_1 + y_0)] \left[\because \frac{41}{42} \simeq 1 \right] \\
& = \frac{3h}{10} [y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6]
\end{aligned}$$

Similarly,

$$\begin{aligned}
\int_{x_0+6h}^{x_0+12h} f(x) dx &= \frac{3h}{10} [y_6 + 5y_7 + y_8 + 6y_9 + y_{10} + 5y_{11} + y_{12}] \\
&\dots\dots\dots \\
&\dots\dots\dots \\
\int_{x_0+(n-6)h}^{x_0+nh} f(x) dx &= \frac{3h}{10} [y_{n-6} + 5y_{n-5} + y_{n-4} + 6y_{n-3} + y_{n-2} + 5y_{n-1} + y_n]
\end{aligned}$$

Adding the above integrals, we get

$$\begin{aligned}
\int_{x_0}^{x_0+nh} f(x) dx &= \frac{3h}{10} [y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + 2y_6 \\
&\quad + 5y_7 + y_8 + 6y_9 + y_{10} + 5y_{11} + 2y_{12} + \dots\dots\dots]
\end{aligned}$$

which is known as Weddle's rule.

3.2.7 Algorithm of Trapezoidal Rule

Step 01. Start of the program.

Step 02. Input Lower limit a

Step 03. Input Upper Limit b

Step 04. Input number of sub intervals n

Step 05. $h=(b-a)/n$

Step 06. sum=0

Step 07. sum=fun(a)+fun(b)

Step 08. for i=1; i<n; i++

Step 09. sum +=2*fun(a+i)

Step 10. End Loop i

Step 11. result =sum*h/2;

Step 12. Print Output result

Step 13. End of Program

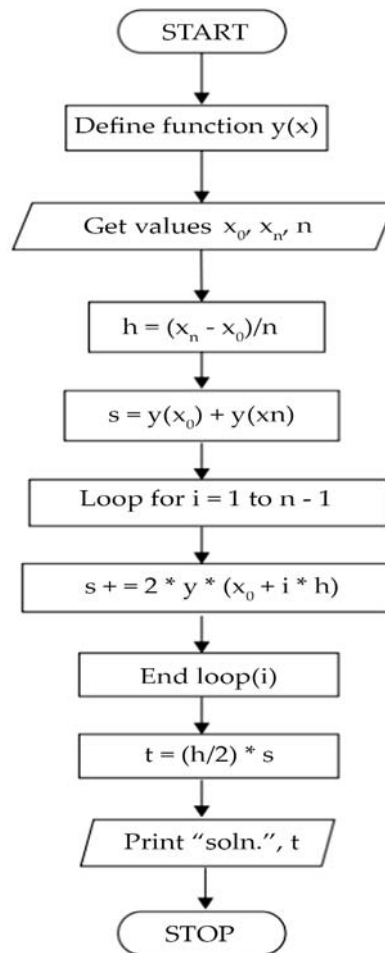
Step 14. Start of Section fun

Step 15. temp = 1/(1+(x*x))

Step 16. Return temp

Step 17. End of Section fun.



Flow-Chart For Trapezoidal Rule*Program to Implement Trapezoidal Method of Numerical Integration*

```
//... HEADER FILES DECLARATION
```

```
# include <stdio.h>
```

```
# include <conio.h>
```

```
# include <math.h>
```

```
# include <process.h>
```

```
# include <string.h>

//... Function Prototype Declaration

float fun(float);

//... Main Execution Thread

void main()

{

//... Variable Declaration Field

//... Floating Type

float result=1;

float a,b;

float h,sum;

//... Integer Type

int i,j;

int n;

//... Invoke Clear Screen Function

clrscr();

//... Input Section

//... Input Range

printf("\n\n Enter the range - ");

printf("\n\n Lower Limit a - ");

scanf("%f",&a);

printf("\n\n Upper Limit b - ");

scanf("%f",&b);

//... Input Number of subintervals

printf("\n\n Enter number of subintervals - ");

scanf("%d",&n);
```



```
//... Calculation and Processing Section
h=(b-a)/n;
sum=0;
sum=fun(a)+fun(b);
for(i=1;i<n;i++)
{
sum+=2*fun(a+i);
}
result=sum*h/2;
//... Output Section
printf("\n\n\n Value of the integral is %6.4f\t",result);
//...Invoke User Watch Halt Function
printf("\n\n\n Press Enter to Exit");
getch();
}
//... Termination of Main Execution Thread
//... Function Body
float fun(float x)
{
float temp;
temp = 1/(1+(x*x));
return temp;
}
//... Termination of Function Body
```

Output

Enter the range -

Lower Limit a - 0

Upper Limit b - 6

Enter number of subintervals - 6

Value of the integral is 1.4108

Press Enter to Exit

3.2.8 Algorithm of Simpson's 3/8th Rule

Step 01. Start of the program.

Step 02. Input Lower limit a

Step 03. Input Upper limit b

Step 04. Input number of sub intervals n

Step 05. $h = (b - a)/n$

Step 06. sum = 0

Step 07. sum = fun(a) + fun (b)

Step 08. for i = 1; i < n; i++

Step 09. if i%3=0:

Step 10. sum += 2*fun(a + i*h)

Step 11. else:

Step 12. sum += 3*fun(a+(i)*h)

Step 13. End of loop i

Step 14. result = sum*3*h/8

Step 15. Print Output result

Step 16. End of Program



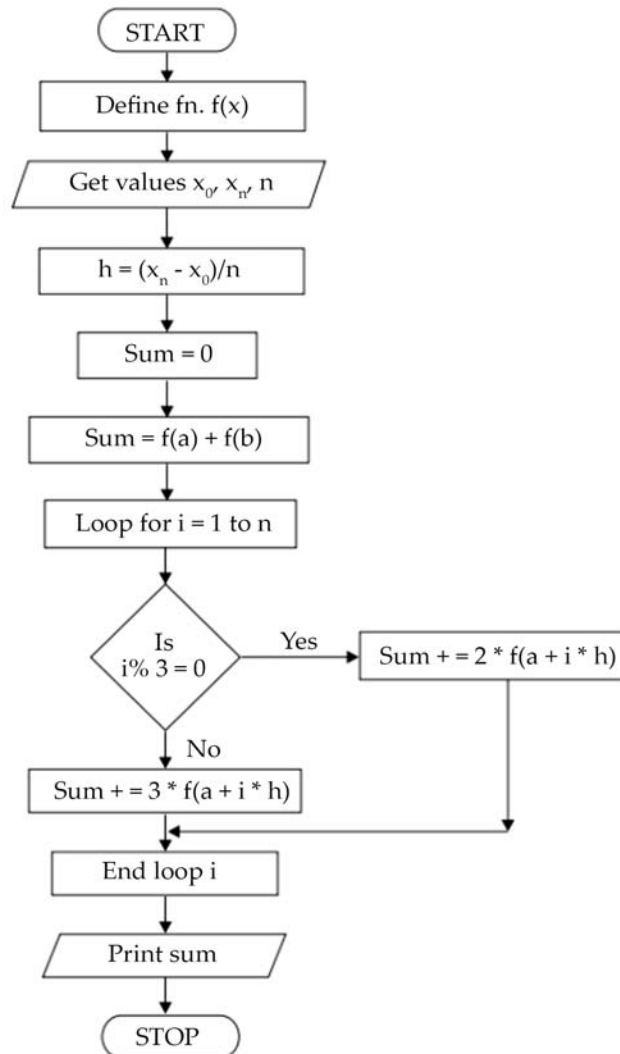
Step 17. Start of Section fun

Step 18. $\text{temp} = 1/(1+(x*x))$

Step 19. Return temp

Step 20. End of section fun

Flow-chart of simpson's 3/8th rule



Program to Implement Simpson's 3/8th Method of Numerical Integration

```
//... HEADER FILES DECLARATION
#include <stdio.h>
#include <conio.h>
#include <math.h>
#include <process.h>
#include <string.h>

//... Function Prototype Declaration
float fun(float);

//... Main Execution Thread
void main()
{
//... Variable Declaration Field
//... Floating Type
float result=1;
float a,b;
float h,sum;
//...Integer Type
int i,j;
int n;
//...Invoke Clear Screen Function
clrscr();
//...Input Section

//...Input Range
printf("\n\n Enter the range - ");
printf("\n\n Lower Limit a - ");
scanf("%f",&a);
printf("\n\n Upper Limit b - ");
scanf("%f",&b);

//...Input Number of Subintervals
printf("\n\n Enter number of subintervals - ");
scanf("%d",&n);

//...Calculation and Processing Section
h=(b-a)/n;
sum=0;
sum=fun(a)+fun(b);
for(i=1;i<n;i++)
{
if(i%3==0)
{
sum+=2*fun(a+i*h)
}
else
{
sum+=3*fun(a+(i)*h);
}
}
result=sum*3*h/8;
```




```
//... Output Section
printf("\n\n\n Value of the integral is
%.6f\t",result);

//... Invoke User Watch Halt Function
printf("\n\n\n Press Enter to Exit");
getch();
}

//... Termination of Main Execution Thread
//... Function Body
float fun(float x)
{
float temp;
temp=1/(1+(x*x));
return temp;
}

//... Termination of Function Body
```

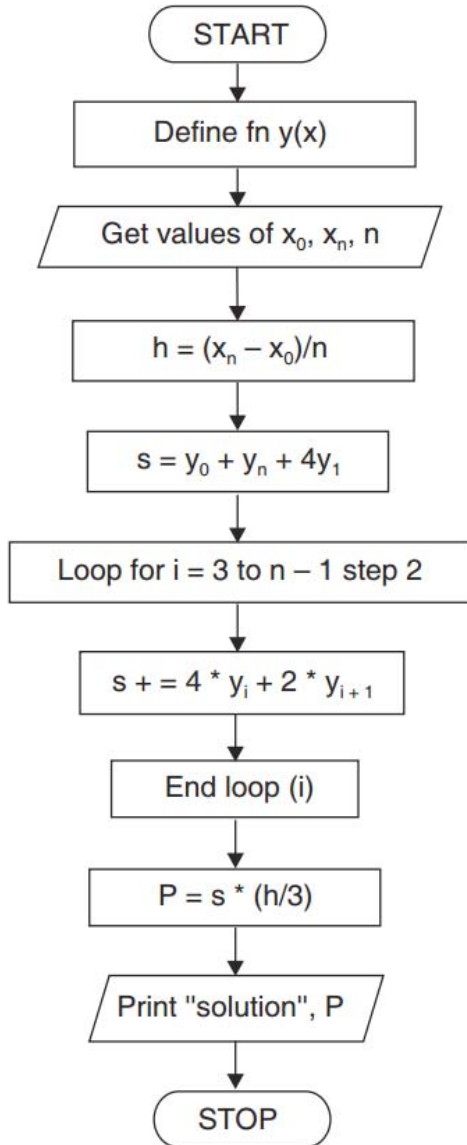
Output

Enter the range -
Lower Limit a - 0
Upper Limit b - 6
Enter number of subintervals - 6
Value of the integral is 1.3571
Press Enter to Exit

3.2.9 Algorithm of Simpson's 1/3rd Rule

- Step 01. Start of the program.
- Step 02. Input Lower limit a
- Step 03. Input Upper limit b
- Step 04. Input number of subintervals n
- Step 05. $h=(b-a)/n$
- Step 06. $sum=0$
- Step 07. $sum=fun(a)+4*fun(a+h)+fun(b)$
- Step 08. for $i=3; i<n; i += 2$
- Step 09. $sum += 2*fun(a+(i-1)*h) + 4*fun(a+i*h)$
- Step 10. End of loop i
- Step 11. $result=sum*h/3$
- Step 12. Print Output result
- Step 13. End of Program
- Step 14. Start of Section fun
- Step 15. $temp = 1/(1+(x*x))$
- Step 16. Return temp
- Step 17. End of Section fun

Flow-Chart of Simpson's 1/3rd Rule



Program to Implement Simpson's 1/3rd Method of Numerical Integration

```
//... HEADER FILES DECLARATION
#include <stdio.h>
```

```
#include <conio.h>
#include <math.h>
#include <process.h>
#include <string.h>
//... Function Prototype Declaration
float fun(float);
//... Main Execution Thread
void main()
{
//...Variable Declaration Field
//... Floating Type
float result=1;
float a,b;
float h,sum;
//... Integer Type
int i,j;
int n;
//... Invoke Clear Screen Function
clrscr();
//... Input Section
//...Input Range
printf("\n\n Enter the range - ");
printf("\n\n Lower Limit a - ");
scanf("%f",&a);
printf("\n\n Upper Limit b - ");
```

```

scanf("%f",&b);
//... Input Number of Subintervals
printf("\n\n Enter number of subintervals
- ");
scanf("%d",&n);
//... Calculation and Processing Section
h=(b-a)/n;
sum=0;
sum=fun(a)+4*fun(a+h)fun(b);
for(i=3;i<n;i+=2)
{
sum+=2*fun(a+(i-1)*h)+4*fun(a+i*h);
}
result=sum*h/3;
//... Output Section
printf("\n\n\n\n Value of the integral is
%6.4f\t",result);
//... Invoke User Watch Halt Function
printf("\n\n\n Press Enter to Exit");
getch();
}
//... Termination of Main Execution Thread
//... Function Body
float fun(float x)
{
float temp;
temp=1/(1+(x*x));
return temp;
}
//... Termination of Function Body

```

Output

Enter the range -

Lower Limit a - 0

Upper Limit b - 6

Enter number of subintervals - 6

Value of the integral is 1.3662

Press Enter to Exit



ROLE MODEL

THOMAS SIMPSON: INVENTOR AND EPO- NYM OF SIMPSON'S RULE

Thomas Simpson, born on August 20, 1710, in Market Bosworth, Leicestershire, England, was the son of a self-taught weaver. Simpson's father naturally expected his son to take up the same profession as his "ol' man". However, with the occurrence of a solar eclipse in 1724, Thomas Simpson turned to "mathematical interests", changing his life forever. After a little help mastering Cocker's Arithmetic and the concepts of algebra, Simpson became an "usher" ("the most distinguished of a group of itinerant lecturers who taught in the London coffee-houses"), which greatly improved his mathematical skills. By 1735, he was able to solve puzzles concerning infinitesimal calculus. In 1737, he wrote the first of his mathematical texts, the first of two volumes of *The Doctrine and Application of Fluxions* (the second volume was finished and both volumes were published in 1750).

Not only did Simpson work on mathematics, but he also delved heavily into probability theory and the concept of approximation and error. Publishing *The Nature and Laws of Chance* and his *Essays* in 1740, Simpson worked on proving that the arithmetic mean of a number of observations was of more analytical value than a single one. Simpson went as far as applying himself to publish a theory on Annuities and Reversions in 1742. This "branch" of mathematics, originated by James Dodson, deals with things such as mortality rates and life insurance. In 1743, he published *Dissertations*, which dealt mostly with planetary and interplanetary physics: subjects like lunar tides and "the force of attraction at the surface of a nearly spherical body". What better subjects were there for Simpson to apply his mathematical skills toward? After all, a solar eclipse "eclipsed" his world of weaving.

Among other works, Simpson published a series of mathematics "texts", *Algebra* (1745), *Geometry* (1747), and *Trigonometry* (1748), "in which he introduced the current abbreviations for the trigonometric functions". Simpson died on May 14, 1761, a very brilliant and well-rounded mathematician, but today, he is usually remembered for his contribution to numerical integration: "Simpson's Rule". Discussed here at the Holistic Numerical Methods Institute, Simpson's Rule "has become a popular and useful special case of the Newton-Cotes formula for approximating an integral".



SUMMARY

1. Consider a function of a single variable $y = f(x)$. If $f(x)$ is defined as an expression, its derivative or integral may often be determined using the techniques of calculus.
2. The error introduced in obtaining derivatives is, in general, much worse than that introduced in determining integrals. It may be observed that any errors in approximating a function are amplified while taking the derivative whereas they are smoothed out in integration.
3. Numerical Differentiation is concerned with the method of finding the successive derivatives of a function at a given argument, using the given table of entries corresponding to a set of arguments, equally or unequally spaced.
4. The criterion is the same as in the case of interpolation problems. In the case of equidistant values of x , if the derivative is to be found at a point near the beginning or the end of the given set of values, Newton's forward or backward difference formula should be used accordingly.
5. Given a set of tabulated values of the integrand $f(x)$, determining the value of $\int_{x_0}^{x_n} f(x) dx$ is called numerical integration.

KNOWLEDGE CHECK

1. The two-segment trapezoidal rule of integration is exact for integrating at most order polynomials.
 - a. first
 - b. second
 - c. third
 - d. fourth
2. The value of $\int_{0.2}^{2.2} xe^x dx$ by the using one-segment trapezoidal rule is most nearly
 - a. 11.672
 - b. 11.807
 - c. 20.099
 - d. 24.119

3. The results obtained by using Simpson's rule will be greater than those obtained by using the trapezoidal rule
- in all cases
 - provided the intervals are small
 - provided the boundary is concave towards the base line
 - provided the boundary is convex towards the base line.
4. The value of integral e^x is evaluated from 0 to 0.4 by the following formula. Which method will give the least error?
- Trapezoidal rule with $h = 0.2$
 - Trapezoidal rule with $h = 0.1$
 - Simpson's 1/3 rule with $h = 0.1$.
 - None of these
5. A function $(ax+b) / ((x-1)(x+4))$ turning value at the point (2, -1). The value of a and b will be
- $a = x, b=1$
 - $a = 3, b = 5$
 - $a = -5, b = 7$
 - $a = -7, b = 8$.
6. The value of $\int_{0.2}^{2.2} xe^x dx$ by the using three-segment trapezoidal rule is most nearly
- 11.672
 - 11.807
 - 20.099
 - 24.119
7. The following data of the velocity of a body is given as a function of time.

Time (s)	0	15	18	22	24
Velocity (m/s)	22	24	37	25	123

The distance in meters covered by the body from $t = s$ 12 to $t = s$ 18 calculated using the trapezoidal rule with unequal segments is

- 162.90
- 166.00
- 181.70
- 436.50



8. The velocity of a body is given by

$$v(t) = 2t, \quad 1 \leq t \leq 5$$

$$= 5t^2 + 3, \quad 5 < t \leq 14$$

where t is given in seconds, and v is given in m/s. Use the two-segment trapezoidal rule to find the distance covered by the body from $t = 2$ to $t = 9$ seconds.

- a. 935.0 m
- b. 1039.7 m
- c. 1260.9 m
- d. 5048.9 m

REVIEW QUESTIONS

1. Find $f'(1.1)$ and $f''(1.1)$ from the following table:

x :	1.0	1.2	1.4	1.6	1.8	2.0
$f(x)$:	0.0	0.1280	0.5540	1.2960	2.4320	4.000.

2. Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using

- Simpson's one-third rule
- Simpson's three-eighth rule
- Trapezoidal rule
- Weddle's rule.

3. Evaluate $\int_1^2 \frac{1}{x} dx$ by Simpson's $\frac{1}{3}$ rd rule with four strips and determine the error by direct integration.

4. Evaluate $\int_4^{5.2} \log_e x dx$ using

- Trapezoidal rule
- Weddle's rule.

5. Use Simpson's rule, taking five ordinates, to find an approximate value of $\int_1^2 \sqrt{x - \frac{1}{x}} dx$ decimal places.

Check your Result

1. (a) 2. (c) 3. (c) 4. (c) 5. (d) 6. (c)
 7. (a) 8. (c)

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Chapter

4

ORDINARY DIFFERENTIAL EQUATIONS

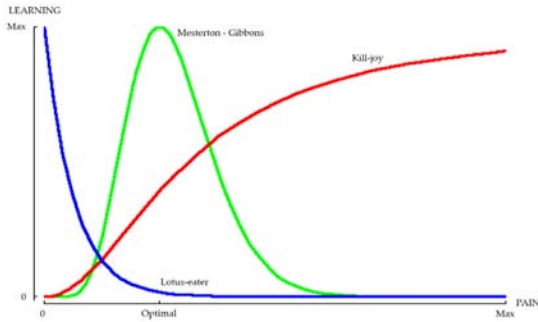
LEARNING OBJECTIVES

After studying this chapter, you will be able to:

1. Evaluate first-order ordinary differential equations
2. Explain the higher order linear differential equations with constant coefficients
3. Describe Cauchy's and Legendre's linear equations
4. Discuss simultaneous first order linear equations with constant coefficients

INTRODUCTION

An ordinary differential equation (ODE) is an equation that involves some ordinary derivatives (as opposed to partial derivatives) of a function. Often, our goal is to solve an ODE, i.e., determine what function or functions satisfy the equation. If you know what the derivative of a function is, how can you find the function itself? You need to find the ant-derivative, i.e., you need to integrate.



For example, if you are given

$$\frac{dx}{dt}(t) = \cos t$$

Then what is the function $x(t)$? Since the anti-derivative of $\cos t$ is $\sin t$, then $x(t)$ must be $\sin t$. Except we forgot one important point: there is always an arbitrary constant that we cannot determine if we only know the derivative.

Therefore, all we can determine from the above equation is that

$$x(t) = \sin t + C$$

For some arbitrary constant C . You can verify that indeed $x(t)$ satisfies the equation

$$\frac{dx}{dt} = \cos t.$$

In general, solving an ODE is more complicated than simple integration. Even so, the basic principle is always integration. Usually, the difficult part is determining what integration we need to do.

The term differential equation was coined by Leibniz in 1676 for a relationship between the two differentials dx and dy for the two variables x and y .

However, soon after the first usage of this term, differential equations quickly became understood as any algebraic or transcendental equation which involved derivatives.

An ordinary differential equation is a relationship between a real variable, (let us say x), a dependent variable (let us call this y), and (possibly many) derivatives of the dependent variable y with respect to x . For example:

$$xy \frac{d^2y}{dx^2} + y \frac{dy}{dx} + e^{3x} = 0$$

is an ordinary differential equation.

A partial differential equation is a relationship between a function of several variables, the partial derivatives of that function, and the independent variables. Usually they are much more difficult to deal with than ordinary differential equations.

$$\frac{\partial^2 u}{\partial x^2}(x, y) + x^2 \frac{\partial^2 u}{\partial y^2}(x, y) = 0$$

is an example of a partial differential equation.

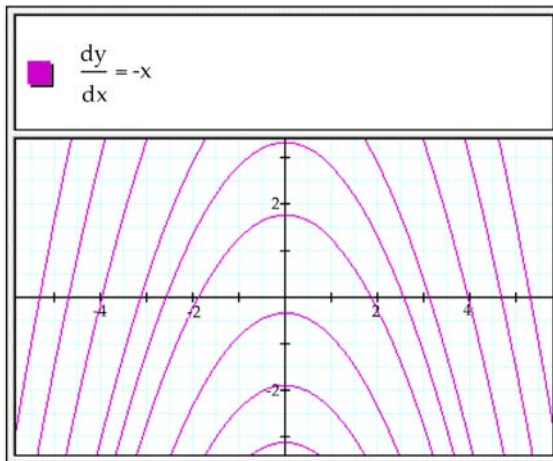


4.1 FIRST-ORDER ORDINARY DIFFERENTIAL EQUATIONS

1 First-order ordinary differential equations have some rather special properties, which result for the most part because they can only contain a limited number of terms. In fact, all linear first-order ordinary differential equations can easily be solved. This is in contrast to higher-order ordinary differential equations that become much more difficult to solve when, for example, they contain variable coefficients. Some methods exist to solve a pretty large class of nonlinear first-order ordinary differential equations, which is generally not the case for higherorder nonlinear equations.

LEARNING OBJECTIVE

1. Evaluate first-order ordinary differential equations



Considers methods to solve first-order ordinary differential equations of the form

$$\frac{dx}{dt}(t) = f(x, t). \quad (1.a)$$

If the first-order ordinary differential equation that must be solved is not of the form of Equation (1.a) it must be transformed into that form, and in the process care must be taken that solutions are neither gained nor lost. Theorem gives conditions for existence and uniqueness of solutions to problems involving equations in the form of Equation (1.a). That as long as $f(x, t)$ is infinitely differentiable in both x and t , then the theorem is satisfied. All the equations considered have the properties necessary for existence and uniqueness of solutions.

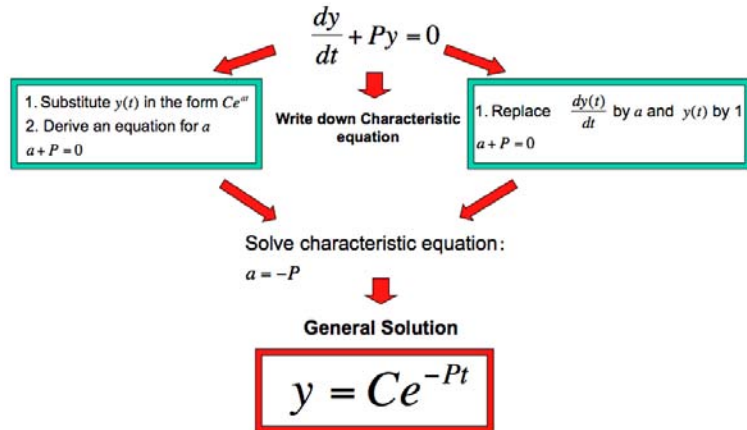
**REMEMBER**

The coverage of the method of undetermined coefficients and solutions to constant-coefficient linear equations of any order and serves as a basis for material.

4.1.1 Homogeneous Constant-Coefficient Linear First-Order Ordinary Differential Equations

Because it is the case that the coefficients of the dependent variable terms in engineering differential equations are often parameters that describe the physical properties of a system, and it is also often the case that such parameters are constant (mass, thermal capacitance, etc.), it is thus often the case that differential equations in engineering have constant coefficients. The first-order, constant-coefficient, linear differential equations.

The following fact regarding ordinary, **constant-coefficient**, linear, homogeneous differential equations of any order and worthy of repeated emphasis.



KEY TERMS

Constant coefficients is a term applied to differential operators, and also some difference operators, to signify that they contain no functions of the independent variables, other than constant functions.

If you remember anything from differential equations, remember the following: ordinary, linear, constant-coefficient, homogeneous differential equations of any order have exponential solutions. To emphasize the fact, let us make it a theorem.

Theorem: Ordinary, linear, constant-coefficient, homogeneous differential equations with dependent variable x and independent variable t have solutions of the form $x = ce^{\lambda t}$ where c and λ are constants.



Proof: Consider an n th-order, ordinary, linear, constant-coefficient, and homogeneous differential equation of the form

$$\alpha_n \frac{d^n x}{dt^n} + \alpha_{n-1} \frac{d^{n-1} x}{dt^{n-1}} + \cdots + \alpha_1 \frac{dx}{dt} + \alpha_0 x = 0. \quad (1.b)$$

To verify the form of the solution, substitute $x = ce^{\lambda t}$ into Equation (1.b):

$$\alpha_n \lambda^n ce^{\lambda t} + \alpha_{n-1} \lambda^{n-1} ce^{\lambda t} + \cdots + \alpha_1 \lambda ce^{\lambda t} + \alpha_0 ce^{\lambda t} = 0.$$

Note that $c = 0$ results in $x(t) = 0$, which is a solution to Equation (1.b), but only satisfies the initial value problem where $x(0) = 0$. For the case where $c \neq 0$, because $ce^{\lambda t}$ is never zero, it is legitimate to divide each side of the equation by it which gives

$$\alpha_n \lambda^n + \alpha_{n-1} \lambda^{n-1} + \cdots + \alpha_1 \lambda + \alpha_0 = 0 \quad (1.c)$$

which is an n th-order polynomial in λ . Because, by the fundamental theorem of algebra, Equation (1.c) has n solutions, there may be, in fact, up to n different solutions of the form $x = e^{\lambda t}$

Homogeneous Equations with Constant Coefficients

We have only worked on first order differential equations. The next step is to investigate second order differential equations. The general second order differential equation has the form

$$y'' = f(t, y, y')$$

The general solution to such an equation is very rough. Instead, we will focus on special cases. In particular, if the differential equation is linear, then it can be written in the form

The general solution to such an equation is very rough. Instead, we will focus on special cases. In particular, if the differential equation is linear, then it can be written in the form

$$P(t)y'' + Q(t)y' + R(t)y = G(t)$$

If $P(t)$ is nonzero, then we can divide by $P(t)$ to get

$$y'' + p(t)y' + q(t)y = g(t)$$

We call a second order linear differential equation *homogeneous* if $g(t) = 0$.

In this method we will be investigating homogeneous second order linear differential equations with constant coefficients. They can be written in the form

$$ay'' + by' + cy = 0$$

Example

$$y'' - y' - 6y = 0$$

Solve

$$y'' - y' - 6y = 0 \quad y(0) = 1 \quad y'(0) = 2$$

As before we seek solutions of the form

$$y = e^{rt}$$

Now calculate derivatives

$$y' = re^{rt} \quad y'' = r^2e^{rt}$$

Substituting into the differential equation gives

$$\begin{aligned} r^2e^{rt} - (re^{rt}) - 6(e^{rt}) \\ = (r^2 - r - 6)e^{rt} = 0 \end{aligned}$$

Now divide by e^{rt} to get

$$r^2 - r - 6 = 0$$

$$(r - 3)(r + 2) = 0$$

We can conclude that two solutions are

$$y_1 = e^{3t} \quad \text{and} \quad y_2 = e^{-2t}$$

We can conclude that

$$y = c_1e^{3t} + c_2e^{-2t}$$

Represents a two dimensional family (vector space) of solutions.

Now use the initial conditions to find that

$$1 = c_1 + c_2$$

We have that

$$y' = 3c_1e^{3t} - 2c_2e^{-2t}$$

Plugging in the initial condition with y' , gives

$$2 = 3c_1 - 2c_2$$

This is a system of two equations and two unknowns. We can use a matrix to arrive at

$$c_1 = 4/5 \quad \text{and} \quad c_2 = 1/5$$

The final solution is

$$y = 4/5 e^{3t} + 1/5 e^{-2t}$$

In general for

$$ay'' + by' + cy = 0$$

we call



$$ar^2 + br + c = 0$$

The characteristic equation for this differential equation. The examples demonstrated how to solve it if we have two distinct real roots.

4.1.2 Second Order Linear Homogeneous Differential Equations with Constant Coefficients

For the most part, we will only learn how to solve second order linear equation with constant coefficients (that is, when $p(t)$ and $q(t)$ are constants).

Since a homogeneous equation is easier to solve compares to its nonhomogeneous counterpart, we start with second order linear homogeneous equations that contain constant coefficients only:

$$a y'' + b y' + c y = 0.$$

Where a , b , and c are constants, $a \neq 0$.

A very simple instance of such type of equations is

$$y'' - y = 0.$$

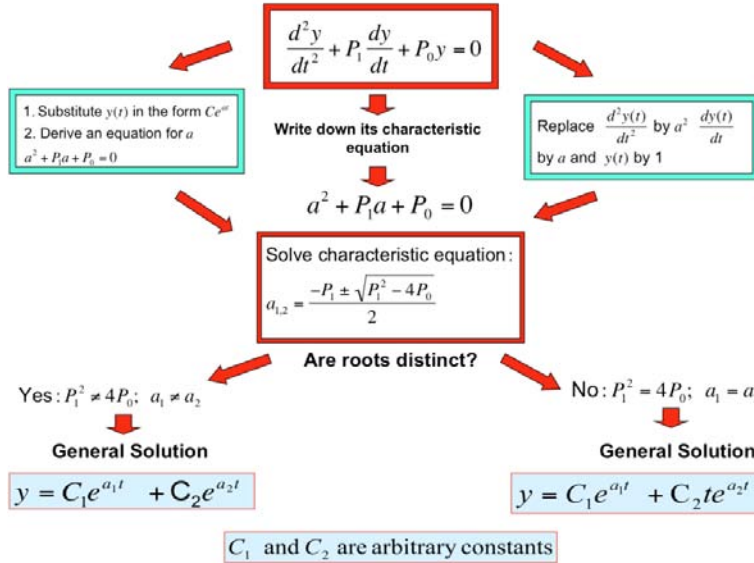
The equation's solution is any function satisfying the equality $y'' = y$. obviously $y_1 = e^t$ is a solution, and so is any constant multiple of it, $C_1 e^t$. Not as obvious, but still easy to see, is that $y_2 = e^{-t}$ is another solution (and so is any function of the form $C_2 e^{-t}$).

It can be easily verified that any function of the form

$$y = C_1 e^t + C_2 e^{-t}$$

will satisfy the equation. In fact, this is the general solution of the above differential equation.

However, while the general solution of $y'' + p(t) y' + q(t) y = 0$ will always be in the form of $C_1 y_1 + C_2 y_2$, where y_1 and y_2 are some solutions of the equation, the converse is not always true. Not every pair of solutions y_1 and y_2 could be used to give a general solution in the form $y = C_1 y_1 + C_2 y_2$. We shall see shortly the exact condition that y_1 and y_2 must satisfy that would give us a general solution of this form.



Fact: The general solution of a second order equation contains two arbitrary constants / coefficients. To find a particular solution, therefore, requires two initial values. The initial conditions for a second order equation will appear in the form: $y(t_0) = y_0$ and $y'(t_0) = y'_0$.



Find the general solution of $y'' - 5y' = 0$.

Solve

There is no need to "guess" an answer here. We actually know a way to solve the equation already. Observe that if we let $u = y'$, then $u' = y''$. Substitute them into the equation and we get a new equation:

$$u' - 5u = 0.$$

This is a first order linear equation with $p(t) = -5$ and $g(t) = 0$.

The integrating factor is $\mu = e^{-5t}$.

$$u(t) = \frac{1}{\mu(t)} \left(\int \mu(t) g(t) dt \right) = e^{5t} \left(\int 0 dt \right) = e^{5t} (C) = C e^{5t}$$

The actual solution y is given by the relation $u = y'$, and can be found by integration:

$$y(t) = \int u(t) dt = \int C e^{5t} dt = \frac{C}{5} e^{5t} + C_2 = C_1 e^{5t} + C_2.$$

The method used in the above example can be used to solve any second order linear equation of the form $y'' + p(t) y' = g(t)$, regardless whether its coefficients are constant or no constant, or it is a homogeneous equation or nonhomogeneous.



4.2 HIGHER ORDER LINEAR DIFFERENTIAL EQUATIONS WITH CONSTANT COEFFICIENTS

 The solutions of linear differential equations with constant coefficients of the third order or higher can be found in similar ways as the solutions of second order linear equations. For an n^{th} order homogeneous linear equation with constant coefficients:

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_2 y'' + a_1 y' + a_0 y = 0, \quad a_n \neq 0.$$

It has a general solution of the form

$$y = C_1 y_1 + C_2 y_2 + \dots + C_{n-1} y_{n-1} + C_n y_n$$

where $y_1, y_2, \dots, y_{n-1}, y_n$ are any n linearly independent solutions of the equation. (Thus, they form a set of fundamental solutions of the differential equation.) The linear independence of those solutions can be determined by their Wronskian, i.e., $W(y_1, y_2, \dots, y_{n-1}, y_n)(t) \neq 0$.

Note 1: In order to determine the n unknown coefficients C_i , each n^{th} order equation requires a set of n initial conditions in an initial value problem:

$$y(t_0) = y_0, y'(t_0) = y'_0, y''(t_0) = y''_0, \text{ and } y^{(n-1)}(t_0) = y^{(n-1)}_0.$$

Note 2: The Wronskian $W(y_1, y_2, \dots, y_{n-1}, y_n)(t)$ is defined to be the determinant of the following $n \times n$ matrix

$$\begin{bmatrix} y_1 & y_2 & \dots & \dots & y_n \\ y'_1 & y'_2 & \dots & \dots & y'_n \\ y''_1 & y''_2 & \dots & \dots & y''_n \\ \vdots & \vdots & & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \dots & \dots & y_n^{(n-1)} \end{bmatrix}$$

Such a set of linearly independent solutions, and therefore, a general solution of the equation, can be found by first solving

LEARNING OBJECTIVE

 Explain the higher order linear differential equations with constant coefficients

the differential equation's characteristic equation:

$$a_n r^n + a_{n-1} r^{n-1} + \dots + a_2 r^2 + a_1 r + a_0 = 0.$$

This is a polynomial equation of degree n ; therefore, it has n real and/or complex roots (not necessarily distinct).

Those necessary n linearly independent solutions can then be found using the four rules is given:

- (i) If r is a distinct real root, then $y = e^{rt}$ is a solution.
- (ii) If $r = \lambda \pm \mu i$ are distinct complex conjugate roots, then $y = e^{\lambda t} \cos \mu t$ and $y = e^{\lambda t} \sin \mu t$ are solutions.
- (iii) If r is a real root appearing k -times, then $y = e^{rt}$, $y = t e^{rt}$, $y = t^2 e^{rt}$, , and $y = t^{k-1} e^{rt}$ are all solutions.
- (iv) If $r = \lambda \pm \mu i$ are complex conjugate roots each appears k -times, then

$$\begin{array}{ll} y = e^{\lambda t} \cos \mu t, & y = e^{\lambda t} \sin \mu t, \\ y = t e^{\lambda t} \cos \mu t, & y = t e^{\lambda t} \sin \mu t, \\ y = t^2 e^{\lambda t} \cos \mu t, & y = t^2 e^{\lambda t} \sin \mu t, \\ \vdots & \vdots \\ y = t^{k-1} e^{\lambda t} \cos \mu t, & \text{and } y = t^{k-1} e^{\lambda t} \sin \mu t, \end{array}$$

are all solutions.

Example: $y^{(4)} - y = 0$

The characteristic equation is $r^4 - 1 = (r^2 + 1)(r + 1)(r - 1) = 0$, which has roots $r = 1, -1, i, -i$. Hence, the general solution is

$$y = C_1 e^t + C_2 e^{-t} + C_3 \cos t + C_4 \sin t.$$

Example: $y^{(5)} - 3y^{(4)} + 3y^{(3)} - y'' = 0$

The characteristic equation is $r^5 - 3r^4 + 3r^3 - r^2 = r^2 (r - 1)^3 = 0$, which has roots $r = 0$ (a double root), and 1 (a triple root). Hence, the general solution is

$$\begin{aligned} y &= C_1 e^{0t} + C_2 t e^{0t} + C_3 e^t + C_4 t e^t + C_5 t^2 e^t \\ &= C_1 + C_2 t + C_3 e^t + C_4 t e^t + C_5 t^2 e^t. \end{aligned}$$

Example: $y^{(4)} + 4y^{(3)} + 8y'' + 8y' + 4y = 0$

The characteristic equation is $r^4 + 4r^3 + 8r^2 + 8r + 4 = (r^2 + 2r + 2)^2 = 0$, which has roots $r = -1 \pm i$ (repeated). Hence, the general solution is

$$y = C_1 e^{-t} \cos t + C_2 e^{-t} \sin t + C_3 t e^{-t} \cos t + C_4 t e^{-t} \sin t.$$



Example: What is a 4th order homogeneous **linear equation** whose general solution is

$$y = C_1 e^t + C_2 e^{2t} + C_3 e^{3t} + C_4 e^{4t} ?$$

The solution implies that $r = 1, 2, 3$, and 4 are the four roots of the characteristic equation. Therefore, $r - 1$, $r - 2$, $r - 3$, and $r - 4$ are its factors. Consequently, the characteristic equation is

$$(r - 1)(r - 2)(r - 3)(r - 4) = 0$$

$$r^4 - 10r^3 + 35r^2 - 50r + 24 = 0$$

Hence, an equation is

$$y^{(4)} - 10y^{(3)} + 35y'' - 50y' + 24y = 0.$$

Every nonzero constant multiple of the above equation also has the same general solution. However, the indicated equation is the only equation in the standard form that has the given general solution.

Theorem 1

Suppose the functions $p_0, p_1, \dots, p_{n-1}$ are all continuous on the open interval I and further suppose that $y_1(t), y_2(t), \dots, y_n(t)$ are all solutions to (1). If $W(y_1, y_2, \dots, y_n)(t) \neq 0$ for every t in I then $y_1(t), y_2(t), \dots, y_n(t)$ form a Fundamental Set of Solutions and the general solution to (1) is,

$$y(t) = c_1 y_1(t) + c_2 y_2(t) + \dots + c_n y_n(t) \quad (1)$$

Recall as well that if a set of solutions form a fundamental set of solutions then they will also be a set of linearly independent functions.

You will close this method off with a quick reminder of how we find solutions to the nonhomogeneous differential equation, $(y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y = g(t))$. You first need the n th order version of a theorem we saw back in the 2nd order material.

Theorem 2

Suppose that $y_1(t)$ and $y_2(t)$ are two solutions to (

KEY TERMS

Linear Equation is an algebraic equation in which each term is either a constant or the product of a constant and (the first power of) a single variable.

$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y = g(t)$) and that $y_1(t), y_2(t), \dots, y_n(t)$ are a fundamental set of solutions to the homogeneous differential equation (1) then,

$$Y_1(t) - y_2(t)$$

is a solution to (1) and it can be written as

$$y_1(t) - y_2(t) = c_1 y_1(t) + c_2 y_2(t) + \dots + c_n y_n(t)$$

Now, just as we did with the 2nd order material if we let $y(t)$ be the general solution to ($y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y = g(t)$) and if we let $y_p(t)$ be any solution to ($y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y = g(t)$) then using the result of this theorem we see that we must have,

$$y(t) = c_1 y_1(t) + c_2 y_2(t) + \dots + c_n y_n(t) + y_p(t) = y_c(t) (+) y_p(t)$$

where, $y_c(t) = c_1 y_1(t) + c_2 y_2(t) + \dots + c_n y_n(t)$ is called the complementary solution and $y_p(t)$ is called a particular solution.

KEY TERMS

Homogeneous is a term in physical chemistry and material science that refers to substances and mixtures which are in a single phase.

4.2.1 Method of Variation of Parameters

The method of variation of parameters applies to solve

$$a(x)y'' + b(x)y' + c(x)y = f(x)$$

Continuity of a, b, c and f is assumed, plus $a(x) \neq 0$. the method is important because it solves the largest class of equations. Specifically included are functions $f(x)$ like $\ln|x|, |x|, e^{x^2}$. OR

The calculations of particular integrals/solutions for some special cases have been studied. Recall that the **homogeneous** part of the equation had constant coefficients. We deal with a useful technique of finding a particular solution when the coefficients of the homogeneous part are continuous functions and the forcing function $f(x)$ (or the non-homogeneous term) is piecewise continuous. Suppose y_1 and y_2 are two linearly independent solutions of

$$y'' + q(x)y' + r(x)y = 0 \quad (2)$$

On I where $q(x)$ and $r(x)$ are arbitrary continuous functions defined on I Then we know that

$$y = c_1 y_1 + c_2 y_2$$

is a solution of (2) for any constants c_1 and c_2 . We now "vary" c_1 and c_2 to functions of x , so that

$$y = u(x)y_1 + v(x)y_2, \quad x \in I \quad (3)$$

is a solution of the equation

$$y'' = q(x)y' + r(x)y = f(x), \text{ on } I, \quad (4)$$

where f is a piecewise continuous function defined on I . The details are given in the following theorem.

Theorem 4:

Method of Variation of Parameters: Let $q(x)$ and $r(x)$ be continuous functions defined on I and let f be a piecewise continuous function on I . Let y_1 and y_2 be two linearly independent solutions of (2) on I . Then a particular solution y_p of (4) is given by

$$y_p = -y_1 \int \frac{y_2 f(x)}{W} dx + y_2 \int \frac{y_1 f(x)}{W} dx, \quad (5)$$

Where $W = W(y_1, y_2)$ is the Wronskian of y_1 and y_2 (Note that the integrals in (5) are the indefinite integrals of the respective arguments.)

Proof. Let $u(x)$ and $v(x)$ be continuously differentiable functions (to be determined) such that

$$y_p = uy_1 + vy_2, \quad x \in I \quad (6)$$

is a particular solution of (4). Differentiation of (6) leads to

$$y'_p = uy'_1 + vy'_2 + u'y_1 + v'y_2 \quad (7)$$

We choose u and v so that

$$u'y_1 + v'y_2 = 0 \quad (8)$$

Substituting (8) in (7), we have

$$y'_p = uy'_1 + vy'_2, \text{ and } y''_p = uy''_1 + vy''_2 + u'y'_2 + v'y'_1 \quad (9)$$

Since y_p is a particular solution of (4), substitution of (6) and (9) in (4), we get

$$u(y''_1 + q(x)y'_1 + r(x)y_1) + v(y''_2 + q(x)y'_2 + r(x)y_2) + u'y'_2 + v'y'_1 = f(x)$$

As y_1 and y_2 are solutions of the homogeneous equation (2), we obtain the condition

$$u'y_1' + v'y_2' = f(x) \quad (10)$$

We now determine u and v from (8) and (10). By using the Cramer's rule for a linear system of equations, we get

$$u' = -\frac{y_2 f(x)}{W} \text{ and } v' = \frac{y_1 f(x)}{W} \quad (11)$$

(note that y_1 and y_2 are linearly independent solutions of (2) and hence the Wronskian, $W \neq 0$ for any $x \in I$). Integration of (11) gives us

$$u = -\int \frac{y_2 f(x)}{W} dx \text{ and } v = \int \frac{y_1 f(x)}{W} dx \quad (12)$$

(Without loss of generality, we set the values of integration constants to zero). Equations (12) and (6) yield the desired results. Thus the proof is complete. Height6pt width 6pt depth 0pt

Before, we move onto some examples, the following comments are useful.

4.3 CAUCHY'S AND LEGENDRE'S LINEAR EQUATIONS

LEARNING OBJECTIVE

3. Describe Cauchy's and Legendre's linear equations



Any equation of the form

$$(a_0 x^n D^n + a_1 x^{n-1} D^{n-1} + a_2 x^{n-2} D^{n-2} + \dots + a_n) y = X(x) \quad (13)$$

Where $a_0 \neq 0$, $a_1, a_2, \dots, a_n$ are constant is called cauchy's homogeneous linear differential equation

Equation (13) reduce to the case of linear DE with constant coefficient by sustainable by transformation of the **independent variable**

$$\text{Let } z = \log x \text{ or } x = e^z \quad \therefore \frac{dz}{dx} = \frac{1}{x}$$



We have $\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} \Rightarrow \frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{1}{x} \Rightarrow x \frac{dy}{dx} = \frac{dy}{dz}$

i.e., $x Dy = D_1 y$ Here let $D = \frac{d}{dx}$ and $D_1 = \frac{d}{dz}$

Now $\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{1}{x} \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dz} \right)$

$$= \frac{1}{x} \left\{ \frac{1}{x} \frac{d^2 y}{dz^2} + \frac{dy}{dz} \cdot \frac{-1}{x^2} \cdot \frac{dx}{dz} \right\} = \frac{1}{x^2} \frac{d^2 y}{dz^2} - \frac{1}{x^3} \cdot \frac{dy}{dz} \cdot x$$

$$= \frac{1}{x^2} (D_1^2 y - D_1 y)$$

$$\therefore x^2 = D^2 y = D_1 (D_1 - D y)$$

Similarly we can show that

$$x^3 Dy = D_1 (D_1 - 1)(D_1 - 2)y$$

$$x^r D^r y = D_1 (D_1 - 1)(D_1 - 2) \dots (D_1 - r + 1)y$$

By making this substitution (13) is reduce to linear deferential equation with constant coefficient of the form

$$(a_0' D_1^n + a_1' D_1^{n-1} + a_2' D_1^{n-2} + \dots + a_n') y = Z(z) \quad (14)$$

The solution of this equation is obtain by any one of the method discuss earlier.

4.3.1 Legendre's Linear Equations

An Equation of the form

$$\{a_0(ax+b)^n D^n + a_1(ax+b)^{n-1} D^{n-1} + \dots + a_n\} y = X(x) \quad (15)$$

Where $a_0 \neq 0, a_1, a_2, \dots, a_n$ are constant is called Legendre's equations

KEY TERMS

Independent Variable is a variable that is manipulated to determine the value of dependent variables.

**REMEMBER**

While applying Boole's rule, the number of sub-intervals should be taken as a multiple of 4.

(15) can be reduced to a linear differential equation with constant coefficient by letting $ax + b = e^z$ or $z = \log(ax + b)$.

$$\text{Now } \frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} \Rightarrow Dy = D_1 y \frac{1}{(ax + b)} \cdot a$$

$$\therefore (ax + b)Dy = aD_1 y \text{ where } D = \frac{d}{dx} \text{ and } D_1 = \frac{d}{dz}$$

$$\text{Similarly } (ax + b)^2 D^2 y = a^2 D_1(D_1 - 1)y$$

$$(ax + b)^r D^r y = a^r D_1(D_1 - 1)(D_1 - 2) \dots (D_1 - r + 1)y.$$

By making the above substitution in (15) it is reduced to a linear equation with constant coefficient

4.4 SIMULTANEOUS FIRST ORDER LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

LEARNING OBJECTIVE

4. Discuss simultaneous first order linear equations with constant coefficients



A simultaneous differential equation is one of the mathematical equations for an indefinite function of one or more than one variables that relate the values of the function. Differentiation of an equation in various orders. Differential equations play an important function in engineering, physics, economics, and other disciplines. This analysis concentrates on linear equations with constant coefficients.

We shall now consider systems of simultaneous linear differential equations which contain a single independent variable and two or more dependent variables. In general, the number of equations will be equal to the number of dependent variables i.e. if there are n dependent variables there will be n equations.

Example 1

The following two-equation system where x and y is dependent variables and t is the independent variable:



$$2\frac{dx}{dt} + \frac{dy}{dt} - 4x - y = e^t$$

$$\frac{dx}{dt} + 3x + y = 0$$

or, equivalently,

$$2(D - 2)x + (D - 1)y = e^t$$

$$(D + 3)x + y = 0$$

Where $D = d/dt$.

Example 2

The following three-equation system where x , y and z are dependent variables and t is the independent variable:

$$\frac{dx}{dt} + \frac{dy}{dt} + y = 1$$

$$\frac{dx}{dt} - \frac{dz}{dt} + 2x + z = 1$$

$$\frac{dy}{dt} + \frac{dz}{dt} + y + 2z = 0$$

or, equivalently,

$$Dx + (D + 1)y = 1$$

$$(D + 2)x - (D - 1)z = 1$$

$$(D + 1)y + (D + 2)z = 0$$

Where $D = d/dt$.

Each equation in a system will be assumed to have constant coefficients and be of the general form

$$f(D)x + g(D)y + \dots + h(D)u = G(t)$$

Where x , y ,, u are the dependent variables, t is the independent variable.

A system of n equations in the n dependent variables x , y ,, u and independent variable t will have a solution consisting of n functions

$$x = x(t)$$

$$y = y(t)$$

.....

.....

$$u = u(t)$$

The method used to solve a system of n equations in n variables is analogous to the procedure used to solve a system of n linear equations in n unknowns in algebra. In algebra, we solve a system of n equations in n unknowns by eliminating unknowns between equations until we obtain an equation containing a single unknown, from which we deduce the value of the unknown. Then we substitute the value of that unknown into other equations to obtain the values of other unknowns. In solving systems of linear differential equations we go through the same type process to obtain an equation containing a single dependent variable. The equation in this single dependent variable will be a linear differential equation with constant coefficients. We then solve this equation, using methods for solving such equations, to obtain an expression for that dependent variable. We then substitute the expression for that variable into another equation to obtain an expression for another variable. As with the algebraic problem, we can also employ determinants.

Theorem: The number of constants in the general solution of a system of equations must equal the sum of the orders of the equations.

Problem 1: Solve the linear system

$$y'' - y + 5v' = x \quad (16)$$

$$2y' - v'' + 4v = 2 \quad (17)$$

Solution: Step 1. Put the equations into operator form.

$$(D^2 - 1)y + 5Dv = x \quad (18)$$

$$2Dy - (D^2 - 4)v = 2 \quad (19)$$

Step 2: Obtain an equation in v alone by eliminating y .

Multiplying 3) through by $2D$ and 4) through by $(D^2 - 1)$ and then subtracting one from the other we obtain

$$[10D^2 + (D^2 - 1)(D^2 - 4)]v = 2Dx - (D^2 - 1)2$$

or

$$(D^4 + 5D^2 + 4)v = 4 \quad (20)$$



Step 3: Solve 5) for v.

Using methods for solving linear differential equations with constant coefficients we find the solution as

$$v = 1 + a_1 \cos x + a_2 \sin x + a_3 \cos 2x + a_4 \sin 2x \quad (21)$$

Step 4: Obtain an equation in y alone.

We could, if we wished, find an equation in y using the same method as we used in Step 2. However, for the purpose of demonstration, we will find y using determinants. Using Cramer's Rule on the system of equations 3) and 4) we get

$$\begin{vmatrix} D^2 - 1 & 5D \\ 2D & -(D^2 - 4) \end{vmatrix} y = \begin{vmatrix} x & 5D \\ 2 & -(D^2 - 4) \end{vmatrix} \quad (22)$$

Or

$$[-(D^2 - 1)(D^2 - 4) - 10D^2] y = 4x$$

Or

$$(D^4 + 5D^2 + 4) y = -4x$$

Note: We must take care to interpret the right member of 7) correctly. It is interpreted as

$$-(D^2 - 4)(x) - 5D(2)$$

Step 5: Solve 8) for y. The solution is

$$y = -x + b_1 \cos x + b_2 \sin x + b_3 \cos 2x + b_4 \sin 2x \quad (23)$$

Now the number of constants in the solution must be four since the number of constants must equal the sum of the orders of the equations in the original system. Since 1) is of order 2 and 2) is of order 2) the sum of the orders is four. Now 6) and 9) must satisfy the original equations rather than just equations 5) and 8) which resulted from the original ones after some manipulations were performed (the technique we used introduced extra constants).

$$x - 2b_1 \cos x - 2b_2 \sin x - 5b_3 \cos 2x - 5b_4 \sin 2x - 5a_1 \sin x + 5a_2 \cos x - 10a_3 \sin 2x + 10a_4 \cos 2x \equiv x \quad (24)$$

Which is an identity in x

Step 7: Equate coefficients of corresponding terms in 10) to obtain the final coefficients.

The fact that 10) is an identity in x demands that

$$-2b_1 + 5a_2 = 0$$

$$-2b_2 - 5a_1 = 0$$

$$-5b_3 + 10a_4 = 0$$

$$-5b_4 - 10a_3 = 0$$

Note: If we were to go through steps 6 and 7 for equation 2) we would get equivalent results.

Step 8: Write down the general solution. It is

$$v = 1 + a_1 \cos x + a_2 \sin x + a_3 \cos 2x + a_4 \sin 2x$$

$$y = -x + \frac{5}{2}a_2 \cos x - \frac{5}{2}a_1 \sin x + 2a_4 \cos 2x - 2a_3 \sin 2x$$



SUMMARY

1. An ordinary differential equation (ODE) is an equation that involves some ordinary derivatives (as opposed to partial derivatives) of a function.
2. First-order ordinary differential equations have some rather special properties, which result for the most part because they can only contain a limited number of terms. In fact, all linear first-order ordinary differential equations can easily be solved.
3. A simultaneous differential equation is one of the mathematical equations for an indefinite function of one or more than one variables that relate the values of the function.
4. It is the case that the coefficients of the dependent variable terms in engineering differential equations are often parameters that describe the physical properties of a system.
5. The method used to solve a system of n equations in n variables is analogous to the procedure used to solve a system of n linear equations in n unknowns in algebra.

KNOWLEDGE CHECK

1. A differential equation is considered to be ordinary if it has
 - a. one dependent variable
 - b. more than one dependent variable
 - c. one independent variable
 - d. more than one independent variable
2. For partial differential equation, if $b^2 - 4ac = 0$ then equation is called
 - a. hyperbolic
 - b. parabolic
 - c. elliptic
 - d. None of these

3. **Boundary condition which include derivative of boundary value is**

- a. Dirichlet boundary condition
- b. Neumann boundary condition
- c. Forced boundary condition
- d. Discrete boundary condition

4. **The differential equation $2\frac{dy}{dx} + x^2y = 2x + 3, y(0) = 5$ is**

- a. linear
- b. nonlinear
- c. linear with fixed constants
- d. undeterminable to be linear or nonlinear

5. **The maximum number of linearly independent solution of $\frac{d^4y}{dx^4} = 0$ with $y(0) = 1$ is**

- a. 4
- b. 3
- c. 2
- d. 1

6. **Which of the following equations is an exact DE?**

- a. A. $(x^2 + 1) dx - xy dy = 0$
- b. B. $x dy + (3x - 2y) dx = 0$
- c. C. $2xy dx + (2 + x^2) dy = 0$
- d. D. $x^2y dy - y dx = 0$

7. **Given**

$$2\frac{dy}{dx} + 3y = \sin 2x, y(0) = 6$$

$y(2)$ most nearly is

- a. 0.17643
- b. 0.29872



c. 0.32046

d. 0.58024

8. The following nonlinear differential equation can be solved exactly by separation of variables.

$$\frac{d\theta}{dt} = -10^{-6}(\theta^2 - 81), \theta(0) = 1000$$

The value of $\theta(100)$ most nearly is

a. -99.99

b. 909.10

c. 1000.32

d. 1111.10

REVIEW QUESTIONS

1. Define the homogeneous equations with constant coefficients.
2. Find the $y'' - y' - 6y = 0$.
3. Find the general solution of $y'' - 5y' = 0$.
4. Define the method of variation of parameters applies to solve:
 $a(x)y'' + b(x)y' + c(x)y = f(x)$.
5. Explain the Cauchy's and Legendre's linear equations
6. Find a 3rd order homogeneous linear equation which has a particular solution

$$y = e^t - 2e^{-2t} + 5te^{-2t}.$$

Check the Result

1. (c)

2. (b)

3. (b)

4. (a)

5. (a)

6. (b)

7. (c)

8. (b)

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Chapter

5

NUMERICAL SOLUTION OF ORDINARY DIFFER- ENTIAL EQUATIONS

LEARNING OBJECTIVES

After studying this chapter, you will be able to:

1. Understand the initial-value and boundary-value problems
2. Define Picard's method of successive approximations
3. Explain the Euler's method
4. Explain the Taylor's method
5. Explain the Runge-kutta methods
6. Discuss about the predictor-corrector methods

INTRODUCTION

A physical situation concerned with the rate of change of one quantity with respect to another gives rise to a differential equation. Consider the first order ordinary differential equation

$$\frac{dy}{dx} = f(x, y) \quad (1)$$

with the initial condition

$$y(x_0) = y_0 \quad (2)$$

Many analytical techniques exist for solving such equations, but these methods

can be applied to solve only a selected class of differential equations. However, a majority of differential equations appearing in physical problems cannot be solved analytically. Thus it becomes imperative to discuss their solution by numerical methods.

In numerical methods, we do not proceed in the hope of finding a relation between variables but we find the numerical values of the dependent variable for certain values of independent variable. It must be noted that even the differential equations which are solvable by analytical methods can be solved numerically as well.

LEARNING OBJECTIVE

1. Understand the initial-value and boundary-value problems

5.1 INITIAL-VALUE AND BOUNDARY-VALUE PROBLEMS

1. Problems in which all the conditions are specified at the initial point only are called initial-value problems. For example, the problem given by equations (1) and (2) is an **initial value problem**.

Problems involving second and higher order differential equations, in which the conditions at two or more points are specified, are called boundary-value problems.

To obtain a unique solution of n^{th} order ordinary differential equation, it is necessary to specify n values of the dependent variable and/or its derivative at specific values of independent variable.

Single Step and Multi-Step Methods

The numerical solutions are obtained step-by-step through a series of equal intervals in the independent variable so that as soon as the solution y has been obtained at $x = x_i$, the next step consists of evaluating y_{i+1} at $x = x_{i+1}$. The methods which require only the numerical value y_i in order to compute the next value y_{i+1} for solving equation (1) given above are termed as single step methods.

KEY TERMS

Initial value problem is an ordinary differential equation together with a specified value, called the initial condition, of the unknown function at a given point in the domain of the solution.



The methods which require not only the numerical value y_i but also at least one of the past values $y_{i-1}, y_{i-2}, \dots$ are termed as multi-step methods.

Comparison of Single-Step and Multi-Step Methods

The single step method has obvious advantages over the multi-step methods that use several past values ($y_n, y_{n-1}, \dots, y_{n-p}$) and that require initial values ($y_1, y_2, \dots, y_n$) that have to be calculated by another method.

5.2 PICARD'S METHOD OF SUCCESSIVE APPROXIMATIONS



Picard was a distinguished Professor of Mathematics at the university of Paris, France. He was famous for his research on the Theory of Functions. Consider the differential equation

$$\frac{dy}{dx} = f(x, y); \quad y(x_0) = y_0 \quad (3)$$

Integrating equation (3) between the limits x_0 and x and the corresponding limits y_0 and y , we get

$$\int_{y_0}^y dy = \int_{x_0}^x f(x, y) dx$$

$$\Rightarrow y - y_0 = \int_{x_0}^x f(x, y) dx$$

$$\text{or,} \quad y = y_0 + \int_{x_0}^x f(x, y) dx \quad (4)$$

In equation (4), the unknown function y appears under the integral sign. This type of equation is called integral equation. This equation can be solved by the method of successive approximations or iterations.

To obtain the first approximation, we replace y by y_0 in the R.H.S. of equation (4). Now, the first approximation is

LEARNING OBJECTIVE

2. Define Picard's method of successive approximations

$$y^{(1)} = y_0 + \int_{x_0}^x f(x, y_0) dx$$

The integrand is a function of x alone and can be integrated.

For a second approximation, replace y_0 by $y^{(1)}$ in $f(x, y_0)$ which gives

$$y^{(2)} = y_0 + \int_{x_0}^x f\{x, y^{(1)}\} dx$$

Proceeding in this way, we obtain $y^{(3)}$, $y^{(4)}$,, $y^{(n-1)}$ and $y^{(n)}$ where

$$y^{(n)} = y_0 + \int_{x_0}^x f\{x, y^{(n-1)}\} dx \text{ with } y(x_0) = y_0$$

As a matter of fact, the process is stopped when the two values of y viz. $y^{(n-1)}$ and $y^{(n)}$ are the same to the desired degree of accuracy.

Picard's method is of considerable theoretical value. Practically, it is unsatisfactory because of the difficulties which arise in performing the necessary integrations. However, each step gives a better approximation of the required solution than the preceding one.

Example

Given the differential eqn.

$$\frac{dy}{dx} = \frac{x^2}{y^2 + 1}$$

with the initial condition $y = 0$ when $x = 0$. Use Picard's method to obtain y for $x = 0.25, 0.5$ and 1.0 correct to three decimal places.

Sol. (a) The given initial value problem is

$$\frac{dy}{dx} = f(x, y) = \frac{x^2}{y^2 + 1}$$

where $y = y_0 = 0$ at $x = x_0 = 0$

We have first approximation,

$$\begin{aligned} y^{(1)} &= y_0 + \int_{x_0}^x f(x, y_0) dx \\ &= 0 + \int_0^x \frac{x^2}{0 + 1} dx = \frac{1}{3}x^3 \end{aligned} \quad (5)$$

Second approximation,

$$\begin{aligned} y^{(2)} &= y_0 + \int_{x_0}^x f\{x, y^{(1)}\} dx \\ &= 0 + \int_0^x \frac{x^2}{\left(\frac{x^3}{3}\right)^2 + 1} dx \\ &= \left[\tan^{-1} \frac{x^3}{3} \right]_0^x = \tan^{-1} \frac{x^3}{3} \\ &= \frac{1}{3}x^3 - \frac{1}{3}\left(\frac{1}{3}x^3\right)^3 + \dots \\ &= \frac{1}{3}x^3 - \frac{1}{81}x^9 + \dots \end{aligned} \quad (6)$$

From (5) and (6), we see that $y^{(1)}$ and $y^{(2)}$ agree to the first term $\frac{x^3}{3}$. To find the range of values of x so that the series with the term $\frac{1}{3}x^3$ alone will give the result correct to three decimal places, we put

$$\frac{1}{81}x^9 \leq .0005$$

which gives,

$$x^9 \leq .0405 \text{ or } x \leq 0.7$$



Hence,

$$y(.25) = \frac{1}{3} (.25)^3 = .005$$

and

$$y(0.5) = \frac{1}{3} (0.5)^3 = .042$$

To find $y(1.0)$, we make use of equation (6) which gives,

$$y(1.0) = \frac{1}{3} - \frac{1}{81} = 0.321.$$

5.2.1 Picard's Method for Simultaneous First Order Differential Equations

Let

$$\frac{dy}{dx} = \phi(x, y, z) \quad \text{and} \quad \frac{dz}{dx} = f(x, y, z)$$

be the simultaneous differential eqns. with initial conditions $y(x_0) = y_0$; $z(x_0) = z_0$. Picard's method gives

$$\begin{aligned} y^{(1)} &= y_0 + \int_{x_0}^x \phi(x, y_0, z_0) dx; & z^{(1)} &= z_0 + \int_{x_0}^x f(x, y_0, z_0) dx \\ y^{(2)} &= y_0 + \int_{x_0}^x \phi\{x, y^{(1)}, z^{(1)}\} dx; & z^{(2)} &= z_0 + \int_{x_0}^x f\{x, y^{(1)}, z^{(1)}\} dx \end{aligned}$$

and so on as successive approximations.

5.3 EULER'S METHOD

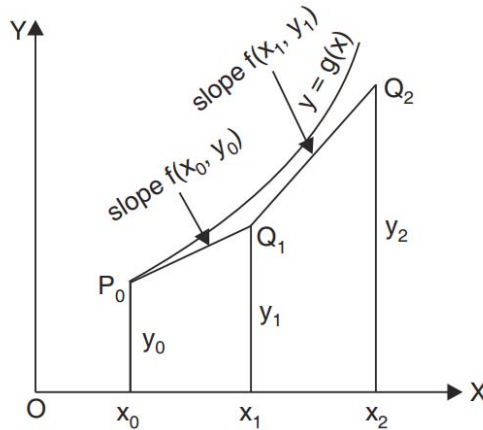


Euler's method is the simplest one-step method and has a limited application because of its low accuracy. This method yields solution of an ordinary differential equation in the form of a set of tabulated values. In this method, we determine the change Δy in y corresponding to small increase in the argument x . Consider the differential equation

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \quad (7)$$

Let $y = g(x)$ be the solution of (7). Let $x_0, x_1, x_2, \dots$ be equidistant values of x .

In this method, we use the property that in a small interval, a curve is nearly a straight line. Thus at the point (x_0, y_0) , we approximate the **curve** by the tangent at the point (x_0, y_0) .



The equation of the tangent at $P_0(x_0, y_0)$ is

$$y - y_0 = \left(\frac{dy}{dx} \right)_{P_0} (x - x_0) = f(x_0, y_0) (x - x_0)$$

$$\Rightarrow y = y_0 + (x - x_0) f(x_0, y_0) \quad (8)$$

This gives the y -coordinate of any point on the tangent. Since the curve is approximated by the tangent in the interval (x_0, x_1) , the value of y on the curve corresponding to $x = x_1$ is given by the above value of y in equation (8) approximately. Putting $x = x_1 (= x_0 + h)$ in equation (8), we get

LEARNING OBJECTIVE

3. Explain the Euler's method

KEY TERMS

Curve is an object similar to a line but that need not be straight. A curve is a generalization of a line, in that curvature is not necessarily zero.

$$y_1 = y_0 + hf(x_0, y_0)$$

Thus Q_1 is (x_1, y_1)

Similarly, approximating the curve in the next interval (x_1, x_2) by a line through $Q_1(x_1, y_1)$ with slope $f(x_1, y_1)$, we get

$$y_2 = y_1 + hf(x_1, y_1)$$

In general, it can be shown that,

$$y_{n+1} = y_n + hf(x_n, y_n)$$

This is called Euler's Formula.

A great disadvantage of this method lies in the fact that if $\frac{dy}{dx}$ changes rapidly over an interval, its value at the beginning of the interval may give a poor approximation as compared to its average value over the interval and thus the value of y calculated from Euler's method may be in much error from its true value. These errors accumulate in the succeeding intervals and the value of y becomes erroneous.

Algorithm of Euler's Method

1. Function $F(x,y)=(x-y)/(x+y)$
2. Input x_0, y_0, h, x_n
3. $n=((x_n-x_0)/h)+1$
4. For $i=1, n$
5. $y=y_0+h \cdot F(x_0, y_0)$
6. $x=x+h$
7. Print x_0, y_0

8. If $x < x_n$ then

$$x_0 = x$$

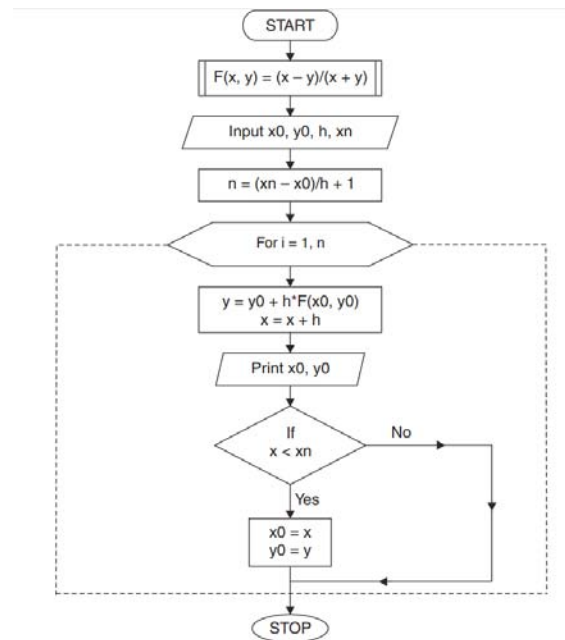
$$y_0 = y$$

ELSE

9. Next i

10. Stop

Flow-Chart of Euler's Method



Program of Euler's Method

```

#include<stdio.h>

#define F(x,y) (x-y)/(x+y)

main ( )
{

```

```

int i,n;

float x0,y0,h,xn,x,y;

printf("\n Enter the values: x0,y0,h,xn:
\n");

scanf ("%f%f%f%f", &x0,&y0,&h,&xn);

n=(xn-x0)/h+1;

for (i=1;i<=n;i++)
{
y=y0+h*F(x0,y0);
x=x0+h;

printf("\n X=%f Y=%f",x0,y0);

if(x<xn)
{
x0=x;
y0=y;
}
}

return;
}

```

Output

Enter the values: x0,y0,h,xn:

0 1 0.02 0.1

X=0.000000 Y=1.000000

X=0.020000 Y=0.980000

X=0.040000 Y=0.960800

X=0.060000 Y=0.942399

X=0.080000 Y=0.924793

X=0.100000 Y=0.907978

Notations used in the Program

(i) x_0 is the initial value of x .

(ii) y_0 is the initial value of y .

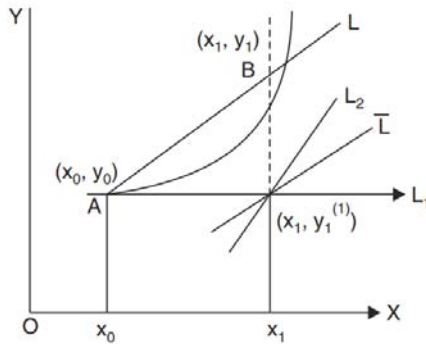
(iii) h is the spacing value of x .

(iv) x_n is the last value of x at which value of y is required.

5.3.1 Modified Euler's Method

The modified Euler's method gives greater improvement in accuracy over the original **Euler's method**. Here the core idea is that we use a line through (x_0, y_0) whose slope is the average of the slopes at (x_0, y_0) and $(x_1, y_1^{(1)})$ where $y_1^{(1)} = y_0 + hf(x_0, y_0)$. This line approximates the curve in the interval (x_0, x_1) .

Geometrically, if L_1 is the tangent at (x_0, y_0) , L_2 is a line through $(x_1, y_1^{(1)})$ of slope $f(x_1, y_1^{(1)})$ and L is the line through $(x_1, y_1^{(1)})$ but with a slope equal to the average of $f(x_0, y_0)$ and $f(x_1, y_1^{(1)})$ then the line L through (x_0, y_0) and parallel to L is used to approximate the curve in the interval (x_0, x_1) . Thus the ordinate of the point B will give the value of y_1 . Now, the equation of the line AL is given by:



$$y_1 = y_0 + (x_1 - x_0) \left[\frac{f(x_0, y_0) + f(x_1, y_1^{(1)})}{2} \right]$$

$$= y_0 + h \left[\frac{f(x_0, y_0) + f(x_1, y_1^{(1)})}{2} \right]$$

A generalized form of Euler's modified formula is

$$y_1^{(n+1)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(n)})] ; n = 0, 1, 2, \dots$$

where $y_1^{(n)}$ is the n th approximation to y_1 .

The above iteration formula can be started by choosing $y_1^{(1)}$ from Euler's formula

$$y_1^{(1)} = y_0 + hf(x_0, y_0)$$

Since this formula attempts to correct the values of y_{n+1} using the predicted value of y_{n+1} (by Euler's method), it is classified as a one-step predictor-corrector method.

Algorithm of Modified Euler's Method

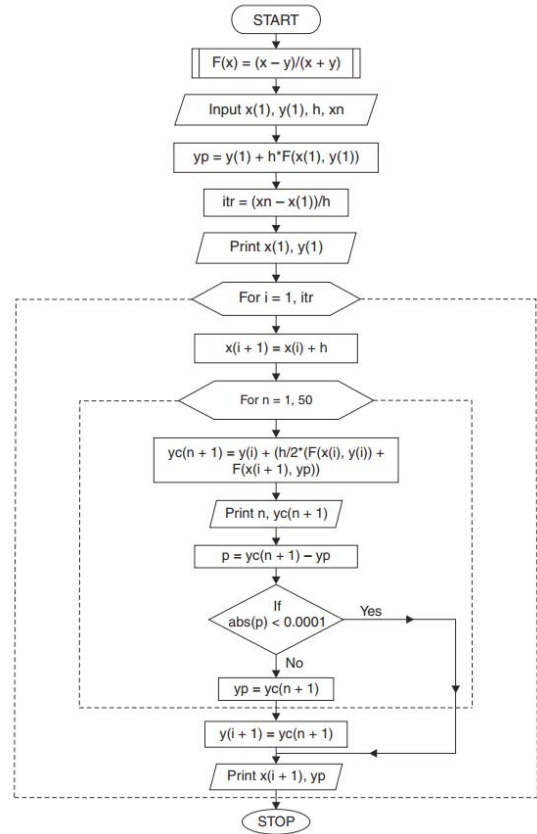
1. Function $F(x) = (x-y)/(x+y)$
2. Input $x(1), y(1), h, x_n$
3. $yp = y(1) + h * F(x(1), y(1))$
4. $itr = (x_n - x(1)) / h$

KEY TERMS

Euler's Method is a form of numerical integration — a way to approximate the solution of a first-order differential equation where the initial point on the solution curve is known, but the shape of the curve is unknown.

5. Print $x(1), y(1)$
6. For $i=1, itr$
7. $x(i+1)=x(i)+h$
8. For $n=1, 50$
9. $yc(n+1)=y(i)+(h/2*(F(x(i), y(i))+F(x(i+1), yp)))$
10. Print $n, yc(n+1)$
11. $p=yc(n+1)-yp$
12. If $abs(p)<.0001$ then
 goto Step 14
 ELSE
 $yp=yc(n+1)$
13. Next n
14. $y(i+1)=yc(n+1)$
15. print $x(i+1), yp$
16. Next i
17. Stop

Flow-Chart of Modified Euler's Method



Program of Modified Euler's Method

```

#include<stdio.h>

#include<math.h>

#define F(x,y) (x-y)/(x+y)

main ()
{
    int i,n,itr ;

```



```

float x[5],y[50],yc[50],h,yp,p,xn;

printf("\n Enter the values:
x[1],y[1],h,xn: \n");

scanf("%f%f%f%f",&x[1],&y[1],&h,&xn);

yp=y[1]+h*F(x[1],y[1]);
itr=(xn-x[1])/h;

printf("\n\n X=%f Y=%f\n",x[1],y[1];
for (i=1;i<=itr;i++)
{
x[i+1]=x[i]+h;
for (n=1;n<=50;n++)
{
yc[n+1]=y[i]+(h/2.0)*(F(x[i],y[i])+F(x[i+1],
yp));
printf("\nN=%d Y=%f",n,yc[n+1]);
p=yc[n+1]-yp;
if(fabs(p)<0.0001)
goto next;
else
yp=yc[n+1];
}
next:
y[i+1]=yc[n+1];
printf("\n\n X=%f Y=%f\n",x[i+1], yp);

```

```

}
return;
}

```

Output

Enter the values: x[1],y[1],h,xn:

0 1 0.02 0.06

X=0.000000 Y=1.000000

N=1 Y=0.980400

N=2 Y=0.980400

X=0.020000 Y=0.980400

N=1 Y=0.961584

N=2 Y=0.961598

X=0.040000 Y=0.961584

N=1 Y=0.943572

N=2 Y=0.943593

X=0.060000 Y=0.943572

Notations used in the Program

- (i) $x(1)$ is an array of the initial value of x .
- (ii) $y(1)$ is an array of the initial value of y .
- (iii) h is the spacing value of x .
- (iv) x_n is the last value of x at which value of y is required.

LEARNING OBJECTIVE

4. Explain the Taylor's method

5.4 TAYLOR'S METHOD

4 Consider the differential equation

$$\left. \begin{aligned} \frac{dy}{dx} &= f(x, y) \\ y(x_0) &= y_0 \end{aligned} \right\} \quad (9)$$

If $y(x)$ is the exact solution of (9) then $y(x)$ can be expanded into a Taylor's series about the point $x = x_0$ as

$$y(x) = y_0 + (x - x_0) y_0' + \frac{(x - x_0)^2}{2!} y_0'' + \frac{(x - x_0)^3}{3!} y_0''' + \dots \quad (10)$$

where dashes denote differentiation with respect to x .

Differentiating (9) successively with respect to x , we get

$$y'' = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} = \frac{\partial f}{\partial x} + f \frac{\partial f}{\partial y} = \left(\frac{\partial}{\partial x} + f \frac{\partial}{\partial y} \right) f \quad (11)$$

$$\begin{aligned} \therefore y''' &= \frac{d}{dx}(y'') = \left(\frac{\partial}{\partial x} + f \frac{\partial}{\partial y} \right) \left(\frac{\partial f}{\partial x} + f \frac{\partial f}{\partial y} \right) \\ &= \frac{\partial^2 f}{\partial x^2} + \frac{\partial f}{\partial x} \frac{\partial f}{\partial y} + f \frac{\partial^2 f}{\partial x \partial y} + f \frac{\partial^2 f}{\partial y \partial x} + f \left(\frac{\partial f}{\partial y} \right)^2 + f^2 \frac{\partial^2 f}{\partial y^2} \end{aligned} \quad (12)$$

and so on.

Putting $x = x_0$ and $y = y_0$ in the expressions for y' , y'' , y''' , and substituting them in equation (10), we get a power series for $y(x)$ in powers of $x - x_0$.

i.e.,

$$\begin{aligned} y(x) &= y_0 + (x - x_0) y_0' + \frac{(x - x_0)^2}{2!} y_0'' \\ &\quad + \frac{(x - x_0)^3}{3!} y_0''' + \dots \end{aligned} \quad (13)$$

Putting $x = x_1 (= x_0 + h)$ in (13), we get

$$y_1 = y(x_1) = y_0 + h y_0' + \frac{h^2}{2!} y_0'' + \frac{h^3}{3!} y_0''' + \dots \quad (14)$$

Here y_0' , y_0'' , y_0''' , can be found by using (9) and its successive differentiations (11) and (12) at $x = x_0$. The series (14) can be truncated at any stage if h is small.

After obtaining y_1 , we can calculate y_1' , y_1'' , y_1''' , from (9)



at $x_1 = x_0 + h$. Now, expanding $y(x)$ by Taylor's series about $x = x_1$, we get

$$y_2 = y_1 + hy'_1 + \frac{h^2}{2!} y''_1 + \frac{h^3}{3!} y'''_1 + \dots$$

Proceeding, we get

$$y_n = y_{n-1} + hy'_{n-1} + \frac{h^2}{2!} y''_{n-1} + \frac{h^3}{3!} y'''_{n-1} + \dots$$

Practically, this method is not of much importance because of its need of **partial derivatives**.

Moreover if we are interested in a better approximation with a small truncation error, the evaluation of higher order derivatives is needed which are complicated in evaluation. Besides its impracticability, it is useful in judging the degree of accuracy of the approximations given by other methods.

We can determine the extent to which any other formula agrees with the Taylor's series expansion. Taylor's method is one of those methods which yield the solution of a differential equation in the form of a power series. This method suffers from a serious disadvantage that h should be small enough so that successive terms in the series diminish quite rapidly.

KEY TERMS

Partial derivative of a function of several variables is its derivative with respect to one of those variables, with the others held constant.

5.4.1 Taylor's Method for Simultaneous I Order Differential Equations

Simultaneous differential equations of the type

$$\frac{dy}{dx} = f(x, y, z) \quad (15)$$

and

$$\frac{dz}{dx} = \phi(x, y, z) \quad (16)$$

with initial conditions $y(x_0) = y_0$ and $z(x_0) = z_0$ can be solved by Taylor's method.

If h is the step-size then

$$y_1 = y(x_0 + h) \quad \text{and} \quad z_1 = z(x_0 + h)$$

Taylor's algorithm for (15) and (16) gives

$$y_1 = y_0 + hy_0' + \frac{h^2}{2!} y_0'' + \frac{h^3}{3!} y_0''' + \dots \quad (17)$$

and

$$z_1 = z_0 + hz_0' + \frac{h^2}{2!} z_0'' + \frac{h^3}{3!} z_0''' + \dots \quad (18)$$

Differentiating (15) and (16) successively, we get $y'', y''', \dots, z'', z''', \dots$ etc. So the values $y_0'', y_0''', \dots$ and $z_0'', z_0''', \dots$ can be obtained.

Substituting them in (17) and (18), we get y_1, z_1 for the next step.

$$y_2 = y_1 + hy_1' + \frac{h^2}{2!} y_1'' + \frac{h^3}{3!} y_1''' + \dots$$

and

$$z_2 = z_1 + hz_1' + \frac{h^2}{2!} z_1'' + \frac{h^3}{3!} z_1''' + \dots$$

Since y_1 and z_1 are known, $y_1', y_1'', y_1''', \dots, z_1', z_1'', z_1''', \dots$ can be calculated. Hence y_2 and z_2 can be obtained. Proceeding in this manner, we get other values of y , step-by-step.

Example

For the differential equation, $\frac{dy}{dx} = -xy^2, y(0) = 2$. Calculate $y(0.2)$ by Taylor's series method retaining four non-zero terms only.

Sol.

Here $x_0 = 0, y_0 = 2$ Also $y' = -xy^2$

Taylor's series for $y(x)$ is given by

$$\begin{aligned} y(x) = y_0 + xy_0' + \frac{x^2}{2} y_0'' + \frac{x^3}{6} y_0''' + \frac{x^4}{24} y_0^{(iv)} \\ + \frac{x^5}{120} y_0^{(v)} + \dots \end{aligned} \quad (19)$$

The values of the derivatives $y_0', y_0'', \dots$, etc. are obtained as follows:

$$\begin{aligned}
 y' &= -xy^2 & y_0' &= -x_0 y_0^2 = 0 \\
 y'' &= -y^2 - 2xyy' & y_0'' &= -2^2 - 0 = -4 \\
 y''' &= -4yy' - 2xy'^2 - 2xyy'' & y_0''' &= 0 \\
 y^{(iv)} &= -6y'^2 - 6y'y'' - 6xy'y'' - 2xyy''' & y_0^{(iv)} &= 48 \\
 y^{(v)} &= -24y'y'' - 8yy''' - 6xy'^2 & y_0^{(v)} &= 0 \\
 &\quad - 8xy'y''' - 2xyy^{(iv)} & & \\
 y^{(vi)} &= -40y'y''' - 30y'^2 - 10yy^{(iv)} - 20xy''y''' & y_0^{(vi)} &= -1440 \\
 &\quad - 10xy'y^{(iv)} - 2xyy^{(v)}. & &
 \end{aligned}$$

We stop here as we shall get four non-zero terms in the Taylor's series (19).

$$\begin{aligned}
 \therefore y(x) &= 2 + \frac{x^2}{2} (-4) + \frac{x^4}{24} (48) + \frac{x^6}{720} (-1440) + \dots \\
 &= 2 - 2x^2 + 2x^4 - 2x^6 + \dots \\
 \therefore y(0.2) &= 2 - 2(0.2)^2 + 2(0.2)^4 - 2(0.2)^6 + \dots \\
 &= 2 - 0.08 + 0.0032 - 0.000128 = 1.923072 \\
 &\approx 1.9231 \quad \text{correct up to four decimal places.}
 \end{aligned}$$

Example

From the Taylor's series, for $y(x)$, find $y(0.1)$ correct to four decimal places if $y(x)$ satisfies $\frac{dy}{dx} = x - y^2$ and $y(0) = 1$. Also find $y(0.2)$.

Sol.

$$\begin{aligned}
 \text{Here } x_0 &= 0, & y_0 &= 1 \\
 y' &= x - y^2 & y_0' &= 0 - 1 = -1 \\
 y'' &= 1 - 2yy' & y_0'' &= 3 \\
 y''' &= -2yy'' - 2y'^2 & y_0''' &= -8 \\
 y^{(iv)} &= -2yy''' - 6y'y'' & y_0^{(iv)} &= 34 \\
 y^{(v)} &= -2yy^{(iv)} - 8y'y''' - 6y''^2 & y_0^{(v)} &= -186 \\
 y^{(vi)} &= -2yy^{(v)} - 10y'y^{(iv)} - 20y''y''' & y_0^{(vi)} &= 1192 \\
 y^{(vii)} &= -2yy^{(vi)} - 12y'y^{(v)} - 50y''y^{(iv)} & y_0^{(vii)} &= -10996 \\
 &\quad - 20y'''^2 & &
 \end{aligned}
 \left. \begin{array}{l} \\ \\ \\ \\ \\ \\ \end{array} \right\} \text{ only for } y(0.2)$$

Using these values, Taylor's series becomes

$$y(x) = 1 - x + \frac{3}{2}x^2 - \frac{4}{3}x^3 + \frac{17}{12}x^4 - \frac{31}{20}x^5 + \dots \quad (20)$$

Put

$x = 0.1$ in (20), we get

$$y(0.1) = 0.91379 \sim 0.9138 \quad (\text{upto four decimal places})$$

To determine $y(0.2)$, we have

$$\begin{aligned} y(x) &= 1 - x + \frac{3}{2}x^2 - \frac{4}{3}x^3 + \frac{17}{12}x^4 - \frac{31}{20}x^5 + \frac{1192}{720}x^6 - \frac{10996}{5040}x^7 + \dots \\ &= 0.8512 \quad (\text{correct to four decimal places}). \end{aligned}$$

5.5 RUNGE-KUTTA METHODS



More efficient methods in terms of accuracy were developed by two German Mathematicians Carl Runge (1856-1927) and Wilhelm Kutta (1867-1944). These methods are well-known as Runge-Kutta methods. They are distinguished by their orders in the sense that they agree with **Taylor's series** solution up to terms of h^r where r is the order of the method.

These methods do not demand prior computation of higher derivatives of $y(x)$ as in Taylor's method. In place of these derivatives, extra values of the given function $f(x, y)$ are used. The fourth order Runge-Kutta method is used widely for finding the numerical solutions of linear or non-linear ordinary differential equations.

Runge-Kutta methods are referred to as single step methods. A class of methods known as Runge-Kutta methods combines the advantage of high order accuracy with the property of being one step.

5.5.1 First Order Runge-Kutta Method

Consider the differential equation

LEARNING OBJECTIVE

5. Explain the Runge-kutta methods

KEY TERMS

Taylor series is a representation of a function as an infinite sum of terms that are calculated from the values of the function's derivatives at a single point.

$$\frac{dy}{dx} = f(x, y); \quad y(x_0) = y_0 \quad (21)$$

Euler's method gives

$$y_1 = y_0 + hf(x_0, y_0) = y_0 + hy_0' \quad (22)$$

Expanding by Taylor's series, we get

$$y_1 = y(x_0 + h) = y_0 + hy_0' + \frac{h^2}{2!} y_0'' + \dots \quad (23)$$

Comparing (22) and (23), it follows that Euler's method agrees with Taylor's series solution up to the term in h . Hence Euler's method is the first order **Runge-Kutta method**.



REMEMBER

The major disadvantage of Runge-Kutta methods is that they use many more evaluations of the derivative $f(x, y)$ to obtain the same accuracy compared with multi-step methods.

5.5.2 Second Order Runge-Kutta Method

Consider the differential equation

$$y' = f(x, y) \text{ with the initial condition } y(x_0) = y_0$$

Let h be the interval between equidistant values of x then in II order Runge-Kutta method, the first increment in y is computed from the formulae

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf(x_0 + h, y_0 + k_1)$$

$$\Delta y = \frac{1}{2} (k_1 + k_2)$$

taken in the given order.

Then,

$$x_1 = x_0 + h$$

$$y_1 = y_0 + \Delta y = y_0 + \frac{1}{2} (k_1 + k_2)$$

In a similar manner, the increment in y for the second interval is computed by means of the formulae,

KEY TERMS

Runge-Kutta methods are a family of implicit and explicit iterative methods, which includes the well-known routine called the Euler Method, used in temporal discretization for the approximate solutions of ordinary differential equations.

$$k_1 = hf(x_1, y_1)$$

$$k_2 = hf(x_1 + h, y_1 + k_1)$$

$$\Delta y = \frac{1}{2}(k_1 + k_2)$$

and similarly for the next intervals.

The inherent error in the second order Runge-Kutta method is of order h^3 .

5.5.3 Third Order Runge-Kutta Method

This method gives the approximate solution of the initial value problem

$$\frac{dy}{dx} = f(x, y); y(x_0) = y_0 \text{ as}$$

where

$$\left. \begin{aligned} y_1 &= y_0 + \delta y \\ \delta y &= \frac{h}{6}(k_1 + 4k_2 + k_3) \end{aligned} \right\} \quad (24)$$

Here,

$$k_1 = f(x_0, y_0)$$

$$k_2 = f\left\{x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right\}$$

$$k_3 = f(x_0 + h, y_0 + k'); \quad k' = hf(x_0 + h, y_0 + k_1)$$

Formula (24) can be generalized for successive approximations. Expression in (24) agrees with Taylor's series expansion for y_1 up to and including terms in h^3 . This method is also known as Runge's method.

5.5.4 Fourth Order Runge-Kutta Method

The fourth order Runge-Kutta Method is one of the most widely used methods and is particularly suitable in cases when the computation of higher derivatives is complicated. Consider the differential equation $y' = f(x, y)$ with the initial condition $y(x_0) = y_0$. Let h be the interval between equidistant values of x , then the first increment in y is computed from the formulae



$$\left. \begin{aligned}
 k_1 &= hf(x_0, y_0) \\
 k_2 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \\
 k_3 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) \\
 k_4 &= hf(x_0 + h, y_0 + k_3) \\
 \Delta y &= \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)
 \end{aligned} \right\} \quad (25)$$

taken in the given order.

Then,

$$x_1 = x_0 + h \text{ and } y_1 = y_0 + \Delta y$$

In a similar manner, the increment in y for the II interval is computed by means of the formulae

$$\begin{aligned}
 k_1 &= hf(x_1, y_1) \\
 k_2 &= hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) \\
 k_3 &= hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) \\
 k_4 &= hf(x_1 + h, y_1 + k_3) \\
 \Delta y &= \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)
 \end{aligned}$$

and similarly for the next intervals.

This method is also simply termed as Runge-Kutta's method.

It is to be noted that the calculations for the first increment are exactly the same as

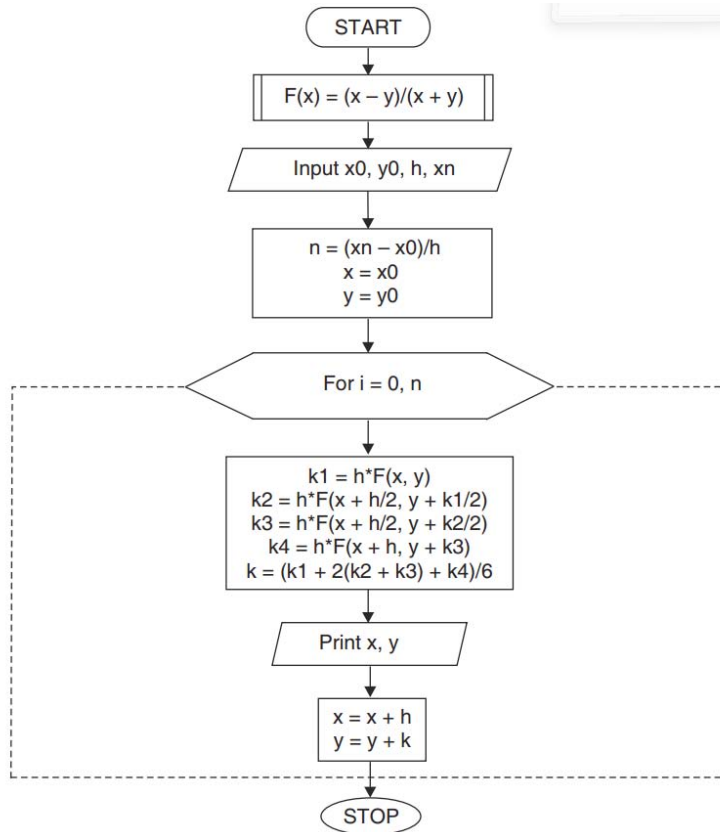
for any other increment. The change in the formula for the different intervals is only in the values of x and y to be substituted. Hence, to obtain Δy for the n th interval, we substitute x_{n-1} , y_{n-1} in the expressions for k_1 , k_2 , etc.

The inherent error in the fourth order Runge-Kutta method is of the order h^5 .

Algorithm of Runge-Kutta Method

1. Function $F(x)=(x-y)/(x+y)$
2. Input x_0, y_0, h, x_n
3. $n=(x_n-x_0)/h$
4. $x=x_0$
5. $y=y_0$
6. For $i=0, n$
7. $k_1=h*F(x, y)$
8. $k_2=h*F(x+h/2, y+k_1/2)$
9. $k_3=h*F(x+h/2, y+k_2/2)$
10. $k_4=h*F(x+h, y+k_3)$
11. $k=(k_1+(k_2+k_3)*2+k_4)/6$
12. Print x, y
13. $x=x+h$
14. $y=y+k$
15. Next i
16. Stop

Flow-Chart of Runge-Kutta Method



Program of Runge-Kutta Method

```

#include<stdio.h>

#define F(x,y) (x-y)/(x+y)

main()
{
    int i,n;

    float x0,y0,h,xn,k1,k2,k3,k4,x,y,k;

    printf("\n Enter the values: x0,y0,h,xn:\n");

```

```
scanf("%f%f%f%f", &x0,&y0,&h,&xn);
n=(xn-x0)/h;
x=x0;
y=y0;
for(i=0;i<=n;i++)
{
    k1=h*F(x,y);
    k2=h*F(x+h/2.0,y+k1/2.0);
    k3=h*F(x+h/2.0,y+k2/2.0);
    k4=h*F(x+h,y+k3);
    k=(k1+(k2+k3)*2.0+k4)/6.0;
    printf("\n X=%f Y=%f", x, y);
    x=x+h;
    y=y+k;
}
return;
}
```

Output

Enter the values: x0,y0,h,xn:

0 1 0.02 0.1

X=0.000000 Y=1.000000

X=0.020000 Y=0.980000

X=0.040000 Y=0.960816

X=0.060000 Y=0.942446

X=0.080000 Y=0.924885

X=0.100000 Y=0.908128

Notations used in the Program

(i) x_0 is the initial value of x .

(ii) y_0 is the initial value of y .

(iii) h is the spacing value of x .

(iv) x_n is the last value of x at which value of y is required.

5.5.5 Runge-Kutta Method for Simultaneous First Order Equations

Consider the simultaneous equations

$$\frac{dy}{dx} = f_1(x, y, z)$$

$$\frac{dz}{dx} = f_2(x, y, z)$$

With the initial condition $y(x_0) = y_0$ and $z(x_0) = z_0$. Now, starting from (x_0, y_0, z_0) , the increments k and l in y and z are given by the following formulae:

$$k_1 = hf_1(x_0, y_0, z_0);$$

$$l_1 = hf_2(x_0, y_0, z_0)$$

$$k_2 = hf_1\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right);$$

$$l_2 = hf_2\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right)$$

$$k_3 = hf_1\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right);$$

$$l_3 = hf_2\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right)$$

$$k_4 = hf_1(x_0 + h, y_0 + k_3, z_0 + l_3);$$

$$l_4 = hf_2(x_0 + h, y_0 + k_3, z_0 + l_3)$$

$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4);$$

$$l = \frac{1}{6}(l_1 + 2l_2 + 2l_3 + l_4)$$

Hence

$$y_1 = y_0 + k, \quad z_1 = z_0 + l$$

To compute y_2, z_2 , we simply replace x_0, y_0, z_0 by x_1, y_1, z_1 in the above formulae.

5.6 PREDICTOR-CORRECTOR METHODS



In Runge-Kutta Methods, we need only the information at (x_i, y_i) to calculate the value of y_{i+1} and no attention is paid to the nature of the solution at the earlier points.

LEARNING OBJECTIVE



6. Discuss about the predictor-corrector methods

To overcome this defect, Predictor-Corrector Methods are useful. The technique of refining an initially crude predicted estimate of y_i by means of a more accurate corrector formula is called, Predictor-Corrector Method.

The modified Euler's Method of solving the initial value problem,

$$y' = f(x, y), y(x_0) = y_0 \quad (26)$$

can be stated as

$$y_1^p = y_0 + hf(x_0, y_0) \quad (27)$$

$$y_1^c = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^p)] \quad (28)$$

Here we predict the value of y_1 by Euler's Method and use it in (28) to get a corrected or improved value. This is a typical case of Predictor-Corrector Method.

We will obtain two important Predictor-Corrector Methods, namely, Milne's Simpson Method and Adams-Moulton (or Adams-Bash Fourth) Method. Both of these methods are of IV order and the error is of order h^5 . These methods make use of four starting values of y , namely, y_0, y_1, y_2 , and y_3 . Hence, these methods are also called as Multi-Step Methods.

5.6.1 Milne's Method

Milne's Method is a simple and reasonably accurate method of solving differential equations numerically. To solve the differential equation $y' = f(x, y)$ by this method, first we get the approximate value of y_{n+1} by predictor formula and then improve this value using a corrector formula. These formula are derived from Newton's Formula.

Newton's Forward Interpolation Formula in terms of y' and u is

$$y' = y_0' + u\Delta y_0' + \frac{u(u-1)}{2} \Delta^2 y_0' + \frac{u(u-1)(u-2)}{6} \Delta^3 y_0' + \frac{u(u-1)(u-2)(u-3)}{24} \Delta^4 y_0' + \dots \quad (29)$$

where

$$u = \frac{x - x_0}{h} \quad \text{or} \quad x = x_0 + uh$$

Now integrating (29) over the interval x_0 to $x_0 + 4h$ (or $u = 0$ to 4), we get

$$\int_{x_0}^{x_0 + 4h} y' dx = h \int_0^4 y' du \quad | \quad \because dx = h du$$

or

$$\begin{aligned}
 y_4 - y_0 &= h \int_0^4 \left[y'_0 + u \Delta y'_0 + \frac{u(u-1)}{2} \Delta^2 y'_0 + \frac{u(u-1)(u-2)}{6} \Delta^3 y'_0 \right. \\
 &\quad \left. + \frac{u(u-1)(u-2)(u-3)}{24} \Delta^4 y'_0 + \dots \right] du \\
 &= h \left(4y'_0 + 8\Delta y'_0 + \frac{20}{3} \Delta^2 y'_0 + \frac{8}{3} \Delta^3 y'_0 + \frac{28}{90} \Delta^4 y'_0 \right)
 \end{aligned}$$

Here, y_0 and y_4 stand for values of y at $x = x_0$ and $x = x_0 + 4h$ respectively. Substituting the values of I, II and III differences, we get

$$\begin{aligned}
 y_4 - y_0 &= h \left(4y'_0 + 8(E-1)y'_0 + \frac{20}{3}(E-1)^2 y'_0 + \frac{8}{3}(E-1)^3 y'_0 + \frac{28}{90} \Delta^4 y'_0 \right) \\
 &= \frac{4h}{3} (2y'_1 - y'_2 + 2y'_3) + \frac{28}{90} h \Delta^4 y'_0
 \end{aligned} \tag{30}$$

or

$$y_4 = y_0 + \frac{4h}{3} (2y'_1 - y'_2 + 2y'_3) + \frac{28}{90} h \Delta^4 y'_0 \tag{31}$$

This is Milne's Predictor (Extrapolation) formula.

It is used to predict the value of y_4 when the value of y_0 , y_1 , y_2 , and y_3 are known.

To obtain the corrector formula, we integrate (29) over the interval x_0 to $x_0 + 2h$ (or $u = 0$ to 2) and consequently.

$$y_2 - y_0 = h \left(2y'_0 + 2\Delta y'_0 + \frac{1}{3} \Delta^2 y'_0 - \frac{1}{90} \Delta^4 y'_0 \right)$$

Expressing the I, II and III differences in terms of the function value by using $D \equiv E - 1$,

we obtain,

$$\begin{aligned}
 y_2 - y_0 &= \frac{h}{3} (y'_0 + 4y'_1 + y'_2) - \frac{h}{90} \Delta^4 y'_0 \\
 \Rightarrow \quad y_2 &= y_0 + \frac{h}{3} (y'_0 + 4y'_1 + y'_2) - \frac{h}{90} \Delta^4 y'_0
 \end{aligned} \tag{32}$$

This is Milne's Corrector Formula

The value of y_4 obtained from (31) and (32) can be put as

$$y_{n+1} = y_{n-3} + \frac{4h}{3}(2y'_{n-2} - y'_{n-1} + 2y'_n) \quad (33)$$

$$y'_{n+1} = y'_{n-1} + \frac{h}{3}(y'_{n-1} + 4y'_n + y'_{n+1}) \quad (34)$$

It is to be noted that we have considered the differences up to the third order because we fit up a **polynomial** of degree four.

The terms containing $\Delta^4 y'_0$ are not used explicitly in the formula, but they give the principal parts of the errors in the two values of y_{n+1} as computed from (33) and (34).

We notice that this error in (34) is of opposite sign to that in (33) but it is very small in magnitude.

So we may take,

$$(y_{n+1})_{\text{exact}} = y_{n+1} + \frac{28}{90} h \Delta^4 y'$$

and

$$(y_{n+1})_{\text{exact}} = y^{(1)}_{n+1} - \frac{h}{90} \Delta^4 y'$$

where y_{n+1} and $y^{(1)}_{n+1}$ denote the predicted and first corrected value of y at $x = x_{n+1}$.

Equating these two values, we get

$$y_{n+1} - y^{(1)}_{n+1} = -\frac{29}{90} h \Delta^4 y' = 29 \delta$$

where $\delta = -\frac{h}{90} \Delta^4 y'$ denotes the principal part of the error in (34). Thus it gives

$$\delta = \frac{1}{29} [y_{n+1} - y^{(1)}_{n+1}]$$

Thus we observe that the error in (34) is $\frac{1}{29}^{\text{th}}$ of the difference between the predicted and corrected values.

KEY TERMS

Polynomial is a mathematical expression consisting of a sum of terms, each term including a variable or variables raised to a power and multiplied by a coefficient.

Algorithm of Milne's Predictor-Corrector Method

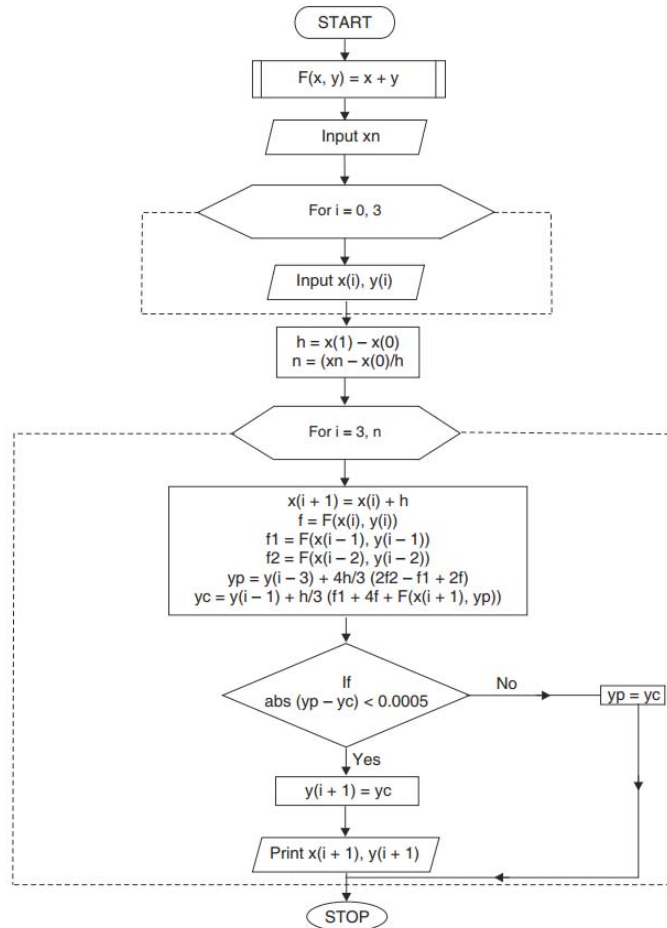
1. Function $F(x,y)=x+y$
2. Input x_n
3. For $i=0,3$
4. Input $x(i),y(i)$
5. Next i
6. $h=x(1)-x(0)$
7. $n=(x_n-x(0))/h$
8. For $i=3,n$
9. $x(i+1)=x(i)+h$
10. $f=F(x(i),y(i))$
11. $f1=F(x(i-1),y(i-1))$
12. $f2=F(x(i-2),y(i-2))$
13. $yp=y(i-3)+4h/3(2f2-f1+2f)$
14. $yc=y(i-1)+h/3(f1+4f+F(x(i+1),yp))$
15. If $\text{abs}(yp-yc)<0.0005$ then

$y(i+1)=yc$
 print $x(i+1), y(i+1)$

ELSE

$yp=yc$
16. Next i
17. Stop

Flow-Chart of Milne's Predictor Corrector Method



Program of Milne's Method

```

#include<stdio.h>
#include<math.h>

#define F(x,y) x+y

main()
{
    int i,n;
    float x[20],y[20],h,f,f1,f2,yp,yc,xn;

```

```

printf("\n Enter the value: xn: ");
scanf("%f",&xn);
printf("\n Enter the value: x[i], y[i]:\n");
for(i=0;i<=3;i++)
scanf("%f%f",&x[i],&y[i]);
h=x[1]-x[0];
n=(xn-x[0]/h;
for(i=3;i<=n;i++)
{
x[i+1]=x[i]+h;
f=F[x[i],y[i]);
f1=F(x[i-1],y[i-1]);
f2=F(x[i-2],y[i-2]);
yp=y[i-3]+4.0*h/3.0*(2.0*f2-f1+2.0*f);
yc=y[i-1]+h/3.0*(f1+4.0*f+F(x[i+1],yp));
printf("\n\nPredicated Y=%f Correctd Y=%f", yp,yc);
If(fabs (yp-yc)<0.00005)
goto next;
yp=yc;
next;
y[i+1]=yc;
printf("\n\n X=%f Y=%f", x[i+1], y[i+1]);
}
return;
}

```



Output

Enter the value: xn: 1

Enter the value: x[i], y[i]:

0.0 0.0

0.2 0.02

0.4 0.0906

0.6 0.2214

Predicted Y=0.423147 Corrected Y=0.429650

X=0.800000 Y=0.429650

Predicted Y=0.721307 Corrected Y=0.718820

X=1.000000 Y=0.718820

Notations used in the Program

(i) x_n is the last value of x at which value of y is required.

(ii) x(i) is an array for prior values of x.

(iii) y(i) is an array for prior values of y.

(iv) y_p is the predicted value of y.

(v) y_c is the corrected value of y.

SUMMARY

1. Problems involving second and higher order differential equations, in which the conditions at two or more points are specified, are called boundary-value problems.
2. The numerical solutions are obtained step-by-step through a series of equal intervals in the independent variable so that as soon as the solution y has been obtained at $x = x_i$, the next step consists of evaluating y_{i+1} at $x = x_{i+1}$.
3. Picard's method is of considerable theoretical value. Practically, it is unsatisfactory because of the difficulties which arise in performing the necessary integrations.
4. Euler's method is the simplest one-step method and has a limited application because of its low accuracy. This method yields solution of an ordinary differential equation in the form of a set of tabulated values.
5. The modified Euler's method gives greater improvement in accuracy over the original Euler's method. Here the core idea is that we use a line through (x_0, y_0) whose slope is the average of the slopes at (x_0, y_0) and $(x_1, y_1^{(1)})$ where $y_1^{(1)} = y_0 + hf(x_0, y_0)$.
6. The major disadvantage of Runge-Kutta methods is that they use many more evaluations of the derivative $f(x, y)$ to obtain the same accuracy compared with multi-step methods. A class of methods known as Runge-Kutta methods combines the advantage of high order accuracy with the property of being one step.
7. Predictor-Corrector Methods are useful. The technique of refining an initially crude predicted estimate of y_i by means of a more accurate corrector formula is called, Predictor-Corrector Method.

KNOWLEDGE CHECK

1. In which of the following method, we approximate the curve of solution by the tangent in each interval.
 - a. Picard's method
 - b. Euler's method
 - c. Newton's method
 - d. Runge Kutta method



2. Given $3\frac{dy}{dx} + y^2 = e^x$, $y(0.3) = 5$, and using a step size of $h=0.3$, the best estimate of $\frac{dy}{dx}(0.9)$ Runge-Kutta 4th order method is most nearly

- a. -1.6604
- b. -1.1785
- c. -0.45831
- d. 2.7270

3. The velocity (m/s) of a parachutist is given as a function of time (seconds) by $V(t)=55.5 \tan(0.17t)$, $t \geq 0$

Using Runge-Kutta 4th order method with a step size of 5 seconds, the distance traveled by the body from $t=2$ to $t=12$ seconds is estimated most nearly as

- a. 341.43 m
- b. 428.97 m
- c. 429.05 m
- d. 703.50 m

4. The convergence of which of the following method is sensitive to starting value?

- a. False position
- b. Gauss seidal method
- c. Newton-Raphson method
- d. All of these

5. Given $3\frac{dy}{dx} + 5y^2 = \sin x$, $y(0.3) = 5$ and using a step size of $h=0.3$, the value of $y(0.9)$ using Runge-Kutta 4th order method is most nearly

- a. 0.25011×10^{40}
- b. -4297.4
- c. -1261.5
- d. 0.88498

6. Find the coefficient of x^2 in the Taylor series about $x=0$ for $F(x) = e^{-x^2}$.

- a. 1/4
- b. -1

- c. $1/2$
 - d. -2
 - e. 1
7. Find the coefficient of x^3 in the Taylor series about $x=0$ for $f(x)=\sin 2x$.
- a. $-2/3$
 - b. $-4/3$
 - c. $4/3$
 - d. $-8/3$
 - e. $2/3$
8. The convergence of which of the following method is sensitive to starting value?
- a. False position
 - b. Gauss seidal method
 - c. Newton-Raphson method
 - d. All of these
9. Which of the following are true statements?
- a. Any function $f(x)$ is equal to its Taylor series for all real x .
 - b. There exist functions $f(x)$ which are equal to their Taylor series for all real x .
 - c. There exist functions $f(x)$ which are equal to their Taylor series for some, but not all, real numbers x .
 - d. A function $f(x)$ can never equal its Taylor series. The Taylor series is only ever an approximation to the function.
10. Taylor's method is one of those methods which yield the solution of a differential equation in the form of a power series.
- a. True
 - b. False

REVIEW QUESTIONS

1. Find $y(0.2)$ if $\frac{dy}{dx} = \log(x + y)$; $y(0) = 1$. Use Picard's method.
2. Find y for $x = 0.2$ and $x = 0.5$ using modified Euler's method, given that $\frac{dy}{dx} = \log_e(x + y)$; $y(0) = 1$
3. Solve by Taylor's method, $y' = x^2 + y^2$, $y(0) = 1$ compute $y(0.1)$.



4. Use the Runge-Kutta Method to approximate y when $x = 0.1$ given that $x = 0$ when

$$y = 1 \text{ and } \frac{dy}{dx} = x + y.$$

5. Prove that the solution of $y' = y$, $y(0) = 1$ by Second Order Runge-Kutta Method yields

$$y_m = \left(1 + h + \frac{h^2}{2}\right)^m.$$

6. Compute the true solution to the problem

$$Y'(t) = -e^{-t}Y(t), Y(0) = 1.$$

Using Euler's method, solve this equation numerically with stepsizes of $h = 0.2, 0.1, 0.05$. Compute the error and relative error using the true solution $Y(t)$.

Check Your Result

- | | | | | | |
|--------|--------|--------|---------|--------|--------|
| 1. (b) | 2. (a) | 3. (a) | 4. (c) | 5. (c) | 6. (b) |
| 7. (b) | 8. (c) | 9. (c) | 10. (a) | | |

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Chapter

6

PARTIAL DIFFERENTIAL EQUATIONS

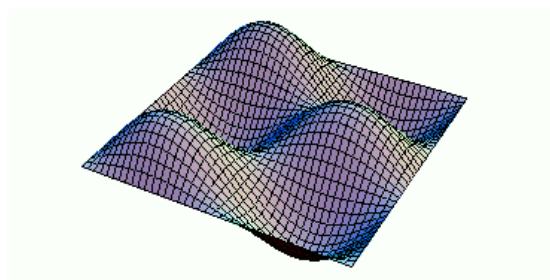
LEARNING OBJECTIVES

After studying this chapter, you will be able to:

1. Understand the concept of partial differential equation
2. Explain linear partial differential equations of first order
3. Describe second order linear partial differential equations

INTRODUCTION

A partial differential equation (PDE) is a mathematical equation that involves two or more independent variables, an unknown function, and partial derivatives of the unknown function with respect to the independent variables. The order of a partial differential equation is the order of the highest derivative involved. A solution (or a particular solution) to a partial differential equation is a function that solves the equation or, in other words, turns it into an identity when substituted into the equation. A solution is called general if it contains all particular solutions of the equation concerned.



A differential equation involving partial derivatives of a dependent variable (one or more) with more than one independent variable is called a partial differential equation. To start with partial differential equations, just like ordinary differential or integral equations, are functional equations. That means that the unknown, or unknowns, we are trying to determine are functions. In the case of partial differential equations (PDE) these functions are to be determined from equations which involve, in addition to the usual operations of addition and multiplication, partial derivatives of the functions.

6.1 CONCEPT OF PARTIAL DIFFERENTIAL EQUATION

LEARNING OBJECTIVE

1. Understand the concept of partial differential equation



A differential equation which involves partial derivatives is called a partial differential equation. For example,

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0 \quad (a)$$

$$\left(\frac{\partial^2 z}{\partial x^2} \right)^3 + x \frac{\partial z}{\partial y} = z. \quad (b)$$

The order of a partial differential equation is the order of the highest partial derivatives in the equation. The degree of a partial differential equation is the degree of the highest order partial derivatives occurring in the equation. Thus, in

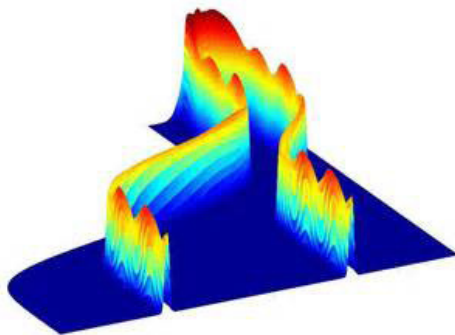
above example, the first equation is first order and the second one is second order with degree three.

Notations: If z is a function of two independent variables x , and y , then we shall use the following notation for the partial derivatives of z :

$$p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}, r = \frac{\partial^2 z}{\partial x^2}, s = \frac{\partial^2 z}{\partial x \partial y}, t = \frac{\partial^2 z}{\partial y^2}$$

6.1.1 Formation of Partial Differential Equation

Partial differential equations can be formed either by the elimination of arbitrary constants or by the elimination of arbitrary functions. If the number of arbitrary constants to be eliminated is equal to the number of independent variables, the partial differential equations that arise are of the first order. If the number of **arbitrary constant** to be eliminated is more than the number of independent variables, the partial differential equations obtained are of second or higher order. If the partial differential equation is obtained by elimination of arbitrary functions, then the order of the partial differential equation is, in general, equal to the number of arbitrary functions eliminated.



Example. Let $z = ax + by + ab$ with a and b are arbitrary constants. Differentiate partially with respect to x and y , we get $p = a$ and $q = b$. From that we get the corresponding partial differential equation is $z = px + qy + pq$.

KEY TERMS

Arbitrary constant is a constant whose value could be assumed to be anything, just so long as it does not depend on the other variables in an equation or expression.

Example. Let $z = (x^2 + a)(y^2 + b)$. Differentiate partially with respect to x and y , we get $p = 2x(y^2 + b)$ and $q = 2y(x^2 + a)$. From that, the corresponding partial differential equation is $pq = 4xyz$.

In practice, there are two methods to form a partial differential equation.

- By the elimination of arbitrary constants.
- By the elimination of arbitrary functions.

(i) Formation of Partial Differential Equations by the elimination of arbitrary constants method:

- Let $f(x, y, z, a, b) = 0$ be an equation containing 2 arbitrary constants 'a' and 'b'.
- To eliminate 2 constants we require at least 3 equations hence we partially differentiate the above equation with respect to (w.r.t) x and w.r.t y to obtain 2 more equations.
- From the three equations we can eliminate the constants 'a' and 'b'.

Method to Solve Problems:

- Step 1: Differentiate the given question first w.r.t x and then w.r.t y .
- Step 2: We know $p = \partial z / \partial x$ and $q = \partial z / \partial y$.
- Step 3: Now find out a and b values in terms of p and q .
- Step 4: Substitute these values in the given question equation.
- Step 5: Hence the final equation (answer) is in terms of p and q

and free of arbitrary constants 'a' and 'b'.

(ii) Formation of Partial Differential Equations by the elimination of arbitrary functions method:

- Here it is the arbitrary function that gets eliminated instead of the arbitrary constants 'a' and 'b'.

Method to Solve Problems:

- Step 1: Differentiate the given question first w.r.t x and then w.r.t y .
- Step 2: We know $p = \partial z / \partial x$ and $q = \partial z / \partial y$.
- Step 3: Now find out the value of the differentiated function (f') from both the equations separately. $[(f') = ?]$
- Step 4: Equate the other side of the differentiated function (f') which is in terms of p in one equation and q in other.
- Step 5: Hence the final equation (answer) is in terms of p and q and free of the arbitrary function.

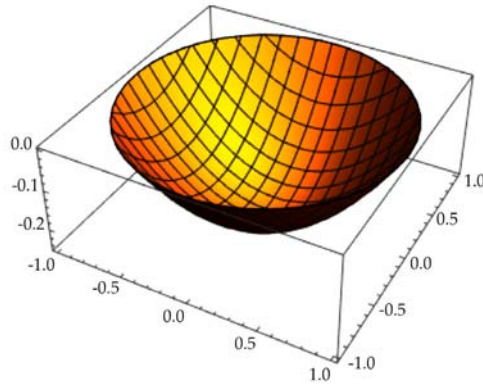
6.1.2 Solutions of Partial Differential Equations

Let $f(x, y, z, p, q) = 0$ be a general first order partial differential equation.

Definition. A solution or integral of a partial differential equation is a relation between the independent and dependent variables which satisfies the partial differential equation. A solution which contains as many arbitrary constants as



there are independent variables is called a complete solution or complete integral. A solution got by giving particular values to the arbitrary constants in a complete solution is called particular solution.

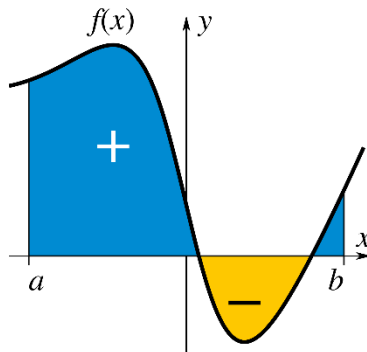


Let $g(x, y, z, a, b) = 0$ be a complete solution. Put $b = \phi(a)$ in the complete solution and eliminate the arbitrary constant a from $g(x, y, z, a, \phi(a)) = 0$, we get a solution involving an arbitrary function ϕ . This solution is called the general solution.

Differentiate the complete solution $g(x, y, z, a, b) = 0$ partially with respect to a and b , and then eliminate a and b we get a solution, that solution is called singular solution.

Equations Solvable by Direct Integration

Those equations which contain only one partial derivative can be solved by direct **integration**. However, in place of the constants of integration, we must use arbitrary functions of the variable kept constant.



Example. We can find a solution of the partial differential equation $\frac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$, by integrating with respect to x and y . If we integrate with respect to x , we get

$$\frac{\partial z}{\partial y} = -\cos x \sin y + f(y)$$

and now respect to y , we get the most general solution as

$$z = \cos x \cos y + \int f(y) dy + \phi(x).$$

We can find the particular solution using initial conditions on z . For example, if z satisfies, $\frac{\partial z}{\partial y} = -2 \sin y$ when $x = 0$. Then, $f(y) = -\sin y$. Thus the solution is

$$z = \cos x \cos y + \cos y + \phi(x).$$

KEY TERMS

Integration is a way of adding slices to find the whole. Integration can be used to find areas, volumes, central points and many useful things.

Non-Linear Partial Differential Equations of First Order

A differential equation involving first order partial derivatives p and q only is called a partial differential equation of the first order. If p and q both occur in the first degree only and are not multiplied together, then it is called a linear partial differential equation of the first order otherwise it called nonlinear partial differential equation of the first order. In this subsection first we study non-linear partial differential equations.

TYPE - I: Equations of the Form $f(p, q) = 0$

Suppose the partial differential equation involving only p and q and no x, y, z those equations come under Type-I. Assume that, $z = ax + by + c$ be a solution of $f(p, q) = 0$. Then $p = a$ and $q = b$, and a and b are connected by the relation $f(a, b) = 0$. Thus the complete solution of the Type-I is $z = ax + \phi(a) + c$.

Problem. Solve: $p = e^q$.

Solution: Let $z = ax + by + c$ be a solution of the given partial differential equation. Differentiate partially with respect to x and y we get, $p = a$ and



$q = b$. Then $a = e^b$. The complete solution is $z = e^b x + by + c$.

TYPE - II: Equations of the Form $z = px + qy + f(p, q)$

This form is called Clairute's form. For finding complete solution, we assume that $z = ax + by + c$ be a solution of $z = px + qy + f(p, q)$. Then $p = a$ and $q = b$ and the complete solution is $z = ax + by + f(a, b)$.

Problem. Solve: $z = px + qy + p^2 + q^2$

The complete solution is $z = ax + by + a^2 + b^2$.

TYPE - III: Equations of the Form $f(z, p, q) = 0$

Let us assume that $z = \phi(u)$, where $u = x+ay$ as a trial solution of the given equation.

Then $p = \frac{dz}{du}$ and $q = a \frac{dz}{du}$. Substituting these values of p and q in the given equation,

we get $f\left(z, \frac{dz}{du}, a \frac{dz}{du}\right) = 0$ which is an ordinary differential equation of the first order. Integrating it, we get the complete solution.

Problem. Solve: $z^2 = p^2 + q^2$.

Solution: Assume that $z = \phi(u)$, where $u = x+ay$ as a trial solution of the given equation. Then

$p = \frac{dz}{du}$ and $q = a \frac{dz}{du}$. Substituting these values of p and q in $z^2 = p^2 + q^2$, we get

$$z^2 = (1 + a^2) \left(\frac{dz}{du} \right)^2.$$

Taking square root both side and separate the variables,

$$du = \sqrt{1 + a^2} \frac{dz}{z},$$

integrating, we get $u = \sqrt{1 + a^2} \log z + c$. i.e., $x + ay = \sqrt{1 + a^2} \log z + c$ as complete solution.

TYPE - IV: Equations of the Form $f(x, p) = g(y, q)$

Equations in which z is absent and the terms involving x and p can be separated from those involving q and y .

As a trial solution, let us put each side of the equation $f(x, p) = g(y, q)$ equal to an arbitrary constant a . Then

$$f(x, p) = g(y, q) = a.$$

Solving these equations for p and q , let $p = F(x)$ and $q = G(y)$. Since

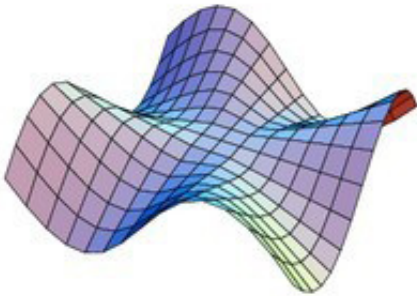
$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = p dx + q dy. \text{ Thus}$$

$$z = \int F(x) dx + G(y) dy + c$$

be a complete solution.

Equations Reducible to Standard Types or Equations of the Form $f(x^m z^k p, y^n z^k q) = 0$

Many non-linear partial differential equations of the first order do not fall under any of the four standard types. However, it is possible to transform the given partial differential equation into one of the standard types by change of variables. i.e., equations of the form $f(x^m z^k p, y^n z^k q) = 0$.



Using the following transformation for x, y and z to convert the given partial differential equation to standard type.

Let

$$\begin{aligned} X &= \begin{cases} x^{1-m}, & \text{if } m \neq 1; \\ \log x, & m = 1. \end{cases} \\ Y &= \begin{cases} y^{1-n}, & \text{if } n \neq 1; \\ \log y, & n = 1. \end{cases} \\ Z &= \begin{cases} z^{1+k}, & \text{if } k \neq -1; \\ \log z, & k = -1. \end{cases} \end{aligned}$$

Problem. Solve: $2x^4 p^2 - yzq - 3z^2 = 0$.

Solution: The given partial differential equation can be written as

$$2 \left(\frac{x^2 p}{z} \right)^2 - \frac{yq}{z} - 3 = 0.$$

Here, $m = 2$, $n = 1$ and $k = -1$. Hence put $X = x^{-1}$, $Y = \log y$ and $Z = \log z$. Then

$$P = \frac{\partial Z}{\partial X} = \frac{\partial Z}{\partial x} \frac{\partial x}{\partial X} = -\frac{x^2 p}{z}$$

And

$$Q = \frac{\partial Z}{\partial Y} = \frac{\partial Z}{\partial y} \frac{\partial y}{\partial Y} = \frac{yq}{z}.$$

Thus the partial differential equation becomes,

$$2P^2 - Q - 3 = 0.$$

This is of the standard Type -I. The complete integral is $Z = aX + bY + c$, where $2a^2 - b - 3 = 0$. Thus $b = 2a^2 - 3$. The complete solution becomes

$$\log z = \frac{a}{x} + (2a^2 - 3) \log y + c$$



6.2 LINEAR PARTIAL DIFFERENTIAL EQUATIONS OF FIRST ORDER

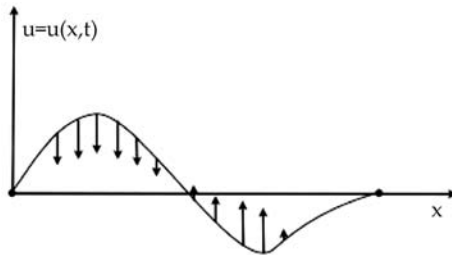
2. A Differential Equation which involves partial derivatives p and q only and no higher order derivatives is called a first order equation. If p and q have the degree one, it is called a linear partial differential equation of first order; otherwise it is called a non-linear partial equation of first order.

LEARNING OBJECTIVE

2. Explain linear partial differential equations of first order

6.2.1 First-order PDEs

First-order partial differential equations can be tackled with the method of characteristics, a powerful tool which also reaches beyond first-order. We will be looking primarily at equations in two variables, but there is an extension to higher dimensions.



Wave Equation with Constant Speed

Consider the first-order wave equation with constant speed:

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0.$$

It responds well to a change of variables:

$$\xi = x + ct \quad \eta = x - ct$$

The chain rule gives us

$$\begin{aligned}\frac{\partial}{\partial x} &= \frac{\partial \xi}{\partial x} \frac{\partial}{\partial \xi} + \frac{\partial \eta}{\partial x} \frac{\partial}{\partial \eta} = \frac{\partial}{\partial \xi} + \frac{\partial}{\partial \eta} \\ \frac{\partial}{\partial t} &= \frac{\partial \xi}{\partial t} \frac{\partial}{\partial \xi} + \frac{\partial \eta}{\partial t} \frac{\partial}{\partial \eta} = c \left(\frac{\partial}{\partial \xi} - \frac{\partial}{\partial \eta} \right)\end{aligned}$$

and so the wave equation is equivalent to

$$2c \frac{\partial u}{\partial \xi} = 0.$$

Integrating gives the general solution $u = F(\eta)$, $u = F(x - ct)$.

But where did we get the change of variables from? The line $x - ct = \text{constant}$ is a line in the x - t plane along which u is constant. This means that if we parameterize this line

$$x = x(r) \quad t = t(r)$$

then moving along the line by changing r will not change u , i.e.

$$\frac{du}{dr} = 0.$$

This is the underlying principle of the characteristic.

Variable Speed

Let's look now at the variable speed case:

$$\frac{\partial u}{\partial t} + c(x, t) \frac{\partial u}{\partial x} = 0.$$

We would like again to find curves along which u is constant. Suppose such a curve is given by $x = x(r)$ and $t = t(r)$. Then, using the chain rule,

$$\frac{du}{dr} = \frac{\partial u}{\partial t} \frac{dt}{dr} + \frac{\partial u}{\partial x} \frac{dx}{dr}.$$

We want this to be zero, which is easily achieved if we make this expression the same as the original linear operator:

$$\frac{dt}{dr} \frac{\partial u}{\partial t} + \frac{dx}{dr} \frac{\partial u}{\partial x} = \frac{\partial u}{\partial t} + c(x, t) \frac{\partial u}{\partial x} = 0.$$

This gives us the two parametric equations governing the shape of the characteristic curve:

$$\frac{dt}{dr} = 1, \quad \frac{dx}{dr} = c(x, r).$$

These are both ODEs and straightforward to solve.

Example. Look at the equation

$$2 \sin \theta \cos 2\phi \frac{\partial u}{\partial \theta} - \frac{\cos \theta \sin 2\phi}{\sin \theta} \frac{\partial u}{\partial \phi} = 0.$$

Suppose that our characteristic is given by $\theta = \theta(r)$, $\phi = \phi(r)$. Then the requirement that u be constant along a characteristic becomes

$$\frac{\partial u}{\partial \theta} \frac{d\theta}{dr} + \frac{\partial u}{\partial \phi} \frac{d\phi}{dr} = 0.$$

A native attempt would be to look at the coupled ODEs

$$\frac{d\theta}{dr} = 2 \sin \theta \cos 2\phi \quad \frac{d\phi}{dr} = -\frac{\cos \theta \sin 2\phi}{\sin \theta}$$

but we can uncouple them if, before we start, we multiply the original equation by $\sin \theta / \cos \theta \cos 2\phi$:

$$\frac{2 \sin^2 \theta}{\cos \theta} \frac{\partial u}{\partial \theta} - \frac{\sin 2\phi}{\cos 2\phi} \frac{\partial u}{\partial \phi} = 0,$$

$$\frac{d\theta}{dr} = \frac{2 \sin^2 \theta}{\cos \theta} \quad \frac{d\phi}{dr} = -\frac{\sin 2\phi}{\cos 2\phi}.$$



Now the equations are decoupled, and solving them in turn gives

$$\sin \theta = -\frac{1}{2r} \quad \sin 2\phi = \exp [C - 2r].$$

Note that we only use a constant of integration in one of these equations; since r is just a parameter, the point $r = 0$ is not defined a priori. Effectively, we are making a change of variables from x, t to r, C . We can invert the transformation:

$$C = \ln \sin 2\phi - \frac{1}{\sin \theta} \quad r = -\frac{1}{2 \sin \theta}$$

and since u is constant on this curve, we can deduce the general solution

$$u = F(C) = F\left(\ln \sin 2\phi - \frac{1}{\sin \theta}\right)$$

More than Two Dimensions

Now suppose we have the PDE

$$\frac{\partial u}{\partial x} + c_1(x, y, z) \frac{\partial u}{\partial y} + c_2(x, y, z) \frac{\partial u}{\partial z} = 0.$$

Again, we look for a curve on which u is constant; being a curve, it can still be described with a single variable r so we set $x = x(r)$, $y = y(r)$ and $z = z(r)$. Then the chain rule gives

$$\frac{du}{dr} = \frac{\partial u}{\partial x} \frac{dx}{dr} + \frac{\partial u}{\partial y} \frac{dy}{dr} + \frac{\partial u}{\partial z} \frac{dz}{dr}$$

and to make this equal to zero we choose

$$\frac{dx}{dr} = 1 \quad \frac{dy}{dr} = c_1(x(r), y(r), z(r)) \quad \frac{dz}{dr} = c_2(x(r), y(r), z(r)).$$

Example. Look at the equation

$$\frac{\partial u}{\partial x} + xy \frac{\partial u}{\partial y} + 2x^2 z \ln y \frac{\partial u}{\partial z} = 0.$$

We set $x = x(r)$, $y = y(r)$ and $z = z(r)$ and the chain rule gives

$$\frac{du}{dr} = \frac{\partial u}{\partial x} \frac{dx}{dr} + \frac{\partial u}{\partial y} \frac{dy}{dr} + \frac{\partial u}{\partial z} \frac{dz}{dr}.$$

To match the three coefficients we set:

$$\begin{aligned} \frac{dx}{dr} &= 1 & x(r) &= r \\ \frac{dy}{dr} &= xy = ry & y(r) &= y_0 \exp[r^2/2] \\ \frac{dz}{dr} &= 2x^2 z \ln y = -r^4 z \ln y_0 & z(r) &= z_0 \exp[-r^5 \ln y_0/5]. \end{aligned}$$

Now we have expressed all points in terms of the three parameters r , y_0 and z_0 and u is independent of r , so the solution is any function of y_0 and z_0 . Reversing the change of variables gives

$$r = x \quad y_0 = y \exp[-x^2/2] \quad z_0 = z \exp[-x^7/10] y^{[x^5/5]}$$

and the full solution is

$$u = F(y \exp[-x^2/2]; z \exp[-x^7/10] y^{[x^5/5]}).$$

Inhomogeneous Case

The characteristic curve is just as fundamental if the equation is not homogeneous, although the function value is no longer constant along characteristics. The method is best seen by example:

$$\frac{\partial u}{\partial t} + 2xt \frac{\partial u}{\partial x} = u$$

over all x , with initial condition (at $t = 0$) $u = x$. We start by finding the characteristic. Here the characteristic is given by

$$\frac{dt}{dr} = 1 \quad t = r \quad \frac{dx}{dr} = 2xr \quad x = x_0 \exp[r^2].$$

This time our two new variables are r and x_0 . Along a specific characteristic we have

$$\frac{du}{dr} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \frac{dx}{dr} = u \quad u = u_0 e^r \quad \text{or more exactly} \quad u = F(x_0) e^r.$$



We now have a one-parameter family of solutions (parameter x_0): on the curve

$$x = x_0 \exp[t^2], \quad u = F(x_0)e^t \quad \text{so} \quad u = F(x \exp[-t^2])e^t.$$

We need to apply the initial conditions to determine the function F . At $t = 0$ we have $x = x_0$ and $u = F(x_0)$ so the initial condition gives $F(x_0) = x_0$:

$$u = x \exp[-t^2]e^t = x \exp[t - t^2].$$

Nonlinear Homogeneous Case

A general first-order homogeneous PDE in two variables can be written as

$$\frac{\partial u}{\partial t} + c(u, x, t) \frac{\partial u}{\partial x} = 0$$

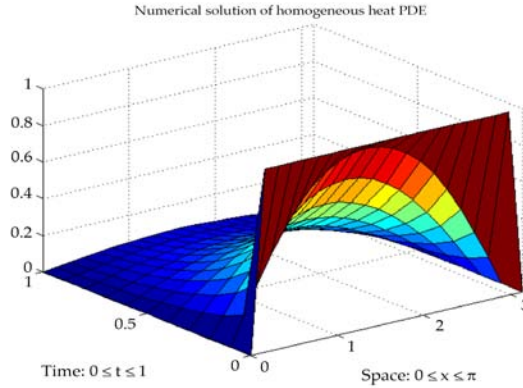
and the method of characteristics still applies (but we expect an implicit solution in general). The characteristic curves are given by

$$\frac{dx}{dt} = c(u, x, t).$$

Again, we will write the curve parametrically as $t = r$, and x some function of r and a constant x_0 (that is, constant for a given characteristic). Along any characteristic we will have

$$\frac{du}{dr} = 0$$

and so u is constant along a characteristic. We can then make this into a solution everywhere by setting $u = F(x_0)$ on the characteristic specified by x_0 .



Since u is constant on our characteristic, the equation of the curve is simply

$$\frac{dx}{dt} = c(F(x_0), x, t),$$

which is a straightforward ODE in x and t . Once we have solved it we have the characteristic curve

$$t = r \quad x = G(x_0, F(x_0), r) = G(x_0, u, r).$$

The implicit form of the solution is now:

$$x = G(x_0, u, t) \quad u = F(x_0)$$

which is a one-parameter family with parameter x_0 . In many cases it is possible to rearrange the first equation to obtain x_0 in terms of u and t ; then substituting this into the second equation gives the more standard form of the implicit solution.

Example. Consider the advection equation

$$\frac{\partial u}{\partial t} + ux^2 \frac{\partial u}{\partial x} = 0.$$

Because it is homogeneous, we expect u to be constant along characteristics: so we parameterize with x_0 (constant on each characteristic) and r (which varies along the characteristic) and we can say $u = F(x_0)$.

Now our characteristic curve becomes

$$\frac{dx}{dt} = ux^2t = F(x_0)x^2t,$$

which we can solve:

$$\int \frac{dx}{x^2} = F(x_0) \int t dt \quad -\frac{1}{x} = \frac{1}{2}F(x_0)t^2 - \frac{1}{x_0}$$

$$x = \frac{2x_0}{2 - x_0F(x_0)t^2}.$$

Thus the characteristic curve and implicit solution are:

$$t = r \quad x = \frac{2x_0}{2 - x_0F(x_0)r^2} \quad u = F(x_0).$$

As above, we can rearrange to get x_0 in terms of x , t and u :

$$x = \frac{2x_0}{2 - x_0ut^2} \quad x_0 = \frac{2x}{2 + uxt^2}$$

and the standard implicit form of the solution is

$$u = F\left(\frac{2x}{2 + uxt^2}\right).$$

Nonlinear Inhomogeneous

We are now looking at the most complex of first-order PDEs: those of the form

$$\frac{\partial u}{\partial t} + c(u, x, t) \frac{\partial u}{\partial x} = f(u, x, t)$$

Characteristics still exist in these systems, and they may have important physical properties (for instance, discontinuities in the derivatives of the solution will propagate along them) but unfortunately, since u itself now varies along a characteristic, we can no longer solve even implicitly in general.

Example. Here is a case where we can achieve a little:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \cos t.$$

In this case we can immediately spot one solution

$$u = A + \sin t$$

The characteristics are defined by

$$\frac{dx}{dt} = u;$$

let us suppose we know the family of curves $x = f(x_0, r)$, $t = r$. Then we have

$$\frac{du}{dr} = \frac{\partial u}{\partial t} \frac{dt}{dr} + \frac{\partial u}{\partial x} \frac{dx}{dr} = \frac{\partial u}{\partial t} + \frac{\partial}{\partial r} f(x_0, r) \frac{\partial u}{\partial x}$$

which gives us the two coupled equations

$$\frac{\partial}{\partial r} f(x_0, r) = u \quad \frac{du}{dr} = \cos r.$$

These are easy to solve in reverse order:

$$u = A(x_0) + \sin r \quad x = A(x_0)r - \cos r + x_0.$$

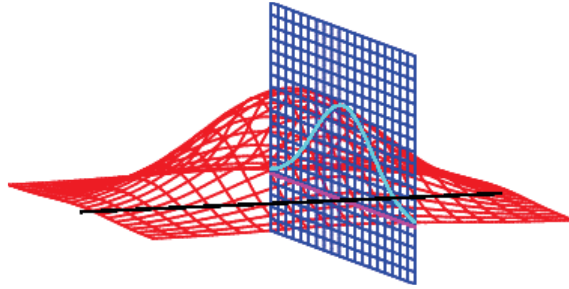
Unfortunately, we cannot extract x_0 explicitly without choosing the function $A(x_0)$; but note that

$$\begin{array}{llll} A(x_0) = \alpha & \implies & x_0 = x + \cos t - \alpha t & u = \alpha + \sin t \\ A(x_0) = x_0/\beta & \implies & x_0 = x + \frac{\beta \cos t}{t + \beta} & u = \frac{x}{\beta} + \frac{\cos t}{t + \beta} + \sin t \end{array}$$

so our “spotted” solution is only one of a family of possible solutions.

6.2.2 Lagrange's Linear Equation

A linear Partial Differential Equation of order one, involving a **dependent variable** and two independent variables x and y , and is of the form $Pp + Qq = R$, where P , Q , R are functions of x , y , z is called Lagrange's Linear Equation.



Solution of the Linear Equation

Consider $Pp + Qq = R$

Now, $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

Case 1: If it is possible to separate variables then, consider any two equations, solve them by integrating. Let the solutions of these equations are $u=a$, $u=b$

$\therefore \phi(u, v) = 0$ is the required solution of given equation.

Case 2: If it is not possible to separate variables then

$$\frac{dx}{P(x,y,z)} = \frac{dy}{Q(x,y,z)} = \frac{dz}{R(x,y,z)}$$

To solve above type of problems we have following methods

- *Method of grouping:* In some problems, it is possible to solve any two of the equations

$$\frac{dx}{P} = \frac{dy}{Q} \quad (\text{or}) \quad \frac{dy}{Q} = \frac{dz}{R} \quad (\text{or}) \quad \frac{dx}{P} = \frac{dz}{R}$$

In such cases, solve the differential equation, get the solution and then substitute in the other differential equation

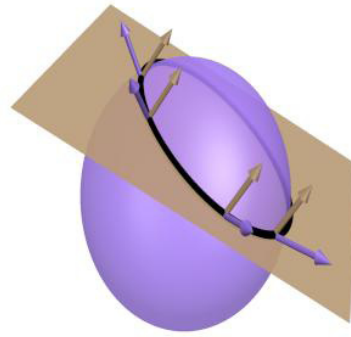
- *Method of Multiplier:* consider $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l dx + m dy + n dz}{0}$

In this, we have to choose l, m, n so that denominator $= 0$. That will give us solution by integrating $l dx + m dy + n dz$.

Lagrange Multiplier

Lagrange multipliers, also called Lagrangian multipliers, can be used to find the extreme of a **multivariate function**

$f(x_1, x_2, \dots, x_n) = 0$ subject to the constraint $g(x_1, x_2, \dots, x_n) = 0$, where f and g are functions with continuous first partial derivatives on the open set containing the curve $g(x_1, x_2, \dots, x_n) = 0$, and $\nabla g \neq 0$ at any point on the curve (where ∇ is the gradient).



KEY TERMS

Multivariable function is a function with several variables. Functions with more than one variable are needed in order to mathematically model complicated physical phenomena.

For an extreme of f to exist on g , the gradient of f must line up with the gradient of g . In the illustration above, f is shown in red, g in blue, and the intersection of f and g is indicated in light blue. The gradient is a horizontal vector (i.e., it has no z -component) that shows the direction that the function increases; for g it is perpendicular to the curve, which is a straight line in this case. If the two gradients are in the same direction, then one is a multiple ($-\lambda$) of the other, so

$$\nabla f = -\lambda \nabla g$$

The two vectors are equal, so all of their components are as well, giving

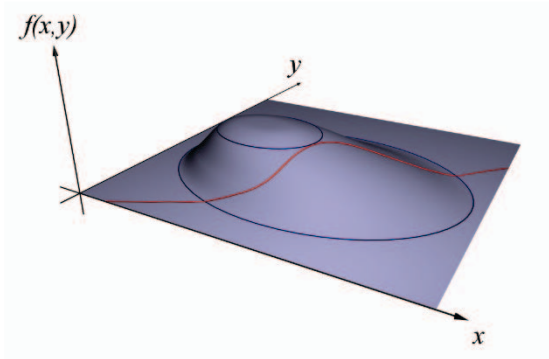
$$\frac{\partial f}{\partial x_k} + \lambda \frac{\partial g}{\partial x_k} = 0$$

for all $k=1,2,\dots,n$, where the constant λ is called the Lagrange multiplier.

The extreme is then found by solving the $n+1$ equations in $n+1$ unknowns, which is done without inverting g , which is why Lagrange multipliers can be so useful.

For multiple constraints $g_1 = 0, g_2 = 0, \dots$

$$\nabla f + \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2 + \dots = 0$$



REMEMBER

A solution of a PDE is generally not unique; additional conditions must generally be specified on the boundary of the region where the solution is defined.

6.3 SECOND ORDER LINEAR PARTIAL DIFFERENTIAL EQUATIONS



We are about to study a simple type of partial differential equations (PDEs): the second order linear PDEs. Recall that a partial differential equation is any differential equation that contains two or more independent variables. Therefore the derivative(s) in the equation are partial derivatives. We will examine the simplest case of equations with 2 independent variables.

A few examples of second order linear PDEs in 2 variables are:

$$\alpha^2 u_{xx} = u_t \quad (\text{one-dimensional heat conduction equation})$$

$$\alpha^2 u_{xx} = u_{tt} \quad (\text{one-dimensional wave equation})$$

$$u_{xx} + u_{yy} = 0 \quad (\text{two-dimensional Laplace/potential equation})$$

In this class we will develop a method known as the method of Separation of Variables to solve the above types of equations.

LEARNING OBJECTIVE

3. Describe second order linear partial differential equations

6.3.1 Classification of Second Order Linear PDEs

Consider the following general second order linear PDE in two independent variables:

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + F u + G = 0, \quad (1)$$

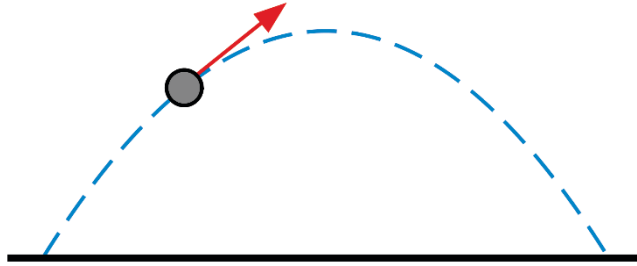
Where A, B, C, D, E, F and G are functions of the independent variables x and y. The equation (1) may be written in the form

$$A u_{xx} + B u_{xy} + C u_{yy} + f(x, y, u_x, u_y, u) = 0, \quad (2)$$

Where

$$u_x = \frac{\partial u}{\partial x}, \quad u_y = \frac{\partial u}{\partial y}, \quad u_{xx} = \frac{\partial^2 u}{\partial x^2}, \quad u_{xy} = \frac{\partial^2 u}{\partial x \partial y}, \quad u_{yy} = \frac{\partial^2 u}{\partial y^2}.$$

Assume that A, B and C are continuous functions of x and y possessing continuous partial derivatives of as high order as necessary.



The classification of PDE is motivated by the classification of second order algebraic equations in two-variables

$$ax^2 + bxy + cy^2 + dx + ey + f = 0. \quad (3)$$

We know that the nature of the curves will be decided by the principal part $ax^2 + bxy + cy^2$ i.e., the term containing highest degree. Depending on the sign of the discriminant $b^2 - 4ac$, we classify the curve as follows:

For the equation to be of second order, a, b, and c cannot all be zero. Define its discriminant to be $b^2 - 4ac$. The properties and behavior of its solution are largely dependent of its type, as classified below.

- If $b^2 - 4ac > 0$, then the equation is called hyperbolic. The wave equation is one such example.
- If $b^2 - 4ac = 0$, then the equation is called parabolic. The heat conduction



equation is one such example.

- If $b^2 - 4ac < 0$, then the equation is called elliptic. The **Laplace equation** is one such example.

With suitable transformation, we can transform (3) into the following normal form

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad (\text{hyperbola}).$$

$$x^2 = y \quad (\text{parabola}).$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (\text{ellipse}).$$

In general, elliptic equations describe processes in equilibrium. While the hyperbolic and parabolic equations model processes which evolve over time.

Example: Consider the one-dimensional damped wave equation

$$9u_{xx} = u_{tt} + 6u_t.$$

It can be rewritten as: $9u_{xx} - u_{tt} - 6u_t = 0$. It has coefficients $a = 9$, $b = 0$, and $c = -1$. Its discriminant is $9 > 0$. Therefore, the equation is hyperbolic.

KEY TERMS

Laplace's equation is a second-order partial differential equation named after Pierre-Simon Laplace who first studied its properties.

6.3.2 Linear PDE with Constant Coefficients

Let us first consider the following general linear second order PDE in two independent variables x and y with constant coefficients:

$$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu + G = 0, \quad (4)$$

where the coefficients A , B , C , D , E , F and G are constants. The nature of the equation (4) is determined by the principal part containing highest partial derivatives i.e.,

$$Lu \equiv Au_{xx} + Bu_{xy} + Cu_{yy}. \quad (5)$$

For classification, we attach a symbol to (5) as $P(x, y) = Ax^2 + Bxy + Cy^2$ (as if we have replaced

x by $\frac{\partial}{\partial x}$ and y by $\frac{\partial}{\partial y}$). Now depending on the sign of the discriminant ($B^2 - 4AC$), the classification of (4) is done as follows:

$$B^2 - 4AC > 0 \implies \text{Eq. (4) is hyperbolic} \quad (6)$$

$$B^2 - 4AC = 0 \implies \text{Eq. (4) is parabolic} \quad (7)$$

$$B^2 - 4AC < 0 \implies \text{Eq. (4) is elliptic} \quad (8)$$

Linear PDE with Variable Coefficients

The above classification of (4) is still valid if the coefficients A , B , C , D , E and F depend on x , y . In this case, the conditions (6), (7) and (8) should be satisfied at each point (x, y) in the region where we want to describe its nature e.g., for elliptic we need to verify

$$B^2(x, y) - 4A(x, y)C(x, y) < 0$$

for each (x, y) in the region of interest. Thus, we classify linear PDE with variable coefficients as follows:

$$B^2(x, y) - 4A(x, y)C(x, y) > 0 \text{ at } (x, y) \implies \text{Eq. (4) is hyperbolic at } (x, y) \quad (9)$$

$$B^2(x, y) - 4A(x, y)C(x, y) = 0 \text{ at } (x, y) \implies \text{Eq. (4) is parabolic at } (x, y) \quad (10)$$

$$B^2(x, y) - 4A(x, y)C(x, y) < 0 \text{ at } (x, y) \implies \text{Eq. (4) is elliptic at } (x, y) \quad (11)$$

Note: Equation. (4) is hyperbolic, parabolic, or elliptic depends only on the coefficients of the second derivatives. It has nothing to do with the first-derivative terms, the term in u , or the nonhomogeneous term.

Example

- $u_{xx} + u_{yy} = 0$ (Laplace equation). Here, $A = 1$, $B = 0$, $C = 1$ and $B^2 - 4AC = -4 < 0$. Therefore, it is an elliptic type.
- $u_t = u_{xx}$ (Heat equation). Here, $A = -1$, $B = 0$, $C = 0$. $B^2 - 4AC = 0$. Thus, it is of parabolic type.
- $u_{tt} - u_{xx} = 0$ (Wave equation). In this case, $A = -1$, $B = 0$, $C = 1$ and $B^2 - 4AC = 4 > 0$. Hence, it is of hyperbolic type.
- $u_{xx} + xu_{yy} = 0$, $x \neq 0$ (Tricomi equation). $B^2 - 4AC = -4x$. Given PDE is hyperbolic for $x < 0$ and elliptic for $x > 0$. This example shows that equations with variable coefficients can change form in the different regions of the domain.



6.3.3 Classification with More Than Two Variables

Consider the second-order PDE in general form:

$$\sum_{i=1}^n \sum_{j=1}^n a_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j} + \sum_{i=1}^n b_i \frac{\partial u}{\partial x_i} + cu + d = 0, \quad (12)$$

where the coefficients a_{ij} , b_i , c and d are functions of $x = (x_1, x_2, \dots, x_n)$ alone and $u = u(x_1, x_2, \dots, x_n)$.

Its principal part is

$$L \equiv \sum_{i=1}^n \sum_{j=1}^n a_{ij} \frac{\partial^2}{\partial x_i \partial x_j}. \quad (13)$$

It is enough to assume that $A = [a_{ij}]$ is symmetric if not, let $\bar{a}_{ij} = \frac{1}{2}(a_{ij} + a_{ji})$ and rewrite

$$L \equiv \sum_{i=1}^n \sum_{j=1}^n \bar{a}_{ij} \frac{\partial^2}{\partial x_i \partial x_j}. \quad (14)$$

Note that $\frac{\partial^2 u}{\partial x_i \partial x_j} = \frac{\partial^2 u}{\partial x_j \partial x_i}$. As in two-space dimension, let us attach a quadratic form P with (14) (i.e., replacing $\frac{\partial u}{\partial x_i}$ by x_i).

$$P(x_1, x_2, \dots, x_n) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j. \quad (15)$$

Since A is a real valued symmetric ($a_{ij} = a_{ji}$) matrix, it is diagonalizable with real eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (counted with their multiplicities). In other words, there exists a corresponding set of orthonormal set of n eigenvectors, say $\sigma_1, \sigma_2, \dots, \sigma_n$ with $R = [\sigma_1, \sigma_2, \dots, \sigma_n]$ as column vectors such that



REMEMBER

In free space the Laplace equation of any electrostatic potential must equal zero since q is zero in free space.

$$R^T A R = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \ddots & \\ & & & & \lambda_n \end{bmatrix} = D \quad (16)$$

We now classify (12) depending on sign of eigenvalues of A:

- If $\lambda_i > 0 \forall i$ or $\lambda_i < 0 \forall i$ then (12) is elliptic type.
- If one or more of the $\lambda_i = 0$ then (12) is parabolic type.
- If one of the $\lambda_i < 0$ or $\lambda_i > 0$, and all the remaining have opposite sign then (12) is said to be of hyperbolic type.

Example

- $\nabla^2 u = u_{xx} + u_{yy} + u_{zz} = 0$. In this case, $\lambda_i = 1 > 0$ for all $i = 1, 2, 3$. It is an elliptic PDE since all eigenvalues are of one sign.
- It is an easy exercise to check that $u_t - \nabla^2 u = 0$ is of parabolic type.
- The equation $u_{tt} - \nabla^2 u = 0$ is of hyperbolic type.

Example

Classify $u_{x_1 x_1} + 2(1 + cx_2)u_{x_2 x_3} = 0$.

To symmetrize, write it as

$$u_{x_1 x_1} + (1 + cx_2)u_{x_2 x_3} + (1 + cx_2)u_{x_3 x_2} = 0$$

i.e., $\partial_x^T A \partial_x - c \partial_{x_2} = 0$, where

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 + cx_2 \\ 0 & 1 + cx_2 & 0 \end{bmatrix} \quad \partial_x = \begin{bmatrix} \partial_{x_1} \\ \partial_{x_2} \\ \partial_{x_3} \end{bmatrix}$$

Eigenvalues are $\lambda_1 = 1$, $\lambda_2 = 1 + cx_2$, $\lambda_3 = -(1 + cx_2)$ and normalized eigenvectors

$$\sigma_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \sigma_2 = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \quad \sigma_3 = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

So

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

Note that $R = R^T = R^{-1}$.

$$R^T A R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 + cx_2 & 0 \\ 0 & 0 & -(1 + cx_2) \end{bmatrix} = D$$

Equation is parabolic if $x_2 = -\frac{1}{c}$ ($c \neq 0$), hyperbolic if $x_2 > -\frac{1}{c}$ and $x_2 < -\frac{1}{c}$. For $c = 0$, $\lambda_1 = \lambda_2 = 1$ and $\lambda_3 = -1$, it is **hyperbolic** type.

KEY TERMS

Hyperbolic is an adjective describing something that resembles or pertains to a hyperbola (a curve), to hyperbole (an overstatement or exaggeration), or to hyperbolic geometry.



ROLE MODEL

PIERRE-SIMON LAPLACE: FRENCH MATHEMATICIAN

Pierre-Simon Laplace was a French mathematician and astronomer who carried out remarkable studies regarding the stability of the solar system and is famously known as the 'French Newton'. He also did pioneering work in mathematics regarding the theory of probability and statistics which influenced a whole new generation of mathematicians. Born in a poor family, his education was financed by neighbors and he was sent to study theology at the age of 16. But, he soon developed a keen interest in mathematics and was subsequently drawn to physics and astronomy. He served as a professor of mathematics for seven years and also published several scientific papers alongside. Laplace successfully accounted for all the observed deviations of the planets from their theoretical orbits by applying Sir Isaac Newton's theory of gravitation to the solar system, and he developed a conceptual view of evolutionary change in the structure of the solar system. He also demonstrated the usefulness of probability for interpreting scientific data and applied his own definition of probability to justify the fundamental mathematical manipulations. He restated and developed the nebular hypothesis of the origin of the solar system and also postulated the existence of black holes along with the notion of gravitational collapse.

Childhood and Early Life

- Pierre-Simon Laplace was born on March 23, 1749, in Beaumont-en-Auge, a village in Normandy, France, to Pierre de Laplace, owner of small farms of Maarquis, and his wife, Marie-Anne Sochon.
- Despite his family's poor financial condition, Laplace was able to receive a good education courtesy his wealthy neighbors. His father wanted him to be ordained in the Roman Catholic Church and thus sent him to Caen University to study theology at the age of 16. But, Laplace developed a keen interest in mathematics.
- At the age of 19, he dropped out of college and moved to Paris where he worked as a professor of mathematics at the École Militaire from 1769 to 1776. During this time, he published several papers regarding integral calculus, mechanics and physical astronomy, which gained him much acclaim all over France.



Career

- In 1771, his early published work focused on differential equations and finite differences. Subsequently, he started to think about the mathematical and philosophical concepts of probability and statistics.
- In 1774, his paper titled 'Mémoire sur la probabilité des causes par les événements' was published. Two years later, he published another paper which further elaborated his statistical thinking and also began his systematic work on celestial mechanics and the stability of the solar system.
- During 1784–87, he worked on the subject of attraction between spheroids which laid the mathematical foundation for the scientific study of heat, magnetism, and electricity.
- In 1796, he published 'Exposition du système du monde (The System of the World)', which included his 'nebular hypotheses'.
- Between 1799 and 1825, he published five volumes of 'Traité de mécanique céleste (Celestial Mechanics)', which summarized the results obtained by his mathematical development and application of the law of gravitation.
- In 1812, he published his comprehensive research in mathematics titled 'Théorie analytique des probabilités (Analytic Theory of Probability)'. Another major work on the theory of probability, titled 'Essai philosophique sur les probabilités (A Philosophical Essay on Probability)' was published in 1814.
- He theorized the existence of black holes, suggesting that there could be massive stars whose gravity is so great that not even light could escape from its surface. He also stated the fact that the speed of sound in air depends on the heat capacity ratio.
- His other discoveries in pure and applied mathematics include Laplace's method for approximating integrals, solution of the linear partial differential equation of the second order, and general proof of the Lagrange reversion theorem.

Major Works

- Pierre-Simon Laplace is highly regarded for his influential five-volume treatise 'Traité de mécanique céleste' (Celestial mechanics; 1799-1825), which developed a strong mathematical understanding of the motion of the heavenly bodies. He established that the small perturbations observed in the orbital motion of the planets will always remain small, constant and self-correcting.
- He formulated Laplace's equation⁰, and pioneered the Laplace transform which appears in many branches of mathematical physics. The Laplacian differential operator, widely used in mathematics, is also named after him.

Awards and Achievements

- In 1806, Laplace was elected a foreign member of the Royal Swedish Academy of Sciences.
- In 1822, he was appointed a Foreign Honorary Member of the American Academy of Arts and Sciences.

Personal Life and Legacy

- In March 1788, he married Marie-Charlotte de Courty de Romanges, a girl from Besançon who was twenty years younger to him. The couple had a son, Charles-Émile, and a daughter, Sophie-Suzanne.
- Pierre-Simon Laplace died on March 5, 1827, in Paris, France, at the age of 77. After he died, his physician, François Magendie, removed Laplace's brain which was later displayed in a roving anatomical museum in Britain.



SUMMARY

1. A partial differential equation (PDE) is a mathematical equation that involves two or more independent variables, an unknown function, and partial derivatives of the unknown function with respect to the independent variables.
2. A solution or integral of a partial differential equation is a relation between the independent and dependent variables which satisfies the partial differential equation. A solution which contains as many arbitrary constants as there are independent variables is called a complete solution or complete integral.
3. A differential equation involving first order partial derivatives p and q only is called a partial differential equation of the first order.
4. A Differential Equation which involves partial derivatives p and q only and no higher order derivatives is called a first order equation. If p and q have the degree one, it is called a linear partial differential equation of first order; otherwise it is called a non-linear partial equation of first order.
5. First-order partial differential equations can be tackled with the method of characteristics, a powerful tool which also reaches beyond first-order.
6. A linear Partial Differential Equation of order one, involving a dependent variable and two independent variables x and y , and is of the form $Pp + Qq = R$, where P, Q, R are functions of x, y, z is called Lagrange's Linear Equation.

KNOWLEDGE CHECK

1. For partial differential equation, if $b^2 - 4ac = 0$ then equation is called
 - a. hyperbolic
 - b. parabolic
 - c. elliptic
 - d. None of these
2. A partial differential equation requires
 - a. exactly one independent variable
 - b. two or more independent variables

- c. more than one dependent variable
- d. equal number of dependent and independent variables

3. The partial differential equation

$$5 \frac{\partial^2 z}{\partial x^2} + 6 \frac{\partial^2 z}{\partial y^2} = xy$$

is classified as

- a. elliptic
- b. parabolic
- c. hyperbolic
- d. none of the above

4. The partial differential equation

$$xy \frac{\partial z}{\partial x} = 5 \frac{\partial^2 z}{\partial y^2}$$

is classified as

- a. elliptic
- b. parabolic
- c. hyperbolic
- d. none of the above

5. The partial differential equation

$$\frac{\partial^2 z}{\partial x^2} - 5 \frac{\partial^2 z}{\partial y^2} = 0$$

is classified as

- a. elliptic
- b. parabolic
- c. hyperbolic
- d. none of the above



6. The following is true for the following partial differential equation used in nonlinear mechanics known as the Korteweg-de Vries equation.

$$\frac{\partial w}{\partial t} + \frac{\partial^3 w}{\partial x^3} - 6w \frac{\partial w}{\partial x} = 0$$

- a. linear; 3rd order
 - b. nonlinear; 3rd order
 - c. linear; 1st order
 - d. nonlinear; 1st order
7. A solution to the partial differential equation.

$$\frac{\partial^2 u}{\partial x^2} = 9 \frac{\partial^2 u}{\partial y^2}$$

is

- a. $\cos(3x - y)$
 - b. $x^2 + y^2$
 - c. $\sin(3x - y)$
 - d. $e^{-3\pi x} \sin(\pi y)$
8. If P_u is upstream pressure and P_d is downstream pressure, Euler number is equal to
- a. $P_d - P_u / \rho V^2$
 - b. $P_u - P_d / \rho V^2$
 - c. $P_u - P_d / \rho V^3$
 - d. $P_u - P_d / \rho V$

REVIEW QUESTIONS

1. Solve the following partial differential equations
 - a. $p^2 + q^2 = 2$.
 - b. $p^2 + p = q^2$.

c. $p = e^q$

2. Eliminate the arbitrary constants a and b from the following equations to form the PDE:

- $ax^2 + by^2 + z^2 = 1;$
- $z = (x^2 + a)(y^2 + b).$

3. Eliminate the arbitrary function f from the following and form the PDE:

- $z = x + y + f(xy);$
- $z = f\left(\frac{xy}{z}\right).$

4. Classify the following equations into hyperbolic, elliptic or parabolic type

a. $5u_{xx} - 3u_{yy} + (\cos x)u_x + e^y u_y + u = 0.$

b. $e^y u_{xx} + e^y u_{yy} - u = 0.$

5. Determine the regions where $u_{xx} - 2x^2 u_{xz} + u_{yy} + u_{zz} = 0$ is of hyperbolic, elliptic and parabolic.

6. Find the partial differential equation of the family of planes, the sum of whose x , y , z intercepts is equal to unity.

7. Solve the following partial differential equations;

- $p + q + pq = 0,$
- $p^2 + 6p + 2q + 4 = 0$
- $p^2 + q^2 = npq$

8. Find the partial differential equation by eliminating arbitrary functions from the following relations

a. $z = f(xy) + g\left(\frac{x}{y}\right)$

b. $f(x + y + z, x^2 + y^2 + z^2) = 0$

c. $lx + my + nz = \phi(x^2 + y^2 + z^2)$

Check Your Result

1. (b)
7. (d)

2. (b)
8. (b)

3. (a)

4. (b)

5. (c)

6. (b)



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Chapter

7

APPROXIMATE ANALYTICAL METHODS

LEARNING OBJECTIVES

After studying this chapter, you will be able to:

1. Discuss about the approximate analysis
2. Explain the graphical analysis: the phase plane
3. Describe least squares method
4. Explain perturbation method
5. Understand the WKB method

INTRODUCTION

Analytical methods use mathematical principles to fully predict the implications of a theory. They can be used to solve an equation in its entirety without any degree of estimation. They stand in contrast with numerical methods, which can only attain an approximate prediction. Analytical methods are the preferred way to determine the outcome of a hypothesis when associated equations are simple and a precise answer is desired. Numerical methods are used when equations are too complex to solve fully.

Mathematicians employ analytical methods when they use the basic principles of algebra to solve an equation. If the equation is sufficiently simple, a full solution can be obtained by manipulating the equation in symbolic form. There is no room for approximation or guesswork—the principles of mathematics always govern which operations can be performed. If the variable in question can be successfully isolated, analytics are the tools that make this possible.

In the equation $y = 2x$, one can solve for x using analytical methods. To isolate the variable x , both sides of the equation must be divided by the number 2. For any value of y , x can be fully determined with relative ease.

In a simple real-world application of this equation, one might hypothesize that the length of a human foot was twice its width: $\text{length} = 2 * \text{width}$. This equation implies, necessarily, that $\text{width} = \frac{1}{2} * \text{length}$. The practical application of the equation may not constitute an accurate theory, but the manipulation of the equation is done using analytical methods. That is, the equation can predict the width of the foot without introducing any approximations.

There are some equations that nobody knows how to solve analytically. For example, many differential equations have no known solutions. A differential equation relates a quantity's rate of change to its value. Unlike an algebraic equation, differential equations must be solved using calculus. Often times, their results can only be approximated.

Numerical methods are used to solve a number of practical problems. Many businesses try to optimize their sales using numerical methods to approximate market conditions. They can try to predict the outcome of various different business strategies, but they cannot generally use analytics. Making analytical predictions, like in the case of the human foot dimensions, would require one or more equations that related different variables. The market is generally very complicated and has too many variables to be modeled in this way.



7.1 APPROXIMATE ANALYSIS

 Approximate analysis is a tool engineers use to estimate the magnitude of member forces and moments, and support reactions in indeterminate structures. To understand approximate methods for vertical loadings it is important to understand the difference between approximate analyses of statically indeterminate structures compared with other methods such as the Force Method for Multiple Degrees of Indeterminacy, Moment Distribution Method for Continuous Beams or Slope-Deflection Method for Continuous Beams. These methods involve obtaining numerical solutions to an indeterminate system through analysis using the physical properties of structural members. The ability to obtain a numerical solution for indeterminate systems means these methods are 'exact' methods. The approximate method relies on assumptions about the behavior of an indeterminate structural system to model it as a determinate one which may be solved. The approximate analysis of structures is an 'inexact' method that is often used in preliminary designs process because of its simplicity, the analysis is less complex with a tradeoff of reduced accuracy.

Approximate analysis is useful in determining (approximately) the forces and moments in the different members and in coming up with preliminary designs. Based on the preliminary design, a more detailed analysis can be conducted and then the design can be refined.

7.1.1 Chaplygin's Method

A method of approximate integration of differential equations. Proposed by S. A. Chaplygin in 1919, the method permits an approximate solution to be found for a differential equation to a given degree of accuracy. It involves the construction of sequences of functions $\{u_n\}$ and $\{v_n\}$ that approximate with continually increasing accuracy the desired solution y of a given differential equation and that fulfill the following relations:

$$u_n \geq u_{n+1} \geq y \geq v_{n+1} \geq v_n$$

LEARNING OBJECTIVE

-  1. Discuss about the approximate analysis



REMEMBER

Approximate analysis is conducted by making realistic assumptions about the behavior of the structure.

**REMEMBER**

The integral of the solution function should be the limit of the integrals of the approximations. However, many functions that can be obtained as limits are not Riemann-integral, and so such limit theorems do not hold with the Riemann integral.

The construction of the sequences $\{u_n\}$ and $\{v_n\}$ is based on Chaplygin's theorem of differential inequalities and is a generalization of Newton's method to the case of differential equations. The rate of convergence is the same as in Newton's method; that is, $u_n - y$ tends to zero like $C/2^{2n}$.

7.1.2 Collocation Method

A **projection method** for solving integral and differential equations in which the approximate solution is determined from the condition that the equation be satisfied at certain given points. For example, for the approximate solution of the integral equation

$$u(t) = \int_a^b K(t, s, u(s)) ds + f(t)$$

one chooses a certain n -parameter family of functions $\phi(t, c_1, \dots, c_n)$ and certain points (collocation knots, collocation nodes) $t_1, \dots, t_n$ on the interval $[a, b]$. The approximate solution $u_n(t) = \phi(t, c_1, \dots, c_n)$ is determined from the conditions

$$u_n(t_i) = \int_a^b K(t_i, s, u_n(s)) ds + f(t_i), \quad i = 1, \dots, n$$

which represent a system of n equations in the unknowns $c_1, \dots, c_n$. If the equation is linear and the approximate solution is sought in the form of a linear combination

$u_n(t) = c_1 \phi_1(t) + \dots + c_n \phi_n(t)$ of the given (so-called coordinate) functions $\phi_1, \dots, \phi_n$, then the system of equations in $c_1, \dots, c_n$ will also be linear.

7.1.3 Dominant Balance

The method is used to identify those terms in an equation that may be neglected in an asymptotic limit. This is most general and powerful procedure for finding approximate solutions to equations. The technique consists of 3 steps:

- We drop all terms that appear small and replace an exact equation with the asymptotic relation.

KEY TERMS

Projection method is an effective means of numerically solving time-dependent incompressible fluid-flow problems.



- We replace the asymptotic expansion with the relation by exchanging $\sim$ sign for an $=$ sign and solve the resulting equation exactly (the solution to this equation automatically satisfies the asymptotic relation although it's certainly not the only function that does so).
- We check that the solution we have obtained is consistent with the approximations made in (1). Even if it is consistent we must still show that the equation obtained by factoring off the dominant balance solution from the exact solution itself has a solution that varies less rapidly than the dominant balance solution. When this happens we conclude that the controlling factor (and not the leading behavior) obtained from the dominant balance relation is the same as that of the exact solution.

Example: Behavior of Taylor series for large x . Consider the **function** $y(x)$ defined by the Taylor series

$$y(x) = \sum_{n=0}^{\infty} \frac{x^n}{(n!)^2} \quad (1)$$

The series converges for all x , however while the series converges very rapidly for small x its convergence is very slow when x is large. If $x = 10$ one needs only 16 terms to calculate $y(x)$ up to 10 significant digits, but for $x = 10000$ about 150 terms are required. There is no analytic function equal to the sum of the series, however, it is possible to describe behavior of $y(x)$ when $x \rightarrow \infty$ using elementary functions.

We start with the observation:

$$y = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n!)(n-1)!} \Rightarrow (xy)' = \sum_{n=1}^{\infty} \frac{x^{n-1}}{[(n-1)!]^2} = y$$

Thus $y(x)$ is the solution to the equation

$$xy + y = y \quad (2)$$

This equation has an irregular singular point at $+\infty$.

We determine the leading behavior of $y(x)$ using the method of dominant balance and start with the substitution $y(x) = eS(x)$. The resulting equation:

$$xS + x(S)' + S = 1 \quad (3)$$

KEY TERMS

Function is a relation between a set of inputs and a set of permissible outputs with the property that each input is related to exactly one output.

We assume $S \ll (S)_2 \Rightarrow xS \ll x(S)_2$. The asymptotic differential equation is

$$x(S)_2 \sim 1 - S \quad (4)$$

Solving it we get $S \sim [-1 \pm (1 + 4x)_{1/2}]/2x$, $x \rightarrow +\infty$. Thus, since x is large $S(x) \sim \pm x_{-1/2}$.

Integrating this asymptotic relation we obtain

$$S(x) \sim \pm 2x_{1/2} \Rightarrow S(x) = \pm 2x_{1/2} + C(x), \quad x \rightarrow +\infty \quad (5)$$

where $C(x) \ll 2x_{1/2}$ (since all the coefficients in Taylor series are positive, $y(x)$ is an increasing function, so we could ignore the negative S). Substituting (5) into (2) and combining terms we get an equation for $C(x)$:

$$xC'' + x(C')^2 + (2x^{1/2} + 1)C' + \frac{1}{2}x^{-1/2} = 0 \quad (6)$$

From $C(x) \ll 2x^{1/2}$ follows that $xC'''(x) \ll \frac{1}{2}x^{-1/2}$ and $x[C'(x)]^2 \ll 2x^{1/2}C'(x)$. As result we obtain the asymptotic differential equation

$$C' \sim -\frac{1}{4}x^{-1}. \quad (7)$$

whose solution is $C(x) = -\frac{1}{4}\ln(x)$

Hence, the leading behavior of $y(x)$ is given by $e^{S(x)+C(x)} = cx^{-1/4}e^{2x^{1/2}}$. The constant c cannot be determined from the local analysis because (2) is homogeneous. However, noticing the similarity between the Taylor series for $y(x)$ and the Bessel function of zeroth order, one could find that $c = \frac{1}{2}\pi^{-1/2}$. One could compute the next factor in the leading term expansion by requesting $S = S(x) + C(x) + D(x)$ and doing a similar analysis.

We may also significantly improve the accuracy of the approximation by considering

$$y(x) = \frac{1}{2}\pi^{-1/2}x^{-1/4}e^{2x^{1/2}}\omega(x), \quad (8)$$

Substituting this expression into (2) yields the differential equation for unknown ω . The equation could be solved using the asymptotic inverse power series expansion (due to the fact that we could represent $\omega(x)$ as $1 + \varepsilon(x)$, where $\varepsilon(x) \ll 1$ when $x \rightarrow +\infty$):

$$\omega(x) \sim \sum_{n=0}^{\infty} a_n x^{-n}, \quad a_0 = 1, \quad x \rightarrow +\infty \quad (9)$$

Substituting this expression into equation instead of w yields a recursive relation for coefficients computed using the same methods as we used for regular power series expansion.



The series (9) are, in general, diverging, but any finite sum of them provides us with the best possible approximation of ω and consequently of y .

7.1.4 Equation Splitting

The idea is simple: *divide et impera* (divide and conquer). It is probably the most important systematic technique for the development of algorithms in general. Suppose we wish to solve the differential equation

$$dz/dt = f(z) \equiv f_1(z) + f_2(z),$$

where each of two differential equations

$$dz/dt = f_1(z)$$

and

$$dz/dt = f_2(z)$$

happen to be completely integrable (i.e. analytically solvable). [Note that $dz/dt = f(z)$ is not solvable just because the two pieces are!]

The way this is done is diagrammed in Figure 7.1. We first note that at any point in **phase space**, the vector field (hence the tangent vector to the solution curve) can be broken up into the two components f_1 and f_2 . A method can then be constructed by stepping forward along first one, then a second, solution curve, each time for a small time step h . That is, we start from some point z_n and solve first the initial value problem.

$$dz/dt = f_1(z), \quad z(0) = z_n$$

KEY TERMS

Phase space refers to the plotting of both a particle's momentum and position on a two dimensional graph.

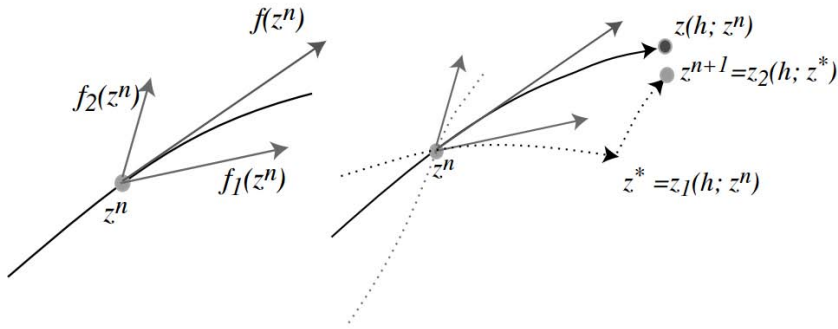


Figure 7.1: Divide and Conquer: An approximation to the solution is constructed by following the solutions of the two parts of the vector field.

for a time h . This means to compute the exact solution of this part of the system, which is possible, since f_1 is assumed integrable. This takes us to a point $z^* = z_1(h; z_n)$, where z_1 represents the exact solution of the differential equations on vector field f_1 .

Next we start from z^* and solve the second differential equation initial value problem

$$dz/dt = f_2(z), \quad z(0) = z^*$$

for a time h . This takes us to the point $z_{n+1} = z_2(h; z^*)$. Denoting the flow maps of the vector fields f_1 and f_2 as $\Phi_{\nu f_1}$ and $\Phi_{\nu f_2}$, we have just computed $z_{n+1} = \Phi_{h/f_2}(\Phi_{h/f_1}(z_n)) = (\Phi_{h/f_2} \circ \Phi_{h/f_1})(z_n)$. Such a method is referred to as a splitting method.

7.1.5 Floquet Theory

Consider the linear periodic system as follows.

$$\dot{x} = A(t)x, \quad A(t+p) = A(t), \quad p > 0,$$

where $A(t) \in \mathbb{C}(\mathbb{R})$.

Lemma 7.1: If C is a $n \times n$ matrix with $\det C \neq 0$, then, there exists a $n \times n$ (complex) matrix B such that $e^B = C$

Proof: For any matrix C , there exists an invertible matrix P , s.t. $P^{-1}CP = J$, where J is a Jordan matrix.

If $e^B = C$, then, $e^{P^{-1}BP} = P^{-1}e^B P = P^{-1}CP = J$. Therefore, it is suffice to prove the result when C is in a canonical form.

Suppose that $C = \text{diag}(C_1, \dots, C_s)$, $C_j = \lambda_j I_j + N_j$, where N_j is nilpotent, that is,

$$N_j = \begin{pmatrix} 0 & 1 & \cdots & 0 \\ 0 & 0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & 0 \end{pmatrix} \text{ with } N_j^{n_j} = O.$$

Since C is invertible for each $\lambda_j \neq 0$.

If we can show that for each C_j there exists B_j s.t. $C_j = e^{B_j} \Rightarrow C = e^B$.

Since $C_j = \lambda_j(I_j + \frac{N_j}{\lambda_j})$, using the expansion of $\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k, |x| < 1$,

we have

$$\begin{aligned} B_j &= \ln C_j = \ln\{\lambda_j(I_j + \frac{N_j}{\lambda_j})\} = I_j \ln \lambda_j + \ln(I_j + \frac{N_j}{\lambda_j}) \\ &= I_j \ln \lambda_j + \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} (\frac{N_j^k}{\lambda_j^k}). \end{aligned}$$

Since $N_j^{n_j} = O$, we actually have

$$B_j = \ln C_j = I_j \ln \lambda_j + \sum_{k=1}^{n_j-1} \frac{(-1)^{k-1}}{k} (\frac{N_j^k}{\lambda_j^k}) = I_j \ln \lambda_j + M_j, \quad j = 1, 2, \dots, s,$$

where $M_j = \sum_{k=1}^{n_j-1} \frac{(-1)^{k-1}}{k} (\frac{N_j^k}{\lambda_j^k})$. Therefore, we have

$$e^{B_j} = \exp\{I_j \ln \lambda_j + M_j\} = \exp\{\ln C_j\} = C_j, \quad j = 1, 2, \dots, s.$$

Let $B = \text{diag}(B_1, \dots, B_s)$, where B_j is defined above. We have the desired result given by

$$e^B = \text{diag}(e^{B_1}, e^{B_2}, \dots, e^{B_s}) = \text{diag}(C_1, C_2, \dots, C_s) = C.$$

Remark: Clearly, B is not unique since $e^{B+2\pi k i I_n} = e^B e^{2\pi k i I_n} = e^B e^{2\pi k i} I_n = e^B e^{2\pi k i} = e^B$ for any integer k .

Theorem: (Floquet Theorem) If $\Phi(t)$ is a fundamental matrix solution of the periodic system $\dot{x} = A(t)x$, then so is $\Phi(t+p)$. Moreover, there exists an invertible matrix $P(t)$ with p -period such that

$$\Phi(t) = P(t)e^{Bt}$$

Proof. Let $\Psi(t) = \Phi(t+p)$. Since $\Psi'(t) = A(t)\Psi(t)$, it follows that

$$\Psi'(t) = \Phi'(t+p) = A(t+p)\Phi(t+p) = A(t)\Psi(t),$$

Hence, $\Psi(t)$ is also a matrix solution. Since $\Phi(t)$ is invertible for all $t \in \mathbb{R}$, so is $\Phi(t+p) \Rightarrow \Psi(t)$ is also a fundamental matrix solution. Therefore, there exists an invertible matrix C (for example, if $\Phi(t)$ satisfies $\Phi(0) = I_n$, then $C = \Phi(p)$!! Depends on solutions. It is a point of difficulty for computation) s.t.

$$\Phi(t+p) = \Phi(t)C \quad \text{for all } t \in \mathbb{R}.$$

By Lemma 7.1, there exists a matrix B such that $e^{Bp} = C$. For such a matrix B , we take $P(t) := \Phi(t)e^{-Bt}$, that is, $\Phi(t) = P(t)e^{Bt}$. Then

$$P(t+p) = \Phi(t+p)e^{-B(t+p)} = \Phi(t)Ce^{-B(t+p)} = \Phi(t)e^{-Bt} = P(t)$$

Therefore $P(t)$ is invertible for all $t \in \mathbb{R}$ and p -periodic. This concludes the proof.

KEY TERMS

Graphical method is a great way to solve a system of equations, and also to check your work if you are solving the system using elimination or substitution.

LEARNING OBJECTIVE

2. Explain the graphical analysis: the phase plane



Phase plane analysis is a graphical method for studying second-order systems. Objective is to gain familiarity of the nonlinear systems through the simple **graphical method**.



7.2.1 Concepts of Phase Plane Analysis

Phase Portraits

The phase plane method is concerned with the graphical study of second-order autonomous systems described by

$$\dot{x}_1 = f_1(x_1, x_2) \quad (10a)$$

$$\dot{x}_2 = f_2(x_1, x_2) \quad (10b)$$

where

x_1, x_2 : states of the system

f_1, f_2 : nonlinear functions of the states

Geometrically, the state space of this system is a plane having x_1, x_2 as **coordinates**. This plane is called phase plane. The solution of (10) with time varies from zero to infinity can be represented as a curve in the phase plane. Such a curve is called a phase plane trajectory. A family of phase plane trajectories is called a phase portrait of a system.

Example: Phase portrait of a mass-spring system

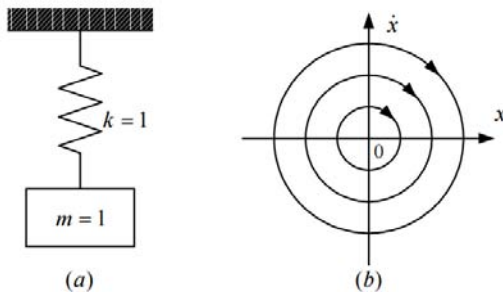


Figure 7.2: A mass-spring system and its phase portrait.

The governing equation of the mass-spring system in Figure 7.2 is the familiar linear second-order differential equation

$$\ddot{x} + x = 0 \quad (11)$$

Assume that the mass is initially at rest, at length x_0 . Then

KEY TERMS

Coordinates are a set of values that show an exact position. On maps and graphs it is common to have a pair of numbers to show where a point is: the first number shows the distance along and the second number shows the distance up or down.

the solution of this equation is

$$x(t) = x_0 \cos(t)$$

$$\dot{x}(t) = -x_0 \sin(t)$$

Eliminating time t from the above equations, we obtain the equation of the trajectories

$$x^2 + \dot{x}^2 = x_0^2$$

This represents a circle in the phase plane. Its plot is given in Figure 7.2.b.

The nature of the system response corresponding to various initial conditions is directly displayed on the phase plane. In the above example, we can easily see that the system trajectories neither converge to the origin nor diverge to infinity. They simply circle around the origin, indicating the marginal nature of the system's stability.

A major class of second-order systems can be described by the differential equations of the form

$$\ddot{x} = f(x, \dot{x}) \quad (12)$$

In state space form, this dynamics can be represented with $x_1 = x$ and $x_2 = \dot{x}$ as follows

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = f(x_1, x_2)$$

Singular Points

A singular point is an equilibrium point in the phase plane. Since an equilibrium point is defined as a point where the system states can stay forever, this implies that $\dot{\mathbf{x}} = \mathbf{0}$, and using (10)

$$\begin{cases} f_1(x_1, x_2) = 0 \\ f_2(x_1, x_2) = 0 \end{cases} \quad (13)$$

For a **linear system**, there is usually only one singular point although in some cases there can be a set of singular points.

Example: A nonlinear second-order system

KEY TERMS

Linear system is a mathematical model of a system based on the use of a linear operator. Linear systems typically exhibit features and properties that are much simpler than the nonlinear case.

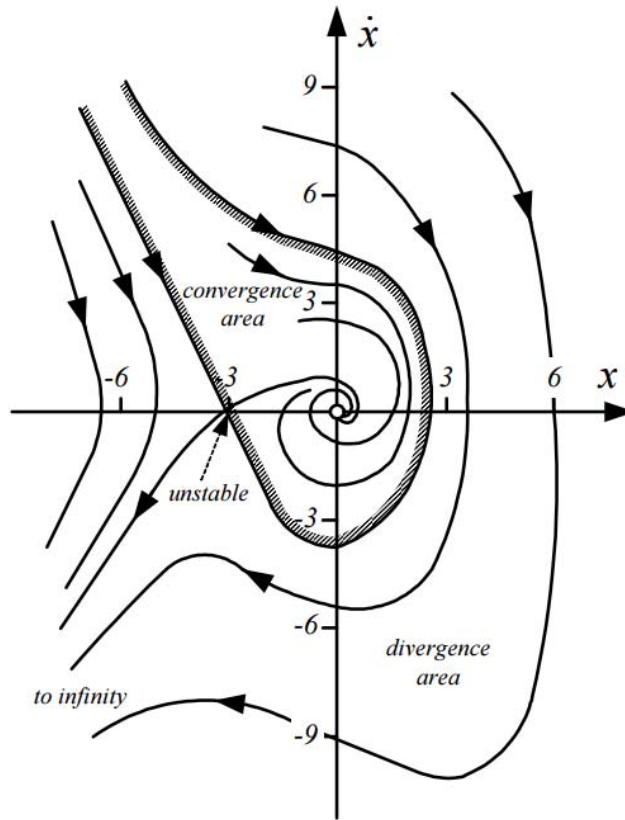


Figure 7.3: A mass-spring system and its phase portrait.

Consider the system $\ddot{x} + 0.6\dot{x} + 3x + x^2 = 0$ whose phase portrait is plot in Figure 7.3.

The system has two singular points, one at $(0, 0)$ and the other at $(-3, 0)$. The motion patterns of the system trajectories in the vicinity of the two singular points have different natures. The trajectories move towards the point $x = 0$ while moving away from the point $x = -3$.

Why an equilibrium point of a second order system is called a singular point? Let us examine the slope of the phase portrait. The slope of the phase trajectory passing through a point (x_1, x_2) is determined by

$$\frac{dx_2}{dx_1} = \frac{f_2(x_1, x_2)}{f_1(x_1, x_2)} \quad (14)$$

where f_1, f_2 are assumed to be single valued functions. This implies that the phase trajectories will not intersect. At singular point, however, the value of the slope is $0/0$,

i.e., the slope is indeterminate. Many trajectories may intersect at such point, as seen from Figure 7.3. This indeterminacy of the slope accounts for the adjective “singular”.

Singular points are very important features in the phase plane. Examining the singular points can reveal a great deal of information about the properties of a system. In fact, the stability of linear systems is uniquely characterized by the nature of their singular points.

Although the phase plane method is developed primarily for second-order systems, it can also be applied to the analysis of first-order systems of the form

$$\dot{x} + f(x) = 0$$

The difference now is that the phase portrait is composed of a single trajectory.

Example: A first-order system

Consider the system $\dot{x} = -4x + x^3$ there are three singular points, defined by $-4x + x^3 = 0$, namely, $x = 0, -2, 2$. The phase portrait of the system consists of a single trajectory, and is shown in Figure 7.4.

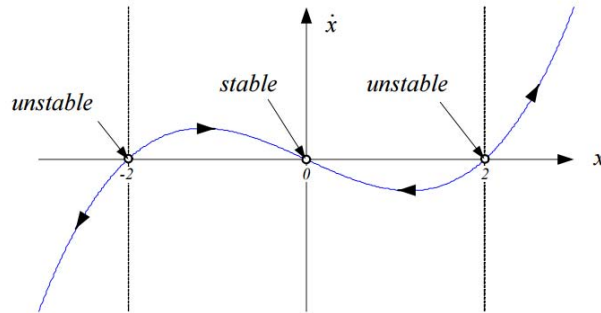


Figure 7.4: Phase trajectory of a first-order system.

The arrows in the figure denote the direction of motion, and whether they point toward the left or the right at a particular point is determined by the sign of $\dot{x}$ at that point. It is seen from the phase portrait of this system that the equilibrium point $x = 0$ is stable, while the other two are unstable.

Symmetry in Phase Plane Portrait

Let us consider the second-order dynamics (12): $\ddot{x} = f(x, \dot{x})$. The slope of trajectories in the phase plane is of the form

$$\frac{dx_2}{dx_1} = -\frac{f(x_1, x_2)}{\dot{x}}$$

Since symmetry of the phase portraits also implies symmetry of the slopes (equal in absolute value but opposite in sign), we can identify the following situations:

$$f(x_1, x_2) = f(x_1, -x_2) \Rightarrow \text{symmetry about the } x_1 \text{ axis.}$$

$$f(x_1, x_2) = -f(x_1, -x_2) \Rightarrow \text{symmetry about the } x_2 \text{ axis.}$$

$$f(x_1, x_2) = -f(-x_1, -x_2) \Rightarrow \text{symmetry about the origin.}$$

7.2.2 Constructing Phase Portraits

There are a number of methods for constructing phase plane trajectories for linear or nonlinear system, such that so-called analytical method, the method of isoclines, the delta method, Lienard's method, and Pell's method.

Analytical Method

There are two techniques for generating phase plane portraits analytically. Both techniques lead to a functional relation between the two phase variables x_1 and x_2 in the form

$$g(x_1, x_2) = 0 \quad (15)$$

where the constant c represents the effects of initial conditions (and, possibly, of external input signals). Plotting this relation in the phase plane for different initial conditions yields a phase portrait.

The first technique involves solving (10) for x_1 and x_2 as a function of time t , i.e. $x_1(t) = g_1(t)$ and $x_2(t) = g_2(t)$, and then, eliminating time t from these equations.

The second technique, on the other hand, involves directly eliminating the time variable, by noting that $\frac{dx_2}{dx_1} = \frac{f_2(x_1, x_2)}{f_1(x_1, x_2)}$ and then solving this equation for a functional relation between x_1 and x_2 . Let us use this technique to solve the mass spring equation again.

Example: Mass-spring system

By noting that $\ddot{x} = (d\dot{x}/dx)/(dx/dt)$, we can rewrite (11) as $\dot{x} \frac{d\dot{x}}{dx} + x = 0$. Integration of this equation yields $\dot{x}^2 + x^2 = x_0^2$.

Most nonlinear systems cannot be easily solved by either of the above two techniques. However, for piece-wise linear systems, an important class of nonlinear systems, this can be conveniently used.

7.2.3 Determining Time from Phase Portraits

Time t does not explicitly appear in the phase plane having x_1 and x_2 as coordinates. We now describe two techniques for computing time history from phase portrait. Both of techniques involve a step-by step procedure for recovering time.

Obtaining time from $\Delta t \approx \Delta x / \dot{x}$

In a short time Δt , the change of x is approximately

$$\Delta x \approx \dot{x} \Delta t \quad (16)$$

where $\dot{x}$ is the velocity corresponding to the increment Δx .

From (16), the length of time corresponding to the increment Δx is $\Delta t \approx \Delta x / \dot{x}$. This implies that, in order to obtain the time corresponding to the motion from one point to another point along the trajectory, we should divide the corresponding part of the trajectory into a number of small segments (not necessarily equally spaced), find the time associated with each segment, and then add up the results. To obtain the history of states corresponding to a certain initial condition, we simply compute the time t for each point on the phase trajectory, and then plots x with respects to t and $\dot{x}$ with respects to t .

Obtaining time from $t \approx \int (1/\dot{x}) dx$

Since $\dot{x} = dx/dt$, we can write $dt = dx/\dot{x}$. Therefore,

$$t - t_0 \approx \int_{x_0}^x (1/\dot{x}) dx$$

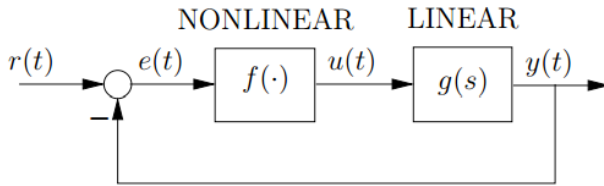
where x corresponding to time t and x_0 corresponding to time t_0 . This implies that, if we plot a phase plane portrait with new coordinates x and $(1/\dot{x})$, then the area under the resulting curve is the corresponding time interval.

7.2.4 Harmonic Balance

The describing function method (also called the method of harmonic balance) uses frequency domain (Fourier series) techniques to investigate limit cycle behavior in **nonlinear systems**. The method involves an approximation, but nevertheless often gives a reliable prediction concerning limit cycle behavior.



The usual context for the method is the following feedback system.



There is a linear system with transfer function $g(s)$ and a memoryless nonlinearity $f(\cdot)$ (e.g. saturation, hysteresis, backlash). For simplicity we will take the case of $r = 0$. We are looking for possible limit cycle (i.e. periodic) behavior in the feedback system. Therefore let us take a trial solution:

$$e(t) = E \sin(\omega t).$$

If the output of the non-linearity is periodic with frequency ω there will be a Fourier series representation:

$$u(t) = U_0 + \sum_{k=1}^{\infty} (U_k \sin(k\omega t) + V_k \cos(k\omega t)).$$

The Fourier coefficients corresponding to the fundamental harmonic are given by

$$U_1 = \frac{\omega}{\pi} \int_0^{2\pi/\omega} f(E \sin \omega t) \sin \omega t \, dt = \frac{1}{\pi} \int_0^{2\pi} f(E \sin \theta) \sin \theta \, d\theta,$$

$$V_1 = \frac{1}{\pi} \int_0^{2\pi} f(E \sin \theta) \cos \theta \, d\theta.$$

We now consider the approximation:

$$u(t) \approx U_1 \sin(\omega t) + V_1 \cos(\omega t)$$

as the input to the linear system. In the steady state, the output of the linear system is:

KEY TERMS

Nonlinear systems cannot be decomposed into parts and reassembled into the same thing, and do not change in proportion to a change in an input.

$$\begin{aligned}
 y(t) &= |g(j\omega)|[U_1 \sin(\omega t + \phi) + V_1 \cos(\omega t + \phi)] \\
 &= \text{Im}[|g(j\omega)|(U_1 + jV_1)e^{j\omega t + \phi}] \\
 &= \text{Im}[g(j\omega)(U_1 + jV_1)e^{j\omega t}]
 \end{aligned}$$

where $\phi = \arg(g(j\omega))$. (We remark that the neglect of the higher harmonics is most likely to be a valid approximation when $g(s)$ behaves like a low pass filter.) In order to satisfy the feedback equations we need to have $y(t) = -e(t)$. We thus obtain

$$\begin{aligned}
 0 &\equiv y(t) + e(t) \\
 \Rightarrow 0 &\equiv \text{Im}[(g(j\omega)(U_1 + jV_1) + E)e^{j\omega t}] = 0 \\
 \Rightarrow 0 &= g(j\omega)(U_1 + jV_1) + E.
 \end{aligned}$$

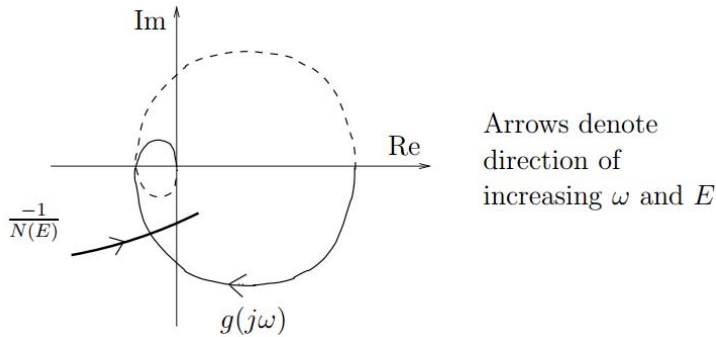
This is the condition for “harmonic balance”, i.e. for the fundamental harmonic to solve the feedback equations. We now define the describing function

$$N(E) = \frac{U_1 + jV_1}{E}.$$

Then the condition for harmonic balance becomes

$$g(j\omega) = \frac{-1}{N(E)}.$$

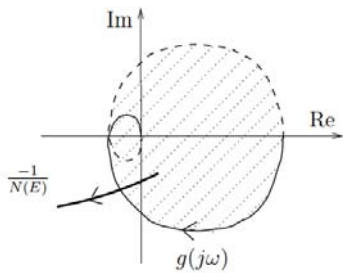
This condition has an interesting graphical interpretation. Let the Nyquist diagram $g(j\omega)$ of the linear system be plotted in the complex plane. On the same diagram plot the locus $-1/N(E)$. Note that these loci can be plotted independently since they are respectively functions of ω and E only. A limit cycle is predicted if there is an intersection of the two loci. Moreover, the intersection point gives an estimate for the frequency ω and amplitude E of the limit cycle.



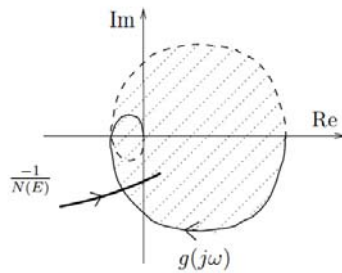
A useful prediction of the stability of limit cycles can be made from the manner of intersection of the two loci. Recall from the Nyquist stability criterion that, for an open



loop stable $g(s)$ in the standard unity gain negative feedback configuration (non-linearity absent of course), the closed loop feedback system is stable providing the critical point $(-1, 0)$ lies to the left of the Nyquist locus. The point $-1/N(E)$ can be thought of as the 'critical point' with respect to the Nyquist locus. Thus, if a small increase of the amplitude E moves the value of $-1/N(E)$ outside of the shaded region below, then a stabilizing effect will be predicted, which will act to reduce the amplitude. Conversely, if an increase of E moves $-1/N(E)$ inside the shaded region, then a destabilizing effect will be predicted.



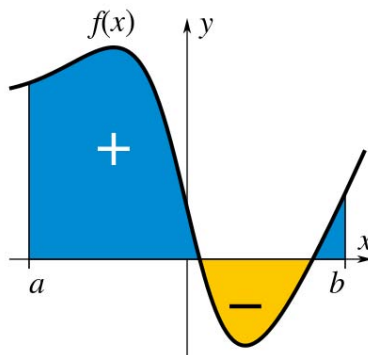
PREDICT STABLE LIMIT CYCLE



PREDICT UNSTABLE LIMIT CYCLE

7.2.5 Integral Methods

An integral assigns numbers to functions in a way that can describe displacement, area, volume, and other concepts that arise by combining infinitesimal data. Integration is one of the two main operations of calculus, with its inverse, differentiation, being the other.



Given a function f of a real variable x and an interval $[a, b]$ of the real line, the definite integral

$$\int_a^b f(x) dx$$

is defined informally as the signed area of the region in the xy -plane that is bounded by the graph of f , the x -axis and the vertical lines $x = a$ and $x = b$. The area above the x -axis adds to the total and that below the x -axis subtracts from the total.

There are two types of integral methods, such as:

- Indefinite Integra
- Definite Integral

Indefinite Integral

The opposite of a derivative is the anti-derivative or the indefinite integral. The indefinite integral of a function $f(x)$ is denoted,

$$\int f(x) dx.$$

It is defined by the property that

$$\frac{d}{dx} \int f(x) dx = f(x).$$

While a function $f(x)$ has a unique derivative if it is differentiable, it has an infinite number of indefinite integrals, each of which differ by an additive constant.

Zero Slope Implies a Constant Function

If the value of a function's derivative is identically zero, $\frac{df}{dx} = 0$, then the function is a constant, $f(x) = c$. To prove this, we assume that there exists a non-constant differentiable function whose derivative

is zero and obtain a contradiction. Let $f(x)$ be such a function. Since $f(x)$ is non-constant, there exist points a and b such that $f(a) \neq f(b)$. By the Mean Value Theorem of differential calculus, there exists a point $\xi \in (a, b)$ such that

$$f'(\xi) = \frac{f(b) - f(a)}{b - a} \neq 0,$$

which contradicts that the derivative is everywhere zero.

Indefinite Integrals Differ by an Additive Constant

Suppose that $F(x)$ and $G(x)$ are indefinite integrals of $f(x)$. Then we have

$$\frac{d}{dx}(F(x) - G(x)) = F'(x) - G'(x) = F(x) - f(x) = 0.$$

Thus we see that $F(x) - G(x) = c$ and the two indefinite integrals must differ by a constant.

For example, we have $\int \sin x dx = -\cos x + c$. While every function that can be expressed in terms of elementary functions, (the exponent, logarithm, trigonometric functions, etc.), has a derivative that can be written explicitly in terms of elementary functions, the same is not true of integrals.

For example, $\int \sin(\sin x) dx$ cannot be written explicitly in terms of elementary functions.

Properties: Since the derivative is linear, so is the indefinite integral. That is,

$$\int (af(x) + bg(x)) dx = a \int f(x) dx + b \int g(x) dx.$$

For each derivative identity there is a corresponding integral identity. Consider

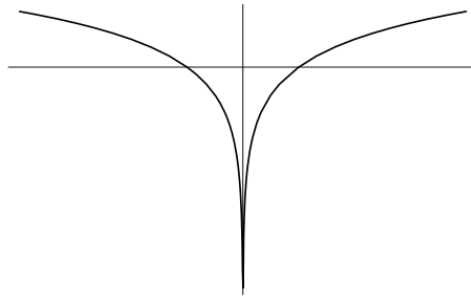


the power law identity, $\frac{d}{dx} (f(x))^a = a(f(x))^{a-1} f'(x)$. The corresponding integral identity is $\int \frac{d}{dx} (f(x))^a = a(f(x))^{a-1} f'(x)$. The corresponding integral identity is

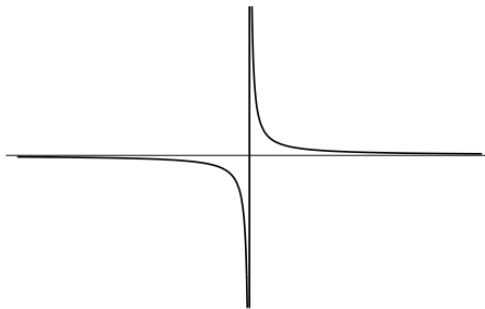
$$\int (f(x))^a f'(x) dx = \frac{(f(x))^{a+1}}{a+1} + c, \quad a \neq -1,$$

where we require that $a \neq -1$ to avoid division by zero. From the derivative of a logarithm, $\frac{d}{dx} \ln(f(x)) = \frac{f'(x)}{f(x)}$, we obtain,

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c.$$



(a)



(b)

Figure 7.5: Plot of $\ln |x|$ and $1/x$.

Note the absolute value signs. This is because $\frac{d}{dx} \ln |x| = \frac{1}{x}$ for $x \neq 0$. In figure 7.5 is a plot of $\ln |x|$ and $\frac{1}{x}$ to reinforce this.

Example: Consider

$$I = \int \frac{x}{(x^2 + 1)^2} dx.$$

We evaluate the integral by choosing $u = x^2 + 1$, $du = 2x dx$.

$$\begin{aligned} I &= \frac{1}{2} \int \frac{2x}{(x^2 + 1)^2} dx \\ &= \frac{1}{2} \int \frac{du}{u^2} \\ &= \frac{1}{2} \frac{-1}{u} \\ &= -\frac{1}{2(x^2 + 1)}. \end{aligned}$$

Example: Consider

$$I = \int \tan x dx = \int \frac{\sin x}{\cos x} dx.$$

By choosing $f(x) = \cos x$, $f'(x) = -\sin x$, we see that the integral is

$$I = - \int \frac{-\sin x}{\cos x} dx = -\ln |\cos x| + c.$$

$$\int f(\xi) d\xi = \int f(g(x))g'(x) dx, \quad (17)$$

since $d\xi = dg = g'(x) dx$. This turns out to be true. To prove it we will appeal to the chain rule for differentiation. Let ξ be a function of x . The chain rule is

$$\frac{d}{dx} f(\xi) = f'(\xi) \xi'(x),$$

$$\frac{d}{dx} f(\xi) = \frac{df}{d\xi} \frac{d\xi}{dx}.$$

We can also write this as

$$\frac{df}{d\xi} = \frac{dx}{d\xi} \frac{df}{dx},$$

or in operator notation,

$$\frac{d}{d\xi} = \frac{dx}{d\xi} \frac{d}{dx}.$$

Now we're ready to start. The derivative of the left side of equation 17 is

$$\frac{d}{d\xi} \int f(\xi) d\xi = f(\xi).$$

Next we differentiate the right side,

Change of Variable

The differential of a function $g(x)$ is $dg = g'(x) dx$. Thus one might suspect that for $\xi = g(x)$,

$$\begin{aligned}
 \frac{d}{d\xi} \int f(g(x))g'(x) dx &= \frac{dx}{d\xi} \frac{d}{dx} \int f(g(x))g'(x) dx \\
 &= \frac{dx}{d\xi} f(g(x))g'(x) \\
 &= \frac{dx}{dg} f(g(x)) \frac{dg}{dx} \\
 &= f(g(x)) \\
 &= f(\xi)
 \end{aligned}$$

to see that it is in fact an identity for $\xi = g(x)$.

Example: Consider

$$\int x \sin(x^2) dx.$$

We choose $\xi = x^2$, $d\xi = 2x dx$ to evaluate the integral.

$$\begin{aligned}
 \int x \sin(x^2) dx &= \frac{1}{2} \int \sin(x^2) 2x dx \\
 &= \frac{1}{2} \int \sin \xi d\xi \\
 &= \frac{1}{2} (-\cos \xi) + c \\
 &= -\frac{1}{2} \cos(x^2) + c
 \end{aligned}$$

Integration by Parts

The product rule for differentiation gives us an identity called integration by parts. We start with the product rule and then integrate both sides of the equation.

$$\frac{d}{dx}(u(x)v(x)) = u'(x)v(x) + u(x)v'(x)$$

$$\int (u'(x)v(x) + u(x)v'(x)) dx = u(x)v(x) + c$$

$$\int u'(x)v(x) dx + \int u(x)v'(x) dx = u(x)v(x)$$

$$\int u(x)v'(x) dx = u(x)v(x) - \int v(x)u'(x) dx$$

The theorem is most often written in the form

$$\int u dv = uv - \int v du.$$

So what is the usefulness of this? Well, it may happen for some integrals and a good choice of u and v that the integral on the right is easier to evaluate than the integral on the left.

Example

Consider $\int x e^x dx$. If we choose $u = x$, $dv = e^x dx$ then integration by parts yields

$$\int x e^x dx = x e^x - \int e^x dx = (x - 1) e^x.$$

Now notice what happens when we choose $u = e^x$, $dv = x dx$.

$$\int x e^x dx = \frac{1}{2} x^2 e^x - \int \frac{1}{2} x^2 e^x dx$$

The integral gets harder instead of easier.

When applying integration by parts, one must choose u and dv wisely. As general rules of thumb:

- Pick u so that u' is simpler than u .
- Pick dv so that v is not more complicated, (hopefully simpler), than dv .

Also note that you may have to apply integration by parts several times to evaluate



some integrals.

Definite Integral

The area bounded by the x axis, the vertical lines $x = a$ and $x = b$ and the function $f(x)$ is denoted with a definite integral,

$$\int_a^b f(x) dx.$$

The area is signed, that is, if $f(x)$ is negative, then the area is negative. We measure the area with a divide-and-conquer strategy. First partition the interval (a, b) with $a = x_0 < x_1 < \dots < x_{n-1} < x_n = b$. Note that the area under the curve on the subinterval is approximately the area of a rectangle of base $\Delta x_i = x_{i+1} - x_i$ and height $f(\xi_i)$, where $\xi_i \in [x_i, x_{i+1}]$. If we add up the areas of the rectangles, we get an approximation of the area under the curve. See figure 7.6

$$\int_a^b f(x) dx \approx \sum_{i=0}^{n-1} f(\xi_i) \Delta x_i$$

As the Δx_i 's get smaller, we expect the approximation of the area to get better. Let $\Delta x = \max_{0 \leq i \leq n-1} \Delta x_i$. We define the definite integral as the sum of the areas of the rectangles in the limit that $\Delta x \rightarrow 0$.

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{n-1} f(\xi_i) \Delta x_i$$

The integral is defined when the limit exists. This is known as the Riemann integral of $f(x)$. $f(x)$ is called the integrand.

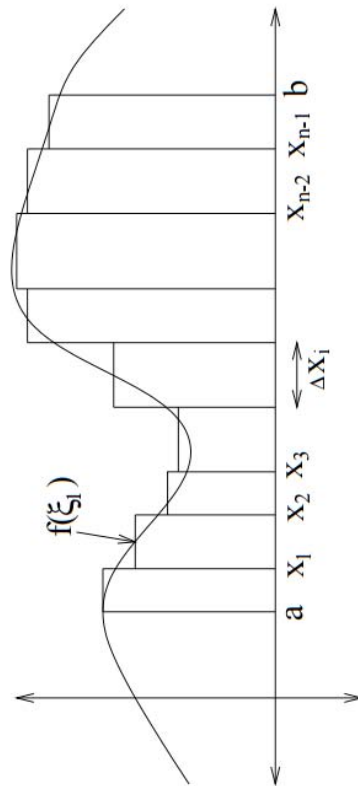


Figure 7.6: Divide-and-Conquer Strategy for Approximating a Definite Integral.

7.3 LEAST SQUARES METHOD



The least square methods (LSM) is probably the most popular technique in statistics. This is due to several factors. First, most common estimators can be casted within this framework. For example, the mean of a distribution is the value that minimizes the sum of squared deviations of the scores. Second, using squares makes LSM mathematically very tractable because the Pythagorean Theorem indicates that, when the error is independent of an estimated quantity, one can add the squared error and the squared estimated quantity. Third, the mathematical tools and algorithms involved in LSM (derivatives, Eigen decomposition, and singular value decomposition) have been well studied for a relatively long time.

LEARNING OBJECTIVE



3. Describe least squares method

LSM is one of the oldest techniques of modern statistics, and even though ancestors of LSM can be traced up to Greek mathematics, the first modern precursor is probably Galileo. The modern approach was first exposed in 1805 by the French mathematician Legendre in a now classic memoir, but this method is somewhat older because it turned out that, after the publication of Legendre's memoir, Gauss (the famous German mathematician) contested Legendre's priority. Gauss often did not published ideas when he thought that they could be controversial or not yet ripe, but would mention his discoveries when others would publish them (the way he did, for example for the discovery of Non-Euclidean geometry). And in 1809, Gauss published another memoir in which he mentioned that he had previously discovered LSM and used it as early as 1795 in estimating the orbit of an asteroid. A somewhat bitter anteriority dispute followed (a bit reminiscent of the Leibniz-Newton controversy about the invention of Calculus), which, however, did not diminish the popularity of this technique.

The use of LSM in a modern statistical framework can be traced to Galton (1886) who used it in his work on the heritability of size which laid down the foundations of correlation and (also gave the name to) regression analysis. The two antagonistic giants of statistics Pearson and Fisher, who did so much in the early development of statistics, used and developed it in different contexts (factor analysis for



Pearson and experimental design for Fisher). Nowadays, the least square method is widely used to find or estimate the numerical values of the parameters to fit a function to a set of data and to characterize the statistical properties of estimates. It exists with several variations: Its simpler version is called **ordinary least squares (OLS)**, a more sophisticated version is called weighted least squares (WLS), which often performs better than OLS because it can modulate the importance of each observation in the final solution. Recent variations of the least square method are alternating least squares (ALS) and partial least squares (PLS).

7.3.1 Functional Fit Example: Regression

The oldest (and still the most frequent) use of OLS was linear regression, which corresponds to the problem of finding a line (or curve) that best fits a set of data points. In the standard formulation, a set of N pairs of observations $\{Y_i, X_i\}$ is used to find a function relating the value of the dependent variable (Y) to the values of an independent variable (X). With one variable and a linear function, the prediction is given by the following equation:

$$\hat{Y} = a + bX. \quad (18)$$

This equation involves two free parameters which specify the intercept (a) and the slope (b) of the regression line. The least square method defines the estimate of these parameters as the values which minimize the sum of the squares (hence the name least squares) between the measurements and the model (i.e., the predicted values). This amounts to minimizing the expression:

$$\mathcal{E} = \sum_i (Y_i - \hat{Y}_i)^2 = \sum_i [Y_i - (a + bX_i)]^2 \quad (19)$$

(where $\mathcal{E}$ stands for “error” which is the quantity to be minimized). The estimation of the parameters is obtained using basic results from calculus and, specifically, uses the property that a quadratic expression reaches its minimum value when its derivatives vanish. Taking the derivative of $\mathcal{E}$

KEY TERMS

Ordinary least squares (OLS) is a method for estimating the unknown parameters in a linear regression model, with the goal of minimizing the sum of the squares of the differences between the observed responses in the given dataset and those predicted by a linear function of a set of explanatory variables.

with respect to a and b and setting them to zero gives the following set of equations (called the normal equations):

$$\frac{\partial \mathcal{E}}{\partial a} = 2Na + 2b \sum X_i - 2 \sum Y_i = 0 \quad (20)$$

and

$$\frac{\partial \mathcal{E}}{\partial b} = 2b \sum X_i^2 + 2a \sum X_i - 2 \sum Y_i X_i = 0. \quad (21)$$

Solving the normal equations gives the following least square estimates of a and b as:

$$a = M_Y - bM_X \quad (22)$$

(with M_Y and M_X denoting the means of X and Y) and

$$b = \frac{\sum (Y_i - M_Y)(X_i - M_X)}{\sum (X_i - M_X)^2}. \quad (23)$$

OLS can be extended to more than one independent variable (using matrix algebra) and to non-linear functions.

The Geometry of Least Squares

OLS can be interpreted in a geometrical framework as an orthogonal projection of the data vector onto the space defined by the independent variable. The projection is orthogonal because the predicted values and the actual values are uncorrelated. This is illustrated in Figure 7.7, which depicts the case of two independent variables (vectors x_1 and x_2) and the data vector (y), and shows that the error vector ($y - \hat{y}$) is orthogonal to the least square ($\hat{y}$) estimate which lies in the subspace defined by the two independent variables.

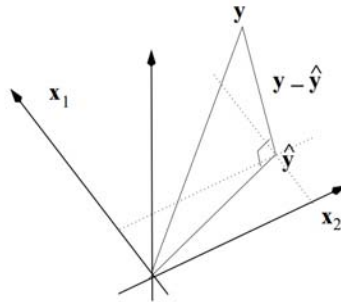


Figure 7.7: The least square estimate of the data is the orthogonal projection of the data vector onto the independent variable subspace.

Optimality of Least Square Estimates

OLS estimates have some strong statistical properties. Specifically when

- The data obtained constitute a random sample from a well-defined population,
- The population model is linear,
- The error has a zero expected value,
- The independent variables are linearly independent, and
- The error is normally distributed and uncorrelated with the independent variables (the so-called homoscedasticity assumption); then the OLS estimate is the best linear unbiased estimate often denoted with the acronym “BLUE” (the 5 conditions and the proof are called the Gauss-Markov conditions and theorem).

In addition, when the Gauss-Markov conditions hold, OLS estimates are also maximum likelihood estimates.

Weighted Least Squares

The optimality of OLS relies heavily on the homoscedasticity assumption. When the data come from different sub-populations for which an independent estimate of the error variance is available, a better estimate than OLS can be obtained using weighted least squares (WLS), also called **generalized least squares (GLS)**. The idea is to assign to each observation a weight that reflects the uncertainty of the measurement. In general, the weight w_i , assigned to the i th observation, will be a function of the variance of this observation, denoted σ_i^2 . A straightforward weighting schema is to define $w_i = \sigma_i^{-1}$ (but other more sophisticated weighted schemes can also be proposed). For the linear regression example, WLS will find the values of a and b minimizing:

$$\mathcal{E}_w = \sum_i w_i (Y_i - \hat{Y}_i)^2 = \sum_i w_i [Y_i - (a + bX_i)]^2. \quad (24)$$

KEY TERMS

Generalized least squares (GLS) is a technique for estimating the unknown parameters in a linear regression model.

Iterative Methods: Gradient Descent

When estimating the parameters of a nonlinear function with OLS or WLS, the standard approach using derivatives is not always possible. In this case, iterative methods are very often used. These methods search in a stepwise fashion for the best values of the estimate. Often they proceed by using at each step a linear approximation of the function and refine this approximation by successive corrections. The techniques involved are known as gradient descent and Gauss-Newton approximations. They correspond to nonlinear least squares approximation in numerical analysis and nonlinear regression in statistics. Neural networks constitutes a popular recent application of these techniques.

7.3.2 Newton's Method

We developed a simple method, bisection, for approximately solving the equation $f(x) = 0$. Unfortunately, this method, while guaranteed to find a solution on an interval that is known to contain one, is not practical because of the large number of iterations that are necessary to obtain an accurate approximate solution. To develop a more effective method for solving this problem of computing a solution to $f(x) = 0$, we can address the following questions:

- Are there cases in which the problem easy to solve, and if so, how do we solve it in such cases?
- Is it possible to apply our method of solving the problem in these “easy” cases to more general cases?

We will see that these questions are useful for solving a variety of problems. For the problem at hand, we ask whether the equation $f(x) = 0$ is easy to solve for any particular choice of the function f . Certainly, if f is a linear function, then it has a formula of the form $f(x) = m(x-a) + b$, where m and b are constants and $m \neq 0$. Setting $f(x) = 0$ yields the equation

$$m(x - a) + b = 0,$$

which can easily be solved for x to obtain the unique solution

$$x = a - \frac{b}{m}.$$

We now consider the case where f is not a linear function. Using Taylor's theorem, it is simple to construct a linear function that approximates $f(x)$ near a given point x_0 . This function is simply the first Taylor polynomial of $f(x)$ with center x_0 .



$$P_1(x) = f(x_0) + f'(x_0)(x - x_0).$$

This function has a useful geometric interpretation, as its graph is the tangent line of $f(x)$ at the point $(x_0, f(x_0))$.

We can obtain an approximate solution to the equation $f(x) = 0$ by determining where the linear function $P_1(x)$ is equal to zero. If the resulting value, x_1 , is not a solution, then we can repeat this process, approximating f by a linear function near x_1 and once again determining where this approximation is equal to zero. The resulting algorithm is Newton's method.

Algorithm (Newton's Method) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function. The following algorithm computes an approximate solution x^* to the equation $f(x) = 0$

Choose an initial guess x_0

for $k = 0, 1, 2, \dots$ do

if $f(x_k)$ is sufficiently small then

$x^* = x_k$

return x^*

end

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

if $|x_{k+1} - x_k|$ is sufficiently small then

$$x^* = x_{k+1}$$

return x^*

end

end

When Newton's method converges, it does so very rapidly. However, it can be difficult to ensure convergence, particularly if $f(x)$ has horizontal tangents near the solution x^* . Typically, it is necessary to choose a starting iterate x_0 that is close to x^* . As the following result indicates, such a choice, if close enough, is indeed sufficient for convergence.

Theorem (Convergence of Newton's Method) Let f be twice continuously differentiable on the interval $[a, b]$, and suppose that $f(c) = 0$ and $f'(c) \neq 0$ for some $c \in [a, b]$. Then there exists a $\delta > 0$ such that Newton's Method applied to $f(x)$ converges to c for any initial guess x_0 in the interval $[c - \delta, c + \delta]$.

Example

We will use of Newton's Method in computing $\sqrt{2}$. This number satisfies the equation $f(x) = 0$ where

$$f(x) = x^2 - 2.$$

Since $f'(x) = 2x$, it follows that in Newton's Method, we can obtain the next iterate x_{n+1} from the previous iterate x_n by

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 2}{2x_n} = x_n - \frac{x_n^2}{2x_n} + \frac{2}{2x_n} = \frac{x_n}{2} + \frac{1}{x_n}.$$

We choose our starting iterate $x_0 = 1$, and compute the next several iterates as follows:

$$x_1 = \frac{1}{2} + \frac{1}{1} = 1.5$$

$$x_2 = \frac{1.5}{2} + \frac{1}{1.5} = 1.41666667$$

$$x_3 = 1.41421569$$

$$x_4 = 1.41421356$$

$$x_5 = 1.41421356.$$

Since the fourth and fifth iterates agree in to eight decimal places, we assume that 1.41421356 is a correct solution to $f(x) = 0$, to at least eight decimal places. The first two iterations are illustrated in Figure 7.8.

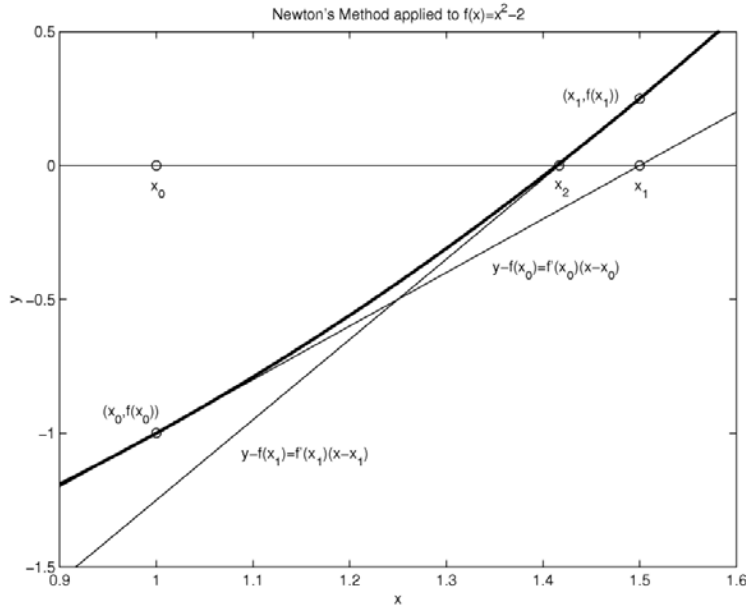


Figure 7.8: Newton's Method applied to $f(x) = x^2 - 2$. The bold curve is the graph of f . The initial iterate x_0 is chosen to be 1. The tangent line of $f(x)$ at the point $(x_0, f(x_0))$ is used to approximate $f(x)$, and it crosses the x -axis at $x_1 = 1.5$, which is much closer to the exact solution than x_0 . Then, the tangent line at $(x_1, f(x_1))$ is used to approximate $f(x)$, and it crosses the x -axis at $x_2 = 1.41\bar{6}$, which is already very close to the exact solution.

Example: Newton's Method can be used to compute the reciprocal of a number a without performing any divisions. The solution, $1/a$, satisfies the equation $f(x) = 0$, where

$$f(x) = a - \frac{1}{x}.$$

Since

$$f'(x) = \frac{1}{x^2},$$

it follows that in Newton's Method, we can obtain the next iterate x_{n+1} from the previous iterate x_n by

$$x_{n+1} = x_n - \frac{a - 1/x_n}{1/x_n^2} = x_n - \frac{a}{1/x_n} + \frac{1/x_n}{1/x_n^2} = 2x_n - ax_n^2.$$

Note that no divisions are necessary to obtain x_{n+1} from x_n . This iteration was actually used on older IBM computers to implement division in hardware. We use this iteration to compute the reciprocal of $a = 12$. Choosing our starting iterate to be

0.1, we compute the next several iterates as follows:

$$x_1 = 2(0.1) - 8(0.1)^2 = 0.08$$

$$x_2 = 2(0.12) - 8(0.12)^2 = 0.0832$$

$$x_3 = 0.0833312$$

$$x_4 = 0.0833333333279$$

$$x_5 = 0.0833333333333.$$

We conclude that 0.0833333333333 is an accurate approximation to the correct solution. Now, suppose we repeat this process, but with an initial iterate of $x_0 = 1$. Then, we have

$$x_1 = 2(1) - 12(1)^2 = -10$$

$$x_2 = 2(-6) - 12(-6)^2 = -1220$$

$$x_3 = 2(-300) - 12(-300)^2 = -17863240$$

It is clear that this sequence of iterates is not going to converge to the correct solution. In general, for this iteration to converge to the reciprocal of a , the initial iterate x_0 must be chosen so that $0 < x_0 < 2/a$. This condition guarantees that the next iterate x_1 will at least be positive. The contrast between the two choices of x_0 are illustrated in Figure 7.9.

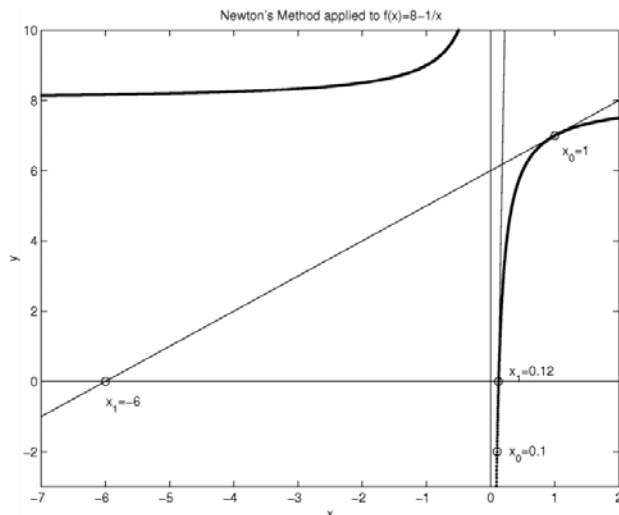


Figure 7.9: Newton's Method used to compute the reciprocal of 8 by solving the equation $f(x) = 8 - 1/x = 0$. When $x_0 = 0.1$, the tangent line of $f(x)$ at $(x_0, f(x_0))$ crosses the x -axis at

$x_1 = 0.12$, which is close to the exact solution. When $x_0 = 1$, the tangent line crosses the x -axis at $x_1 = -6$, which causes searching to continue on the wrong portion of the graph, so the sequence of iterates does not converge to the correct solution.

7.4 PERTURBATION METHOD

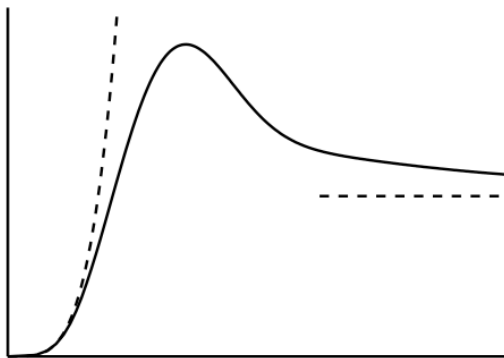
4 Perturbation methods are methods which rely on there being a dimensionless parameter in the problem that is relatively small: $\varepsilon \ll 1$. The most common example you may have seen before is that of high-Reynolds number fluid mechanics, in which a viscous boundary layer is found close to a solid surface. Note that in this case the standard physical parameter Re is large: our small parameter is $\varepsilon = Re^{-1}$.

LEARNING OBJECTIVE

4. Describe least squares method

A Real Research Example

This comes from my own research². The parameter varied is wavelength: and for both very long waves (wavenumber $k \ll 1$) and very short waves ($k^{-1} \ll 1$) the system is much simplified. The long-wave case, in particular, gives very good insight into the physics of the problem. If we look at the plot of growth rate of the instability against wavenumber (inverse wavelength):



We can see good agreement between the perturbation method solutions (the dotted lines) and the numerical calculations (solid curve): this kind of agreement gives confidence in the numeric in the middle region where perturbation methods cannot help.

Method of Averaging

Consider a weakly nonlinear oscillation problem

$$\ddot{x} + \varepsilon h(x, \dot{x}) + x = 0, \varepsilon \ll 1.$$

The effect of the perturbation is to cause the parameters of the unperturbed system to vary with time. If the perturbation is small, the variation of the parameters within one period of the unperturbed motion will also be small. Thus, the perturbation theory is based on the premise that during small intervals of time the perturbed system moves along a route, which has the same functional form as the unperturbed system but with parameters changing with time. The latter variation may be periodic or secular. In the latter case, the change in a parameter is cumulative so that no matter how small this change may be over a small interval of time, eventually the perturbed route will differ greatly from the unperturbed route. The periodic variations in the parameters, on the other hand, do not materialize over long intervals of time because they can be eliminated by averaging the perturbation effects over the unperturbed period!

Assume that this system has at least one periodic solution, so that its phase diagram consists of either a limit cycle or a center. Such an equation is in a sense “close” to the linear equation

$$\ddot{x} + x = 0$$

and the paths will be close to being circular. One may use this fact in constructing approximate solutions. In order to determine periodic solutions, in particular, we assume the existence of a nearly circular closed path in the phase diagram. This method calculates the average value of the amplitude of a slowly-varying oscillation over a cycle of the path. It can calculate approximate periods of limit cycles and the shape of the spiral paths around limit cycles, or it can calculate amplitude-frequency relations in the case of a center.



7.5 WKB METHOD

 WKB method is a method for finding approximate solutions to linear differential equations with spatially varying coefficients. It is typically used for a semiclassical calculation in quantum mechanics in which the wave function is recast as an exponential function, semiclassically expanded, and then either the amplitude or the phase is taken to be slowly changing.

We apply the WKB method to approximate solutions to equations of the form

$$\varepsilon^2 y'' + q(x)y = 0, \quad \varepsilon \ll 1, \quad (25)$$

$$y'' + q(\varepsilon x)^2 y = 0, \quad \varepsilon \ll 1, \quad (26)$$

and

$$-y'' + q(x)y = \lambda^2 p(x)y, \quad \lambda \gg 1. \quad (27)$$

Example: The wave function $\Psi(x, t)$ in one space dimension satisfies the Schrödinger equation

$$i\hbar \Psi_t = -\frac{\hbar^2}{2m} \Psi_{xx} + V(x)\Psi,$$

where V is the potential, m the mass and $\hbar = h/2\pi$, h being Planck's constant. We separate variables by setting

$$\Psi(x, t) = \phi(t)y(x).$$

We set the x and t terms equal to the constant E and obtain the equation

$$\frac{\hbar^2}{2m} y'' + (E - V(x))y = 0, \quad (28)$$

which is the time-independent Schrödinger equation. Set

LEARNING OBJECTIVE

 Understand the WKB method

$$q(x) = E - V(x),$$

and

$$\frac{\hbar^2}{2m} = \varepsilon^2$$

to obtain an equation of the form (25).

The Nonoscillatory Case: If $q(x) < 0$ over the interval of interest, set $q(x) = -k(x)^2$, where $k(x) > 0$. The equation becomes

$$\varepsilon^2 y'' - k(x)^2 y = 0. \quad (29)$$

Were $k(x) \equiv k_0$, a real constant, then (29) would have linearly independent, nonoscillatory solutions of the form $\exp(\pm k_0 x / \varepsilon)$. This suggests the change of variable

$$y = e^{\frac{u(x)}{\varepsilon}}. \quad (30)$$

The function u satisfies the equation

$$\varepsilon u'' + u'^2 - k(x)^2 = 0.$$

We set $u' = v$ to get

$$\varepsilon v' + v^2 - k(x)^2 = 0. \quad (31)$$

Plug the regular perturbation expansion

$$v = v_0 + \varepsilon v_1 + \cdots$$

into (31):

$$\varepsilon(v_0 + \varepsilon v_1 + \cdots)' + (v_0 + \varepsilon v_1 + \cdots)^2 - k(x)^2 = 0.$$

Matching powers of ε gives the equations

$$O(1) : v_0^2 - k(x)^2 = 0,$$

$$O(\varepsilon) : 2v_0v_1 = -v_0'.$$

From the $O(1)$ equation we obtain

$$v_0(x) = \pm k(x).$$

Put this in the $O(\varepsilon)$ equation to get

$$v_1(x) = -\frac{k'(x)}{2k(x)}.$$

Thus v has the expansion

$$v(x) = \pm k(x) - \varepsilon \frac{k'(x)}{2k(x)} + O(\varepsilon^2). \quad (32)$$

And since $v = u'$,

$$u(x) = \pm \int_{\xi}^x k(z) dz - \frac{\varepsilon}{2} \ln \frac{k(x)}{k(\xi)} + O(\varepsilon^2), \quad (33)$$

where ξ is arbitrary. Thus

$$\begin{aligned} y_{\pm}(x) &= e^{\frac{u(x)}{\varepsilon}} \\ &= \left[\frac{k(\xi)}{k(x)} \right]^{\frac{1}{2}} e^{\pm \frac{1}{\varepsilon} \int_{\xi}^x k(z) dz} e^{O(\varepsilon)} \\ &= \left[\frac{k(\xi)}{k(x)} \right]^{\frac{1}{2}} e^{\pm \frac{1}{\varepsilon} \int_{\xi}^x k(z) dz} (1 + O(\varepsilon)). \end{aligned} \quad (34)$$

In (34), we have approximations of two linearly independent solutions to (29), one with the plus sign and the other the minus. Thus, to leading order, any solution y to (29) will have the WKB approximation

$$y(x) \approx y_a(x) = \frac{c_1}{\sqrt{k(x)}} e^{\frac{1}{\varepsilon} \int_{\xi}^x k(z) dz} + \frac{c_2}{\sqrt{k(x)}} e^{-\frac{1}{\varepsilon} \int_{\xi}^x k(z) dz}, \quad (35)$$

for constants c_1 and c_2 .

Example: Find the WKB approximation to the solution of the equation

$$\begin{cases} \varepsilon^2 y'' - (1+x)^2 y = 0, & \text{for } x > 0, \varepsilon \ll 1, \\ y(0) = 1, \\ y(\infty) = 0. \end{cases}$$

Take $\xi = 0$. The WKB approximation has the form

$$y_a(x) = \frac{c_1}{\sqrt{1+x}} e^{\frac{1}{\varepsilon} \int_0^x (1+z) dz} + \frac{c_2}{\sqrt{1+x}} e^{-\frac{1}{\varepsilon} \int_0^x (1+z) dz}. \quad (36)$$

The second boundary condition forces us to take $c_1 = 0$. The first boundary condition then implies that $c_2 = 1$. Hence, to leading order,

$$\begin{aligned} y(x) &\approx y_a \\ &= \frac{1}{\sqrt{1+x}} e^{-\frac{1}{\varepsilon} \int_0^x (1+z) dz} \\ &= \frac{1}{\sqrt{1+x}} e^{-\frac{1}{\varepsilon} \left(x + \frac{x^2}{2}\right)}. \end{aligned} \quad (37)$$

The Oscillatory Case: When $q(x) > 0$ over the interval of interest, we set $q(x) = k(x)^2$, where $k(x) > 0$. The equation becomes

$$\varepsilon^2 y'' + k(x)^2 y = 0. \quad (38)$$

Were $k(x) \equiv k_0$, a real constant, then (38) would have oscillatory solutions of the form $\exp(\pm i k_0 x / \varepsilon)$. This suggests the change of variable

$$y = e^{\frac{i u(x)}{\varepsilon}}, \quad (39)$$

for some real-valued function $u(x)$. The same analysis as the foregoing yields WKB approximations to linearly independent solutions to equation (38):

$$y_{\pm}(x) = \frac{1}{\sqrt{k(x)}} e^{\pm \frac{i}{\varepsilon} \int_{\xi}^x k(z) dz}. \quad (40)$$

To finish the derivation, use the Coates-Euler formula $e^{i\theta} = \cos \theta + i \sin \theta$, to rewrite $y_{\pm}$ terms of sines and cosines. Then use these to form the the WKB approximation to the general solution to (38):

$$y(x) \approx y_a(x) = \frac{c_1}{\sqrt{k(x)}} \sin\left(\frac{1}{\varepsilon} \int_{\xi}^x k(z) dz\right) + \frac{c_2}{\sqrt{k(x)}} \cos\left(\frac{1}{\varepsilon} \int_{\xi}^x k(z) dz\right), \quad (41)$$

for constants c_1 and c_2 . We can multiply and divide the above expression by

$$A = (c_1^2 + c_2^2)^{\frac{1}{2}},$$

and then rewrite it as

$$\frac{A}{\sqrt{k(x)}} \cos \left(\frac{1}{\varepsilon} \int_{\xi}^x k(z) dz - \phi \right), \quad (42)$$

where

$$\phi = \arctan \frac{c_1}{c_2}$$

is the phase.

SUMMARY

1. Analytical methods are the preferred way to determine the outcome of a hypothesis when associated equations are simple and a precise answer is desired. Numerical methods are used when equations are too complex to solve fully.
2. Approximate analysis is a tool engineers use to estimate the magnitude of member forces and moments, and support reactions in indeterminate structures.
3. Approximate analysis is useful in determining (approximately) the forces and moments in the different members and in coming up with preliminary designs.
4. A projection method for solving integral and differential equations in which the approximate solution is determined from the condition that the equation be satisfied at certain given points.
5. Phase plane analysis is a graphical method for studying second-order systems. Objective is to gain familiarity of the nonlinear systems through the simple graphical method.
6. The least square methods (LSM) is probably the most popular technique in statistics. This is due to several factors.
7. Perturbation methods are methods which rely on there being a dimensionless parameter in the problem that is relatively small: $\varepsilon \ll 1$.
8. WKB method is a method for finding approximate solutions to linear differential equations with spatially varying coefficients. It is typically used for a semiclassical calculation in quantum mechanics in which the wave function is recast as an exponential function, semiclassically expanded, and then either the amplitude or the phase is taken to be slowly changing.

KNOWLEDGE CHECK

1. Which of the following filter we use in least square design methods?
 - a. All zero
 - b. All pole
 - c. Pole-zero
 - d. Any of the mentioned
2. Which of the following are cascaded in this method?
 - a. $H_d(z)$, $H(z)$
 - b. $1/H_d(z)$, $1/H(z)$



- c. $1/H_d(z)$, $H(z)$
 - d. $H_d(z)$, $1/H(z)$
3. Point on graph where equations intercept crosses x-coordinate is called
- a. y-intercept
 - b. x-intercept
 - c. a-intercept
 - d. b-intercept
4. Graph of linear equation with two variables is plotted in
- a. two dimensions
 - b. one dimension
 - c. four dimensions
 - d. three dimensions
5. Slope of linear equation straight line is quantified with help of
- a. zero ordinate
 - b. negative integers
 - c. positive integers
 - d. real numbers
6. Jacobi's method is also known as
- a. Displacement method
 - b. Simultaneous displacement method
 - c. Simultaneous method
 - d. Diagonal method
7. The least squares method can also be used in a pole-zero approximation for $H_d(z)$.
- a. True
 - b. False
8. In which of the following condition we can use the desired response $h_d(n)$?
- a. $n < M$
 - b. $n=M$
 - c. $n > M$
 - d. None of the mentioned

REVIEW QUESTIONS

1. Explain Chaplygin's method and collocation method.
2. Discuss concepts of phase plane analysis.
3. Find a function whose graph has a relative minimum when $x = 1$ and a relative maximum when $x = 4$.
4. Evaluate $\int (7x^2 + 3\cos(x) - e^x) dx$
5. Evaluate the definite integral using integration by parts with Way 1.

$$\int_2^4 e^{2x} x^2 dx$$
6. Consider the time-independent Schrödinger equation (4). If $E > V(x)$, we have the oscillatory case. Since $\hbar$ is small, we can apply the WKB method obtain the approximate solution

$$y_a(x) = \frac{A}{(E - V(x))^{\frac{1}{4}}} \cos\left(\frac{\sqrt{2m}}{\hbar} \int_{\xi}^x \sqrt{E - V(z)} dz - \phi\right).$$
7. What do you understand by dominant balance?
8. Discuss about the integral methods.

Check your Result

1. (b) 2. (d) 3. (b) 4. (b) 5. (d) 6. (b)
 7. (a) 8. (c)

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Chapter

8

TESTING OF HYPOTHESIS

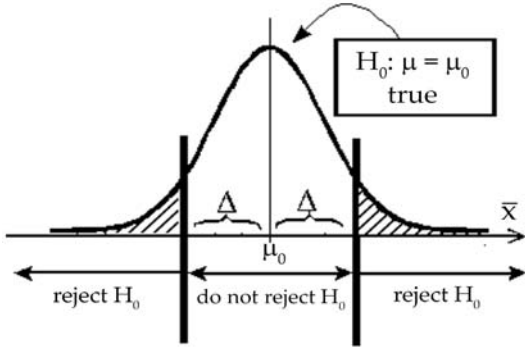
LEARNING OBJECTIVES

After studying this chapter, you will be able to:

1. Overview of hypothesis testing
2. Discuss about sampling
3. Explain the reliability of the estimates
4. Describe the Test of Significance for Large Samples
5. Define the statistical significance (T-Test)
6. Understand Chi-Square (χ^2) Test

INTRODUCTION

A hypothesis test is a statistical test that is used to determine whether there is enough evidence in a sample of data to infer that a certain condition is true for the entire population.



A hypothesis test examines two opposing hypotheses about a population:

the null hypothesis and the alternative hypothesis. The null hypothesis is the statement being tested. Usually the null hypothesis is a statement of “no effect” or “no difference”. The alternative hypothesis is the statement you want to be able to conclude is true.

Based on the sample data, the test determines whether to reject the null hypothesis. You use a p-value, to make the determination. If the p-value is less than or equal to the level of significance, which is a cut-off point that you define, then you can reject the null hypothesis. A common misconception is that statistical hypothesis tests are designed to select the more likely of two hypotheses. Instead, a test will remain with the null hypothesis until there is enough evidence (data) to support the alternative hypothesis.

8.1 OVERVIEW OF HYPOTHESIS TESTING

LEARNING OBJECTIVE

1. Overview of hypothesis testing

1. A statistical hypothesis is an assumption about a population parameter. This assumption may or may not be true. Hypothesis testing refers to the formal procedures used by statisticians to accept or reject **statistical hypotheses**.

A hypothesis test is a statistical method that uses sample data to evaluate a hypothesis about a population parameter. So, the procedure is to first state a hypothesis about a population parameter, such as μ . The next step is to collect sample data that would address the hypothesis. And then to determine the extent to which the sample data are consistent with the hypothesis about the parameter.

8.1.1 Statistical Hypotheses

The best way to determine whether a statistical hypothesis is true would be to examine the entire population. Since that is often impractical, researchers typically examine a random sample from the population. If sample data are not consistent with the statistical hypothesis, the hypothesis is rejected.

There are two types of statistical hypotheses.

- **Null hypothesis.** The null hypothesis, denoted by H_0 , is usually the hypothesis that sample observations result purely from chance.
- **Alternative hypothesis.** The alternative hypothesis, denoted by H_1 or H_a , is the hypothesis that sample observations are influenced by some non-random cause.

For example, suppose we wanted to determine whether a coin was fair and balanced. A null hypothesis might be that half the flips would result in Heads and half, in Tails. The alternative hypothesis might be that the number of Heads and Tails would be very different. Symbolically, these hypotheses would be expressed as

$$H_0: P = 0.5$$

$$H_a: P \neq 0.5$$

Suppose we flipped the coin 50 times, resulting in 40 Heads and 10 Tails. Given this result, we would be inclined to reject the null hypothesis. We would conclude, based on the evidence, that the coin was probably not fair and balanced.

8.1.2 Hypothesis Tests

Statisticians follow a formal process to determine whether to reject a **null hypothesis**, based on sample data. This process, called hypothesis testing, consists of four steps.

- State the hypotheses. This involves stating the null and alternative hypotheses. The hypotheses are stated in such a way that they are mutually exclusive. That is, if one is true, the other must be false.

KEY TERMS

Statistical hypothesis is a hypothesis concerning the parameters or from of the probability distribution for a designated population or populations, or, more generally, of a probabilistic mechanism which is supposed to generate the observations.



REMEMBER

Hypothesis test is a rule that tells us for which sample values we should decide to accept H_0 as true and for which sample values we should decide to reject H_0 and accept H_a as true.

KEY TERMS

Null hypothesis is a hypothesis that says there is no statistical significance between the two variables in the hypothesis.

- Formulate an analysis plan. The analysis plan describes how to use sample data to evaluate the null hypothesis. The evaluation often focuses around a single test statistic.
- Analyze sample data. Find the value of the test statistic (mean score, proportion, t statistic, z-score, etc.) described in the analysis plan.
- Interpret results. Apply the decision rule described in the analysis plan. If the value of the test statistic is unlikely, based on the null hypothesis, reject the null hypothesis.

8.1.3 Decision Errors

Two types of errors can result from a hypothesis test.

- **Type I error.** A Type I error occurs when the researcher rejects a null hypothesis when it is true. The probability of committing a Type I error is called the significance level. This probability is also called alpha, and is often denoted by α .
- **Type II error.** A Type II error occurs when the researcher fails to reject a null hypothesis that is false. The probability of committing a Type II error is called Beta, and is often denoted by β . The probability of *not* committing a Type II error is called the Power of the test.

8.1.4 Decision Rules

The analysis plan includes decision rules for rejecting the null hypothesis. In practice, statisticians describe these decision rules in two ways - with reference to a P-value or with reference to a region of acceptance.

- **P-value.** The strength of evidence in support of a null hypothesis is measured by the P-value. Suppose the test statistic is equal to S . The P-value is the probability of observing a test statistic as extreme as S , assuming the null hypothesis is true. If the P-value is less than the significance level, we reject



the null hypothesis.

- **Region of acceptance.** The region of acceptance is a range of values. If the test statistic falls within the region of acceptance, the null hypothesis is not rejected. The region of acceptance is defined so that the chance of making a Type I error is equal to the significance level.

The set of values outside the region of acceptance is called the region of rejection. If the test statistic falls within the region of rejection, the null hypothesis is rejected. In such cases, we say that the hypothesis has been rejected at the α level of significance. These approaches are equivalent. Some statistics texts use the P-value approach; others use the region of acceptance approach.

8.1.5 One-Tailed and Two-Tailed Tests

A test of a statistical hypothesis, where the region of rejection is on only one side of the sampling distribution, is called a one-tailed test.

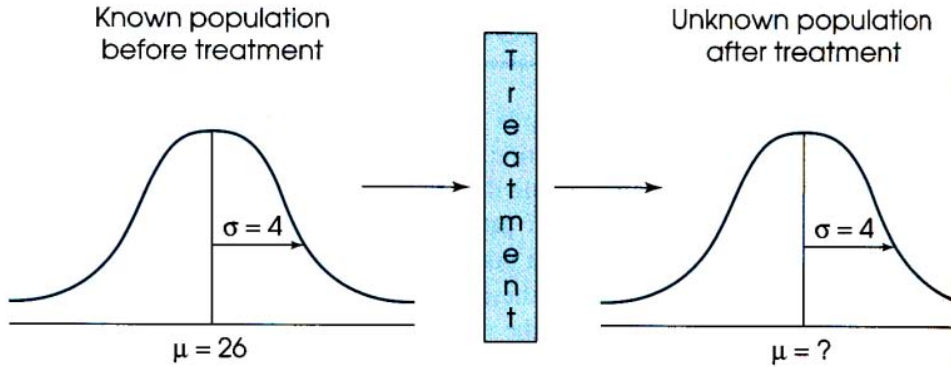
Suppose the null hypothesis states that the mean is less than or equal to 10. The alternative hypothesis would be that the mean is greater than 10. The region of rejection would consist of a range of numbers located on the right side of sampling distribution; that is, a set of numbers greater than 10.



A test of a statistical hypothesis, where the region of rejection is on both sides of the sampling distribution, is called a two-tailed test. For example, suppose the null hypothesis states that the mean is equal to 10. The alternative hypothesis would be that the mean is less than 10 or greater than 10. The region of rejection would consist of a range of numbers located on both sides of sampling distribution; that is, the region of rejection would consist partly of numbers that were less than 10 and partly of numbers that were greater than 10.

Here's an illustration of the logic of null hypothesis significance testing (NHST). Suppose that a population of 2-year-old children has a mean weight $\mu = 26$ pounds and $\sigma = 4$. If you were to administer a treatment (handling) to

every member of the population, what would happen to the scores? The effect of treatment is modeled as an additive constant (adding either plus or minus constant). If you recall the impact of an additive constant on variability, you will realize that the standard deviation would stay the same. If the treatment has no effect, then the treatment constant would be zero, and the population mean would be unchanged. Schematically, the situation is illustrated below:

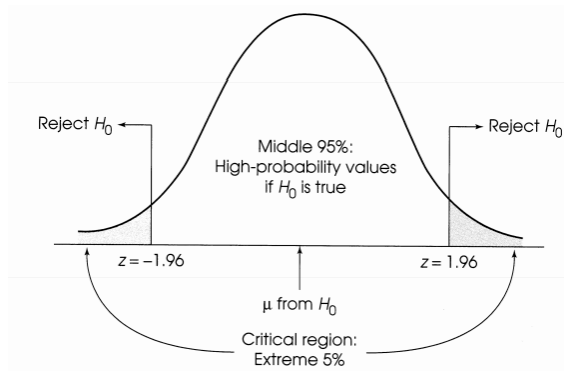


So, the question becomes, “Was the treatment effect zero, or was it greater than zero?” To test that question, we would typically construct a testable statistical hypothesis, called the null hypothesis (H_0). In this case, $H_0: \mu = 26$. But, of course, we cannot treat and measure every member of the population. Instead, we will take a sample (e.g., $n = 16$) and give them extra handling. If the mean weight of the sample is near 26 pounds, then the handling treatment will likely be considered to be ineffective. To the extent that the mean weight of the sample is much larger (or smaller) than 26, then we would be inclined to think that the handling treatment was effective. The crucial question is, “How much must the mean weight differ from 26 pounds to convince us that the treatment was effective?”

The conventional way of determining the extent to which the treatment was effective is by establishing a level of significance or alpha level. What an alpha level represents is one’s willingness to say that a sample mean was not drawn from the population with H_0 true, when in fact it was. In other words, what level of error are you willing to tolerate? (This particular kind of error is called a Type I Error, but more on that a bit later on.) Another way of thinking about alpha level is that it determines the weird or unusual scores that would occur in the distribution when H_0 is true. The typical alpha level is .05 ($\alpha = .05$), which means that the “definition” of weird is a score that would occur so infrequently as to lie in the lower or upper 2.5% of the sampling distribution. Alternatively, you could think that you are willing to conclude that if your sample mean falls in the lower or upper 2.5% of the distribution when H_0 is true, you would be better off concluding that H_0 is false. As you may recall, the z-scores that determine



the lower and upper .025 of a **normal distribution** are -1.96 and $+1.96$, as illustrated in the figure below.



After you have determined your null and alternative hypotheses, and your significance level (typically .05), you are ready to collect your sample and determine the sample mean. With the sample mean, you are now in a position to compute a z-score:

$$z = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}} = \frac{\text{obtained difference}}{\text{difference due to chance}}$$

Finally, given your z-score, you can make a decision about the null hypothesis. If the sample mean leads to a z-score that would fall in the critical region, you would decide to reject H_0 . If the sample mean leads to a z-score that does not fall in the critical region, you would fail to reject, or retain, H_0 .

8.1.6 Uncertainty and Errors in Hypothesis Testing

One of the reasons that we never talk about “proving” anything in science is that we recognize the ubiquity of uncertainty. Because we can never know the “truth” independent of our research, we can never be certain of our conclusions. Thus, when you decide to reject H_0 , you need to be aware that H_0 could really be false, in which case you have made a correct rejection. It is also possible, however, that rejected H_0 and it is really true. If so, you have made an error. We call that

KEY TERMS

Normal distribution is an arrangement of a data set in which most values cluster in the middle of the range and the rest taper off symmetrically toward either extreme.

error a Type I error. You should recognize that the significance level you choose is an expression of tolerance for a Type I error.

When you decide to retain H_0 , it may well be that H_0 is true, which is a correct retention. It is also possible, however, that H_0 is false. In that case retaining H_0 would be an error—a Type II error. You can summarize the four possibilities in a table:

		Actual Situation	
		H_0 True	H_0 False
Experimenter's Decision	Reject H_0	Type I Error	Correct Rejection
	Retain H_0	Correct Retention	Type II Error

Note that if you make the decision to reject H_0 , you cannot make a Type II Error.

Assumptions underlying hypothesis tests with z-scores

You will actually see very few hypothesis tests using z-scores. That's because doing so requires that one knows σ , which is a highly unusual circumstance. Beyond that major assumption, there are other assumptions of note:

- Participants are randomly sampled.
- Observations are independent of one another. Random sampling will typically produce observations that are independent of one another.
- The value of σ will not be affected by the treatment. Even in the rare circumstances under which we might actually know σ , it is also crucial that any treatment we use has no impact on σ , otherwise that statistic will be thrown off.
- The sampling distribution of the mean is normal. Of course, with increasingly large sample size (approaching infinity), the sampling distribution will become normal.

Effect Size

Some people have expressed reservations about null hypothesis significance testing. A crucial question that is not addressed by a significance test is the size of the treatment effect. That effect can be assessed by a number of measures, one of which is Cohen's d . You compute this measure of effect size as follows:

$$d = \frac{\text{mean difference}}{\text{standard deviation}}$$

A value of d between 0 and 0.2 is considered a small effect, between 0.2 and 0.8 is considered a medium effect, and a value of d greater than 0.8 is considered a large effect.



Statistical Power

The power of a test is the probability that the test will correctly reject a false null hypothesis. It represents the opposite of a Type II error, also called a β error. Thus, power is described as $1 - \beta$.

Several factors can influence power, but for now, you should think of the impact of treatment effect on power. In the example of the impact of prenatal alcohol on birth weight, if the alcohol had a greater impact, the mean weight of the sample would be smaller still (leading to an even larger z-score).

8.2 SAMPLING

 2. The statistician is often confronted with the problem of discussing a universe of which he cannot examine every member, i.e., of which complete enumeration is impracticable. For example, if we want to have an idea of the average per capita income of the people of a country, enumeration of every earning individual in the country is a very difficult task. Naturally, the question arises: What can be said about a universe of which we can examine only a limited number of members? This question is the origin of the Theory of **Sampling**.

A finite subset of a universe is called a sample. A sample is thus a small portion of the universe. The number of individuals in a sample is called the sample size. The process of selecting a sample from a universe is called sampling.

The theory of sampling is a study of relationship existing between a population and samples drawn from the population. The fundamental object of sampling is to get as much information as possible of the whole universe by examining only a part of it. An attempt is thus made through sampling to give the maximum information about the parent universe with the minimum effort. Sampling is quite often used in our day-to-day practical life.

LEARNING OBJECTIVE

 2. Discuss about sampling

KEY TERMS

Sampling is a process used in statistical analysis in which a predetermined number of observations are taken from a larger population.



In a shop we assess the quality of sugar, rice, or any other commodity by taking only a handful of it from the bag and then decide whether to purchase it or not. A housewife normally tests the cooked products to find if they are properly cooked and contain the proper quantity of salt or sugar, by taking a spoonful of it.

8.2.1 Parameters of Statistics

The statistical constants of the population such as mean, the variance, etc. are known as the parameters. The statistical concepts of the sample from the members of the sample to estimate the parameters of the population from which the sample has been drawn is known as statistic. A parameter is a numerical quantity that describes some characteristic of a population. Parameters are often estimated since their value is generally unknown, especially when the population is large enough that it is impossible or impractical to obtain measurements for all observations. Parameters are normally represented by Greek letters. The most common parameters are the population mean and variance, represented by the Greek letters m and s^2 , respectively.

A statistic is a quantitative value that is calculated from the observations in a sample. They are usually represented by lowercase English letters with other symbols. The sample mean and variance, two of the most common statistics derived from samples, are denoted by the symbols $\bar{x}$ and s^2 , respectively. Population mean and variance are denoted by μ and σ^2 , while those of the samples are given by $\bar{x}$, s^2 .

8.2.2 Standard Error

The standard deviation of the sampling distribution of a statistic is known as the standard error (S.E.). It plays an important role in the theory of large samples and it forms a basis of the testing of hypothesis. If t is any statistic, for

large sample. $z = \frac{t - E(t)}{S.E(t)}$ is normally distributed with mean 0 and variance unity.

For large sample, the standard errors of some of the well-known statistic are listed below:



n —sample size; σ^2 —population variance; s^2 —sample variance; p —population proportion ; $Q = 1 - p$; n_1, n_2 —are sizes of two independent random samples.

Number	Statistic	Standard error
1.	$\bar{x}$	$\sigma/\sqrt{n}$
2.	s	$\sqrt{\sigma^2/2n}$
3.	Difference of two sample means $\bar{x}_1 - \bar{x}_2$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
4.	Difference of two sample standard deviation $s_1 - s_2$	$\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}$
5.	Difference of two sample proportions $p_1 - p_2$	$\sqrt{\frac{P_1Q_1}{n_1} + \frac{P_2Q_2}{n_2}}$
6.	Observed sample proportion p	$\sqrt{PQ/n}$

8.2.3 Test of Significance

An important aspect of the sampling theory is to study the test of significance which will enable us to decide, on the basis of the results of the sample, whether

- the deviation between the observed sample statistic and the hypothetical parameter value or
- the deviation between two sample statistics is significant or might be attributed due to chance or the fluctuations of the sampling

For applying the tests of significance, we first set up a hypothesis which is a definite statement about the population parameter called Null hypothesis denoted by H_0 .

Any hypothesis which is complementary to the null hypothesis (H_0) is called an Alternative hypothesis denoted by H_1 .

For example, if we want to test the null hypothesis that the population has a specified mean μ_0 , then we have

$$H_0: \mu = \mu_0$$

Alternative hypothesis will be

- $H_1: \mu \neq \mu_0$ ($\mu > \mu_0$ or $\mu < \mu_0$) (two tailed alternative hypothesis).
- $H_1: \mu > \mu_0$ (right tailed alternative hypothesis (or) single tailed).
- $H_1: \mu < \mu_0$ (left tailed alternative hypothesis (or) single tailed).

Hence alternative hypothesis helps to know whether the test is two tailed test or one tailed test.

A region corresponding to a statistic t , in the sample space S which amounts to rejection of the null hypothesis H_0 is called as critical region or region of rejection. The region of the sample space S which amounts to the acceptance of H_0 is called acceptance region.

8.2.4 Level of Significance

The probability of the value of the variate falling in the critical region is known as level of significance. The probability α that a random value of the statistic t belongs to the critical region is known as the level of significance.

$$P(t \in \omega | H_0) = \alpha$$

i.e., the level of significance is the size of the type I error or the maximum producer's risk.

8.3 RELIABILITY OF THE ESTIMATES



The data presented, estimates for the population as a whole although they are based on a sample survey. As such, they are prone to errors, which may be of two types.

LEARNING OBJECTIVE



3. Explain the reliability of the estimates

8.3.1 Sampling Errors

Sampling errors occur because only a sample of the population is investigated. The particular sample used in this survey is one of a large number of possible samples of the same size that could have been selected using the same sampling method. It is clear that estimates based on different samples will differ from one another, and almost all of them will differ from the value obtained had a complete census been taken – “the census value”.

- The estimate X' is the value, estimated from the specific sample of this survey, of the corresponding value X that would have been obtained had a complete census been conducted.



- The sampling error $\sigma(X')$ is a measure of the variability between the different values of the estimate that would have been obtained from all possible samples of the same size and derived by the same sampling method and the census value obtained under the same conditions of data collection.

The main aim of the sampling theory is to draw a valid conclusion about the population parameters on the basis of the sample results. In doing this we may commit the following two types of errors:

Type I Error. When H_0 is true, we may reject it.

$$P(\text{Reject } H_0 \text{ when it is true}) = P(\text{Reject } H_0 / H_0) = \alpha$$

α is called the size of the type I error also referred to as producer's risk.

Type II Error. When H_0 is wrong we may accept it $P(\text{Accept } H_0 \text{ when it is wrong}) = P(\text{Accept } H_0 / H_1) = \beta$. β is called the size of the type II error, also referred to as consumer's risk.

For larger samples corresponding to the statistic t , the variable $z = \frac{t - E(t)}{S.E(t)}$ is normally distributed with mean 0 and variance 1. The value of z given above under the null hypothesis is known as test statistic.

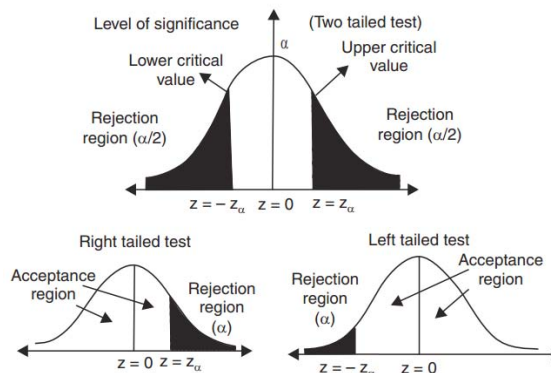
The critical value of z_α of the test statistic at level of significance α for a two-tailed test is given by

$$P(|z| > z_\alpha) = \alpha \quad (1)$$

i.e., z_α is the value of z so that the total area of the critical region on both tails is α . Since the normal curve is symmetrical, from equation (1), we get

$$p(z > z_\alpha) + p(z < -z_\alpha) = \alpha; \text{ i.e., } 2p(z > z_\alpha) = \alpha; p(z > z_\alpha) = \alpha/2$$

i.e., the area of each tail is $\alpha/2$.



The critical value z_α is that value such that the area to the right of z_α is $\alpha/2$ and the area to the left of $-z_\alpha$ is $\alpha/2$.

In the case of the one-tailed test, $p(z > z_\alpha) = \alpha$ if it is right-tailed; $p(z < -z_\alpha) = \alpha$ if it is left-tailed.

The critical value of z for a single-tailed test (right or left) at level of significance α is same as the critical value of z for two-tailed test at level of significance 2α .

Using the equation, also using the normal tables, the critical value of z at different levels of significance (α) for both single tailed and two tailed test are calculated and listed below. The equations are

$$p(|z| > z_\alpha) = \alpha; p(z > z_\alpha) = \alpha; p(z < -z_\alpha) = \alpha$$

Level of significance			
	1% (0.01)	5% (0.05)	10% (0.1)
Two tailed test	$ z_\alpha = 2.58$	$ z = 1.966$	$ z = 0.645$
Right tailed	$z_\alpha = 2.33$	$z_\alpha = 1.645$	$z_\alpha = 1.28$
Left tailed	$z_\alpha = -2.33$	$z_\alpha = -1.645$	$z_\alpha = -1.28$

8.3.2 Steps in Testing of Statistical Hypothesis

The steps in testing of statistical hypothesis are given below as:

- **Step 1.** Null hypothesis. Set up H_0 in clear terms.
- **Step 2.** Alternative hypothesis. Set up H_1 , so that we could decide whether we should use one tailed test or two tailed test.
- **Step 3.** Level of significance. Select the appropriate level of significance in advance depending on the reliability of the estimates.
- **Step 4.** Test statistic. Compute the test statistic $z = \frac{t - E(t)}{S.E(t)}$ under the null hypothesis.
- **Step 5.** Conclusion. Compare the computed value of z with the critical value z_α at level of significance (α).

If $|z| > z_\alpha$, we reject H_0 and conclude that there is significant difference. If $|z| < z_\alpha$, we accept H_0 and conclude that there is no significant difference.



8.3.3 Non Sampling Errors

It is a general assumption in the sampling theory that the true value of each unit in the population can be obtained and tabulated without any errors. In practice, this assumption may be violated due to several reasons and practical constraints. This results in errors in the observations as well as in the tabulation. Such errors which are due to the factors other than sampling are called non-sampling errors.

The non-sampling errors are unavoidable in census and surveys. The data collected by complete enumeration in census is free from sampling error but would not remain free from non-sampling errors. The data collected through sample surveys can have both – sampling errors as well as non-sampling errors. The non-sampling errors arise because of the factors other than the inductive process of inferring about the population from a sample.

In general, the sampling errors decrease as the sample size increases whereas non-sampling error increases as the sample size increases. In some situations, the non-sampling errors may be large and deserve greater attention than the sampling error.

In any survey, it is assumed that the value of the characteristic to be measured has been defined precisely for every population unit. Such a value exists and is unique. This is called the true value of the characteristic for the population value. In practical applications, data collected on the selected units are called survey values and they differ from the true values. Such difference between the true and observed values is termed as the observational error or response error. Such an error arises mainly from the lack of precision in measurement techniques and variability in the performance of the investigators.

Sources of non-sampling errors:

Non sampling errors can occur at every stage of planning and execution of survey or census. It occurs at planning stage, field work stage as well as at tabulation and computation stage. The main sources of the non-sampling errors are

- lack of proper specification of the domain of study and scope of investigation,
- incomplete coverage of the population or sample,
- faulty definition,
- defective methods of data collection and
- tabulation errors

More specifically, one or more of the following reasons may give rise to non-sampling errors or indicate its presence:

- The data specification may be inadequate and inconsistent with the objectives of the survey or census.

- Due to imprecise definition of the boundaries of area units, incomplete or wrong identification of units, faulty methods of enumeration etc., the data may be duplicated or may be omitted.
- The methods of interview and observation collection may be inaccurate or inappropriate.
- The questionnaire, definitions and instructions may be ambiguous.
- The investigators may be inexperienced or not trained properly.
- The recall errors may pose difficulty in reporting the true data.
- The scrutiny of data is not adequate.
- The coding, tabulation etc. of the data may be erroneous.
- There can be errors in presenting and printing the tabulated results, graphs etc.
- In a sample survey, the non-sampling errors arise due to defective frames and faulty selection of sampling units.

These sources are not exhaustive but surely indicate the possible source of errors. Non-sampling errors may be broadly classified into three categories.

- **Specification errors:** These errors occur at planning stage due to various reasons, e.g., inadequate and inconsistent specification of data with respect to the objectives of surveys/census, omission or duplication of units due to imprecise definitions, faulty method of enumeration/interview/ambiguous schedules etc.
- **Ascertainment errors:** These errors occur at field stage due to various reasons e.g., lack of trained and experienced investigations, recall errors and other type of errors in data collection, lack of adequate inspection and lack of supervision of primary staff etc.
- **Tabulation errors:** These errors occur at tabulation stage due to various reasons, e.g., inadequate scrutiny of data, errors in processing the data, errors in publishing the tabulated results, graphs etc.

Ascertainment errors may be further sub-divided into

- Coverage errors owing to over-enumeration or under-enumeration of the population or the sample, resulting from duplication or omission of units and from the non-response.
- Content errors relating to the wrong entries due to the errors on the part of investigators and respondents.

Same division can be made in the case of tabulation error also. There is a possibility of missing data or repetition of data at tabulation stage which gives rise to coverage errors and also of errors in coding, calculations etc. which gives rise to content errors.



Treatment of non-sampling errors:

Some conceptual background is needed for the mathematical treatment of non-sampling errors.

Total error

Difference between the sample survey estimate and the parametric true value being estimated is termed as total error.

Sampling error

If complete accuracy can be ensured in the procedures such as determination, identification and observation of sample units and the tabulation of collected data, then the total error would consist only of the error due to sampling, termed as sampling error.

Measure of sampling error is mean squared error (MSE). The MSE is the difference between the estimator and the true value and has two components:

- square of sampling bias.
- sampling variance.

If the results are also subjected to the non-sampling errors, then the total error would have both sampling and non-sampling error.

Total bias:

The difference between the expected value and the true value of the estimator is termed as total bias. This consists of sampling bias and non-sampling bias.

Non-sampling bias: For the sake of simplicity, assume that the two following

steps are involved in the randomization:

- for selecting the sample of units and
- for selecting the survey personnel.

Let $\hat{\bar{Y}}_{sr}$ be the estimate of population mean $\bar{Y}$ based on s^{th} sample of units supplied by the r^{th} sample of the survey personnel. The conditional expected value of $\hat{\bar{Y}}_{sr}$ taken over the second step of randomization for a fixed sample of units is

$$E_r(\hat{\bar{Y}}_{sr}) = \hat{\bar{Y}}_{so},$$

which may be different from $\hat{\bar{Y}}_{sr}$ based on true values of the units in the sample. The expected value of $\hat{\bar{Y}}_{so}$ over the first step of randomization gives

$$E_s(\hat{\bar{Y}}_{so}) = \bar{Y}^*,$$

which is the value for which an unbiased estimator can be had by the specified survey process. The value $\bar{Y}^*$ may be different from true population mean $\bar{Y}$ and the total bias is given as

$$\text{Bias}_t(\hat{\bar{Y}}_{sr}) = \bar{Y}^* - \bar{Y}.$$

The sampling bias is given by

$$\text{Bias}(\hat{\bar{Y}}) = E_s(\hat{\bar{Y}}_s) - \bar{Y}.$$

The non-sampling bias is

$$\begin{aligned}
 Bias_r(\hat{\bar{Y}}_{sr}) &= Bias_t(\hat{\bar{Y}}_{sr}) - Bias_s(\hat{\bar{Y}}_s) \\
 &= \bar{Y}^* - E_s(\hat{\bar{Y}}_s) \\
 &= E_s(\hat{\bar{Y}}_{so} - \hat{\bar{Y}}_s)
 \end{aligned}$$

which is the expected value of the non-sampling deviation.

In case of complete enumeration, there is no sampling bias and the total bias consists only of non-sampling bias. In case of sample surveys, the total bias consists only of the non-sampling bias.

The non-sampling bias in a census can be estimated by surveying a sample of units in the population using better techniques of data collection and compilation than those adopted under general census condition. The surveys are called post-enumeration surveys, which are usually conducted just after the census for studying the quality of census data, may be used for this purpose.

In a large scale sample survey, the ascertainment bias can be estimated by resurveying a sub-sample of the original sample using better survey techniques.

Another method of checking survey data is to compare the values of the units obtained in the two surveys and to reconcile the discrepant figures by further investigation. This method of checking is termed reconciliation (check) surveys.

Non-sampling variance:

The MSE of $\hat{\bar{Y}}_{sr}$ based on s^{th} sample of units and supplied by r^{th} sample of the survey personnel is

$$MSE(\hat{\bar{Y}}_{sr}) = E_{sr}(\hat{\bar{Y}}_{sr} - \bar{Y})^2$$

where $\bar{Y}$ is the true value being estimated. This takes into account both the sampling and the non-sampling errors, i.e.,

$$\begin{aligned}
 MSE(\hat{\bar{Y}}_{sr}) &= Var(\hat{\bar{Y}}_{sr}) + [Bias(\hat{\bar{Y}}_{sr})]^2 \\
 &= E(\hat{\bar{Y}}_{sr} - \bar{Y}^*)^2 + (\bar{Y}^* - \bar{Y})^2
 \end{aligned}$$

where $\bar{Y}^*$ is the expected value of the estimator taken over both steps of randomization.

Taking the variance over the two steps of randomization, we get

$$\begin{aligned}
 Var_{sr}(\hat{\bar{Y}}_{sr}) &= Var_s \left[E_r(\hat{\bar{Y}}_{sr}) \right] + E_s \left[Var_r(\hat{\bar{Y}}_{sr}) \right] \\
 &= Var_s \left[\hat{\bar{Y}}_{so} \right] + E_s \left[E_r(\hat{\bar{Y}}_{sr} - \hat{\bar{Y}}_{so})^2 \right]
 \end{aligned}$$

$\downarrow$

sampling
variance

$\downarrow$

non-sampling
variance

Note that

$$\hat{\bar{Y}}_{sr} - \hat{\bar{Y}}_{so} = (\hat{\bar{Y}}_{sr} - \hat{\bar{Y}}_{so} - \hat{\bar{Y}}_{or} + \bar{Y}^*) + (\hat{\bar{Y}}_{or} - \hat{\bar{Y}}^*)$$

Where $\hat{\bar{Y}}_{or} = E_s(\hat{\bar{Y}}_{sr})$.

$$E(\hat{\bar{Y}}_{sr} - \hat{\bar{Y}}_{so})^2 = E_{sr}(\hat{\bar{Y}}_{sr} - \hat{\bar{Y}}_{so} - \hat{\bar{Y}}_{or} + \bar{Y}^*)^2 + E_r(\hat{\bar{Y}}_{or} - \hat{\bar{Y}}^*)^2$$

$\downarrow$

Interaction between
sampling and
non-sampling errors

$\downarrow$

Variance
between
survey personnel

The MSE of an estimator consists of,

- sampling variance,
- interaction between the sampling and the non-sampling errors,
- variance between survey personnel and
- square of the sum of sampling and non-sampling biases

In complete census, the MSE is composed of only the non-sampling variance and square of the non-sampling bias.

Non-response error:

The non-response error may occur due to refusal by respondents to give information or the sampling units may be inaccessible. This error arises because the set of units getting excluded may have characteristic so different from the set of units actually surveyed as to make the results biased. This error is termed as non-response error since it arises from the exclusion of some of the anticipated units in the sample or population. One way of dealing with the problem of non-response is to make all the efforts to collect information from a sub-sample of the units not responding in the first attempt.

Measurement and control of errors:

Some suitable methods and adequate procedures for control can be adopted before initiating the main census or sample survey. Some separate programs for estimating the different types of non-sampling errors are also required. Some such procedures are as follows:

- Consistency checks:
Certain items in the questionnaires can be added which may serve as a check on the quality of collected data. To locate the doubtful observations, the data can be arranged in increasing order of some basic variable. Then they can be plotted against each sample unit. Such graph is expected to follow a certain pattern and any deviation from this pattern would help in spotting the discrepant values.
- Sample check:
An independent duplicate census or sample survey can be conducted on a comparatively smaller group by trained and experienced staff. If the sample is properly designed and if the checking operation is efficiently carried out, then it is possible to detect the presence of non-sampling errors and to get an idea of their magnitude. Such procedure is termed as method of sample check.
- Post-census and post-survey checks:

It is a type of sample check in which a sample (or subsample) is selected of the units covered in the census (or survey) and re-enumerate or re-survey it by using better trained and more experienced survey staff than those involved in the main investigation. This procedure is called as post-survey check or post-census. The effectiveness of such check surveys can be increased by

- re-enumerating or re-surveying immediately after the main census to avoid recall error
 - Taking steps to minimize the conditioning effect that the main survey may have on the work of the check-survey
- External record check:
Take a sample of relevant units from a different source, if available, and to check whether all the units have been enumerated in the main investigation and whether there are discrepancies between the values when matched. The list from which the check-sample is drawn for this purpose, need not be a complete one.
 - Quality control techniques:
The use of tools of statistical quality control like control chart and acceptance sampling techniques can be used in assessing the

quality of data and in improving the reliability of final results in large scale surveys and census.

- Study or recall error:
Response errors arise due to various factors like the attitude of respondents towards the survey, method of interview, skill of the investigators and recall errors. Recall error depends on the length of the reporting period and on the interval between the reporting period and data of survey. One way of studying recall error is to collect and analyze data related to more than one reporting period in a sample (or sub-sample) of units covered in the census or survey.
- Interpenetrating sub-samples:
The use of interpenetrating sub-sample technique helps in providing an appraisal of the quality of information as the interpenetrating sub-samples can be used to secure information on non-sampling errors such as differences arising from differential interviewer bias, different methods of eliciting information etc. After the sub-samples have been surveyed by different groups of investigators and processed by different team of workers at the tabulation stage, a comparison of the final estimates based on the sub-samples provides a broad check on the quality of the survey results.



8.4 TEST OF SIGNIFICANCE FOR LARGE SAMPLES

 If the sample size $n > 30$, the sample is taken as large sample. For such sample we apply normal test, as Binomial, Poisson, chi square, etc. are closely approximated by normal distributions assuming the population as normal.

Under large sample test, the following are the important tests to test the significance:

- Testing of significance for single proportion.
- Testing of significance for difference of proportions.
- Testing of significance for single mean.
- Testing of significance for difference of means.
- Testing of significance for difference of standard deviations.

LEARNING OBJECTIVE

 4. Describe the Test of Significance for Large Samples

8.4.1 Testing of Significance for Single Proportion

This test is used to find the significant difference between proportion of the sample and the population. Let X be the number of successes in n independent trials with constant probability P of success for each trial.

$E(X) = nP$; $V(X) = nPQ$; $Q = 1 - P = \text{Probability of failure}$.

Let

$p = X/n$ called the observed proportion of success.

$$E(p) = E(X/n) = \frac{1}{n} E(x) = \frac{np}{n} = p; E(p) = p$$

$$V(p) = V(X/n) = \frac{1}{n^2} V(X) = \frac{1(PQ)}{n} = PQ/n$$

$$S.E.(p) = \sqrt{\frac{PQ}{n}}; z = \frac{p - E(p)}{S.E.(p)} = \frac{p - p}{\sqrt{PQ/n}} \sim N(0, 1)$$

This z is called test statistic which is used to test the significant difference of sample and population proportion.

1. The probable limit for the observed proportion of successes are $p \pm z_\alpha \sqrt{PQ/n}$, where z_α is the significant value at level of significance α .
2. If p is not known, the limits for the proportion in the population are $p \pm z_\alpha \sqrt{pq/n}$, $q = 1 - p$
3. If α is not given, we can take safely 3σ limits.

Hence, the confidence limits for observed proportion p are $p \pm \sqrt{\frac{PQ}{n}}$

The confidence limits for the population proportion p are $p \pm \sqrt{\frac{pq}{n}}$.

Example. A coin was tossed 400 times and the head turned up 216 times. Test the hypothesis that the coin is unbiased.

Solution. H_0 : The coin is unbiased i.e., $P = 0.5$.

H_1 : The coin is not unbiased (biased); $P \neq 0.5$

Here $n = 400$; X = Number of success = 216

p = proportion of success in the sample
 $\frac{X}{n} = \frac{216}{400} = 0.54$

Population proportion = $0.5 = P$; $Q = 1 - P = 1 - 0.5 = 0.5$.

Under H_0 , test statistic $z = \frac{p - P}{\sqrt{PQ/n}}$

$$|z| = \left| \frac{0.54 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{400}}} \right| = 1.6$$

we use a two-tailed test.

Since $|z| = 1.6 < 1.96$

i.e., $|z| < z_\alpha$, z_α is the significant value of z at 5% level of significance. i.e., the coin is unbiased is $P = 0.5$.

8.4.2 Testing of Significance for Difference of Proportions

Consider two samples X_1 and X_2 of sizes n_1 and n_2 , respectively, taken from two different populations. Test the significance of the difference between the sample proportion p_1 and p_2 . The test statistic under the null hypothesis H_0 , that there is no significant difference between the two sample proportions, we have

$$z = \frac{p_1 - p_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}, \text{ where } P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

And

$$Q = 1 - P.$$

Example

Before an increase in excise duty on tea, 800 people out of a sample of 1000 persons were found to be tea drinkers. After an increase in the duty, 800 persons were known to be tea drinkers in a sample of 1200 people. Do you think that there has been a significant decrease in the consumption of tea after the increase in the excise duty?

Solution. Here

$$n_1 = 800, n_2 = 1200$$

$$p_1 = \frac{X_1}{n_1} = \frac{800}{1000} = \frac{4}{5}; p_2 = \frac{X_2}{n_2} = \frac{800}{1200} = \frac{2}{3}$$

$$P = \frac{p_1 n_1 + p_2 n_2}{n_1 + n_2} = \frac{X_1 + X_2}{n_1 + n_2} = \frac{800 + 800}{1000 + 1200} = \frac{8}{11}; Q = \frac{3}{11}$$

Null hypothesis H_0 : $p_1 = p_2$, i.e., there is no significant difference in the consumption of tea before and after increase of excise duty.

$$H_1: p_1 > p_2 \text{ (right-tailed test)}$$

The test statistic

$$z = \frac{p_1 - p_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.8 - 0.6666}{\sqrt{\frac{8}{11} \times \frac{3}{11} \left(\frac{1}{1000} + \frac{1}{1200} \right)}} = 6.842.$$

Since the calculated value of $|z| > 1.645$ also $|z| > 2.33$, both the significant values of z at 5% and 1% level of significance. Hence H_0 is rejected, i.e., there is a significant decrease in the consumption of tea due to increase in excise duty.

8.4.3 Testing of Significance for Single Mean

To test whether the difference between sample mean and population mean is significant or not.

Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a large population $X_1, X_2, \dots, X_N$ of size N with mean μ and variance σ^2 .

$\therefore$ the standard error of mean of a random sample of size n from a population with variance σ^2 is $\sigma/\sqrt{n}$.

To test whether the given sample of size n has been drawn from a population with mean μ , i.e. to test whether the difference between the sample mean and the population mean is significant. Under the null hypothesis, there is no difference between the sample mean and population mean.

The test statistic is $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$, where σ is the standard deviation of the population.

If σ is not known, we use the test statistic $z = \frac{\bar{X} - \mu}{s/\sqrt{n}}$, where s is the standard deviation of the sample.

Example. A normal population has a mean of 6.8 and standard deviation of 1.5. A sample of 400 members gave a mean of 6.75. Is the difference significant?

Solution. H_0 : There is no significant difference between $\bar{x}$ and μ .

H_1 : There is significant difference between $\bar{x}$ and μ .

Given $\mu = 6.8$, $\sigma = 1.5$, $\bar{x} = 6.75$ and $n = 400$

$$|z| = \left| \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \right| = \left| \frac{6.75 - 6.8}{1.5/\sqrt{400}} \right| = |-0.67| = 0.67$$

As the calculated value of $|z| < z_{\alpha} = 1.96$ at 5% level of significance, H_0 is accepted, i.e., there is no significant difference between $\bar{x}$ and μ .

8.4.4 Test of Significance for Difference of Means of Two Large Samples

Let $\bar{x}_1$ be the mean of a sample of size n_1 from a population with mean μ_1 , and variance σ_1^2 . Let $\bar{x}_2$ be the mean of an independent sample of size n_2 from another population with mean μ_2 and variance σ_2^2 . The test statistic is given by

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Under the null hypothesis that the samples are drawn from the same population where $\sigma_1 = \sigma_2 = \sigma$, i.e., $\mu_1 = \mu_2$ the test statistic is given by

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

1. If σ_1 , σ_2 are not known and $\sigma_1 \neq \sigma_2$ the test statistic in this case is

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$



2. If σ is not known and $\sigma_1 = \sigma_2$. We use $\sigma^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}$ to calculate σ ;

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Example. The average income of persons was 210 with a standard deviation of 10 in a sample of 100 people. For another sample of 150 people, the average income was 220 with a standard deviation of 12. The standard deviation of incomes of the people of the city was 11. Test whether there is any significant difference between the average incomes of the localities.

Solution. Here $n_1 = 100$, $n_2 = 150$, $x_1 = 210$, $x_2 = 220$, $s_1 = 10$, $s_2 = 12$.

Null hypothesis. The difference is not significant, i.e., there is no difference between the incomes of the localities.

$$H_0: \bar{x}_1 = \bar{x}_2, \quad H_1: \bar{x}_1 \neq \bar{x}_2$$

Under H_0 ,

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{210 - 220}{\sqrt{\frac{10^2}{100} + \frac{12^2}{150}}} = -7.1428 \quad \therefore |z| = 7.1428.$$

As the calculated value of $|z| > 1.96$, the significant value of z at 5% level of significance, H_0 is rejected i.e., there is significant difference between the average incomes of the localities.

8.4.5 Test of Significance for the Difference of Standard Deviations

If s_1 and s_2 are the standard deviations of two independent samples, then under the null hypothesis $H_0: \sigma_1 = \sigma_2$, i.e., the



REMEMBER

A sample is a smaller group of members of a population selected to represent the population. In order to use statistics to learn things about the population, the sample must be random.

$$z = \frac{s_1 - s_2}{\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}},$$

sample standard deviations do not differ significantly, the statistic where σ_1 and σ_2 are population standard deviations.

When population standard deviations are not known, then

$$z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}}$$

Example. Random samples drawn from two countries gave the following data relating to the heights of adult males:

	Country A	Country B
Mean height (in inches)	67.42	67.25
Standard deviation	2.58	2.50
Number in samples	1000	1200

- Is the difference between the means significant?
- Is the difference between the standard deviations significant?

Solution. Given: $n_1 = 1000$, $n_2 = 1200$, $x_1 = 67.42$; $x_2 = 67.25$, $s_1 = 2.58$, $s_2 = 2.50$

Since the samples size are large we can take $\sigma_1 = s_1 = 2.58$; $\sigma_2 = s_2 = 2.50$.

(i) Null hypothesis: $H_0 = \mu_1 = \mu_2$ i.e., sample means do not differ significantly.

Alternative hypothesis: $H_1: \mu_1 \neq \mu_2$ (two tailed test)

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{67.42 - 67.25}{\sqrt{\frac{(2.58)^2}{1000} + \frac{(2.50)^2}{1200}}} = 1.56.$$

Since $|z| < 1.96$ we accept the null hypothesis at 5% level of significance.

(ii) We set up the null hypothesis.

$H_0: \sigma_1 = \sigma_2$ i.e., the sample standard deviations do not differ significantly. Alternative hypothesis: $H_1 = \sigma_1 \neq \sigma_2$ (two tailed)

$\therefore$ The test statistic is given by

$$z = \frac{s_1 - s_2}{\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}} = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}}$$

($\because \sigma_1 = s_1, \sigma_2 = s_2$ for large samples)

$$= \frac{2.58 - 2.50}{\sqrt{\frac{(2.58)^2}{2 \times 1000} + \frac{(2.50)^2}{2 \times 1200}}} = \frac{0.08}{\sqrt{\frac{6.6564}{2000} + \frac{6.25}{2400}}} = 1.0387$$

Since $|z| < 1.96$ we accept the null hypothesis at 5% level of significance.



REMEMBER

A t-test is most commonly applied when the test statistic would follow a normal distribution if the value of a scaling term in the test statistic were known.

8.5 STATISTICAL SIGNIFICANCE (T-TEST)

5 A t-test's statistical significance indicates whether or not the difference between two groups' averages most likely reflects a "real" difference in the population from which the groups were sampled. The t-test assesses whether the means of two groups are *statistically* different from each other. This analysis is appropriate whenever you want to compare the means of two groups, and especially appropriate as the analysis for the posttest-only two-group randomized experimental design.

LEARNING OBJECTIVE

5. Define the Statistical Significance (T-Test)

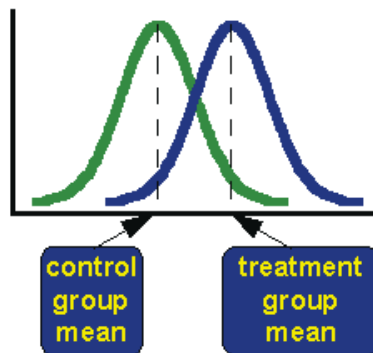


Figure 8.1: Idealized distributions for treated and comparison group posttest values.

Figure 8.1 shows the distributions for the treated (blue) and control (green) groups in a study. Actually, the figure shows the idealized distribution -- the actual distribution would usually be depicted with a histogram or bar graph. The figure indicates where the control and treatment group means are located. The question the t-test addresses is whether the means are statistically different.

What does it mean to say that the averages for two groups are statistically different? Consider the three situations shown in Figure 8.2. The first thing to notice about the three situations is that the difference between the means is the same in all three. But, you should also notice that the three situations do not look the same -- they tell very different stories. The top example shows a case with moderate variability of scores within each group. The second situation shows the high variability case. The third shows the case with low variability. Clearly, we would conclude that the two groups appear most different or distinct in the bottom or low-variability case. Why? Because there is relatively little overlap between the two bell-shaped curves. In the high variability case, the group difference appears least striking because the two bell-shaped distributions overlap so much.

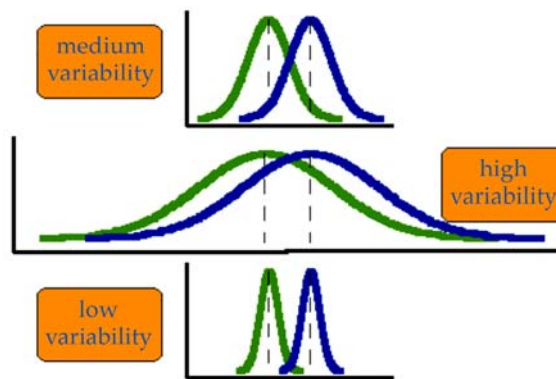


Figure 8.2: Three scenarios for differences between means.

This leads us to a very important conclusion: when we are looking at the differences between scores for two groups, we have to judge the difference between their means relative to the spread or variability of their scores. The t-test does just this.

8.5.1 Statistical Analysis of the t-test

The formula for the t-test is a ratio. The top part of the ratio is just the difference between the two means or averages. The bottom part is a measure of the variability or dispersion of the scores. This formula is essentially another example of the signal-

to-noise metaphor in research: the difference between the means is the signal that, in this case, we think our program or treatment introduced into the data; the bottom part of the formula is a measure of variability that is essentially noise that may make it harder to see the group difference. Figure 8.3 shows the formula for the t-test and how the numerator and denominator are related to the distributions.

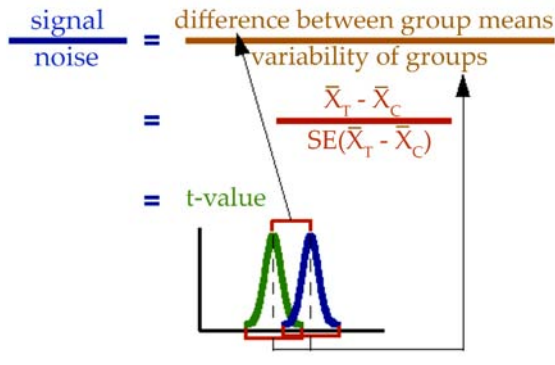


Figure 8.3: Formula for the t-test.

The top part of the formula is easy to compute -- just find the difference between the means. The bottom part is called the standard error of the difference. To compute it, we take the variance for each group and divide it by the number of people in that group. We add these two values and then take their square root. The specific formula is given below:

$$SE(\bar{X}_T - \bar{X}_C) = \sqrt{\frac{\text{var}_T}{n_T} + \frac{\text{var}_C}{n_C}}$$

Remember, that the variance is simply the square of the standard deviation.

The final formula for the t-test is shown in below:

$$t = \frac{\bar{X}_T - \bar{X}_C}{\sqrt{\frac{\text{var}_T}{n_T} + \frac{\text{var}_C}{n_C}}}$$

The t-value will be positive if the first mean is larger than the

KEY TERMS

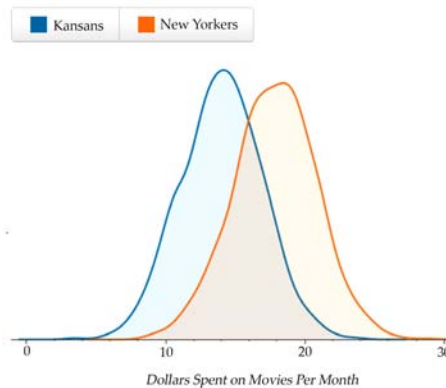
Generalized least squares (GLS) is a technique for estimating the unknown parameters in a linear regression model.

second and negative if it is smaller. Once you compute the t-value you have to look it up in a table of significance to test whether the ratio is large enough to say that the difference between the groups is not likely to have been a chance finding. To test the significance, you need to set a risk level (called the alpha level). In most social research, the “rule of thumb” is to set the alpha level at .05. This means that five times out of a hundred you would find a statistically significant difference between the means even if there was none (i.e., by “chance”). You also need to determine the degrees of freedom (df) for the test. In the t-test, the degrees of freedom is the sum of the persons in both groups minus 2. Given the alpha level, the df, and the t-value, you can look the t-value up in a standard table of significance (available as an appendix in the back of most statistics texts) to determine whether the t-value is large enough to be significant. If it is, you can conclude that the difference between the means for the two groups is different (even given the variability). Fortunately, statistical computer programs routinely print the significance test results and save you the trouble of looking them up in a table.

The t-test, one-way Analysis of Variance (ANOVA) and a form of regression analysis are mathematically equivalent and would yield identical results.

Example

Let’s say you are interested in whether the average New Yorker spends more than the average Kansan per month on movies.



You ask a sample of 3 people from each state about their movie spending. You might observe a difference in those averages (like \$14 for the average Kansan and \$18 for the average New Yorker). But that difference is not *statistically significant*; it could easily just be random luck of which 3 people you randomly sampled that makes one group appear to spend more money than the other. If instead you ask 300 New Yorkers and 300 Kansans and still see a big difference, that difference is less likely to be caused by the sample being unrepresentative.



Note that if you asked 300,000 New Yorkers and 300,000 Kansans, the result would likely be statistically significant even if the difference between the groups was only a penny. The t-test's effect size complements its statistical significance, describing the magnitude of the difference, whether or not the difference is statistically significant.

8.5.2 Student's t-Distribution

This t-distribution is used when sample size is ≤ 30 and the population standard deviation is unknown.

t-statistic is defined as $t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t(n - 1 \text{ d.f.})$ d.f. = degrees of freedom

Where

$$s = \sqrt{\frac{\sum (X - \bar{X})^2}{n - 1}}$$

The t-Table

The t-table given at the end is the probability integral of **t-distribution**. The t-distribution has different values for each degrees of freedom and when the degrees of freedom are infinitely large, the t-distribution is equivalent to normal distribution and the probabilities shown in the normal distribution tables are applicable.

Applications of t-Distribution

Some of the applications of t-distribution are given below:

- To test if the sample mean ($\bar{X}$) differs significantly from the hypothetical value μ of the population mean.
- To test the significance between two sample means.
- To test the significance of observed partial and multiple correlation coefficients.

KEY TERMS

t-Distribution is a theoretical probability distribution. It is symmetrical, bell-shaped, and similar to the standard normal curve.

Critical Value of t

The critical value or significant value of t at level of significance α degrees of freedom γ for two tailed test is given by

$$P[|t| > t_{\gamma}(\alpha)] = \alpha$$

$$P[|t| \leq t_{\gamma}(\alpha)] = 1 - \alpha$$

The significant value of t at level of significance α for a single tailed test can be determined from those of two-tailed test by referring to the values at 2α .

8.5.3 Test I: t-Test of Significance of the Mean of a Random Sample

To test whether the mean of a sample drawn from a normal population deviates significantly from a stated value when variance of the population is unknown.

H_0 : There is no significant difference between the sample mean $\bar{x}$ and the population mean μ , i.e., we use the statistic

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}} \quad \text{where } \bar{X} \text{ is mean of the sample.}$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \text{ with degrees of freedom } (n-1).$$

At a given level of significance α_1 and degrees of freedom $(n-1)$. We refer to t -table t_{α} (two-tailed or one-tailed). If calculated t value is such that $|t| < t_{\alpha}$ the null hypothesis is accepted. $|t| > t_{\alpha}$, H_0 is rejected.

Fiducial Limits of Population Mean

If t_{α} is the table of t at level of significance α at $(n-1)$ degrees of freedom.

$$\left| \frac{\bar{X} - \mu}{s/\sqrt{n}} \right| < t_{\alpha} \text{ for acceptance of } H_0.$$

$$\bar{x} - t_{\alpha} s/\sqrt{n} < \mu < \bar{x} + t_{\alpha} s/\sqrt{n}$$

95% confidence limits (level of significance 5%) are $\bar{X} \pm t_{0.05} s/\sqrt{n}$

99% confidence limits (level of significance 1%) are $\bar{X} \pm t_{0.01} s/\sqrt{n}$

Instead of calculating s , we calculate S for the sample.

Since

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$\therefore S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \quad | \quad \because (n-1)s^2 = nS^2$$

Example. A random sample of size 16 has 53 as mean. The sum of squares of the derivation from mean is 135. Can this sample be regarded as taken from the population having 56 as mean? Obtain 95% and 99% confidence limits of the mean of the population.

Solution. H_0 : There is no significant difference between the sample mean and hypothetical population mean.

$$H_0: \mu = 56; \quad H_1: \mu \neq 56 \quad (\text{Two-tailed test})$$

$$t: \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t(n-1 \text{ difference})$$

Given:

$$\bar{X} = 53, \mu = 56, n = 16, \Sigma(X - \bar{X})^2 = 135$$

$$s = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n-1}} = \sqrt{\frac{135}{15}} = 3; t = \frac{53 - 56}{3/\sqrt{16}} = -4$$

$$|t| = 4 \text{ . d.f.v. } = 16 - 1 = 15.$$

$$t_{0.05} = 1.753.$$

Since $|t| = 4 > t_{0.05} = 1.753$, the calculated value of t is more than the table value. The hypothesis is rejected. Hence, the sample mean has not come from a population having 56 as mean.

95% confidence limits of the population mean

$$= \bar{X} \pm \frac{s}{\sqrt{n}} t_{0.05} = 53 \pm \frac{3}{\sqrt{16}} (1.725) = 51.706; 54.293$$

99% confidence limits of the population mean

$$= \bar{X} \pm \frac{s}{\sqrt{n}} t_{0.01}, = 53 \pm \frac{3}{\sqrt{16}} (2.602) = 51.048; 54.951.$$

8.5.4 Test II: t-Test for Difference of Means of Two Small Samples (From a Normal Population)

This test is used to test whether the two samples $x_1, x_2, \dots, x_{n_1}, y_1, y_2, \dots, y_{n_2}$ of sizes n_1, n_2 have been drawn from two normal populations with mean μ_1 and μ_2 respectively, under the assumption that the population **variance** are equal. ($\sigma_1 = \sigma_2 = \sigma$).

H_0 : The samples have been drawn from the normal population with means μ_1 and μ_2 , i.e., $H_0: \mu_1 = \mu_2$.

Let $\bar{X}, \bar{Y}$ be their means of the two samples.

Under this H_0 the test of statistic t is given by

$$t = \frac{(\bar{X} - \bar{Y})}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2 \text{ difference})$$

1. If the two sample's standard deviations s_1, s_2 are given, then we have

$$s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}.$$

KEY TERMS

Variance is the expectation of the squared deviation of a random variable from its mean, and it informally measures how far a set of (random) numbers are spread out from their mean.

$$t = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{s_1^2 + s_2^2}{n-1}}}$$

2. If $n_1 = n_2 = n$, $\frac{\bar{X} - \bar{Y}}{\sqrt{\frac{s_1^2 + s_2^2}{n-1}}}$ can be used as a test statistic.
3. If the pairs of values are in some way associated (correlated) we cannot use the test statistic as given in Note 2. In this case, we find the differences of the associated pairs of values and apply for single mean

$$\text{i.e., } t = \frac{\bar{X} - \mu}{s/\sqrt{n}} \text{ with degrees of freedom } n - 1.$$

The test statistic is

$$t = \frac{\bar{d}}{s/\sqrt{n}}$$

Or

$$t = \frac{\bar{d}}{s/\sqrt{n-1}}, \text{ where } \bar{d} \text{ is the mean of paired difference.}$$

$$\text{i.e., } d_i = x_i - y_i$$

$$\bar{d}_i = \bar{X} - \bar{Y}, \text{ where } (x_i, y_i) \text{ are the paired data } i = 1, 2, \dots, n.$$



REMEMBER

If using Student's original definition of the t-test, the two populations being compared should have the same variance (testable using F-test, Levene's test, Bartlett's test, or the Brown-Forsythe test; or assessable graphically using a Q-Q plot).

8.5.5 Snedecor's Variance Ratio Test or F-Test

In testing the significance of the difference of two means of two samples, we assumed that the two samples came from the same population or populations with equal variance. The object of the F-test is to discover whether two independent estimates of population variance differ significantly or whether the two samples may be regarded as drawn from the normal populations having the same variance. Hence before applying the t-test for the significance of the difference of two means, we have to test for the equality of population variance by using the F-test.

Let n_1 and n_2 be the sizes of two samples with variance s_1^2 and s_2^2 . The estimate of the population variance based on these samples is $s_1^2 = \frac{n_1 s_1^2}{n_1 - 1}$ and $s_2^2 = \frac{n_2 s_2^2}{n_2 - 1}$. The degrees of freedom of these estimates are $v_1 = n_1 - 1$, $v_2 = n_2 - 1$.

To test whether these estimates, s_1^2 and s_2^2 , are significantly different or if the samples may be regarded as drawn from the same population or from two populations with same variance σ^2 , we set-up the null hypothesis

$$H_0: \sigma_1^2 = \sigma_2^2 = \sigma^2,$$

i.e., the independent estimates of the common population do not differ significantly.

To carry out the test of significance of the difference of the variances we calculate the test statistic $F = \frac{s_1^2}{s_2^2}$, the Numerator is greater than the Denominator, i.e., $s_1^2 > s_2^2$.

If the calculated value of F exceeds $F_{0.05}$ for $(n_1 - 1)$, $(n_2 - 1)$ degrees of freedom given in the table, we conclude that the ratio is significant at 5% level.

We conclude that the sample could have come from two normal population with same variance.

The assumptions on which the F-test is based are:

- The populations for each sample must be normally distributed.
- The samples must be random and independent.
- The ratio of σ_1^2 to σ_2^2 should be equal to 1 or greater than 1. That is why we take the larger variance in the Numerator of the ratio.

Applications.

F-test is used to test

- Whether two independent samples have been drawn from the normal populations with the same variance σ^2 .
- Whether the two independent estimates of the population variance are homogeneous or not.

Example. Two random samples drawn from 2 normal populations are as follows:

A	17	27	18	25	27	29	13	17
B	16	16	20	27	26	25	21	

Test whether the samples are drawn from the same normal population.



Solution. To test if two independent samples have been drawn from the same population we have to test (i) equality of the means by applying the t-test and (ii) equality of population variance by applying F-test.

Since the t-test assumes that the sample variances are equal, we shall first apply the F-test.

F-test. 1. Null hypothesis $H_0: \sigma_1^2 = \sigma_2^2$ i.e., the population variance do not differ significantly.

Alternative hypothesis. $H_1: \sigma_1^2 \neq \sigma_2^2$

$$\text{Test statistic: } F = \frac{s_1^2}{s_2^2}, \text{ (if } s_1^2 > s_2^2 \text{)}$$

Computations for s_1^2 and s_2^2

X_1	$X_1 - \bar{X}_1$	$(X_1 - \bar{X}_1)^2$	X_2	$X_2 - \bar{X}_2$	$(X_2 - \bar{X}_2)^2$
17	- 4.625	21.39	16	- 2.714	7.365
27	5.735	28.89	16	- 2.714	7.365
18	- 3.625	13.14	20	1.286	1.653
25	3.375	11.39	27	8.286	68.657
27	5.735	28.89	26	7.286	53.085
29	7.735	54.39	25	6.286	39.513
13	- 8.625	74.39	21	2.286	5.226
17	- 4.625	21.39			

$$\bar{X}_1 = 21.625; n_1 = 8; \Sigma(\bar{X}_1 - \bar{X}_1)^2 = 253.87$$

$$\bar{X}_2 = 18.714; n_2 = 7; \Sigma(X_2 - \bar{X}_2)^2 = 182.859$$

$$s_1^2 = \frac{\Sigma(X_1 - \bar{X}_1)^2}{n_1 - 1} = \frac{253.87}{7} = 36.267;$$

$$s_2^2 = \frac{\Sigma(X_2 - \bar{X}_2)^2}{n_2 - 1} = \frac{182.859}{6} = 30.47$$

$$F = \frac{s_1^2}{s_2^2} = \frac{36.267}{30.47} = 1.190.$$

The table value of F for $v_1 = 7$ and $v_2 = 6$ degrees of freedom at 5% level is 4.21. The calculated value of F is less than the tabulated value of F. $\therefore H_0$ is accepted. Hence we conclude that the **variability** in two populations is same.

t-test: Null hypothesis. $H_0: \mu_1 = \mu_2$ i.e., the population means are equal.

Alternative hypothesis. $H_1: \mu_1 \neq \mu_2$

Test of statistic

$$s^2 = \frac{\Sigma(X_1 - \bar{X}_1)^2 + \Sigma(X_2 - \bar{X}_2)^2}{n_1 + n_2 - 2} = \frac{253.87 + 182.859}{8 + 7 - 2} = 33.594$$

$$\therefore s = 5.796$$

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{21.625 - 18.714}{5.796 \sqrt{\frac{1}{8} + \frac{1}{7}}} = 0.9704 \sim t(n_1 + n_2 - 2) \text{ difference}$$

The tabulated value of t at 5% level of significance for 13 difference is 2.16.

The calculated value of t is less than the tabulated value. H_0 is accepted, i.e., there is no significant difference between the population mean. i.e., $\mu_1 = \mu_2$.

$\therefore$ We conclude that the two samples have been drawn from the same normal population.

KEY TERMS

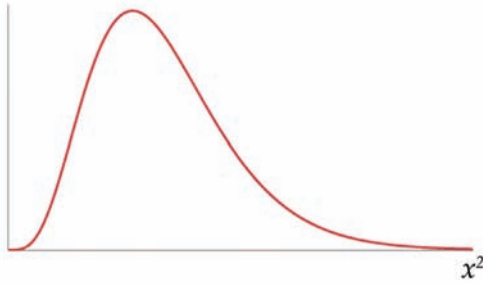
Variability is the extent to which data points in a statistical distribution or data set diverge from the average, or mean, value as well as the extent to which these data points differ from each other.

8.6 CHI-SQUARE (χ^2) TEST

LEARNING OBJECTIVE

6. Understand Chi-Square (χ^2) Test

6. Chi-square is a statistical test commonly used to compare observed data with data we would expect to obtain according to a specific hypothesis. For example, if, according to Mendel's laws, you expected 10 of 20 offspring from a cross to be male and the actual observed number was 8 males, then you might want to know about the "goodness to fit" between the observed and expected. Were the deviations (differences between observed and expected) the result of chance, or were they due to other factors. How much deviation can occur before you, the investigator, must conclude that something other than chance is at work, causing the observed to differ from the expected? The chi-square test is always testing what scientists call the null hypothesis, which states that there is no significant difference between the expected and observed result.



When a coin is tossed 200 times, the theoretical considerations lead us to expect 100 heads and 100 tails. But in practice, these results are rarely achieved. The quantity χ^2 (a Greek letter, pronounced as chi-square) describes the magnitude of discrepancy between theory and observation. If $\chi = 0$, the observed and expected frequencies completely coincide. The greater the discrepancy between the observed and expected frequencies, the greater is the value of χ^2 . Thus χ^2 affords a measure of the correspondence between theory and observation.

If O_i ($i = 1, 2, \dots, n$) is a set of observed (experimental) frequencies and E_i ($i = 1, 2, \dots, n$) is the corresponding set of expected (theoretical or hypothetical) frequencies, then, χ^2 is defined as

$$\chi^2 = \sum_{i=1}^n \left[\frac{(O_i - E_i)^2}{E_i} \right]$$

where $\sum O_i = \sum E_i = N$ (total frequency) and **degrees of freedom** (difference) = $(n - 1)$.

- If $\chi^2 = 0$, the observed and theoretical frequencies agree exactly.
- If $\chi^2 > 0$ they do not agree exactly.

KEY TERMS

Degrees of freedom are the number of values in a study that have the freedom to vary. They are commonly discussed in relationship to various forms of hypothesis testing in statistics, such as a chi-square.

Degrees of Freedom

While comparing the calculated value of χ^2 with the table value, we have to determine the degrees of freedom.

If we have to choose any four numbers whose sum is 50, we can exercise our independent choice for any three numbers only, the fourth being 50 minus the total of the three

numbers selected. Thus, though we were to choose any four numbers, our choice was reduced to three because of one condition imposed. There was only one restraint on our freedom and our degrees of freedom were $4 - 1 = 3$. If two restrictions are imposed, our freedom to choose will be further curtailed and degrees of freedom will be $4 - 2 = 2$.

In general, the number of degrees of freedom is the total number of observations less the number of independent constraints imposed on the observations. Degrees of freedom (difference) are usually denoted by ν (the letter 'nu' of the Greek alphabet).

Thus, $\nu = n - k$, where k is the number of independent constraints in a set of data of n observations.

Applications

χ^2 test is one of the simplest and the most general test known. It is applicable to a very large number of problems in practice which can be summed up under the following heads:

- as a test of goodness of fit.
- as a test of independence of attributes.
- as a test of homogeneity of independent estimates of the population variance.
- as a test of the hypothetical value of the population variance s^2 .
- as a list to the homogeneity of independent estimates of the population correlation coefficient.

Conditions for Applying χ^2 Test

Following are the conditions which should be satisfied before χ^2 test can be applied:

- (a) N , the total number of frequencies should be large. It is difficult to say what constitutes largeness, but as an arbitrary figure, we may say that N should be at least 50, however small the number of cells.
- (b) No theoretical cell-frequency should be small. Here again, it is difficult to say what constitutes smallness, but 5 should be regarded as the very minimum and 10 is better. If small theoretical frequencies occur (i.e., < 10), the difficulty is overcome by grouping two or more classes together before calculating $(O - E)$. It is important to remember that the number of degrees of freedom is determined with the number of classes after regrouping.
- (c) The constraints on the cell frequencies, if any, should be linear.

8.6.1 The χ^2 Distribution

For large sample sizes, the sampling distribution of χ^2 can be closely approximated by a continuous curve known as the chi-square distribution. The probability function of χ^2 distribution is given by

$$f(\chi^2) = c(\chi^2)^{(\nu/2-1)} e^{-\chi^2/2}$$

where $e = 2.71828$, ν = number of degrees of

freedom; c = a constant depending only on v .

Symbolically, the degrees of freedom are denoted by the symbol v or by difference and are obtained by the rule $v = n - k$, where k refers to the number of independent constraints.

In general, when we fit a binomial distribution the number of degrees of freedom is one less than the number of classes; when we fit a Poisson distribution the degrees of freedom are 2 less than the number of classes, because we use the total frequency and the arithmetic mean to get the parameter of the Poisson distribution. When we fit a normal curve the number of degrees of freedom are 3 less than the number of classes, because in this fitting we use the total frequency, mean and standard deviation.

If the data is given in a series of “ n ” numbers then degrees of freedom = $n - 1$.

In the case of Binomial distribution difference = $n - 1$

In the case of Poisson distribution difference = $n - 2$

In the case of Normal distribution difference = $n - 3$.

8.6.2 χ^2 Test As A Test of Goodness of Fit

χ^2 test enables us to ascertain how well the theoretical distributions such as Binomial, Poisson or Normal etc. fit empirical distributions, i.e., distributions obtained from sample data. If the calculated value of χ^2 is less than the table value at a specified level (generally 5%) of significance, the

fit is considered to be good, i.e., the divergence between actual and expected frequencies is attributed to fluctuations of simple sampling. If the calculated value of χ^2 is greater than the table value, the fit is considered to be poor.

8.6.3 Chi-Square Test for Independence

This section explains how to conduct a chi-square test for independence. The test is applied when you have two categorical variables from a single population. It is used to determine whether there is a significant association between the two variables.

For example, in an election survey, voters might be classified by gender (male or female) and voting preference (Democrat, Republican, or Independent). We could use a chi-square test for independence to determine whether gender is related to voting preference.

When to Use Chi-Square Test for Independence

The test procedure described in this section is appropriate when the following conditions are met:

- The sampling method is simple random sampling.
- The variables under study are each categorical.
- If sample data are displayed in a contingency table, the expected frequency count for each cell of the table is at least 5.

This approach consists of four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results.

State the Hypotheses

Suppose that Variable A has r levels, and Variable B has c levels. The null hypothesis states that knowing the level of Variable A does not help you predict the level of Variable B. That is, the variables are independent.

- H_0 : Variable A and Variable B are independent.
- H_a : Variable A and Variable B are not independent.

The alternative hypothesis is that knowing the level of Variable A *can* help you predict the level of Variable B. Support for the alternative hypothesis suggests that the variables are related; but the relationship is not necessarily causal, in the sense that one variable «causes» the other.

Formulate an Analysis Plan

The analysis plan describes how to use sample data to accept or reject the null hypothesis. The plan should specify the following elements.

- **Significance level.** Often, researchers choose significance levels equal to 0.01, 0.05, or 0.10; but any value between 0 and 1 can be used.
- **Test method.** Use the chi-square test for independence to determine whether there is a significant relationship between

two categorical variables.

Analyze Sample Data

Using sample data, find the degrees of freedom, expected frequencies, test statistic, and the P-value associated with the test statistic. The approach described in this section is illustrated in the sample problem at the end of this section.

- **Degrees of freedom.** The degrees of freedom (DF) is equal to:

$$DF = (r - 1) * (c - 1)$$

where r is the number of levels for one categorical variable, and c is the number of levels for the other categorical variable.

- **Expected frequencies.** The expected frequency counts are computed separately for each level of one categorical variable at each level of the other categorical variable. Compute $r * c$ expected frequencies, according to the following formula.

$$E_{r,c} = (n_r * n_c) / n$$

where $E_{r,c}$ is the expected frequency count for level r of Variable A and level c of Variable B, n_r is the total number of sample observations at level r of Variable A, n_c is the total number of sample observations at level c of Variable B, and n is the total sample size.

- **Test statistic.** The test statistic is a chi-square random variable (χ^2) defined by the following equation.

$$\chi^2 = \sum [(O_{r,c} - E_{r,c})^2 / E_{r,c}]$$

where $O_{r,c}$ is the observed frequency count at level r of Variable A and level c of



Variable B, and $E_{r,c}$ is the expected frequency count at level r of Variable A and level c of Variable B.

- **P-value.** The P-value is the probability of observing a sample statistic as extreme as the test statistic. Since the test statistic is a chi-square, use the Chi-Square Distribution Calculator to assess the probability associated with the test statistic. Use the degrees of freedom computed above.

Interpret Results

If the sample findings are unlikely, given the null hypothesis, the researcher rejects the null hypothesis. Typically, this involves comparing the P-value to the significance level, and rejecting the null hypothesis when the P-value is less than the significance level.

Test Your Understanding

Problem

A public opinion poll surveyed a simple random sample of 1000 voters. Respondents were classified by gender (male or female) and by voting preference (Republican, Democrat, or Independent). Results are shown in the contingency table below.

	Voting Preferences			Row total
	Republican	Democrat	Independent	
Male	200	150	50	400
Female	250	300	50	600
Column total	450	450	100	1000

Is there a gender gap? Do the men's voting preferences differ significantly from the women's preferences? Use a 0.05 level of significance.

Solution

The solution to this problem takes four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results. We work through those steps below:

- **State the hypotheses.** The first step is to state the null hypothesis and an alternative hypothesis.

H_0 : Gender and voting preferences are independent.

H_a : Gender and voting preferences are not independent.

- **Formulate an analysis plan.** For this analysis, the significance level is 0.05. Using sample data, we will conduct a chi-square test for independence.
- **Analyze sample data.** Applying the chi-square test for independence to sample data, we compute the degrees of freedom, the expected frequency counts, and the chi-square test statistic. Based on the chi-square statistic and the degrees of freedom, we determine the P-value.

$$DF = (r - 1) * (c - 1) = (2 - 1) * (3 - 1) = 2$$

$$E_{r,c} = (n_r * n_c) / n$$

$$E_{1,1} = (400 * 450) / 1000 = 180000/1000 = 180$$

$$E_{1,2} = (400 * 450) / 1000 = 180000/1000 = 180$$

$$E_{1,3} = (400 * 100) / 1000 = 40000/1000 = 40$$

$$E_{2,1} = (600 * 450) / 1000 = 270000/1000 = 270$$

$$E_{2,2} = (600 * 450) / 1000 = 270000/1000 = 270$$

$$E_{2,3} = (600 * 100) / 1000 = 60000/1000 = 60$$

$$\chi^2 = \sum [(O_{r,c} - E_{r,c})^2 / E_{r,c}]$$

$$\chi^2 = (200 - 180)^2/180 + (150 - 180)^2/180 + (50 - 40)^2/40$$

$$+ (250 - 270)^2/270 + (300 - 270)^2/270 + (50 - 60)^2/60$$

$$\chi^2 = 400/180 + 900/180 + 100/40 + 400/270 + 900/270 + 100/60$$

$$\chi^2 = 2.22 + 5.00 + 2.50 + 1.48 + 3.33 + 1.67 = 16.2$$

where DF is the degrees of freedom, r is the number of levels of gender, c is the number of levels of the voting preference, n_r is the number of observations from level r of gender, n_c is the number of observations from level c of voting preference, n is the number of observations in the sample, $E_{r,c}$ is the expected frequency count when gender is level r and voting preference is level c , and $O_{r,c}$ is the observed frequency count when gender is level r voting preference is level c .

The P-value is the probability that a chi-square statistic having 2 degrees of freedom is more extreme than 16.2.

- We use the Chi-Square Distribution Calculator to find $P(\chi^2 > 16.2) = 0.0003$.
- **Interpret results.** Since the P-value (0.0003) is less than the significance level (0.05), we cannot accept the null hypothesis. Thus, we conclude that there is a relationship between gender and voting preference.





ROLE MODEL

WILLIAM SEALY GOSSET: AN ENGLISH STATISTICIAN

William Sealy Gosset (13 June 1876 – 16 October 1937) was an English statistician. He published under the pen name Student, and developed the Student's t-distribution.

Life and Career

Born in Canterbury, England to Agnes Sealy Vidal and Colonel Frederic Gosset, Gosset attended Winchester College before reading chemistry and mathematics at New College, Oxford. Upon graduating in 1899, he joined the brewery of Arthur Guinness & Son in Dublin, Ireland.

As an employee of Guinness, a progressive agro-chemical business, Gosset applied his statistical knowledge – both in the brewery and on the farm – to the selection of the best yielding varieties of barley. Gosset acquired that knowledge by study, by trial and error, and by spending two terms in 1906–1907 in the biometrical laboratory of Karl Pearson. Gosset and Pearson had a good relationship. Pearson helped Gosset with the mathematics of his papers, including the 1908 papers, but had little appreciation of their importance. The papers addressed the brewer's concern with small samples; biometricians like Pearson, on the other hand, typically had hundreds of observations and saw no urgency in developing small-sample methods.

Another researcher at Guinness had previously published a paper containing trade secrets of the Guinness brewery. To prevent further disclosure of confidential information, Guinness prohibited its employees from publishing any papers regardless of the contained information. However, after pleading with the brewery and explaining that his mathematical and philosophical conclusions were of no possible practical use to competing brewers, he was allowed to publish them, but under a pseudonym ("Student"), to avoid difficulties with the rest of the staff. Thus his most noteworthy achievement is now called Student's, rather than Gosset's, t-distribution.

Gosset had almost all his papers including *The probable error of a mean* published in Pearson's journal *Biometrika* under the pseudonym *Student*. It was, however, not Pearson but Ronald A. Fisher who appreciated the importance of Gosset's small-sample work, after Gosset had written to him to say *I am sending you a copy of Student's Tables as you are the only man that's ever likely to use them!*. Fisher believed that Gosset had effected a "logical revolution". Fisher introduced a new form of Student's statistic,

denoted t , in terms of which Gosset's statistic was $z = \frac{t}{\sqrt{n-1}}$. The t -form was adopted because it fit in with Fisher's theory of degrees of freedom. Fisher was also responsible for applications of the t -distribution to regression analysis.

Although introduced by others, Studentized residuals are named in Student's honour because, like the problem that led to Student's t -distribution, the idea of adjusting for estimated standard deviations is central to that concept.

Gosset's interest in the cultivation of barley led him to speculate that the design of experiments should aim not only at improving the average yield but also at breeding varieties whose yield was insensitive to variation in soil and climate, i.e. robust. This principle only appeared in the later thought of Ronald Fisher, and then in the work of Genichi Taguchi during the 1950s.

In 1935, Gosset left Dublin to take up the position of Head Brewer, in charge of the scientific side of production, at a new Guinness brewery at Park Royal in northwestern London. He died two years later in Beaconsfield, England, of a heart attack.

Gosset was a friend of both Pearson and Fisher, a noteworthy achievement, for each had a massive ego and a loathing for the other.^[4] He was a modest man who once cut short an admirer with the comment that "Fisher would have discovered it all anyway."



SUMMARY

1. A hypothesis test is a statistical test that is used to determine whether there is enough evidence in a sample of data to infer that a certain condition is true for the entire population.
2. A hypothesis test examines two opposing hypotheses about a population: the null hypothesis and the alternative hypothesis. The null hypothesis is the statement being tested.
3. The null hypothesis, denoted by H_0 , is usually the hypothesis that sample observations result purely from chance.
4. A test of a statistical hypothesis, where the region of rejection is on only one side of the sampling distribution, is called a one-tailed test.
5. A test of a statistical hypothesis, where the region of rejection is on both sides of the sampling distribution, is called a two-tailed test.
6. The power of a test is the probability that the test will correctly reject a false null hypothesis. It represents the opposite of a Type II error, also called a β error. Thus, power is described as $1 - \beta$.
7. The theory of sampling is a study of relationship existing between a population and samples drawn from the population. The fundamental object of sampling is to get as much information as possible of the whole universe by examining only a part of it.
8. The statistical constants of the population such as mean, the variance, etc. are known as the parameters. The statistical concepts of the sample from the members of the sample to estimate the parameters of the population from which the sample has been drawn is known as statistic.
9. Sampling errors occur because only a sample of the population is investigated. The particular sample used in this survey is one of a large number of possible samples of the same size that could have been selected using the same sampling method.
10. A t-test's statistical significance indicates whether or not the difference between two groups' averages most likely reflects a "real" difference in the population from which the groups were sampled.

KNOWLEDGE CHECK

1. A statement about a population developed for the purpose of testing is called:
 - a. Hypothesis
 - b. Hypothesis testing
 - c. Level of significance
 - d. Test-statistic
2. Any hypothesis which is tested for the purpose of rejection under the assumption that it is true is called:
 - a. Null hypothesis
 - b. Alternative hypothesis
 - c. Statistical hypothesis
 - d. Composite hypothesis
3. A statement about the value of a population parameter is called:
 - a. Null hypothesis
 - b. Alternative hypothesis
 - c. Simple hypothesis
 - d. Composite hypothesis
4. A hypothesis may be classified as:
 - a. Simple
 - b. Composite
 - c. Null
 - d. All of the above
5. The range of test statistic-Z is:
 - a. 0 to
 - b. -1 to +1
 - c. 0 to ∞
 - d. $-\infty$ to $+\infty$

Check Your Result

1. (a) 2. (c) 3. (c) 4. (d) 5. (d)

REVIEW QUESTIONS

1. Define the uncertainty and errors in hypothesis testing. Also explain the assumptions underlying hypothesis tests with z-scores.
2. A machine is producing bolts a certain fraction of which are defective. A random sample of 400 is taken from a large batch and is found to contain 30 defective bolts. Does this indicate that the proportion of defectives is larger than that claimed by the manufacturer if the manufacturer claims that only 5% of his product are defective? Find 95% confidence limits of the proportion of defective bolts in batch.
3. Two independent sample of sizes 7 and 6 had the following values:

Sample A	28	30	32	33	31	29	34
Sample B	29	30	30	24	27	28	

Examine whether the samples have been drawn from normal populations having the same variance.

4. What do you understand by Chi-square (χ^2) test?
5. Define the Chi-square test for independence.

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